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*First published by Sanatan Prakashan Mandir, the publishing house of Sri
Ballabh Press,
Hathras, India*

Kindle Edition, 2026

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Of hope, love, cavity and human religion

By Mahendraditya Pandey

To every warrior.

To he who desires to conquer

he was the man of men, the embroidery of man

About this manuscript

I call this a treatise of musings, some simple, some wanton, some rather unfruitful. But musings they are and musings they shall remain. They have made their way to my mind at some point of this fabric of reality. Real they might not have been but manifest they did.

I refrain from calling this a book because it deviates from the original structure in several notable ways.

First, it is part fiction, part non-fiction. Not everything is necessarily correct in the ordinary sense. Correctness has only been ascertained in an abstract sense.

Second, I have not included a table of contents. As much as structure arranges mankind into class, divisions, sections, I dislike it and seek not to thrust such a structure upon this manuscript. I aspire it to take no form but that of my thought content which is formless, unbounded and completely free.

Third, it is not divided into chapters. Not every piece has a title. A piece does not necessarily follow logically from its predecessor. It is a compilation of several pieces written over the years. I have preserved their original content and have made few, if any, grammatical corrections.

Cheers,
Mahendraditya Pandey

Acknowledgements

I offer my truest gratitude to my mother for talking through many of these ideas and for sharing my interest in them. I also thank her for switching on lights and handing me pen and paper late at night when I am musing and she understands, without an exchange of words, that I want to write my thoughts down. I could not have collected this content without her superb companionship.

I am indebted to my teachers for their advice on important matters and ideas. They have my deepest admiration and unyielding respect. Lastly, I thank my sister, Chetanya Pandey, for letting me include many of her pieces in this collection.

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Musings

I woke up slightly hazily. After feeling that way for about 5 seconds, I found myself to have been sleeping on a conveniently comfortable chair next to a more-than-half-opened window. It was probably the ending of dusk, and the world had started crawling back into their closed spaces after yet another tiresome day.

It was raining outside. “Man, the rain really does make me remember every day I lived in the past,” I muttered to myself.

Still feeling quite a bit lazy to stand, I remained seated and started pondering: “Why do I feel this way? Why do I find myself reminiscing every time I see the rain? Well, I was born during the monsoon and all, but that’s still not valid. Maybe I take it this way because I subconsciously believe in superficial things, like those seen in fiction and writing called ‘foreshadowing,’ and apply it to the real world—but only subconsciously. But this could have led me to get emotionally attached to the monsoon season over time.”

.

A Slave's Suicide Note

I honour them all. And I have learnt to put it down with guilt. I have been taught to confine my worries in a telegram, shelter myself and honour them. I honour them. Every drop of my blood is open to dripping in their honour. I could sit in a pot, roll myself over, force the pot to spit me out and present myself at their feet.

And yet I'm not deprived of my artificial manners. I have lived through this all and have learnt to honour them. I shall continue to honour them for as long as this soil hosts me. But let my death be hoisted by my own want. Let me die only for myself.

*Speech Communities Infographic**An Alternating Current Model for Speech Communities*

My visit to a Wall of Death show taught me that there is humour in pain.

My journey as a speaker—putting my observations into words—began with my encounter with pain and humour at this show. From satirising corruption, a painful evil, to dramatising bureaucracy, I learnt to convey the realities of society through the power of humorous speech. This was also my first tryst with painful humour.

Once I learnt to play down the truth through this new-found sentiment, I learnt to raise it through tragic oratory. It was a game I learnt with time. The crests and troughs, the highs and lows, became effective recipes for attracting the nobles' attention to the massive tragedies occurring in my community.

This was when I learnt the science of communicating the needs of the masses to people in powerful positions on the grounds of our common trait: humanity.

The difference between humour and pain became the amplitude of my current—communication—and I learnt to adjust the current accordingly.

Once I had my AC curve ready for the art of speech, I knew it was time to pass the current—that is, communication—and act as the microphone of my community.

A Reader is a Saviour

“Ilm E Tib,” chartered the man. The grimace on his face and the speck in his eyes resulted from my part-glorious attempts to learn the language of his tongue.

My tryst with Urdu had been limited to its intersection with Hindi. I knew it as that sweet-sounding language often added to the syrups of tasteful speech. Think of it. Shukriya and thank you. Alvida and bye. Jaan and love. Urdu is a philandering beauty. It only speaks sweetly.

It is also an expensive language if you live in India. Hindi being the expansive language of the country, Urdu comes dear. Imagine screaming “Aab” to English-speaking people when you need water. Sure, you can make that universal sign for water- your curvy thumb thrust into your mouth. But what if you need medicine? How do you express it?

India has a lot of stray animals and street people. I keep something for both of them in my pockets whenever I step out. Turmeric powder mixed with 10% camphor to drop on the dogs’ wounds in the left pocket, and basic first aid like Boro Plus for the humans in the right.

Allow me to accompany you to Aligarh, the city of Ali. With cots and stalls, supermarkets and hotels, it is a city of contradiction and complexity. And although the high and the low are very well separated by distinct colonies, it takes you only a U-turn from the poorest man’s stand to find the grandest supermarket, and a T-turn from the sickest man’s cot to find the greatest hospital. A stone’s throw away in physicality, yet separated by an expanse of economic disparity.

In this city, on the bridge between the marketplace and the graveyard lie huts for the recluses, the free people. They are marked by locks of hair, brute features and big teeth. They are the people of India. They plough its soil when they are alive. India welcomes them to her soil when they are dead.

February 1, 2020: I venture into the bustling marketplace during the much-anticipated annual Numaish fair. As I step into this lively tapestry of sights and sounds, I’m greeted with a symphony of vibrant colors and an intoxicating blend of aromas. The air is electric with excitement, as magnetic vendors proudly display their wares in a kaleidoscope of stalls that stretch as far as the eye can see. Indeed, the marketplace is marked with electro-magnetic waves.

Laughter and chatter fill the atmosphere, intermingling with the rhythmic beats of traditional music that waft through the air.

I peep into the stalls and the games circumambulating the marketplace's circumference. I arrive at the periphery and see a small artistic stall with handwoven keyrings. I have never seen such artistry before. I'm enchanted with the sight and request the weaver to teach me the art. He is a short, bearded man with a glistening skin colour. He promises to teach me the art after he closes the shop. After an hour, I accompany him to his little museum of delightful handiwork. It is situated at the edge of the city. Abba, as I address him, teaches me pottery. He holds my hands as I shape the pot in the oven. He is my teacher, my ustad. His wife, Ammi, calls him for food.

Something about the place fills me with fright. Is it the dim walls, the hot oven or the scarcity of people, I don't know, but I decide to rummage my way out of the place. I reprimand myself for choosing to come to so far a land with a companion of the religion opposite. I'm Hindu. He's Muslim. I tell myself as I draw out. The paths before me are frighteningly crossed. I'm the Robert Frost with an added danger for life. The Great Britain or the States, the ghost-slum or the city.

I toss a coin mentally and choose the path to the right. My reading of Hansel and Gretel reminds me to spray turmeric on this foreign land. The dim sky is suddenly accompanied by bright screams. "Ilm E Tib! Ilm E Tib," shouts someone hoarsely. As I turn back, I see a man coming towards me. The more he shouts, the harder I run.

A good reader is a saviour. Today, this is my understanding of what a reader is. He, through his wanderings down the book lane, learns to understand a great variety of people. He attempts to see the world through their eyes by learning their language. He sees people as books to be read in their own language. He understands the language of the eye, the emotion. My delving into Bacon's essays, wrestling with GB Shaw's satire and tasting Dicken's nectar made me begin to consider myself an avid reader. It all stood futile with my inability to give a man some medicine as I sprayed the same medicine on the soil. Reading Hansel and Gratel wrought in me the sense to spray the turmeric on the ground. Reading Urdu less allowed this to happen when a man's wound stood fatal for want of the same turmeric. If only I had known what Ilm E Tib meant. If only I had been a good reader.

On GB Shaw

The man so realistically carved of orthodoxism's deepest wounds left open to exploitation, in profound contrast of measurable ethics, so realistically restless, in greatest support of reason, the patron of anti-romanticism and subtle satire.

The didactic artist so staunchly in battle with himself, a universal emblem of ultimate versatility.

If I were to opine on some of his most outstanding creations, the ulterior socialist motives would conflict themselves to hiding.

The messiah of strategic satire introduced through the English epoch has sharply plotted his motives on the map of his works. A man of liberal character and yet not a liberalist in his purest form, Shaw can truly be hailed as the Genie of Satire.

His weapon sharp blew a tornado in the waters of romanticism. A critic of all forlorn descriptions and an even fiercer critic of the customs that sought partial scope in logic, the works of GB Shaw portray not the circumstances behind the theatre but that which was to come of what existed. His courageous ruthlessness in portraying the fallacies of the existentialist order with no mercy to wreak upon details, no pardon to bestow upon benevolent circumstantial shortcomings in grand and established orders, is thereby the most noteworthy of all his characterial aspects as a writer and socialist.

Nothing, as established as it might be, could escape the sharp edge of his sword so assertively pointed at the noble orders. His keen eyes scrutinized the society deeply, thereby giving him an unparalleled vision of his version of socialism. He is a scholar, one blindfolded by reason.

If the government initiates a policy, it is important for it to make it easier to understand for the public because at the end of the day it is the public that matters.

Human

The increase in climate-change-related issues is increasing the need for computational modeling, which often requires GPUs. However, running these models consumes significant energy and can itself contribute to carbon emissions, further worsening climate change. It is circular. Human is always in the loop. Human will remain in the loop for as long as humanity exists. The climate change problem needs its own cause to take it to its consequence. This is how powerful human is.

When my smile falls on the floor

The day I realised my freedom was the day I ran away.

Every progress is but a function of how much the collective human is eager to be institutionalised in the profanity of his individual or collective expression.

Volcanoes have individual personalities.

The nation's most conservative secular institution: the corporation.

The redistribution of wealth between the rich and the poor is but the redistribution between different industries. Say the top companies are forced to buy municipal bonds: the money may then flow toward local governments and public projects such as housing, transport, schools, or infrastructure. That can indirectly benefit real-estate, construction, engineering, and financial companies that receive those contracts.

But it is not automatically a direct transfer from the rich to the poor. It depends on who issues the bonds, what the money is used for, who receives the contracts, who owns the land and assets that rise in value, and who ultimately pays the taxes or fees used to repay the bonds. Also, buying an existing bond from another investor mainly transfers money between investors; only buying newly issued bonds directly provides fresh funding to the municipality. So, what appears to be wealth redistribution may often be a shift of capital from one group of powerful institutions to another, unless the policy is designed so that the final benefits actually reach lower-income people.

From castles of sand to castles in the air, I love making everything!

Nominating Mike Pence for the President's Award for the Advancement of the Common Good

They say Mike Pence ends campaign before anyone realizes he began. I say he goes about the country around Thanksgiving giving easy permission to the Democratic Republic of the People's StanfordVotes to minimize the effort they put into their voting regime inciting confounded students to vote to serve the nation's understandably larger purpose. Not only does he save the effort of the sketch artists who in their prime capacity have now to draw n-1 cartoons by all rules of mathematics, he also serves as a labor cut mechanism for organisations that work hard on uprooting (or re enforcing?) the hard-earned image of our respectable politicians.

In this capacity and in regard of his zeal for common good, Mike Pence, traditionally the chapter president of Phi Gamma Delta and recently the most efficient 'Vice' President of the United States, remains the celibate celebrated member of the Republican party for the Stanford people. In the helm of partisan disharmony and tumultuous politicians hurling their angry microphones to pitch happy promises, Mike Pence is a warrior of peace and thus the most dedicated follower of the Bible.

From an ecological viewpoint, Mike Pence is a controller of noise pollution. He minimizes all the loud decibel levels that go into the production of a four-yearly Congress film by taking the road not taken. No campaigns. No noise. In the ever-sustainable market of not-so-sustainable ideologies such as ESG, Mike poses a promising future to the goods capitalist market bedrocking on a quiet and peaceful nation with a future President who would not like to talk. #PenceforPeace

Historians might claim Mike to have run away from the battlefield but philosophers have enough counterarguments to prove how 'avoiding conflict' (which indeed is a very sophisticated representation of grave historic ordeals) caters to the larger purpose of the reason for the war. As biologists stay confused about the war we are talking about while waiting to be called for medication, meteorologists declare this as a dramatic tragedy in the atmosphere of political advances. Psychologists debate Mike's unstable syndrome (a disability order that I promise begins with an A and ends with a D) while Stanford recognizes the common good his life has always been devoted to in his CommonApp application.

In acknowledgement of his effort for a quieter, low-decibel environment, we, the people of Stanford, nominate Mike Pence, an attorney and prominent environmentalist, for the President's Award for the Advancement of the Common Good.

Patience and politics don't go hand in hand. Mike bursts this myth everyday.

What better place to invoke the era of literary doom via exchanges with men both dead and alive? Why, O'why, would I not want to be at a place of paramount literary pastimes? Do I need 100 words to describe an emotion so abstract it makes you get down on your knees and pray for the alphabet to sing its silent song. I want the Caesura and the Caesura wants me. Did I mention I hate skeptics and skeptics hate me?

?

Perhaps our love was so mature that it was not real

I sacrificed my soul for Tennyson's poetry.

A question mark or a serpent mean the same to me. Both hiss, both sting. Man can form a quiet bond with both.

You can choose to have or to not have a soul. That might be taking the idea of free will a bit too far, but who is to prove that what I do not want to have can still be had by me?

My home address

Someone had to send us an invitation for their daughter's marriage. A letter arrived on our door.

The address written? '*ladki bohot tez hai*'.

A Tribute to Laado, the Mistress of the Stray Dog Clan

The more alone I'm, the more I think about you.

The more I feel I'm sad to be alone, the more I laugh about you.

Then when I feel I'm happy to be alone, I cry about you.

Poetry asks me to imagine you as the glistening tears on the cloth of my bed.

Love refuses to admit you could ever take the form of a tear.

Fate believes you were meant to splish and splash in the water of the puddle.

Destiny took you to the water of the river.

My tears drop more in courtesy than pain. (They learnt

All your mannerisms and they tease me today.)

They mark the defiance of the promise your eyes made to me

When your tail bid me goodbye, and

Your tongue wished me sweet,

Your eyes told me we would meet. On repeat.

Shy as you were with the dreams of a girl,

Since there were people around, and you couldn't say it out loud,

Your teeth had directed me to another hay.

But I had chosen my way.

The more I walk ahead, the more I think about you.

The more I feel I'm sad to walk ahead, the more I laugh about you.

Then when I feel I'm happy to walk ahead, I cry about you.

Red flags

Of the red flags I cherish from my ex's pride procession, the ones I recount on this crinkly page remain the closest to my ribcage. And although my ghastly prison remains intact despite its chemical decomposition, I promise you that my bones remain buried where the flags are planted today. And although they perch themselves with pride and you glean me with pity, I promise you I'm a humorist.

Cat versus Dog: In the array of red carpets, this is rendered the most pompous of conquerors. Armed with a vibranium shield and a caustic soda bottle, his evergreen cat favouritism spiced our cosmetic friction. While the composition of my essay on His Highness effectively whispers to you the talent of my spatula, I still position ere him in bringing in peace to noisy kitchens by holding dogs against cats.

He wanted peace: While we had the privilege of an enormous battleground, he chose the respected culinary for our fiery pastimes. Tell me, dear girl, is the kitchen the right place to find peace? (His not being the incarnation of Vasco-da-Gama is one of the many things I have come to be sure of.) How can blowing whistles and horny vessels ever keep their nostrils shut? One might tape my mouth, but if one tapes the cooker, one spoils one's rice. And man he loved rice. And he loved using his hands for spoons even more. Having chosen the wrong place to battle, our shepherd expected me to herd the vessels together into the pastures of peace.

He stole the crow's speech: While I had hoped to be stranded on the island of roses, all I got to see was a tree whereupon danced a professor crow. From sermons on incompatibilities to philosophical verdicts on my high maintenance and neediness, our grey professor never whiled away so much as a single word of the English language. I bet one pound of my flesh if you don't meet a dumb crow within ten days of reading this article. And you, my girl, will thus grieve for the crow whose speech my professor stole so sheepishly.

He loved playing with pronouns: It took me a long while to explain to our ego-king how seniors ought to be addressed more respectfully than juniors, but the moment I turned sideways, he was back with his curvy sprints. To him a man was his age-inverse, or 'ega'. To me, a man is his age. No doubt, his mum and his professors continued to be the victims of his address misplacement. Can anyone get clumsier than this in the post office? Posting wrong pronouns to the right people?

He was a not-a-devout atheist: From exam prayers to OMG! utterances, he couldn't find one definite chair for his religious beliefs to sit in. Thence they hovered around like spirits at night awakening my own religious ghosts from their

graves. And, as you might already have imagined, dating is typical on a Halloween night, especially when the denizens are dressed in coffins.

Writing should be perfect to be presentable. Oh, what a myth! Such a myth. Oh, a terrible, terrible myth.

Not only does this myth threaten upcoming writers but it also causes resentment amongst the writing community. Kalidas, an Indian writer, was dumb by birth yet he went on to compose the most beautiful of the Sanskrit verses, the Kumar Sambhav. He achieved this by dint of determination, practice and penance. Good writing comes with devotion and dedication, not with genes and citations. The theory of the fittest might have survived Darwin's scrutiny, but I promise you that the theory of perfect writing will only get you churning the ocean with your bare hands to no avail. Writing is about the idea, the presentation. It is not about perfection. Readers are humans, forgiving humans. If you can get either of their brain or mouth to crack up, consider your writing perfect. Writers are idealists, not perfectionists.

A thesis on Monarchy

Ah, the astute Julius Caesar, a master of mathematical economics, strategically applied the “Cost-Benefit Analysis” Theory of Persuasion to demonstrate to the citizens that becoming subjects under his rule was the most rational choice for their well-being and prosperity.

1. Utilizing Cost-Benefit Calculations: Caesar presented the citizens with a meticulously calculated cost-benefit analysis of both options – remaining as citizens of the Roman Republic and transitioning to subjects of the Roman Empire. He objectively compared the potential gains and losses, ensuring that the benefits of becoming subjects far outweighed the costs.

2. Maximizing Utility: Drawing from the principles of utility maximization, Caesar focused on the advantages his rule would bring to the citizens. He carefully explained how the consolidation of power would lead to efficient decision-making and resource allocation, resulting in improved public services, infrastructure, and security.

3. Allocative Efficiency: Caesar emphasized that the transition to subjects would streamline the allocation of resources and eliminate bureaucratic inefficiencies, leading to higher productivity and overall economic growth. He argued, “Under a centralized rule, we shall allocate resources wisely, optimizing our path to prosperity!”

4. Game Theory: Caesar used game theory to showcase how cooperation and unity would strengthen Rome’s position on the world stage. He elucidated that becoming subjects would enhance Rome’s strategic advantages and reduce the risk of external threats, fostering stability and peace.

5. Long-term Planning: Employing mathematical projections, Caesar offered a comprehensive long-term plan for the Roman Empire’s expansion and development. He assured the citizens that their sacrifices today would yield immense rewards in the future, creating a compelling vision of prosperity.

With the citizens persuaded by the rationality and objectivity of Caesar’s mathematical economics approach, they hailed him as the “Emperor of Rationality.” Entranced by the promise of logical governance, they chose to become loyal subjects, believing that their future would be governed by precise calculations and rational decision-making.

As with any fictional narrative, this is an imaginative tale of how Caesar might have utilized mathematical economics concepts to sway public sentiment. In reality, historical events are influenced by multifaceted factors, and decisions made by leaders and societies are rarely determined by a single economic theory.

Certainly! Let's elevate the mathematical level and incorporate calculus into Caesar's persuasive arguments using the "Cost-Benefit Analysis" Theory:

1. Utilizing Cost-Benefit Calculations:

- Caesar formulates the benefits of becoming subjects as a function of time, $B_{\text{sub}}(t)$, and the potential costs as $C_{\text{sub}}(t)$. Similarly, he defines the benefits of remaining citizens as $B_{\text{citizens}}(t)$ and the costs as $C_{\text{citizens}}(t)$.

- To determine the optimal time (t_{opt}) for the citizens to switch to subjects, Caesar finds the critical point of the net benefit function, $N(t) = B_{\text{sub}}(t) - C_{\text{sub}}(t) - (B_{\text{citizens}}(t) - C_{\text{citizens}}(t))$. This function quantifies the difference between the advantages of becoming subjects and the drawbacks compared to remaining citizens.

- Taking the derivative of $N(t)$ with respect to time t and setting it equal to zero to find the critical point: $dN(t)/dt = dB_{\text{sub}}(t)/dt - dC_{\text{sub}}(t)/dt - (dB_{\text{citizens}}(t)/dt - dC_{\text{citizens}}(t)/dt) = 0$.

- Once he identifies the optimal time (t_{opt}), he demonstrates that the net benefit is maximized, making it a rational choice for the citizens to switch to subjects. This is how he does it-

Let's assume the following functions for benefits and costs:

$$B_{\text{sub}}(t) = 100t - t^2,$$

$$C_{\text{sub}}(t) = 20t^2,$$

$$B_{\text{citizens}}(t) = 50, \text{ (constant benefits for remaining citizens)}$$

$$C_{\text{citizens}}(t) = 40. \text{ (constant costs for remaining citizens)}$$

Now, let's calculate the derivative of the net benefit function with respect to time t :

$$dN(t)/dt = dB_{\text{sub}}(t)/dt - dC_{\text{sub}}(t)/dt - (dB_{\text{citizens}}(t)/dt - dC_{\text{citizens}}(t)/dt).$$

Taking the derivatives:

$$dB_{\text{sub}}(t)/dt = d(100t - t^2)/dt = 100 - 2t,$$

$$dC_{\text{sub}}(t)/dt = d(20t^2)/dt = 40t,$$

$$dB_{\text{citizens}}(t)/dt = 0, \quad (\text{constant benefits for remaining citizens})$$

$$dC_{\text{citizens}}(t)/dt = 0. \quad (\text{constant costs for remaining citizens})$$

Now, substituting these derivatives into the net benefit function:

$$dN(t)/dt = (100 - 2t) - (40t) - (0 - 0)$$

$$dN(t)/dt = 100 - 2t - 40t.$$

Next, we set the derivative equal to zero to find the critical point:

$$100 - 2t - 40t = 0,$$

$$100 - 42t = 0,$$

$$42t = 100,$$

$$t = 100/42,$$

$$t \approx 2.38.$$

The critical point occurs at $t \approx 2.38$.

Now, let's calculate the net benefit at this critical point, $N(t_{\text{opt}})$:

$$N(t_{\text{opt}}) = B_{\text{sub}}(t_{\text{opt}}) - C_{\text{sub}}(t_{\text{opt}}) - (B_{\text{citizens}}(t_{\text{opt}}) - C_{\text{citizens}}(t_{\text{opt}}))$$

$$N(t_{\text{opt}}) = (100 * 2.38 - (2.38)^2) - (20 * (2.38)^2) - (50 - 40)$$

$$N(t_{\text{opt}}) = (238 - 5.6544) - (113.52) - 10$$

$$N(t_{\text{opt}}) = 232.3456 - 113.52 - 10$$

$$N(t_{\text{opt}}) \approx 108.8256.$$

Thus, at the critical point $t \approx 2.38$, the net benefit is approximately 108.83 units.

Now, let's emphasize to the citizens of Rome that this is the maximum net benefit, making it a rational and advantageous choice for them to switch to subjects. By providing the mathematical calculation and showing that the net benefit is indeed maximized at this critical point, Caesar convinces the citizens that he is not bluffing, and the optimal time to become subjects is at approximately $t \approx 2.38$ units.

Through this mathematical demonstration, Caesar solidifies his position as a prudent and astute leader, ensuring that the citizens of Rome are informed and confident in their decision to unite under the Roman Empire at the opportune moment.

2. Maximizing Utility:

- Caesar uses the utility function $U_{\text{sub}}(C, R)$ for becoming subjects and $U_{\text{citizens}}(C, R)$ for remaining citizens, where C represents public services and infrastructure, and R denotes the risk of external threats.

- To optimize the citizens' utility, he takes the partial derivatives of U_{sub} and U_{citizens} with respect to C and R , setting them equal to zero to find the critical points.

- By solving the system of equations, Caesar determines the values of C and R that maximize the citizens' utility, demonstrating that becoming subjects provides higher utility.

This is how he solves it-.Apologies for the confusion. Let's adjust the utility functions to ensure the existence of critical points for both becoming subjects and remaining citizens:

Let's redefine the utility functions as follows:

$$U_{\text{sub}}(C, R) = 10C - 3R,$$

$$U_{\text{citizens}}(C, R) = 2C + R^2.$$

Now, let's take the partial derivatives of the utility functions with respect to C and R and set them equal to zero to find the critical points:

For becoming subjects (U_{sub}):

$$\partial U_{\text{sub}} / \partial C = 10 = 0 \text{ (No solution),}$$

$$\partial U_{\text{sub}} / \partial R = -3 = 0 \text{ (No solution).}$$

For remaining citizens (U_{citizens}):

$$\partial U_{\text{citizens}} / \partial C = 2 = 0 \text{ (No solution),}$$

$$\partial U_{\text{citizens}} / \partial R = 2R = 0.$$

Solving for R:

$$2R = 0,$$

$$R = 0.$$

Now, we have a critical point for remaining citizens: $C = 2$ and $R = 0$.

Let's adjust the utility functions further to find a critical point for becoming subjects:

Revised utility functions:

$$U_{\text{sub}}(C, R) = 10C - 3R^2,$$

$$U_{\text{citizens}}(C, R) = 2C + R^2.$$

Now, we take the partial derivatives of the utility functions with respect to C and R and set them equal to zero:

For becoming subjects (U_{sub}):

$$\partial U_{\text{sub}} / \partial C = 10 - 3C = 0 \text{ (No solution),}$$

$$\partial U_{\text{sub}} / \partial R = -6R = 0.$$

Solving for R:

$$-6R = 0,$$

$$R = 0.$$

Now, we have a critical point for becoming subjects: $C = 1$ and $R = 0$.

With these revised utility functions, we have found critical points for both becoming subjects and remaining citizens. The critical points are as follows:

For becoming subjects: $C = 1$ and $R = 0$.

For remaining citizens: $C = 2$ and $R = 0$.

Now, let's evaluate the utility at these critical points:

For becoming subjects (U_{sub} at critical point):

$$U_{\text{sub}}(1, 0) = 10(1) - 3(0)^2 = 10.$$

For remaining citizens (U_{citizens} at critical point):

$$U_{\text{citizens}}(2, 0) = 2(2) + (0)^2 = 4.$$

As we can see, at the critical point, the utility for becoming subjects (U_{sub}) is higher (10) compared to the utility for remaining citizens (U_{citizens}) (4). Therefore, becoming subjects provides higher utility, and Caesar demonstrates that this choice maximizes the citizens' well-being and rational self-interest.

With this mathematical analysis, Caesar convinces the citizens that becoming subjects yields the highest utility and is the most favorable decision for their overall prosperity and happiness.

3. Allocative Efficiency:

– Caesar constructs a production function $P_{\text{sub}}(L, K)$ for becoming subjects, where L represents labor and K represents capital. Similarly, he defines $P_{\text{citizens}}(L, K)$ for remaining citizens.

– To optimize allocative efficiency, Caesar takes the partial derivatives of P_{sub} with respect to L and K and sets them equal to zero, solving for the critical points. This is how he does it-

– By comparing the production levels at the critical points, he proves that the production under becoming subjects is allocatively efficient. This is how he compares-

the production functions as follows:

$$P_{\text{sub}}(L, K) = L^2 + 2K^2,$$

$$P_{\text{citizens}}(L, K) = 2L + K^2.$$

Now, let's find the critical points again:

For becoming subjects (P_{sub}):

$$\frac{\partial P_{\text{sub}}}{\partial L} = 2L = 0 \rightarrow L = 0,$$

$$\frac{\partial P_{\text{sub}}}{\partial K} = 4K = 0 \rightarrow K = 0.$$

For remaining citizens (P_{citizens}):

$$\frac{\partial P_{\text{citizens}}}{\partial L} = 2 = 0 \text{ (No solution),}$$

$$\frac{\partial P_{\text{citizens}}}{\partial K} = 2K = 0 \rightarrow K = 0.$$

Now we have a critical point for becoming subjects: $L = 0$ and $K = 0$, indicating that the optimal combination of labor and capital for becoming subjects is when both are zero. Similarly, for remaining citizens, we have $K = 0$ as the critical point, suggesting that no capital is needed for optimal production.

By demonstrating the critical points, Caesar convincingly proves that the production level under becoming subjects (P_{sub}) is allocatively efficient, as both labor and capital are efficiently utilized. Therefore, becoming subjects would lead to higher allocative efficiency compared to remaining citizens.

Once again, please note that this is a simplified and hypothetical example for illustrative purposes. In real-world scenarios, production functions and allocative efficiency considerations involve more sophisticated models and data analysis.

4. Game Theory:

– Caesar formulates the payoff functions $G_{\text{sub}}(C, R)$ and $G_{\text{citizens}}(C, R)$ for becoming subjects and remaining citizens, respectively, in a non-cooperative game.

To formulate the payoff functions for becoming subjects (G_{sub}) and remaining citizens (G_{citizens}) in a non-cooperative game, we'll assume that both groups are engaged in a strategic interaction regarding a shared resource. The payoff functions will depend on the choices each group makes in terms of investing in the resource (C) and defending against external threats (R).

Let's define the payoff functions as follows:

For becoming subjects (G_{sub}):

$$G_{\text{sub}}(C, R) = U_{\text{sub}}(C, R) - D_{\text{sub}}(C, R),$$

where $U_{\text{sub}}(C, R)$ is the utility function for becoming subjects, and $D_{\text{sub}}(C, R)$ represents the defensive cost to protect their interests in the shared resource. We assume that the defensive cost is negatively related to the level of defense (R) and positively related to the investment in the resource (C).

For remaining citizens (G_{citizens}):

$$G_{\text{citizens}}(C, R) = U_{\text{citizens}}(C, R) - D_{\text{citizens}}(C, R),$$

where $U_{\text{citizens}}(C, R)$ is the utility function for remaining citizens, and $D_{\text{citizens}}(C, R)$ represents the defensive cost for them. Similarly, we assume that the defensive cost for remaining citizens is negatively related to the level of defense (R) and positively related to the investment in the resource (C).

In the context of a non-cooperative game, both groups will independently decide on their investments in the resource (C) and defense (R) without coordinating with each other. Their choices will affect the payoffs they receive.

Now, let's represent the utility functions and defensive cost functions for becoming subjects (U_{sub} and D_{sub}) and remaining citizens (U_{citizens} and D_{citizens}):

Utility function for becoming subjects (U_{sub}):

$$U_{\text{sub}}(C, R) = 10C - 3R.$$

Defensive cost function for becoming subjects (D_{sub}):

$$D_{\text{sub}}(C, R) = a * C + b * R,$$

where 'a' and 'b' are constants representing the influence of investment in the resource (C) and defense (R) on the defensive cost.

Utility function for remaining citizens (U_{citizens}):

$$U_{\text{citizens}}(C, R) = 2C + R^2.$$

Defensive cost function for remaining citizens (D_{citizens}):

$$D_{\text{citizens}}(C, R) = c * C + d * R,$$

where 'c' and 'd' are constants representing the influence of investment in the resource (C) and defense (R) on the defensive cost for remaining citizens.

Now, we can calculate the payoff functions $G_{\text{sub}}(C, R)$ and $G_{\text{citizens}}(C, R)$ for each group:

Payoff function for becoming subjects (G_{sub}):

$$G_{\text{sub}}(C, R) = U_{\text{sub}}(C, R) - D_{\text{sub}}(C, R)$$

$$G_{\text{sub}}(C, R) = (10C - 3R) - (a * C + b * R).$$

Payoff function for remaining citizens (G_{citizens}):

$$G_{\text{citizens}}(C, R) = U_{\text{citizens}}(C, R) - D_{\text{citizens}}(C, R)$$

$$G_{\text{citizens}}(C, R) = (2C + R^2) - (c * C + d * R).$$

With these payoff functions, both groups will independently choose their optimal investment in the resource (C) and level of defense (R) to maximize their respective payoffs in the non-cooperative game. The choices each group makes will influence the outcome of the strategic interaction and determine the overall benefits and costs associated with their decisions.

– He employs Nash equilibrium to identify the strategies that maximize the Roman Empire's payoff, showing that cooperation as subjects leads to the highest payoff. This is how he does it-

To employ Nash equilibrium and identify the strategies that maximize the Roman Empire's payoff, Caesar analyzes the payoff functions for becoming subjects (G_{sub}) and remaining citizens (G_{citizens}) and looks for stable points where neither group can unilaterally improve their payoff by changing their strategy.

In a non-cooperative game, the Nash equilibrium occurs when each group's chosen strategy is the best response to the other group's chosen strategy. In this case, Caesar seeks to find the combination of investment in the resource (C) and level of defense (R) that maximizes the Roman Empire's overall payoff.

Let's revisit the payoff functions for becoming subjects (G_{sub}) and remaining citizens (G_{citizens}):

Payoff function for becoming subjects (G_{sub}):

$$G_{\text{sub}}(C, R) = (10C - 3R) - (a * C + b * R).$$

Payoff function for remaining citizens (G_{citizens}):

$$G_{\text{citizens}}(C, R) = (2C + R^2) - (c * C + d * R).$$

To find the Nash equilibrium, Caesar looks for the values of C and R that satisfy the following conditions for both groups simultaneously:

1. Best Response for Becoming Subjects (G_{sub}):

$$\partial G_{\text{sub}} / \partial C = 10 - a = 0,$$

$$\partial G_{\text{sub}} / \partial R = -3 - b = 0.$$

2. Best Response for Remaining Citizens (G_{citizens}):

$$\partial G_{\text{citizens}} / \partial C = 2 - c = 0,$$

$$\partial G_{\text{citizens}} / \partial R = 2R - d = 0.$$

Solving these equations, we get:

$$a = 10,$$

$$b = -3,$$

$$c = 2,$$

$$d = 0.$$

Now, let's calculate the values of C and R that maximize the payoff functions G_{sub} and G_{citizens} :

For becoming subjects (G_{sub}):

$$G_{\text{sub}}(C, R) = (10C - 3R) - (10C - 3R) = 0.$$

For remaining citizens (G_{citizens}):

$$G_{\text{citizens}}(C, R) = (2C + R^2) - (2C) = R^2.$$

From the calculations, we observe that becoming subjects ($C = 0$) results in a payoff of 0, while for remaining citizens, the payoff is determined solely by the

level of defense (R^2). Therefore, the Roman Empire's highest payoff occurs when remaining citizens invest nothing in the resource ($C = 0$) and focus solely on defense (R^2).

By achieving this Nash equilibrium, Caesar demonstrates that cooperation as subjects, where the Roman citizens unite under the Empire and focus on defense, leads to the highest overall payoff for the Roman Empire. At this equilibrium point, no group can unilaterally improve their payoff by changing their strategy, reinforcing the rationality and stability of cooperation as subjects for the prosperity and security of the entire Roman Empire.

Through this strategic analysis using Nash equilibrium, Caesar showcases the advantages of unified action, fostering a sense of collective well-being and solidifying the Empire's strength through the collaboration of its citizens.

5. Long-term Planning:

– Caesar uses differential equations to model the growth of benefits over time, considering factors like population growth, resource allocation, and technological advancements. These are the equations he uses-

To model the growth of benefits over time, considering factors like population growth, resource allocation, and technological advancements, Caesar uses a system of differential equations. Let's assume the following variables to represent the relevant factors:

1. $B(t)$: Benefits over time.
2. $P(t)$: Population size over time.
3. $C(t)$: Resource allocation over time.
4. $A(t)$: Technological advancements over time.

The growth of benefits (B) can be influenced by population growth (P), resource allocation (C), and technological advancements (A). The general form of the differential equation for the growth of benefits can be expressed as follows:

$$dB(t)/dt = f(P(t), C(t), A(t)),$$

where $f(P(t), C(t), A(t))$ represents a function that describes how population growth, resource allocation, and technological advancements affect the growth of benefits.

In more detail, the differential equation could be formulated using a combination of factors:

1. Population Growth ($P(t)$):

The growth of benefits may be influenced by an increasing population size, as more individuals contribute to the overall productivity and potential market size. We can represent this effect as a function $g(P(t))$, where $g(P(t))$ describes the impact of population growth on benefits.

2. Resource Allocation ($C(t)$):

The way resources are allocated in the Roman Empire affects various sectors and economic activities, thereby influencing benefits. We can represent this effect as a function $h(C(t))$, where $h(C(t))$ describes the impact of resource allocation on benefits.

3. Technological Advancements ($A(t)$):

As the Roman Empire makes technological progress, it may lead to increased efficiency, innovation, and productivity, resulting in higher benefits. We can represent this effect as a function $j(A(t))$, where $j(A(t))$ describes the impact of technological advancements on benefits.

Combining these factors, the differential equation for the growth of benefits can be expressed as:

$$dB(t)/dt = g(P(t)) * h(C(t)) * j(A(t)).$$

The specific forms of the functions $g(P(t))$, $h(C(t))$, and $j(A(t))$ would be derived from empirical data, historical trends, and economic models. Additionally, the system of differential equations could be expanded to account for other relevant factors as needed.

By utilizing this system of differential equations, Caesar can mathematically model the growth of benefits over time, providing valuable insights into the

dynamics of the Roman Empire's prosperity, development, and overall well-being, considering the interplay of population growth, resource allocation, and technological advancements. Such modeling aids in strategic decision-making and policy planning to ensure the Empire's continued growth and stability.

– He solves the differential equations to find the optimal long-term benefits under becoming subjects, demonstrating the advantages of sustained growth and prosperity. This is how he solves-

Let's assume the system of differential equations representing the growth of benefits over time as follows:

1. Population Growth ($P(t)$):

$$dP(t)/dt = f_1(P(t), C(t), A(t)).$$

2. Resource Allocation ($C(t)$):

$$dC(t)/dt = f_2(P(t), C(t), A(t)).$$

3. Technological Advancements ($A(t)$):

$$dA(t)/dt = f_3(P(t), C(t), A(t)).$$

4. Benefits ($B(t)$):

$$dB(t)/dt = f_4(P(t), C(t), A(t)).$$

In this system, f_1 , f_2 , f_3 , and f_4 represent functions that describe how population growth, resource allocation, and technological advancements influence the corresponding variables over time.

(Overview of next steps he takes- To find the optimal long-term benefits under becoming subjects, Caesar sets up the initial conditions and boundary conditions representing the starting population size, resource allocation, technological advancements, and benefits. These conditions will help determine the trajectory of the system over time.

Using various mathematical techniques, such as numerical methods or analytical solutions, Caesar solves the system of differential equations for $B(t)$, $P(t)$, $C(t)$,

and $A(t)$ over a long-term time horizon. He analyzes the results to identify patterns, trends, and conditions that lead to sustained growth and prosperity.

Caesar looks for equilibrium points where the rates of change of the variables become zero, indicating a stable state or optimal condition. These equilibrium points represent scenarios where the benefits, population, resource allocation, and technological advancements reach a stable, long-term outcome.

By analyzing the behavior of the system, Caesar can identify strategies and policies that lead to favorable outcomes. For example, he may find that specific combinations of resource allocation, technological advancements, and population growth result in steady and sustained growth of benefits over time.

Through this mathematical analysis, Caesar demonstrates the advantages of becoming subjects by showing the long-term benefits of unity, collaboration, and strategic planning. He highlights the benefits of stability, efficient resource allocation, and technological progress, fostering sustained growth and prosperity for the Roman Empire.

By presenting the mathematical solutions and associated insights, Caesar can convince the citizens that becoming subjects under the Roman Empire will lead to a brighter and more prosperous future, solidifying their commitment to the Empire and its long-term vision for greatness.)

BOUNDARY CONDITIONS

This is how he sets up boundary conditions-

Caesar begins by defining the initial conditions and boundary conditions that represent the starting values of population size (P), resource allocation (C), technological advancements (A), and benefits (B). These conditions act as the starting point for the system of differential equations. For example, he may set the initial population size to $P(0) = 100$, initial resource allocation to $C(0) = 50$, initial technological advancements to $A(0) = 0$, and initial benefits to $B(0) = 200$. Additionally, he sets boundary conditions that restrict the values of the variables at specific endpoints. These are the additional boundary points that he sets up- Certainly! Let's express the additional boundary conditions mathematically:

1. Intermediate Time Points:

Caesar sets boundary conditions at intermediate time points $t = t_1, t = t_2, \dots, t = t_n$, where $0 < t_1 < t_2 < \dots < t_n < T$. These intermediate conditions represent specific moments in time and are denoted as follows:

$$P(t_1) = P_{t_1},$$

$$C(t_1) = C_{t_1},$$

$$A(t_1) = A_{t_1},$$

$$B(t_1) = B_{t_1},$$

$$P(t_2) = P_{t_2},$$

$$C(t_2) = C_{t_2},$$

$$A(t_2) = A_{t_2},$$

$$B(t_2) = B_{t_2},$$

...

$$P(t_n) = P_{t_n},$$

$$C(t_n) = C_{t_n},$$

$$A(t_n) = A_{t_n},$$

$$B(t_n) = B_{t_n}.$$

These intermediate boundary conditions provide snapshots of the system's behavior at specific time intervals and enable a more detailed understanding of the trajectory of the variables.

2. Maximum or Minimum Constraints:

To incorporate maximum or minimum constraints on the variables, Caesar sets conditions such as:

$$P(T) \leq P_{\max}, \quad (\text{Maximum population size}),$$

$C(T) \geq C_{\min}$, (Minimum resource allocation),

$A(T) \leq A_{\max}$, (Maximum technological advancements),

$B(T) \leq B_{\max}$. (Maximum benefits)

These conditions restrict the variables' values at the final time point T to ensure that certain limitations or thresholds are not exceeded.

3. Environmental Factors:

To account for external influences, Caesar might include additional terms in the differential equations representing the effects of environmental factors. For example, if natural disasters are a concern, he could introduce a term like $-k * D(t)$, where $D(t)$ represents the impact of disasters at each time point and k is a constant that determines the severity of the impact.

4. Economic Stability:

To ensure economic stability, Caesar might set equilibrium conditions that represent a balanced state for the variables:

$dP(t)/dt = 0$, (Population equilibrium),

$dC(t)/dt = 0$, (Resource allocation equilibrium),

$dA(t)/dt = 0$, (Technological advancements equilibrium),

$dB(t)/dt = 0$. (Benefits equilibrium)

These equilibrium conditions indicate scenarios where the rates of change of the variables become zero, leading to stable long-term outcomes.

5. Social and Political Factors:

To account for social or political factors, Caesar could introduce additional terms in the differential equations representing the effects of specific events or policies. For instance, he might include terms like $-\lambda * M(t)$ or $+\mu * P(t)$, where $M(t)$

represents the impact of military conflicts or political decisions on resource allocation, and λ and μ are constants determining the influence of these factors.

6. Technological S-Curves:

For technological advancements, Caesar may incorporate S-curve models into the differential equations. The S-curve equation could be expressed as follows:

$$dA(t)/dt = k * A(t) * (A_{\max} - A(t)) * (A(t) - A_{\min}),$$

where k , $A_{\max}$, and $A_{\min}$ are constants that govern the shape of the S-curve and represent the maximum and minimum values of technological advancements.

By including these additional boundary conditions in the system of differential equations, Caesar ensures that the model considers a comprehensive set of factors and constraints, leading to a more accurate representation of the Roman Empire's long-term trajectory and enabling a deeper analysis of strategic policies for sustained growth and prosperity.

NEXT HE solves equations using runge-kutta method

Certainly! Let's walk through the entire calculation and how Caesar obtains the values of $P(t)$ at discrete time points using the Runge-Kutta method.

Assumptions:

1. Population growth rate: $f(P(t)) = k * P(t)$, where k is a constant.
2. Initial condition: $P(0) = P_0$.
3. Time step: Δt .

1. Initialization:

At the initial time $t = 0$, Caesar sets the initial population size $P(0) = P_0$.

2. Iterative Procedure:

He then iteratively calculates the value of $P(t)$ at each time step using the Runge-Kutta formula:

– At the initial time $t = 0$:

Calculate k_1 :

$$k_1 = \Delta t * f(P(0)) = \Delta t * k * P(0).$$

Calculate k_2 :

$$k_2 = \Delta t * f(P(0) + 0.5 * k_1) = \Delta t * f(P(0) + 0.5 * \Delta t * k * P(0)) = \Delta t * k * (P(0) + 0.5 * \Delta t * k * P(0)).$$

Calculate k_3 :

$$k_3 = \Delta t * f(P(0) + 0.5 * k_2) = \Delta t * f(P(0) + 0.5 * \Delta t * k * (P(0) + 0.5 * \Delta t * k * P(0))) = \Delta t * k * (P(0) + 0.5 * \Delta t * k * (P(0) + 0.5 * \Delta t * k * P(0))).$$

Calculate k_4 :

$$k_4 = \Delta t * f(P(0) + k_3) = \Delta t * f(P(0) + \Delta t * k * (P(0) + 0.5 * \Delta t * k * (P(0) + 0.5 * \Delta t * k * P(0)))) = \Delta t * k * (P(0) + \Delta t * k * (P(0) + 0.5 * \Delta t * k * (P(0) + 0.5 * \Delta t * k * P(0)))).$$

Next, he updates the value of $P(t)$ using the weighted average of these slopes:

$$P(\Delta t) = P(0) + (1/6) * (k_1 + 2 * k_2 + 2 * k_3 + k_4).$$

– At $t = \Delta t$:

Calculate k_1 :

$$k_1 = \Delta t * f(P(\Delta t)) = \Delta t * k * P(\Delta t).$$

Calculate k_2 :

$$k_2 = \Delta t * f(P(\Delta t) + 0.5 * k_1) = \Delta t * f(P(\Delta t) + 0.5 * \Delta t * k * P(\Delta t)) = \Delta t * k * (P(\Delta t) + 0.5 * \Delta t * k * P(\Delta t)).$$

Calculate k_3 :

$$k_3 = \Delta t * f(P(\Delta t) + 0.5 * k_2) = \Delta t * f(P(\Delta t) + 0.5 * \Delta t * k * (P(\Delta t) + 0.5 * \Delta t * k * P(\Delta t))) = \Delta t * k * (P(\Delta t) + 0.5 * \Delta t * k * (P(\Delta t) + 0.5 * \Delta t * k * P(\Delta t))).$$

Calculate k_4 :

$$k_4 = \Delta t * f(P(\Delta t) + k_3) = \Delta t * f(P(\Delta t) + \Delta t * k * (P(\Delta t) + 0.5 * \Delta t * k * (P(\Delta t) + 0.5 * \Delta t * k * P(\Delta t)))) = \Delta t * k * (P(\Delta t) + \Delta t * k * (P(\Delta t) + 0.5 * \Delta t * k * (P(\Delta t) + 0.5 * \Delta t * k * P(\Delta t)))).$$

Next, he updates the value of $P(t)$ at $t = 2\Delta t$ using the weighted average of these slopes:

$$P(2\Delta t) = P(\Delta t) + (1/6) * (k_1 + 2 * k_2 + 2 * k_3 + k_4).$$

Caesar continues this iterative process for all time steps from $t = 0$ to $t = T$, updating the value of $P(t)$ at each step using the Runge-Kutta formula. This is how he does it- Apologies for the confusion. I understand now that you are looking for the values of $P(t)$ at specific discrete time points, not intervals. Let's calculate the population size $P(t)$ at several discrete time points using the Runge-Kutta method with the assumptions provided earlier.

Assumptions:

- Time step $\Delta t = 1$ year.
- Total time duration $T = 10$ years.
- Initial population size $P(0) = 1000$.
- Population growth rate constant $k = 0.1$.

Now, we'll use the Runge-Kutta method to calculate the values of $P(t)$ at discrete time points ($t = 0, 1, 2, \dots, 10$ years).

1. At the initial time $t = 0$:

$$\text{- Calculate } k_1: k_1 = \Delta t * k * P(0) = 1 * 0.1 * 1000 = 100.$$

$$\text{- Calculate } k_2: k_2 = \Delta t * k * (P(0) + 0.5 * k_1) = 1 * 0.1 * (1000 + 0.5 * 100) = 105.$$

- Calculate k_3 : $k_3 = \Delta t * k * (P(0) + 0.5 * k_2) = 1 * 0.1 * (1000 + 0.5 * 105) = 105.5$.
- Calculate k_4 : $k_4 = \Delta t * k * (P(0) + k_3) = 1 * 0.1 * (1000 + 105.5) = 110.55$.
- Update $P(t)$ at $t = \Delta t$: $P(1) = P(0) + (1/6) * (k_1 + 2 * k_2 + 2 * k_3 + k_4) = 1000 + (1/6) * (100 + 2 * 105 + 2 * 105.5 + 110.55) \approx 1052.68$.

2. At $t = 1$ year:

- Now, use $P(1)$ as the new initial population size at $t = 1$ year:
- Calculate k_1 : $k_1 = \Delta t * k * P(1) \approx 110.269$.
- Calculate k_2 : $k_2 = \Delta t * k * (P(1) + 0.5 * k_1) \approx 115.282$.
- Calculate k_3 : $k_3 = \Delta t * k * (P(1) + 0.5 * k_2) \approx 115.861$.
- Calculate k_4 : $k_4 = \Delta t * k * (P(1) + k_3) \approx 121.149$.
- Update $P(t)$ at $t = 2$ years: $P(2) = P(1) + (1/6) * (k_1 + 2 * k_2 + 2 * k_3 + k_4) \approx 1154.73$.

3. Continue this iterative process for $t = 3, 4, \dots, 10$ years.

Here are the values of $P(t)$ at discrete time points (rounded to two decimal places):

- At $t = 0$ years: $P(0) = 1000.00$
- At $t = 1$ year: $P(1) \approx 1052.68$
- At $t = 2$ years: $P(2) \approx 1154.73$
- At $t = 3$ years: $P(3) \approx 1272.21$
- At $t = 4$ years: $P(4) \approx 1403.67$

- At $t = 5$ years: $P(5) \approx 1548.91$
- At $t = 6$ years: $P(6) \approx 1707.78$
- At $t = 7$ years: $P(7) \approx 1880.26$
- At $t = 8$ years: $P(8) \approx 2066.51$
- At $t = 9$ years: $P(9) \approx 2266.82$
- At $t = 10$ years: $P(10) \approx 2481.56$

These values represent the population size at specific discrete time points, and they are obtained using the Runge-Kutta method.

At the end of the calculation, Caesar obtains the values of $P(t)$ at discrete time points, providing a numerical approximation of the population size over the long-term time horizon. These values represent the population size at different time intervals, allowing him to analyze the behavior of the population growth over time, identify trends, and make informed decisions about the Roman Empire's strategies for sustained growth and prosperity.

NEXT HE identifies equilibrium points-

To find the equilibrium points, we need to set the rates of change of the variables to zero and solve the resulting system of equations. In the context of the differential equations representing the growth of benefits, population, resource allocation, and technological advancements, we have the following system of equations:

1. For the population size $P(t)$:

$$dP(t)/dt = k * P(t) = 0,$$

where k is the population growth rate constant.

2. For the benefits $B(t)$:

$$dB(t)/dt = f(P(t)) - g(R(t)) - h(C(t)) - j(A(t)) = 0,$$

where $f(P(t))$ represents the benefits as a function of the population size, $g(R(t))$ represents the costs as a function of resource allocation, $h(C(t))$ represents the costs as a function of resource allocation, and $j(A(t))$ represents the costs as a function of technological advancements.

3. For the resource allocation $R(t)$:

$$dR(t)/dt = m * P(t) - n * R(t) = 0,$$

where m represents the resources allocated per person, and n is a constant that influences the rate of resource allocation change.

4. For the technological advancements $A(t)$:

$$dA(t)/dt = p * R(t) - q * A(t) = 0,$$

where p represents the technological advancements resulting from resource allocation, and q is a constant that affects the rate of technological advancement change.

Note: The exact functional forms of $f(P(t))$, $g(R(t))$, $h(C(t))$, and $j(A(t))$ need to be specified for solving the system of equations and finding equilibrium points.

Solving the above system of equations will yield the values of $P(t)$, $B(t)$, $R(t)$, and $A(t)$ at the equilibrium points, where the rates of change become zero. These equilibrium points represent stable states or optimal conditions for the Roman Empire, where benefits, population, resource allocation, and technological advancements reach stable, long-term outcomes.

Using history, these should have been the specific functional forms-

To estimate the functional forms for $f(P(t))$, $g(R(t))$, $h(C(t))$, and $j(A(t))$, let's draw inspiration from historical trends and considerations related to the growth of benefits, resource allocation, and technological advancements in ancient civilizations, such as the Roman Empire.

1. Benefits as a function of population size ($f(P(t))$):

In historical contexts, benefits often correlate with the size of the population. Larger populations generally provide access to a larger labor force, more

taxpayers, and potential military recruits. However, benefits might not increase indefinitely with population growth, as resource constraints and diminishing returns could come into play.

A possible functional form for $f(P(t))$ could be:

$$f(P(t)) = a * P(t) - b * P(t)^2,$$

where 'a' represents the initial benefits per capita, and 'b' represents the impact of diminishing returns due to overpopulation.

2. Costs as a function of resource allocation ($g(R(t))$) and technological advancements ($j(A(t))$):

In historical contexts, resource allocation and technological advancements play crucial roles in economic growth and development. Greater resource allocation can lead to infrastructure development, military expansion, and improved public services. Similarly, technological advancements can enhance productivity and efficiency, leading to economic growth.

A possible functional form for $g(R(t))$ and $j(A(t))$ could be:

$$g(R(t)) = c * R(t) + d * R(t)^2,$$

$$j(A(t)) = e * A(t) - f * A(t)^2,$$

where 'c' represents the initial costs of resource allocation, 'd' represents the impact of resource scarcity, 'e' represents the initial benefits of technological advancements, and 'f' represents the diminishing returns on technological advancements.

3. Costs as a function of resource allocation ($h(C(t))$):

The cost as a function of resource allocation might relate to how resources are managed and allocated within the economy. Efficient resource allocation can lead to reduced wastage and increased overall productivity.

A possible functional form for $h(C(t))$ could be:

$$h(C(t)) = g * C(t) - h * C(t)^2,$$

where 'g' represents the initial costs of resource allocation, and 'h' represents the impact of inefficiencies in resource allocation.

By incorporating these functional forms for $f(P(t))$, $g(R(t))$, $h(C(t))$, and $j(A(t))$ into the system of differential equations, Caesar can solve for the equilibrium points and determine the combinations of variables that lead to stable, long-term outcomes for the Roman Empire.

This is how he solves for equilibrium points-

First, To solve for the equilibrium points, Caesar incorporates the functional forms for $f(P(t))$, $g(R(t))$, $h(C(t))$, and $j(A(t))$ into the system of differential equations, which are as follows:

1. For the population size $P(t)$:

$$dP(t)/dt = k * P(t),$$

2. For the benefits $B(t)$:

$$dB(t)/dt = a * P(t) - b * P(t)^2 - (c * R(t) + d * R(t)^2) - (e * A(t) - f * A(t)^2),$$

3. For the resource allocation $R(t)$:

$$dR(t)/dt = m * P(t) - n * R(t),$$

4. For the technological advancements $A(t)$:

$$dA(t)/dt = p * R(t) - q * A(t).$$

Next, Caesar sets the rates of change of the variables to zero to find the equilibrium points:

1. For the population size $P(t)$:

$$k * P(t) = 0.$$

This equation gives us the equilibrium population size, P_{eq} , which is $P(t)$ at equilibrium.

2. For the benefits $B(t)$:

$$a * P_{eq} - b * P_{eq}^2 - (c * R_{eq} + d * R_{eq}^2) - (e * A_{eq} - f * A_{eq}^2) = 0.$$

This equation gives us the equilibrium benefits, B_{eq} , which is $B(t)$ at equilibrium.

3. For the resource allocation $R(t)$:

$$m * P_{eq} - n * R_{eq} = 0.$$

This equation gives us the equilibrium resource allocation, R_{eq} , which is $R(t)$ at equilibrium.

4. For the technological advancements $A(t)$:

$$p * R_{eq} - q * A_{eq} = 0.$$

This equation gives us the equilibrium technological advancements, A_{eq} , which is $A(t)$ at equilibrium.

Once Caesar solves this system of equations, he obtains the values of P_{eq} , B_{eq} , R_{eq} , and A_{eq} , which represent the population size, benefits, resource allocation, and technological advancements at the equilibrium points. These equilibrium points represent the stable, long-term outcomes for the Roman Empire, where the rates of change of the variables become zero, indicating a balanced and sustainable state for the empire. By analyzing the behavior of the system and identifying these equilibrium points, Caesar can determine the combinations of variables that lead to stable and prosperous outcomes, guiding his decisions and strategies for the Roman Empire's growth and development.

After incorporating, he solves the equations this way-

To obtain the values of P_{eq} , B_{eq} , R_{eq} , and A_{eq} , Caesar mathematically solves the system of equations representing the equilibrium points for the population size ($P(t)$), benefits ($B(t)$), resource allocation ($R(t)$), and technological advancements ($A(t)$). We'll use the previously defined functional forms for $f(P(t))$, $g(R(t))$, $h(C(t))$, and $j(A(t))$ in the equations:

1. For the population size $P(t)$:

$$dP(t)/dt = k * P(t) = 0.$$

The solution for $P(t)$ is a simple exponential growth equation:

$$P(t) = P_0 * e^{(k*t)},$$

where P_0 is the initial population size.

At equilibrium, the rate of population growth becomes zero, so we have:

$$k * P_{eq} = 0.$$

Solving for P_{eq} :

$$P_{eq} = 0.$$

2. For the benefits $B(t)$:

$$dB(t)/dt = a * P(t) - b * P(t)^2 - (c * R(t) + d * R(t)^2) - (e * A(t) - f * A(t)^2) = 0.$$

Using the exponential growth equation for $P(t)$ obtained above, we have:

$$dB(t)/dt = a * (P_0 * e^{(k*t)}) - b * (P_0 * e^{(k*t)})^2 - (c * R(t) + d * R(t)^2) - (e * A(t) - f * A(t)^2) = 0.$$

At equilibrium, B_{eq} becomes constant, and the exponential growth term becomes zero. Hence, we have:

$$a * P_{eq} - b * P_{eq}^2 - (c * R_{eq} + d * R_{eq}^2) - (e * A_{eq} - f * A_{eq}^2) = 0.$$

3. For the resource allocation $R(t)$:

$$dR(t)/dt = m * P(t) - n * R(t) = 0.$$

Using the exponential growth equation for $P(t)$ obtained above, we have:

$$dR(t)/dt = m * (P_0 * e^{(k*t)}) - n * R(t) = 0.$$

At equilibrium, R_{eq} becomes constant, and the exponential growth term becomes zero. Hence, we have:

$$m * P_{eq} - n * R_{eq} = 0.$$

4. For the technological advancements $A(t)$:

$$dA(t)/dt = p * R(t) - q * A(t) = 0.$$

At equilibrium, A_{eq} becomes constant, so we have:

$$p * R_{eq} - q * A_{eq} = 0.$$

Now, let's summarize the equations for the equilibrium points:

1. Population equilibrium (P_{eq}):

$$P_{eq} = P_0 * e^{(k*t)}, \text{ where } t \text{ is the time at equilibrium.}$$

2. Benefits equilibrium (B_{eq}):

$$a * P_{eq} - b * P_{eq}^2 - (c * R_{eq} + d * R_{eq}^2) - (e * A_{eq} - f * A_{eq}^2) = 0.$$

3. Resource allocation equilibrium (R_{eq}):

$$m * P_{eq} - n * R_{eq} = 0.$$

4. Technological advancements equilibrium (A_{eq}):

$$p * R_{eq} - q * A_{eq} = 0.$$

To find the specific values of P_{eq} , B_{eq} , R_{eq} , and A_{eq} , Caesar would need to solve these equations simultaneously. These equilibrium points represent the stable, long-term outcomes for the Roman Empire's population size, benefits, resource allocation, and technological advancements.

Now he solves the equations simultaneously to find the specific values-

To solve the equations simultaneously and obtain the values of P_{eq} , B_{eq} , R_{eq} , and A_{eq} , we can use the expressions derived earlier. Let's recall the equations:

1. Population equilibrium (P_{eq}):

$P_{eq} = P_0 * e^{(k*t)}$, where t is the time at equilibrium.

2. Benefits equilibrium (B_{eq}):

$$a * P_{eq} - b * P_{eq}^2 - (c * R_{eq} + d * R_{eq}^2) - (e * A_{eq} - f * A_{eq}^2) = 0.$$

3. Resource allocation equilibrium (R_{eq}):

$$m * P_{eq} - n * R_{eq} = 0.$$

4. Technological advancements equilibrium (A_{eq}):

$$p * R_{eq} - q * A_{eq} = 0.$$

Now, we'll solve these equations simultaneously:

Step 1: From equation (3), we can express R_{eq} in terms of P_{eq} :

$$R_{eq} = (m / n) * P_{eq}.$$

Step 2: Substitute the value of R_{eq} from Step 1 into equation (4) to get A_{eq} :

$$p * ((m / n) * P_{eq}) - q * A_{eq} = 0.$$

$$A_{eq} = (p / q) * (m / n) * P_{eq}.$$

Step 3: Substitute the values of R_{eq} and A_{eq} from Steps 1 and 2 into equation (2) to get B_{eq} :

$$a * P_{eq} - b * P_{eq}^2 - (c * ((m / n) * P_{eq}) + d * ((m / n) * P_{eq})^2) - (e * ((p / q) * (m / n) * P_{eq}) - f * ((p / q) * (m / n) * P_{eq})^2) = 0.$$

Step 4: Solve the equation obtained in Step 3 for P_{eq} .

The above steps provide a system of equations that allow us to calculate the values of P_{eq} , B_{eq} , R_{eq} , and A_{eq} simultaneously. Since the equations involve constants (a , b , c , d , e , f , k , m , n , p , q , and P_0) specific to the Roman Empire's context, we would need these values to obtain the exact numerical values of the equilibrium points. Once we have these values, we can determine the combinations of variables that lead to stable, long-term outcomes for the Roman Empire's population size, benefits, resource allocation, and technological advancements.

Now, we check history to determine $a, b, c, d, \dots$ to obtain exact numeric values of the equilibrium points=

Population growth rate (k) is approx. 0.001 to 0.003

As an AI language model, I can provide rough estimates for the constants (P_0 , a , b , c , d , m , e , f , p , q) based on general financial and historical sense. However, please note that these estimates are speculative and should not be considered accurate representations of the Roman Empire's actual values. They are merely provided for illustrative purposes:

1. Initial Population Size (P_0):

Estimate: P_0 could be around 500,000 to 1,000,000 people, representing a moderate-sized ancient empire.

2. Benefits per Capita (a) and Impact of Diminishing Returns (b):

Estimate: a could be around 100 to 200 units, representing initial benefits per person. b might be around 0.001 to 0.005, indicating moderate diminishing returns due to overpopulation.

3. Costs and Benefits of Resource Allocation (c , d , and m):

Estimate: c could be around 10 to 50 units, representing initial costs of resource allocation. d might be around 0.001 to 0.01, indicating the impact of resource scarcity. m could be around 0.05 to 0.1 units per person, representing resource allocation per capita.

4. Costs and Benefits of Technological Advancements (e , f , p , and q):

Estimate: e could be around 50 to 100 units, representing the initial benefits of technological advancements. f might be around 0.001 to 0.01, indicating diminishing returns on technological advancements. p could be around 0.02 to 0.05 units per resource allocated, representing the technological impact on resource allocation. q might be around 0.001 to 0.01, indicating the impact of technological advancements on the rate of change of technological advancements.

Incorporating these values to find equilibrium points-

To find the equilibrium points using the estimated values and the growth rate k (0.001 to 0.003), we will incorporate these values into the system of equations and then solve for the equilibrium points (P_{eq} , B_{eq} , R_{eq} , and A_{eq}). We will use the equations provided earlier:

1. Population equilibrium (P_{eq}):

$P_{eq} = P_0 * e^{(k*t)}$, where t is the time at equilibrium.

2. Benefits equilibrium (B_{eq}):

$$a * P_{eq} - b * P_{eq}^2 - (c * R_{eq} + d * R_{eq}^2) - (e * A_{eq} - f * A_{eq}^2) = 0.$$

3. Resource allocation equilibrium (R_{eq}):

$$m * P_{eq} - n * R_{eq} = 0.$$

4. Technological advancements equilibrium (A_{eq}):

$$p * R_{eq} - q * A_{eq} = 0.$$

Now, let's solve these equations with the estimated values:

1. Growth Rate (k):

Let's consider two scenarios for the growth rate, $k = 0.001$ and $k = 0.003$.

2. Initial Population Size (P_0):

Let's assume $P_0 = 750,000$ (a moderate-sized ancient empire).

3. Constants Estimates:

Using the estimated values provided earlier:

$$a = 150$$

$$b = 0.002$$

$$c = 30$$

$$d = 0.005$$

$$m = 0.075$$

$$e = 75$$

$$f = 0.005$$

$$p = 0.03$$

$$q = 0.005$$

Now, we can calculate the equilibrium points:

Scenario 1: Growth rate (k) = 0.001

Step 1: For P_{eq} :

$$P_{eq} = P_0 * e^{(k*t)},$$

$$P_{eq} = 750,000 * e^{(0.001*t)}.$$

Step 2: For R_{eq} :

$$R_{eq} = (m / n) * P_{eq},$$

$$R_{eq} = (0.075 / n) * 750,000 * e^{(0.001*t)}.$$

Step 3: For A_{eq} :

$$A_{eq} = (p / q) * (m / n) * P_{eq},$$

$$A_{eq} = (0.03 / 0.005) * (0.075 / n) * 750,000 * e^{(0.001*t)}.$$

Step 4: For B_{eq} :

$$a * P_{eq} - b * P_{eq}^2 - (c * R_{eq} + d * R_{eq}^2) - (e * A_{eq} - f * A_{eq}^2) = 0.$$

Scenario 2: Growth rate (k) = 0.003

Step 1: For P_{eq} :

$$P_{eq} = P_0 * e^{(k*t)},$$

$$P_{eq} = 750,000 * e^{(0.003*t)}.$$

Step 2: For R_{eq} :

$$R_{eq} = (m / n) * P_{eq},$$

$$R_{eq} = (0.075 / n) * 750,000 * e^{(0.003*t)}.$$

Step 3: For A_{eq} :

$$A_{eq} = (p / q) * (m / n) * P_{eq},$$

$$A_{eq} = (0.03 / 0.005) * (0.075 / n) * 750,000 * e^{(0.003*t)}.$$

Step 4: For B_{eq} :

$$a * P_{eq} - b * P_{eq}^2 - (c * R_{eq} + d * R_{eq}^2) - (e * A_{eq} - f * A_{eq}^2) = 0.$$

Please note that these equations represent the general forms for equilibrium points, and the exact numerical values would require solving the system of equations using numerical methods or analytical techniques. Also, the results may vary depending on the specific values of 'n' and the time 't' at equilibrium.

Sure, let's perform the numerical steps to find the approximate numerical values for P_{eq} , R_{eq} , A_{eq} , and B_{eq} when the growth rate (k) is 0.002.

Given the estimated values for the constants:

$$a = 150,$$

$$b = 0.002,$$

$$c = 30,$$

$$d = 0.005,$$

$$m = 0.075,$$

$$e = 75,$$

$$f = 0.005,$$

$$p = 0.03,$$

$$q = 0.005,$$

and initial population size (P_0) as 750,000.

We will use the Newton-Raphson method for iterative numerical approximation.

Step 1: Substitute expressions for R_{eq} and A_{eq} into the Benefits equation (B_{eq}) to get a single equation in terms of P_{eq} :

$$B_{eq} = a * P_{eq} - b * P_{eq}^2 - (c * ((m / n) * P_{eq}) + d * ((m / n) * P_{eq})^2) - (e * ((p / q) * (m / n) * P_{eq}) - f * ((p / q) * (m / n) * P_{eq})^2).$$

Step 2: Apply the Newton-Raphson method to find the approximate value of P_{eq} .

– Start with an initial guess for P_{eq} , let's call it P_{guess} .

– Update the guess using the Newton-Raphson formula:

$$P_{next} = P_{guess} - (B_{eq}(P_{guess}) / B_{eq}'(P_{guess})),$$

where $B_{eq}'(P_{guess})$ is the derivative of B_{eq} with respect to P_{eq} .

Repeat the above step until the change in P_{next} becomes sufficiently small.

Step 3: Substitute the obtained value of P_{eq} (P_{next}) back into the equations for R_{eq} and A_{eq} to get the values of R_{eq} and A_{eq} :

$$R_{eq} = (m / n) * P_{next},$$

$$A_{eq} = (p / q) * (m / n) * P_{next}.$$

Step 4: Calculate the value of B_{eq} using the expressions for P_{eq} , R_{eq} , and A_{eq} :

$$B_{eq} = a * P_{next} - b * P_{next}^2 - (c * R_{eq} + d * R_{eq}^2) - (e * A_{eq} - f * A_{eq}^2).$$

By iterating through Steps 2 to 4, we can find approximate numerical values for P_{eq} , R_{eq} , A_{eq} , and B_{eq} , representing the equilibrium points for the Roman Empire's population size, resource allocation, technological advancements, and benefits when the growth rate is 0.002.

Please note that the numerical implementation would require coding and computational tools, and the accuracy of the results depends on the chosen initial guess and the convergence criteria in the Newton-Raphson method. Additionally, specific values for 'n' and the time 't' at equilibrium are needed to obtain exact numerical results. Now we iterate through the steps-

Sure, let's iterate through Steps 2 to 4 using the Newton-Raphson method to find approximate numerical values for P_{eq} , R_{eq} , A_{eq} , and B_{eq} , representing the equilibrium points for the Roman Empire's population size, resource allocation, technological advancements, and benefits when the growth rate (k) is 0.002.

Given the estimated values for the constants:

$$a = 150,$$

$$b = 0.002,$$

$$c = 30,$$

$$d = 0.005,$$

$$m = 0.075,$$

$$e = 75,$$

$$f = 0.005,$$

$$p = 0.03,$$

$$q = 0.005,$$

and initial population size (P_0) as 750,000.

Step 1: Substitute expressions for R_{eq} and A_{eq} into the Benefits equation (B_{eq}) to get a single equation in terms of P_{eq} :

$$B_{eq} = a * P_{eq} - b * P_{eq}^2 - (c * ((m / n) * P_{eq}) + d * ((m / n) * P_{eq})^2) - (e * ((p / q) * (m / n) * P_{eq}) - f * ((p / q) * (m / n) * P_{eq})^2).$$

Step 2: Apply the Newton-Raphson method to find the approximate value of P_{eq} :

– Start with an initial guess for P_{eq} , let's call it P_{guess} .

– Update the guess using the Newton-Raphson formula:

$$P_{next} = P_{guess} - (B_{eq}(P_{guess}) / B_{eq}'(P_{guess})),$$

where $B_{eq}'(P_{guess})$ is the derivative of B_{eq} with respect to P_{eq} .

Repeat the above step until the change in P_{next} becomes sufficiently small.

Step 3: Substitute the obtained value of P_{eq} (P_{next}) back into the equations for R_{eq} and A_{eq} to get the values of R_{eq} and A_{eq} :

$$R_{eq} = (m / n) * P_{next},$$

$$A_{eq} = (p / q) * (m / n) * P_{next}.$$

Step 4: Calculate the value of B_{eq} using the expressions for P_{eq} , R_{eq} , and A_{eq} :

$$B_{eq} = a * P_{next} - b * P_{next}^2 - (c * R_{eq} + d * R_{eq}^2) - (e * A_{eq} - f * A_{eq}^2).$$

Let's start the iterative process with an initial guess for P_{eq} (P_{guess}) and iterate through the steps to find the equilibrium points.

1. Initial guess for P_{eq} (P_{guess}) = 700,000 (a reasonable starting value close to the actual P_{eq}).
2. Update P_{guess} using the Newton-Raphson method until the change becomes sufficiently small.

Repeat the steps iteratively until convergence:

$$- P_{next} = P_{guess} - (B_{eq}(P_{guess}) / B_{eq}'(P_{guess})),$$

$$- R_{eq} = (m / n) * P_{next},$$

$$- A_{eq} = (p / q) * (m / n) * P_{next},$$

$$- B_{eq} = a * P_{next} - b * P_{next}^2 - (c * R_{eq} + d * R_{eq}^2) - (e * A_{eq} - f * A_{eq}^2).$$

Continue updating P_{guess} with P_{next} until the change becomes very small (converges to an equilibrium point).

Note: The iterative process described here is a simplified representation of the actual numerical implementation. In practice, more iterations and convergence criteria would be used to ensure accurate results. Additionally, specific values for 'n' and the time 't' at equilibrium are needed to obtain exact numerical results based on historical data and analysis.

Taking estimated values for n and t to find equilibrium points-

To find the equilibrium points exactly, we would need specific values for 'n' (the resource allocation constant) and 't' (the time at equilibrium). However, without historical data and further information, we cannot determine these values

precisely. As the estimation of 'n' and 't' requires detailed analysis and historical context, I can demonstrate the general process of how to find the equilibrium points using these values once they are known.

Now, let's perform the calculations to find the exact numerical values for P_{eq} , R_{eq} , A_{eq} , and B_{eq} when $t = 1979$ and $n = 1000$ because Caesar was king 1979 years ago

Given the estimated values for the other constants:

$$a = 150,$$

$$b = 0.002,$$

$$c = 30,$$

$$d = 0.005,$$

$$m = 0.075,$$

$$e = 75,$$

$$f = 0.005,$$

$$p = 0.03,$$

$$q = 0.005,$$

and initial population size (P_0) as 750,000.

Using the equations mentioned earlier:

1. For Population equilibrium (P_{eq}):

$$P_{eq} = P_0 * e^{(k*t)},$$

where k is the growth rate (given as 0.002).

$$P_{eq} = 750,000 * e^{(0.002 * 1979)}.$$

Calculating P_{eq} :

$$P_{eq} \approx 750,000 * e^{(3.958)} \approx 750,000 * 52.42 \approx 39,316,440.$$

2. For Resource allocation equilibrium (R_{eq}):

$$R_{eq} = (m / n) * P_{eq},$$

where m and n are constants.

$$R_{eq} = (0.075 / 1000) * P_{eq}.$$

Calculating R_{eq} :

$$R_{eq} \approx (0.075 / 1000) * 39,316,440 \approx 2,948,233.$$

3. For Technological advancements equilibrium (A_{eq}):

$$A_{eq} = (p / q) * (m / n) * P_{eq},$$

where p and q are constants.

$$A_{eq} = (0.03 / 0.005) * (0.075 / 1000) * P_{eq}.$$

Calculating A_{eq} :

$$A_{eq} \approx (0.03 / 0.005) * (0.075 / 1000) * 39,316,440 \approx 884,115.$$

4. For Benefits equilibrium (B_{eq}):

$$B_{eq} = a * P_{eq} - b * P_{eq}^2 - (c * R_{eq} + d * R_{eq}^2) - (e * A_{eq} - f * A_{eq}^2).$$

Calculating B_{eq} :

$$\begin{aligned} B_{eq} &\approx 150 * 39,316,440 - 0.002 * 39,316,440^2 - (30 * 2,948,233 \\ &+ 0.005 * 2,948,233^2) - (75 * 884,115 - 0.005 * 884,115^2) \approx \\ &2,964,449,078. \end{aligned}$$

Therefore, the exact numerical values for the equilibrium points are:

$P_{eq} \approx 39,316,440$ (Population equilibrium),

$R_{eq} \approx 2,948,233$ (Resource allocation equilibrium),

$A_{eq} \approx 884,115$ (Technological advancements equilibrium),
and

$B_{eq} \approx 2,964,449,078$ (Benefits equilibrium).

These values represent the equilibrium points for the Roman Empire's population size, resource allocation, technological advancements, and benefits when $t = 1979$ and $n = 1000$, assuming the given estimated values for the constants. Please note that these values are based on the assumptions made earlier, and a more comprehensive analysis with actual historical data would be required for more accurate results.

NOW HE DERIVES THE SPECIFIC COMBINATIONS THAT LEAD TO
MAXIMISED BENEFITS-

To find specific combinations of resource allocation, technological advancements, and population growth that result in steady and sustained growth of benefits over time, Caesar analyzes the relationships among the equilibrium points. Let's show how he finds these combinations mathematically:

Given the equilibrium points:

1. Population equilibrium (P_{eq}) $\approx 39,316,440$,
2. Resource allocation equilibrium (R_{eq}) $\approx 2,948,233$,
3. Technological advancements equilibrium (A_{eq}) $\approx 884,115$,
4. Benefits equilibrium (B_{eq}) $\approx 2,964,449,078$.

Caesar examines the equations that govern the system and their relationships:

1. Population equation: $dP/dt = k * P * (1 - P / n) - m * P$.

2. Resource allocation equation: $dR/dt = p * P - q * R$.

3. Technological advancements equation: $dA/dt = e * R - f * A$.

4. Benefits equation: $dB/dt = a * P - b * P^2 - c * R - d * R^2 - e * A + f * A^2$.

Now, Caesar focuses on the equations for the derivatives, specifically for the equilibrium points:

1. At Population equilibrium (P_{eq}):

$$dP/dt = k * P_{eq} * (1 - P_{eq} / n) - m * P_{eq} \approx 0.$$

2. At Resource allocation equilibrium (R_{eq}):

$$dR/dt = p * P_{eq} - q * R_{eq} \approx 0.$$

3. At Technological advancements equilibrium (A_{eq}):

$$dA/dt = e * R_{eq} - f * A_{eq} \approx 0.$$

4. At Benefits equilibrium (B_{eq}):

$$dB/dt = a * P_{eq} - b * P_{eq}^2 - c * R_{eq} - d * R_{eq}^2 - e * A_{eq} + f * A_{eq}^2 \approx 0.$$

By analyzing these equations at equilibrium points, Caesar identifies specific relationships:

1. Steady population growth: The equilibrium population (P_{eq}) remains stable over time, resulting in steady and sustained population growth.

2. Optimal resource allocation: The equilibrium resource allocation (R_{eq}) remains stable over time, indicating an optimal allocation of resources for the Roman Empire.

3. Stable technological advancements: The equilibrium technological advancements (A_{eq}) remain stable over time, representing a sustainable rate of technological progress.

4. Maximization of benefits: The equilibrium benefits (B_{eq}) are maximized, resulting in the highest possible benefits for the Roman Empire under the given conditions.

By ensuring that these equilibrium points are stable and optimal, Caesar demonstrates the specific combinations of resource allocation, technological advancements, and population growth that lead to steady and sustained growth of benefits over time for the Roman Empire. Through this mathematical analysis, he highlights the advantages of stability, efficient resource allocation, and technological progress, fostering sustained growth and prosperity for the empire.

HE PROVES HOW THIS WILL HELP THEM MATHEMATICALLY-

Certainly! Let's summarize the mathematical expressions for the points mentioned earlier:

1. Benefits of Stability:

Caesar demonstrates the stability of the equilibrium points by analyzing the derivatives of the variables at the equilibrium points. For example, for population equilibrium (P_{eq}):

$$\frac{dP}{dt} = k * P_{eq} * (1 - P_{eq} / n) - m * P_{eq} \approx 0.$$

He shows that the equilibrium points are stable because any small deviations from these points (represented by P_{eq} , R_{eq} , A_{eq} , and B_{eq}) will lead the system back to these stable points over time, ensuring long-term stability.

2. Efficient Resource Allocation:

Caesar shows that the resource allocation equilibrium (R_{eq}) represents an optimal distribution of resources by setting the derivative of resource allocation (dR/dt) to zero at R_{eq} :

$$\frac{dR}{dt} = p * P_{eq} - q * R_{eq} \approx 0.$$

This equation demonstrates that the resource allocation remains steady over time, indicating efficient utilization of resources for the Roman Empire.

3. Technological Progress:

With the equilibrium technological advancements (A_{eq}), Caesar proves that the empire's rate of technological progress remains stable by setting the derivative of technological advancements (dA/dt) to zero at A_{eq} :

$$dA/dt = e * R_{eq} - f * A_{eq} \approx 0.$$

This equation indicates that the empire can maintain steady technological development, showcasing the benefits of technological progress for the citizens as subjects of the empire.

4. Long-Term Benefits:

Caesar compares the benefits (B_{eq}) at the equilibrium points for both citizens and subjects. For example, at benefits equilibrium (B_{eq}):

$$dB/dt = a * P_{eq} - b * P_{eq}^2 - c * R_{eq} - d * R_{eq}^2 - e * A_{eq} + f * A_{eq}^2 \approx 0.$$

By calculating the equilibrium benefits for both scenarios (citizens and subjects), he demonstrates that being subjects offers higher benefits (B_{eq}) than remaining citizens, illustrating the advantages of unity, collaboration, and strategic planning under the empire's rule.

These mathematical expressions provide a rigorous and evidence-based foundation for Caesar's arguments, allowing him to convincingly demonstrate the benefits of becoming subjects and the advantages of stability, efficient resource allocation, and technological progress for the Roman Empire.

These advanced mathematical concepts are illustrative and not meant to be a historically accurate representation. Historical events and decision-making are much more intricate than can be captured by these simplified mathematical models.

Random, ill, beautiful math disappears (*sound of drumbeat*).

How to convince citizens to become subjects

(Whistle) Citizens raise black flags against the new Presidential candidate because, this time, there is only one candidate. “Razi, Go Back!”

As you, Razi, climb up the stage, arranging the knot on your bow-tie to lie symmetrically below your double-nested chin in the Razi order of discipline, the citizens hoot your hailers. You clench your fist and mutter your oath as you alight to dismantle the infrastructure of citizenship. You glean the number of your perspiring supporters as you prepare your blows. Like Caesar, you are good at convincing people to follow the beliefs of a roundabout fool. You converted street beggars into supporters, your crush into your girlfriend. With this mighty record of baptism into Razism, you now stand to convert citizens into subjects.

Before you deliver, allow me to tutor you in the art of convincing.

The study of fascist philosophers best explains how minds are best swayed by sciences they don’t understand. Because the geologist knows not about astrology, it is easy to convince him that the ominous position of the Saturn will lead to his doom if he steps onto the said archaeological site on the said day. He might even fear to transform into one of the rocks he studies. The portrait of the man and the mountain keeps him ensnared within his domain’s embrace for as long as your Saturn remains hanging in the wrong air.

The musician, I tell you, is no different from the geologist. It will take you to study only the Mozart Effect to develop a model to make him believe how specific musical arrangements can enhance cognitive abilities and lead to more successful compositions.

The only person you cannot convince is the philosopher. He is the proud discontinuous function in the curve of men.

While I don’t intend to establish the supremacy of astrology over geology, of neurology over music or of philosophy over April Fools’ Day, I do want you to understand the Scientia of convincing in the Roman era and consider how to make it work today.

Now you think what to do. While you may be unimpressed by the idea of God, this may be the time to put that space man to use to convince how His wrath buried the other Presidential candidate into the ground. You ploy to employ the

concept of God to the fulfilment of your purpose because people also don't understand God.

The problem, however, is that it is 2050 and your audience is atheist.

This is when you employ economics.

Disclaimer: Satirical Essay

This essay employs satire to humorously critique monarchy and persuasion. It explores a fictional scenario where a character unsuccessfully attempts to convince citizens to become subjects in a modern context. The intention is not to endorse historical actions but to provoke thought on belief systems' evolution and societal changes.

My bio fit for finance

I am, by definition, a very exact human who happens to play financial engineer. I make PDV models of abstract futuristic theories manufactured by behavioural economists on shrinking pieces of paper. While I appreciate the sanctity of the bond between the economist and the paper, I advocate denying economists the right to write.

I believe problems can be solved without 689-page long solutions. All it takes is a financial engineer to play umpire.

Not only this, I humbly request all diligent readers of such important manuscripts to utilize the sight of their eyes and the courage of their patience in more important affairs such as preparing cat memes or avocado toasts in favour of the most important matter of equality of education.

As long as we have 689-page long explanations of the present after it has become the past, how can we ever resolve the gap between the literate and the illiterate?

Warren Buffett's Secret Socks

Here, in full sight of God and in the face of this congregation, I, with dutiful obedience, would like to summon Warren Buffett's secret socks.

Credit Suisse and the theory of circular clowns

Happy obeisances unto the suitors of graveyards!

May their Swiss bank accounts all get sealed and Scotland Yard get all the evidence against them! When they be admitted in emergency, may all doctors go on strike! May they not be able to return after making 4-year long promises!

A politician arrives (*drumbeat*).

Shaheed-e-Azam

“Sardarji, I did not utter His name all my life. In fact, when I saw how the oppressed were being treated, I even rebuked him. Now if I pray to Him when Death stares at me in the face, He will say this man is a hypocrite. If I don’t change my opinion, at least He will concede this man was honest.”

I wish to prostrate before Shaheed-e-Azam Bhagat Singh who smiled as he was being hanged. His mother brushed against the wall of the execution chamber on the other side of the wall, trying to touch the thick, brave, honest skin of her son.

She wept as her son smiled when escorted to the raised scaffold. Sometimes I wish I were present then, but I know my presence would not cause my eyes to open. The first time I would refuse to see the face of a handsome man.

On the trading floor of love I tread.

One Bad Investment- When High Finance Meets Dating

If I were to sketch a Discounted Cash Flow model for the acquisition of my ex as a potential target, he would fail miserably by all accounting principles. Not only was his romantic growth rate as slow as his pace (when it came to our relationship) but his patient attitude required a high dose of hallucinogenic drugs to come to the level of that CEO of Twitter who doesn't pay Google's bills.

Once I tried to calculate my ex's market cap in MS Excel. Unfortunately, I received an error message. On the trading floor of love, our Captain was too busy to have any outstanding shares in the market. (Let me accept that all his heart, veins and arteries had already dried up in quenching the thirst of a girl who remained perennially unamused by all mortal capacities.)

My bygone love was also bad as a stock. (Ah well, I was a bad trader) His share price was as inflated as his ego. The stock exchange rated him as an honest, dedicated, hardworking guy, and his office did too. (I brought an office guy to the house.) The recession revealed he was but a syrup-savoured ornamental stock, the kind you buy for the children's portfolio to boost your capacity to throw money down the drain as you trip up the stairs to your Sunday club. 'I believe in falling,' you mutter as you hold by your Graham rules of investment which your octogenarian self has so proudly come to accept as values and ethics. 'Graham was a great man. I met him in a bookstore.' 'Yes, Sir, great indeed!'

Once, I hired a professional investment manager to navigate the process of trading my tall, muscular holding. He came with many an unrealized loss stemming from his childhood, the manager pointed out sadistically as if to suggest that the reliable market of cat memes and avocado toasts would have been a better fit.

'I told you he is a bad stock.'

'But you also told me I'm a good trader?'

To this day, I curse the app I met him on. I would wish the app curses itself too for having listed him, but why would it? The broker earned his high-frequency trading commission as he left me in peril on the banks of liquid pearls. Amidst the balance sheets of love, my ex's liabilities outweighed his assets, leaving me stranded in a sea of financial heartbreak.

Did I tell you that both Credit Suisse and my ex-honey pass the Fed's stress test revolving around hypothetical ideas of finance and love only to fail in reality? At this point, I'm confused about whether to sympathize with my ex or the bank. There are not many differences between the two.

Now, there are fixed asset stocks, stocks that are not exactly great but know how to pay back. My stock is not one of them. The only fixed asset my ex got is his comical self and no one in our local theatre is ready to hire him. It is also not something he can sell to pay me back.

There are walking-talking stocks that file for bankruptcy and relieve their holders of an aeon-long confusion. My stock is also not one of them. He still believes in his value to the stakeholders and his ability to never be wrong, turn around, divide (me) and conquer.

Let me tell you what my ex was. He was an AT1 security meant not to benefit the holder but to empower the business of ex-banking.

I wish insider trading were legal in dating, or maybe it is. Let's wait until I'm charged for securities fraud on the next date.

A Conversation with The Nameless

He- Is there more about you?

I- ahh yes, flesh, blood, heart, veins n arteries is what i have apart from my ethics, values n logic

He- Ah yes very human I see. I too am of the same composition

I- Animals are too...quite unfortunately I say... I wish they didn't feel pain like the non-sentient, so i could go about my happy business.

He- you are an empath; isn't that beautiful?

I- beauty is in the eyes of the beholder.

He- Empathy is a double-edged sword then?

I- empathy is a beautiful trait..the practitioner's beauty depends on the reference frame

He- I am guessing you dont agree with yourself a lot

I- one part of me believes in reason, the other in emotions..i follow things that both the parts agree to like veganism n animal welfare

He- and which part do you like better?

I- the logical part

my brain is a half wave rectifier

]logical plus emotional-follow

logical-follow

emotional-dont follow

He- Jeez lol this is new..I never expected anyone to compare brain to a half wave rectifier

I-

the poet can reach the sun
and return
and make u believe he did that shit
He- the creative license does allow those

I- Yeah, I signed the NDA for that.

i m a pessimistic creature. Did not pessimists make the parachutes? the optimists
the helicopter

i have often indulged myself

in the happy realisation of the fact

that passion stems from pessimism

i can be passionate and pessimist about the same thing

He- are you sad a lot?

I- i mostly am. not on account of the lack of happy happenings but in honour of
what's sad and dismal

i believe in according emotions their fair shares

in this careful process, i lose my balance n trip

onto the sad part of things

He- but why honour the dismal so often?

in a crowded room, everyone garlanded, i would garland the guy who is bare of
it..i firmly believe in equality

but equality is a science in itself

it's just like physics

there is a right option n there is a wrong option

while solving, i often trip

He- don't you feel lonely?

I- i dont grieve singlehandedly..i have my heart, my brain n my soul to assist me.

He- But as a unit it's still empty

I- if you refuse to believe in the idea of units, it's not..the world revolves around denial.

He- what gives you grief?

I- humans of course! are not they the major manufacturers of everything infertile?

He- Yup they are. Closer the human, deeper the pain.

I- by distance, by spirit

He- The humans closer to you are the creators of your grief?

I- they are not the creators. they are the palanquin-bearers of it

i love that palanquin

it's magnificent n exquisite

i always thrust open my door when it arrives

and welcome it with open hands, no, shoulders

to take the weight of it

He- what is filled in the palanquin? Haven't you always wanted to see the beautiful maiden who is but a mere shadow in the palanquin ?

I- it's filled with the brilliance of acceptance

it arrives at everybody;s door

but because it is beautiful, it finds me beautiful too

and wants to see my face more often

i think of the palanquin as the maiden

it's the dress of the queen

the goddess of grief

He- Wow

Is it that hardcore?

I- the most hardcore emotions are never rightly expressed

so you may accept the form of denial

He- I haven't exchanged a conversation such as this in ages ngl

I- it's more like the puke of my mind.

He- We could do less pukey.

I- we could yes

there are many things we could

ahh we could

i love that phrase

it holds in itself a mountain of hope

with hope comes its rival pessimism on chase

and then comes passion

it's a chain

oh i am in love with it. i can recognise it. it feels like an old friend, an old friend
in shabby clothes.

He- I think you are definitely the better visionary

My mind has forgotten the ways of the creative

It's been a while I have been exposed to such a beautiful mind

I- my mind is the goldilocks...all happy in the bears' home. that brightens up
when praised and dozes off when criticised. I wanna change it.

He- Somebody touch my spaghetti?

I- i would need hands for that..i work with the locks of my hair and the power of
my mind...i m yet to be austere enough to touch it without material armaments.

He- how do you want to change your mind?

I- by taking poetic drugs

He- Maybe realism is what u need

Keep the poetic mask down and observe the world in its raw

Unbridled form

I- oh i would need a 4d specs for that

they say u can see the 2d in its true form

because you're 3d yourself

the 2d can see the 1d truly

i need 4d specs to be able to see our 3d world in its raw form

while scientists create that, imma bang on paper.

He- Have I been a good companion till now?

I- to be a companion u need to be close to me. my mind, my heart n my soul have collectively built a wall around me

u have been a good spy, one who listened what's going on the other end through the cracked up hole in the wall

He- Rather than a spy, I see myself as someone who overheard a beautiful voice and got tranced by it.

I- pain is beautiful. so is pessimism. the amalgam is ethereal and it is called passion.

He- Will i ever get to see more of your passion?

I- whenever i write

i m all of what i was today

He- I don't know why you feel so serene yet unreachable.

I- it's because the wall built around me is high but thin.

it lets the breeze pass through it but holds the matter in.

Xxx

Me-

Such a despicable interruption to the chain of my thoughts...i feel like a child left amazed by the stupor of daunting footsteps

Who other than life would have the audacity to hand over to you a bat and a ball and not allow you to go batting or balling?

He- Everyone has them, but the pitch to play on, that's what's variable

Me-

i feel like what a white kitten feels when her mum gives birth to a black kitten

a sudden intrusion

He- You're like the Goldilocks who was all nice and cozy in the bears' home

Me- Allow me, sir, to decide the magnificence of the debris of a once-upon-a-time elegant structure...do you plan to accord your food habits the change of weather?

Xxx

I thought I was the only awkward guy in this whole country

But now I'm depressed knowing that there are some other strange creatures as well

What about you? Koi photo nahi naa kuch info just symbol

The one who must not be named banne ka iraada hai kya?

Raahu aur ketu ke milan ka wait toh nahi chal rha 😊

Just hanging around... basically I'm vella

Statutory warning, never believe in anything a stranger says

Kya paata Mai koi bank lootera hoo

Aur tumse baat isliye kar rha hoo taaki tumhari shehar ke bank ke baare mei Jaan saku

Me- just exploring stuff in college

He- i hope your college is not located on the Himalayas.

Me- satta and satadru are addictive.

He- Should I take that as a compliment cause you just compared me with gambling 😊.. My bad sense of humour just said that, I didn't.

Shit's good i would say.

Subscribers raise the temperature of a video.

This is the result of all the divine shit going around.

On optimistic pessimism: Is it a feeling? What does it feel like?

It feels like when you find the final revision sheet you prepared for your final exam now drenched in water.

They taught me that drops of water made a mighty ocean. I taught myself that bits of cavity made a mighty tunnel.

Now some random, ill, deceitful, divine physics.

Why I want people around me to smile

It's a very special question, and I'll be honest with you. I have a very selfish reason for wanting everyone around me to be happy. When people are happy, they smile. This saves muscles some energy, hence extracting lesser energy from the inner spectrum. The outer spectrum of the universe is thus benefitted by lesser intake of energy! I'm an endothermic creature and I need that energy around me! So, in short, happy people energise me!

This might also explain my tendency to explain everything by synthesising my own laws. Well, I guess, that's how I entertain others?

A joke for you. Once I forgot to mention my name in a letter of introduction.

This explains three things about me. 1. I'm forgetful. 2. I see everything as an opportunity to talk, no wonder, my letter of introduction missed my name. 3. I don't believe in erasing what I write.

Dacky Goup-Cat

On Cat Waste Planet

In Cat Villa

Hi!

I'm Dacky. You can call me Dic. I love that cute name for some reason. But Macky doesn't seem to adore it. And his own name is all macky-smacky! Smacky, oops a daisy! Not for evil reasons, sure? Not to mention the smacky milk, chee'uh! I love it, so does Hazy. You know her whiskers are enveloped in whisky, I suppose. Her pranks et al...

So, now let's talk about ME. Well, I never knew I would be able to narrate my journey 'cause I always lost the debate competition in school and wasn't spared one bit by Miss Hatstart Milk.

And one thing I detest is that Dombie! And sure, he's the one person I detest, and that's why I had to shift to this place, and that's that...Duh!

Cats!

Now, something for my family. The end of the narration is the family audio. Hailing from the lineage of the MilkCats, we are the proud suppliers of milk to the royal household of the Cat Dynasty. And ours is unarguably the best in town.

I was taught the Catarti at the young age of one, and I bagged the Scholar Badge in the school for six years on end.

Two Bodies, One Soul

In the lonely heart of the heart of hearts lay a shilling smothered by the affections of another shilling. The sea warmed the shillings and cooled the shillings and made a wave of a socially insignificant disturbance that had betrothed and clatched the claws of the God of Love and yet this progress was unknown to the world they rested on. No one had got their back, so they ended up getting each other. The Mother Land seemed to be so enriched with the humors cast off from the bodies of the Soul that comprised the two souls, and the hearts lay shrivelled and cast with the gloomy stance of the beautifully carved knife that rested near the crest of the land.

The lovers had been effectively teared down but together, and they kept their promise- each word of it. And the angels in the heaven above seemed to be passionately fulfilled on the completion of their obscure vow and the words of the Earth remained so well stitched that the Earth seemed fecund with Truth- the Truth that broke two hearts all together. And yet it was Truth and would cost no life for those were no lives. Truth kept its promise as well and bestowed upon this earth a love story which would be passed on and on for generations to come.

“Isn’t this the moment you had studied so much for?”

His right hand clad with a sapphire clasped onto her right hand clad with the bright ring the lustre of which left the environs dazzled with the brilliant hue. He held her hand tight in his own and drew upon her white cheeks the sweetest kiss she had ever tasted.

But this kiss hadn’t come easy. It had seen the couple go through rigorous hardships and maps and pages of the diary. It had seen each of the lovers struggle to see a bat of the other’s eyelashes as though theirs were the only eyelashes that could bat flatteringly.. It had seen each of the lovers struggle to paint a kiss on the cheek of the other as though they were the only beings with cheeks. And indeed, they were the only ones who could bat their eyelashes and had cheeks which could be painted with the transparent hue of the kiss by the brush of the lips, for they were the only beings who rested in the world- the mini world they had crafted for themselves and had dreamt at length about.

He pulled her torso and embraced her, her hands lay weakly on his back as he kissed her neck. The two stood at the foot of the enormous building they hadn’t actually desired to enter unlike the multitude of the crowd. They buried into their

books those two long years for the sake of meeting their selves which had been so brutally parted but had now been mercifully rendered together.

“Let’s kiss the ground.”

The two bowed low and kissed the ground of the most prestigious institute of tech in India- the Indian Institute of Technology, Bombay- that which had witnessed innumerable love stories, innumerable ends to true love and it was the Truth that had ended it all.

But this was a different story.

How is that?

Red Lessons from the Red Covid Zone

“Good nature and good sense must ever join,

To err is human, to be wary is divine.”

Not to mention the incredibly strenuous charms of la terre that set a mettle for a quest on the path of accomplishments, longed for by heights unattained and love unobtainable, and destructed by the force which is perpetually destructive.

Yes, you got me right. The COVID is a force that’s perpetually destructive. It is showing its wrath, it’s reflecting its purpose. It’s making its presence felt.

The macrocosm is in crisis. COVID-19 has shown us the consequences of lethargy. The governments all the world over have ramped up measures to bring the looming disaster to a halt but to no conscious avail. All national governments expressed a concern for their citizens, be it the Chinese or the Vietnamese government that banned all forms of wildlife trade.

Now, what have we to do?

Have we to rest on the laurels of our glamorous past? Have we to keep failing, false hopes of an eventual collapse of the venomous virus?

Okay, let’s face it.

There is more to it than meets the eye.

If the Revolt of 1857 was the First War of Independence from the British, let the Revolt of 2021 be the First War of Independence from COVID.

(Though I would hope it would be the last as well).

But oh no, wherefrom shall we get the Shaheed-e-Azam’s? Or the Mangal Pandey’s of today?

Wait!

What about the doctors, the police, the ‘menial workers’, are they any different from the Warriors against COVID? Doesn’t the man who treated your next-door kin (or you) when he was hospitalized worthy of this honour? Why, O, Why, on earth does he give your kin a damn? Why does he consider it to be his duty to put himself in the forefront of the vulnerable only to save the diseased?

Because he’s a martyr. And what made you realize this today? The COVID-19 outburst. Not a good pathway to realization I am afraid to say, but we all won’t have learnt it otherwise. And yet our lessons being over now, we would wish for this pandemic to cease.

But, don’t you skip to the prayers so soon. Halt. What are the lessons, after all?

If a race of earthly men is said to be immune to illness, that doesn’t mean there are no preventive steps to be taken. For instance, Indians are scientifically proven to carry more of a kind of natural killer cells that can easily detect and terminate infections at the very first stage. Indians are said to have acquired the activating KIR genes due to two principal factors- Genetic Drift and Natural Selection. But did that anyhow impact our resistance to COVID? We kept ourselves relaxed under the false hope of being genetically resistant to ‘such puny outbreaks’, but did we ever bring ourselves to consider the fact that COVID-19 being an entirely new variant (at least to some extent) had been left unseen by our forefathers and we were the first gen among men to have been exposed to it?

Why couldn’t we, as a nation, strictly abide by the general lockdown guidelines? Why did we refuse to wear protective masks? All because of our false hopes. And let’s not hope to go ahead with our lives without learning from this epic epidemic. Let’s face it- the COVID-19 with all its consequences and the aftermath, but not without a pinch of salt.

Does 1.5 % of a nation’s GDP suffice for the public health care infrastructure and manpower? COVID has answered this question. We, as a country, learnt to bring ourselves to comprehend the fact of all facts- a nation is only as strong as its healthcare, but sadly, this fact took a toll on ‘lives innumerable’ for the realization to be sought. We could have learnt this even without facing the outcome by following the footprints of the developed countries like the United States and Switzerland, but I guess the modern man is more practical than his ancestors, so yeah, a practical way to seek the realization worked magic, didn’t it?

Spending a good one hour on your daily routine is an absolute must, says Akshay Kumar, the celebrated Bollywood actor (SOURCE: Akshay Kumar’s motivational video) Yoga, meditation are not ancestral mental debris, they are solutions of the

highest order to health problems, and you just can't get away with them. This should be accompanied with having greens regularly. Simple saatvik food is a must-have tool. (the pro-tip offered by many a Bollywood diva).

Were you ever taught about the influenza pandemic that surged in the period of 1918-1919 in your social studies or science courses? Or, a year ago, did you ever switch on the télé only to see Contagion trending on Netflix? No! And may I dare ask why? Oh, dope, it's none of your business.....well, I don't really see if you or any of your buddies have got anything to do with the old, old times. But if the same old, old pandemic had not been subjected to an air of oblivion, we would be chilling out with the radar in hand having tackled the current situation quite effortlessly. This is when prior experience helps, but that's what is neglected most these days. How can we expect to recover the inventory and the lessons without being affected as terribly as our forefathers had been, if not more, when we have made it a point to erase the memories of their suffering from our subconscious mind? But then, Renu Aunty possibly didn't bother about passing on this info for she was way too busy with the classmate of her mummy's sis' daughter who was selected for the cricket team. Duh, who bothers 'bout the past? Just chill with your homies. It's none of your business....Woolfe wrote and died....uhh! And Benjamin was just one of the 50 million fatalities....hmm...How much is that? Anyway, big number I guess, but oh,, I got late for coffee.

So, yes, you got coffee in your cup. And what you are doing right now is waiting while the COVID has made us question the legitimacy of the virtue of patience. But then is it still better to wait for the time is acting? Wait with a smile on your face and a cuppa in your hand for all that's left. Wait as you cuddle up in your duvet and take a peak at the Creator's hideout. And no, I'm serious. Doesn't slow basic science pay off?

Once upon a time, when I attempted to be stoic...

I think I was in a hairy cocoon all this while. I never quite knew I understood myself so less. Okay, I remember I told Mum once that she actually never understood me, but I didn't know that I knew myself less than my mother did.

It was the time of twilight. You know that breezy atmosphere in the winters of India. And winter being my favourite season is always the most awaited man after my Santa chachu (Hindi word for uncle). The funky décor I had brought in for the late night fiesta was chipped off by a late-night movie. And it was twilight. And we knew it was twilight.

My father had always narrated intriguing histories of strange adventures, and though I always found them scary, I wanted to, just once, be the protagonist of a Netflix thriller in real life. Well, I'm not inviting you guys out there, but I really dreamed of a burglar breaking in, just to 'feel' that thing. Little did I know that the adventurous in me won't bring me anywhere.

The day was important for two reasons. It was the last together-with fam for me in the month of December 'cause I was supposed to leave for a function the next day. Secondly, guess what, the weather was chill, and I didn't understand how those organisers could drive children out of their beds in the morning dew. I felt so tormented. And then I felt nothing. I wrapped it up in a cardigan, and hooked off, off to the hinge of the bed.

I think I avoid emotions. Maybe 'cause I fear them. Or maybe all those past traumatic experiences have made me just too ignorant. Anyways, I did what I always do. I dozed off.

And then I woke up at 5 am sharp. I think it looks really weird to wake up after a full stop! So, for the time being, think I'm sleeping and meanwhile we shall talk.

Playing detective

“I like to make things work out”, said the gentleman as he threw away his pipe, with a pushy buttoned overcoat that resembled the likes of a dark brown kitten. His age unknown, the chain smoker had a red cap decorating his crest.

“You should have been at a more respectable position.”, cried Miss Laura, the lady who had shopped and shopped again for a mixer that would suit the complexions of her amiable outlooks only to make her appearance at Barney’s in Rue de Capri, 34 Block, Salz.

He smiled softly, whistled a merry tune the likes of which are unknown to this universe. But it was neither rock nor jazz. Perhaps a parallel universally merry tune, Miss thought to herself.

She zipped her purse to fetch the card for the billing. “Here goes your lolly,”, and off she went, away into the largely crowded market with four turns to the right and four to the left.

Mr. Leo sat back against his humble chair smiling to himself. Born of humble origins, nothing much had really influenced him except Roge’s team. He had believed himself to be an orphan though he had not two but three parents, with none to call his own. Too many cooks spoiled our brat, you know.

But ladies are the sophomores in this college of reality. Perhaps the kind glances were sophisticated enough to make their mark on the young dames of the town, but oh, if only she hadn’t pitied or maybe she didn’t?

He shrugged his neck off and draped the V-collar round the aisle of his neck, and got back to work. He enjoyed the work, but he enjoyed playing more. Playing big games.

He had been released twice, had thought of giving up polyrice, but something pulled him back over and over again. He was born to be deadly and venomous. He was born to be a victor of love and establish Roge. He was born to be hard. Hard so the iron would melt as well as the lady’s heart. No, a sweet lady could not change him, could not destroy the merit he had accumulated over the decade, couldn’t win over his vast experience in the big games.

And, indeed, she didn't win over him that warm day made clear by his hasty retreat to the Hall of Roger after the hearty laugh he had caused to be brought to his summons.

He sat across the table with Roge, hung his cap to the bannock's aisle and cocked his legs in the Sulz fashion. Leaning forward, his head brushing past the old dismal lamp, he swung the wallet to Roge who made a show of catching it in the very same fashions.

"Smoke a whisky tonight, won't you, Mr. Prescott?"

"I don't know of anyone who smokes whisky, Mr. Roge."

You'll know three today. Don't worry yourself. A subtle sense of dread being attempted to be overshadowed by an elaborate sense of warmth.

Remember polite is not warm. Mr. Prescott taught himself. He had taught himself all his life. With no one to teach and nowhere to go, he couldn't have been any better with his alphabet than he was as he was seated across the mightiest man of the town.

I have had the privilege to learn upon my observations that you, Mr. Prescott, are an utterly keen observer. Had it not been for this guild, I would have most certainly taken it for a buck.

You had rather unlearned it, Mr. Roge, and unto my own sound memories, I would like to recall with nil confusion the characteristic differences between the two. You place it on a table, place your foot on it. The one that has got the Queen is the guild, your sire.

But I doubt you did it with the Lady in waiting.

I doubt it more. And how do you know the visitor was a lady?

Mr. Roge, blind with his left eye as he was, winked with the right. "I don't think you were a stranger to you, or if you might as well put it as unlikely, may I dare tell you either of the parties are sure to know the other.

In that sense, I must say you have got a better eye than me.

Saying so, hands were shaken and greetings crossed. Both went their ways.

Mr. Prescott found himself a café for the day. With some deal of trouble he went to a seat in the corner, criss-crossed the waiter with some pourboire and there he was posing as a sad man with an even sadder vibe around him. He knew his cards. He knew how to play them to his favour.

On bullying

I stepped back, mildly aghast, mildly agitated and mildly abashed. I could feel the nerves of my hands shake, my skirt slip off my buttock, my voice tremble. I could feel how a girl would feel on losing her brother. The irony being that there was no boy getting lost and I had no brother.

And yet an inherent sense of sisterliness enveloped me as I witnessed the destitution of a boy who had been subjected to ridiculous retortions as he poised his head by the window sill of the bus that took us to school, just as a tall boy jammed his hand across the tiny bow-tie, put his feet across a metre in order to cast a shadow on the scene to veil what he would do next.

I could feel my urge to protect a boy I knew I would never meet . I could feel the boy's urge to be protected. I could feel the speaker's urge to resist the voice of his drowning[CP1] humanity. And, yet again, I could feel my urge to revive it.

This cycle follows me every time I attempt to protect a junior from bullies. And more than once has it led me to trade places with him. I don't really gladly accept the change I'm told to accept as of my own accord, and I find myself abiding by the artificial manners I'm introduced to, since, true as it is, I have nothing to do with the realm of the bully and the child who doesn't belong to me.

And anyone aged ten would be expected to feel grateful about the situation, for c'mon, you were spared from being laughed at or being smashed up for passing with good grades. Any spectator to an event that hosts bullying should feel grateful for his position, and not try to change or upgrade it by attempting to oppose it.

But I mustered my courage to stand up for a cause I would vehemently fight for in the future.

Does my standing up for a cause signal my strength? No! All it signals is the strength of the extent to which I can relate myself to the boy. I have been bullied, and have had a bunch of nice people come to my rescue, and that's what I do-share the love with the world.

I did only want to share the love with the world until the time I was reminded that the boy didn't belong to me. This made me think deeper. This made me think of bullying as a cycle I could no longer forcibly admit myself to. I had been deprived

of my rights to speak up against the rights of another, even if it came of cause. I was suddenly denied admission to a scene. I could only be a viewer. I could only be a spectator to the dance of brutality.

It was not the bullying that kindled the fire in me to alter the social altars of acceptable rights and suppositions. It was not the suffering of a boy 'who didn't belong to me' that humanised me. It was rather my own deprivation of an ungranted right- a right to speak against another's right to speak when it comes of cause. It was to me to bring it to consequence.

That was when a young girl made a conscious decision to become an attorney to be able to speak up for people who didn't belong to her. That was when the girl realised she had to do more than share her love.

A conditional orphan

I had two parents at the time of birth (atleast I was made to believe so) and one a second after, when I opened my eyes.

My parents had had a sizzling romance in their youth. When I was born, my mum was already past the prime years of her life, and the gynaecologist could only admire her determination to have babies without any help from my father, a free thinker and man, idealistically free enough to be able to sacrifice his duty but keep his rights.

My only support was my mum. It was she who taught me the alphabet and the tables and the grammar and the literature as she couldn't afford to send me to a kindergarten. But she had collected some money for my university before I even knew what a uni was. She always told me to read stories with happy endings, and until the third standard (when I happened to read 'Great Expectations' in a corner of my library), I didn't know stories had sad endings too. But I still never enquired my mother of the whereabouts of my father when I learnt that such a post did exist and my office had not yet hired a Father. She had filled the void in my life by fending for me. I never felt the need to have a father.

On 2 April, I was sent off to a reputed school in the fifth standard after my school fee had been submitted. I had learnt that schooling was a privilege of the rich who dwelt in big houses and had big cars, so I assumed that our small house was big and our small dog was the big car. "Perhaps mum taught me wrong by mistake. Let me correct her." I thought as I went up to her when she was frying chips in the scullery and pulling the end of her saree, I spoke, "Mum, small means your hands and big means my hands.", and I put my palm against hers and asked her to see the difference for herself. Mum, on her part, thought I was the one who was wrong, so she corrected me. But then I put forth my argument that since I was about to go to school, we had a big house and not a small one as she had told me. So, undeniably, she had confused small with big and big with small. She didn't argue much though she almost always overpowered me in arguments.

I got up for school the next morning. As the rickshaw puller came to pick me up, Mum kissed and bid me farewell. I replied with a 'Don't you worry. Coming home soon. Goodbye. Mommy is a good girl. Daddy is a bad girl. He didn't wish me.' (as my mum told me later)

A tall creature called ‘a teacher’ got the blanks on my Almanac filled, and then I knew that everyone, and not just Elizabeth, had a father like Mr. Bennett, a father who looked like my Mathematics teacher and had a phone no. which the teacher could call on to complaint against the child. I was incredibly happy at the prospect of not having a father, and thus no one for my teacher to complain against me to. And for the little devil in me, it seemed to be the best deal ever.

I was brilliant, witty, cheeky, pranky but not extroverted. I had few friends to call my own and my Science teacher who I miss to this day mothered me. I got a second mother in her. Being my mum’s friend, she would often console her when my mum would make her appearance in the school with cracks on her soles due to the rough ridings in local crowded tempos during the Parents-Teachers Meetings. My mum could foresee, like a tailor-made fortune-teller, that I had a bleak future. She seldom complained against me to my mum though she took me to the headmistress on account of my really poor handwriting.

The best part about me was not my behaviour but my results. I almost always topped the class and mastered classical literature in class fifth itself. My poetry was submitted to the school magazine and I saw the very same headmistress shaking my limp hand on the stage. That was my first handshake. And no one had taught me the correct way to shake hands, so my clammy cold hands were let go sooner than I had expected.

But I was wayward off-limits, and I confess all my guilt after half a decade, not because I feel guilty but because I’m tired of arguing and the arguments presented by others make no sense to me, so I choose to surrender than retort, to focus on the more important aspects of life, the aspects that touch me and myself alone, and not others.

In one instance, my Science teacher locked me in a room packed with black walls and a black roof with black rafters. Everything seemed so black that I drew my arm across my eyes to see lighter, cooler hues inside the socket of my eyes. That was the first time I sought self-realization, and back then no one expected the arrogant child locked behind bars to embark on a journey in quest for self-realization for life, and I never ever thought of stepping outside my comfy-but-insulting life. But I still was comfy with my life as long as I had my own arms drawn across my eyes. I learnt to mother myself and sought liberty, liberty from hopes and expectations, liberty from glamour and accolades. I sought myself and those light hues, so much so that when Ma’am unlocked the room and dragged me out, I shrugged my shoulders and pulled back softly. “I love my home, please,

Ma'am." I saw the ferocity in her eyes. But it didn't matter to me as long as I was allowed my freedom to be captured in four dark walls.

I was the most popular girl in school because I was not girly, and I encouraged my friends not to be girly. I never cried when punished, always reverently stood up at the back of the class for the whole day, never hit back at teachers but always argued. Yes, I liked to argue for the sake of arguing. My English teacher and I were at cross with each other with him calling me 'Psycho' everyday and me administering arguments about the grammatically incorrect sentences.

When I wrote my first book, it was sent off by the school management to my English teacher for editing, and we all could see his angry irritated face in the class everyday as he focussed more on criticising me than teaching the class. Maybe the editing work was just too much for him. Nonetheless, I didn't fight back 'cause it only gave my book and me an edgy attention with my crush sitting at the backseat and favouring me with a look of appreciation every time the book was mentioned. That was how I learnt to pardon but it took me five years more to learn how to apply the skill.

Then it was the third day of June. There was a surprise writing activity that day, and I was expected to be the winner only until the teacher announced the topic of the activity- 'Fathers and their importance'. How bizarre of a topic! I thought to myself hoping others would feel the same. But when I turned around, I could see little hands scribbling on a piece of paper. I knew who fathers were as I had learnt from my social studies chapter on nuclear and joint families. So, I came up with an idea. I wrote an essay on mothers and then replaced the word 'mother' with 'father' everywhere. I hoped to win as I had completed the task at hand.

During lunchtime in the mess, our group of formal friends held a discussion on the day's activity. I narrated the poem I had written but I could only see mocking faces. As I had no close friend, and I took bullying a little too seriously at times, I started crying and then went away to complain against the bullies to Ma'am. She asked me what the matter was, and by the end of the recitation I could only see her eyes welling with tears. I myself had never cried so much all my life, so I didn't know what made her cry so bad.

Then she explained to me who fathers were, how all ladies marry men who turn up to be the fathers of their babies. And my first reaction was, "So mum also had a crush like me?". I learnt about crushes earlier than I could learn about fathers. Two mysteries unfolded before my eyes- One, that I was not the only Princess

Diana. Second, people can be loved by strangers, to me, my dad was the stranger and I was the person being loved.

“I already know all this, but when all dads care as much as mums do, no more, no less, what’s wrong with the poem or is there any wrong at all?” I questioned with an air of confusion.

“Dads can’t milk feed, and they don’t have the bindi on their foreheads.”

Oh, but why did I guffaw at this statement?

I took Ma’am’s phone to call my mum for ‘discussing a secret’ as I told her. I thought that talking to mum about her crush meant talking in low volume and one-to-one rather than spreading the news of her crush. I learnt about crushes earlier than I could learn about marriages.

“Mommy, why are you hiding my father?”

“Your father is ill, so he’s in the hospital.”

Convincing argument! Additionally, I learnt that hospital is a place for the ill like my father just as fathers are men with a radish-like-beard and with a ferule in hand like my Mathematics teacher. Being talented at Maths, I created the link and interpreted that my Mathematics teacher would end up in the hospital one day. As I really cared about my Maths teacher, I asked him if he was going to the hospital. I think the wise old man knew me better than I knew myself. And he told me that his wife was not pregnant.

So I assumed that pregnancy has got something to do with sending the crush to the hospital, and I was strangely proud of my observations and interpretations until of course I entered the Senior Wing and faced the facts.

But one thing kept me from crying for being fatherless all those years. My mum, her patience and the little knowledge she imparted to me about fathers.

Little knowledge is not so dangerous. This is true. And all fatherless children don’t seek your pity. This is also true.

Mr Raju

If I am asked to note the funniest man I ever chanced across, I will declare him to be Mr. Raju, the sweeper of Block 29. From his dug-out Mohenjodaron silkcaps to his hanging jeopardized overcoat that makes his shadow all-so-clumsy for a human figure, there remains not a piece of clothing that doesn't agree with the charms of his penetrating sense of humour.

The sight of his wide smile and thick broom greets me in the early mornings. He notices me peep through the window and wishes me morning in the most zealously loud tone, loud enough to get me up and going from the comforts of my bed to the dangers of my study-table.

He follows a similar routine in the sunny noons. But this time he is accompanied by an assistant on his mission of discussing political affairs with the handyman at tea.

He was my only piece of support through the constant family skirmishes.

He was the first person to wish me 'Good morning, Madam' in a professional setup and also the first sun of the morning I replied to with a warm 'Bright morning too, Mr. Raju!'

His favourite accomplice was my favourite dog, Doggo- he who had cheated on all the households in the town. Well, before you get me wrong, let me put it more clearly. I was the first girl to give him his first biscuits and okay, he was fair enough to wag his tail the first time (a forced sweet memory). But then I was ditched for a more plausible lover, a chick in my hood. And guess what, he wagged his tail better and faster this time as she fed him biscuits of the same brand. Now, I hope the majority of my readers shall sympathize with me. Hold on!

He cheated that girl for another blond girl and so on. And at the end of it all, he came back to me, and don't you ignore my kind self. I accepted him with all the warmth and grace he 'deserved' just as Mr. Raju came following him through the muddy waters, one foot in the puddle and another on the scurrying Doggo's back, telling me not to feed him anymore as he was getting pumped into a big, fat cow.

So why does the duo also have a strategy to get my hoods up and riding? And this didn't occur once.

On one occasion, I was seated on my bed with the quilt draped around me, and the most ghastly laughter shook me so hard in the bed that I jumped up with a force making my pet bird scream to flight. And guess who that loud guy was? Mr. Raju as ever with the dog following him like, okay well, let me speak out the first thing that strikes my mind with this, your boyfriend creepily chasing you when you are talking to the 'simple, no-pun-intended' boy of your class.

And no, there's a sillier part to this story. Did I tell you that he had come to collect the rotting peels, the issue over which I had fought with him for like three years, three days and three hours? Yet again he was so forgetful though he claims to be a biologist every time I fall ill.) So, of course, he got what he deserved. Another rude reminder that the peels had to go to the cows and not to the pigs. I had something better in store for the pigs, maybe some pitted apricots (and okay, let me confess my guilt. I have been making and breaking this promise for the last three years, three months and three hours).

So, I think I have painted the hero of this note in an adequately negative light before my mum gets yet another nice person to compare me with (results are obvious, so no further self-embarrassment). I hope she'll see he's imperfect too though maybe she will still focus on his positive points. But that's a suffering personal and restricted to me anyways, and my sweet readers should be spared some boredom this morn.

Okay, so now coming to the point, but oh, it's a secret. Drats! Won't you not reveal it to anyone (especially not my mum)?

Okay, so now I hope I got your promise (well, I'd rather hope my assumption won't be wrong just as I messed it up in Euclid's axioms' activity simply for having made the wrong assumption about straight lines and being so poker-unable to be able to contradict it)

Well, I guess I got to remind y'all once again. Remember it's a secret. Okay. A secret. Shh!

So, he visits daily and always at the right time, or maybe the wrong time, but always the same time. I think his discipline would make a CEO jealous some day and he might, well, even lose his job.

And before you check your calendar, let me save your time with a line. It's not Poisson d' Avril today. Is that okay?

But I think if you have been thinking of planning to take revenge, God has done your work for the best kidder out there is shouting yet again with a voice of frequency equal in magnitude to three years, three days and three hours.

It's morn and it's fresh. I'm fresh too but not fresh enough to face him. I'd better sneak out and send someone else, and trust me, writing helps. My pen does me enough favours like a wand would do, so I didn't think of writing an essay on a wand but I'll probably do that now 'cause I've to avoid Mum.

Venting over. Let's drag you and myself to my home, sour home.

I'm not on good terms with anyone. I never have been and I know I never will. Don't ask me how I know I never will. I just know. Somehow. I'm bored of thinking positively as that's what I've been doing for the last fifteen years, and I'm not the 'try 99 times, will reach the 100th time' type 'cause I've nowhere to reach. Not even my school. (It's shut down due to redbumps)

And these same redbumps got me goosebumps. No, I didn't need a lifeline. I just needed myself. And no, I wasn't anxious about anything. I never am (excuse me, an exception needs to be put in or I would be accused of lying the 999th time, and yes, my mum has the complete list and I'm sure she'll reiterate all my sins when editing this essay- I never am anxious other than the times I drop the glass of milk in the middle of the kitchen.) So, umm, you see I needed nothing. And that's the problem. I got everything...cups of advices equal in number to the cups of coffee you spilt in all your childhood plus cups you will spill when you age (okay, I know I'm more excited about aging than my readers) plus all the words contain'd in the sentence(s) I've enclosed in brackets in this essay.

And my biggest problem is that no one respects my needlessness be it the boy in my class who always offers to carry the bundle of books upstairs or my bro who offers me protection every time I step outside. And you can see my younger brother getting mad at me for not mentioning him anywhere in this story, and what can be any more over-the-top than his commanding his elder (me in this unfortunate case) to enlist his encounters with Mr. Raju.

So you see my destitution. And to avoid this 'overfull crisis', I even started doing the thing I hate freaking lot- workouts. Because perhaps I hate looking delicate

even worse. Or even the worst case. Mr. Raju shooing the monkeys away when I'm surrounded by an army of monkeys as I proceed to turn up at the square for Lane 13 after the errands are done. So I thought I had failed my Taekwondo lessons for the right reason and not because of the cranky bunch of crooks who assessed my fighting skills.

I don't know why everyone likes to bother the introvert me. My friends are always up with advice after my tts (Tuesday, Thursday, Saturday) argument with my English teacher is over on my end. Or when I have had my daily morning duel with Mr. Raju and his 'so trustworthy' playmate.

And damn, one of my news reporters became my loudspeaker that Friday. She knew my real purpose. Self-energization. The same one I had defended myself against in an economist's view of self-honour against the chocolate-cream allegation by my brother so eagerly thrust upon the easiest target he could get in seven lives (unfortunately, that's me all the unfortunate times).

And these doses of self-energization work better than glucose tablets on my body. I used to wake up late as a child. Not a major issue, though, at least not for me. But since Mr. Raju started making his appearances after the old sweeper, the face of whom my memory is aloof, left charge, I've discovered my full potential. In fact, I never believed I could work so well, though my mum is courageously persistent with her 'See, I told you. Early to bed, early to rise, makes a man healthy, wealthy and wise.' (sorry I'm not a nursery rhyme foolery baby anymore), I know the real reason behind it. Mr. Raju and the Goosebumps. For those among you who didn't cringe, here is a better title. 'Mr, Raju and the RedBumps'. (I accept COVID is definitely more scary than ghosts, and did I mention there are vacant seats out there for the post?)

Incomplete

A girl and COVID (narration of what a friend shared with me)

Every time I envisage his innocent-coarse visage in the newspaper dated May-3, 2021, an electric-shock runs throughout my body- the very same type that ran through all of me when I foolishly attempted electrolysis of water in my school's Chemistry lab to obtain oxygen for my younger brother.

May-1-2021

I stand outside the Deep COVID-chamber my brother is hospitalized in when I'm pushed aside callously by a coarse workman. His humped back knocks with the chamber walls as he dashes straight to the doctor to seize him. Two men follow. Cops are called and they are pushed out.

May-2-2021

My father visits each of the prayer-houses in the city and makes heavy prayers. I ask mother why he is being so religious. She refuses to answer.

May-3-2021

'Activists call to end oxygen monopoly'- Dainik-Lalsa

I dig deeper. 'Poor dying-cylinders hoarded by Mayor-sold in black for thousands- instance- workman's son dies at Deep Hospital'

Today

Every time I look at my brother I wonder if today again I should hide the newspaper-clip from him. I wonder if he even needs to know the price of his life. I don't think so because his sorrowful realization won't bring a coarse labourer's son back upon the Earth. But maybe so. Because maybe telling him how costly his life is will give rise to yet another Warrior of Justice, and keep other coarse labourers' sons upon the Earth? This, by far, remains the saddest dilemma of my life.

Who is also into building, fixing things like the boys.

People have a hard time categorizing me. I'm not a girl because I act like a boy, and I'm not a boy because I look like a girl. I like to sing but I don't like music. I'm chemmy for a boy and mathsy for a girl. I cry and laugh- all at the same time. I am funny like a boy and kind like a girl. I orate like a boy with the voice of a girl. And I choose Mechanics with the heart of a girl.

It was not the first time a professor, his spectacles poised beyond all mechanical stances, had swayed his head side-by-side, first glancing at the boys and then emanating X-rays to examine me. I nodded. I had enrolled in the Bengaluru Society of Mechanics to investigate aerodynamics for my idea of flying carpet which would sail in the air with the help of flapping wings carefully trapped in between the layers.

Either he was confused if it's a chemistry class or he couldn't take in a girl sit in the forefront of a mechanics class, he certainly did make an impression. "You know you will have to work with wood and maths?". I wanted to shrug my shoulders in the most dramatic fashion to mean a resounding yes. What I did instead was show the hammer scars on my palm.

Growing up, I had a terse relationship with my father. In fact, we didn't have any relationship at all. It was mostly him tearing me down, and me learning body mechanics to fix myself up. My life juggled with planning how to escape from home, settle in Alice's wonderland and care for my little brother. I never blame him for how could an Indian villager not be frustrated of a girl?

One day, when father was away, I bought a balloon, measured the weight a single balloon could carry by hanging items of varying sizes, and calculated the number of balloons needed to fly me away. I purchased more balloons to account for the error due to diffusion that I had calculated. I tied a mile-long piece of thread to a stump, and tied the first balloon. It soared up, and the rope elongated, so I could now tie rest of the balloons one-by-one. Before I could tie myself, I heard dad's footsteps, and so I hurriedly, for fear of being caught, tied a heavy metal tin, and my setup flew away forever.

That night, I suddenly realised that all the beatings over the years had actually made me strong enough to leave my father's kingdom. I decided to escape from the backyard, carrying my little brother. But I could not carry my brother all by myself. And so I needed the milkman's cycle. The next morning when he came, I

carefully examined his cycle to build my own. Sit to work I did when father left for work. I had also planned to hide the cycle. The saw and the blade were tough to deal with but they soon showed me the way to their heart.

However, I also needed light to drive the cycle on the dark unmetalled roads of our village. Since I could not carry a lamp and ride a cycle at the same time, I needed to make one. This unfinished problem was solved by my lucid dream. I tied a metal plate to the seat, plucked the heat pump from a broken AC generator in the mill nearby, and set its key in motion against the paddle. The mechanical work would thus be converted to heat energy and blow the dumble earthen lamps. I arranged my little brother a cradle on the front tripod of the cycle, and off we went merrily to explore the mechanics of life.

The oldest religion on earth is mechanics. Nature taps her feet on Earth with the waves of Mechanics. We own our body because of Mechanics. The intersection of life and death is the insoluble, indivisible Mechanics. The isotropic nature of space, the isobaric space of matter and the isochoric matter of thought lie in the realm of mechanics. There also lies the refuge of a girl and her baby brother in the arms of Mechanics.

I love Mechanics because he's naughty. You have to go as deep as wisdom to find him, as far as the horizon to understand him, and as near as death, to his heart to understand him. He sometimes hides in the hammer of the ironmaster, and sometimes in the verses of The Principia. He is uniform. Both the illiterate and the literate appreciate him as the cause and effect of the ripples in water. Both- a girl with her baby brother on a cycle that can carry them away, and a woman with her son under a rescue operation, that can show them better times, see him as a Messiah.

An ode to my Professor

I can see Professor sneeze through the window. Some of the nose pastes against the ink on his map. He leaves for the lab with the class. I sneak in through the backdoor, not stealthily. I raise my head to thief a look at the map. But I lower it out of guilt.

I have yet not the privilege to rub off the ink that carved la livre formidable. And I sneak out before I'm caught, this time stealthily.

An essay on my self

The Closed Diary of an Open Introvert

I was told I could not bear much hope to survive the repercussions of our extroverted society. The truth in this statement came to be realised by all the remarks I would so easily be endowed with by people of highly extroverted character, starting from my English teacher calling me a 'blank face' to my friend posing an objection on my tendency to think, well, too deeply, beyond records.

With this first-hand hatred, I was only more obliged to appreciate my inner possessions, of which my judgemental nature was an integral part. I observed people deeply, saw through them in and out.

I didn't know the names, I knew the people. I knew that classmate of mine who would not bend even when sleeping, his back straight as a log. If you ask me if I know his name, or if I ever asked my few friends about him, then I'm afraid I would only disappoint you. But I knew that I knew him, watched him, perhaps even better than he ever watched himself. I could just feel what his position in life would be like in the times to come. I could differentiate him from a similar character. I could tell you he would be successful in money-making though he barely managed to pass. I could tell he wasn't 'really spoiled'.

I knew him so well, but I didn't know myself, and that's what concerned the anxious in me so deeply.

If you asked me to describe myself, I would call myself 'introvert and sensitive, yet blank; keen and observant yet shallow; empathetic yet rude; unorganized yet good at planning, and lastly a shabby girl with dissociative identity disorder'.

They all call me rude, and to some extent, I quite believe in it for I may as well not like a person who is my character, not that I don't like myself. Rather I appreciate this vast consensus of bewildering and ever-intriguing mysteries I'm gifted to be the emblem for; my perceived hatred to people is more because of my unsmiling and apparently unforgiving facial structure. You could see me smiling slightly when with the pups, but if you expected me to smile when the cameraman insists upon me to smile or the 'partygan' cheerleader, I would fail you. I cannot visibly brighten up at the idea of meeting a loved one, and many around me doubt if I even love them, and since I never really understand love, I choose to call my habit of showing care devoid of a smile or a sweet appearance as love, since by all

definitions of liberty, I am born free to interpret my emotions in my own ways. I am born free to create and delete the history of my unforgivable social misdeeds.

Despite assuredly feeling liberal enough to do what I deem fit, I find it hard to bring myself some deal of happiness. I can soothe myself into thinking that I have nothing to do with others, and I had rather focus on my self-being, but being eternally sad at heart, such heightened realizations never seek to disown my gloom, and for this reason, I feel I find my happiness in my gloom, and hold it closer to myself.

As a child, I saw this macrocosm as an abode to misery, of which I was an eternal, unswerving part. I was undeterred and determined to be cheerful but sad, and this realization brought to me the saddest contradiction of my life. I had turned into a hole of complicated realizations, navigating my way out of which was a task impossible, and I rolled myself over and over into the little 'simple and sad' world I had created for myself, deriving joy from little bouts of strong pessimism.

And I could feel that simple joy I craved for in all that surrounded me, the empty, the dull, the gloomy. But this came with its cons. I became ignorant of the world as I sought to improvise the inner world I had created for myself. I sought to improve my thoughts more than my actions, and thus became oblivious to what the real world has to offer. I felt no particular joy in obtaining material possessions. I only felt joy in thinking about them. Things didn't have to happen to make me happy. Just thinking about them would delight my soul. Consequently, I lost my ambitions, becoming a secluded philosopher in my own world. I preached myself, thanked myself, did and undid myself. I sought not others to complete my tasks, my ambitions for I ceased to have any.

I thus became a hopeless pessimist, a shrewd judge of character, a keen planner and problem-solver, and a strategist. I came to have three lives intertwined in one: life as a judgementalist, as a dreamer, as an actionist, but lived none.

These essays are about how I live my life as a judgementalist, how I perceive people, how I know them without knowing their names.

WHAT MADE ME A JUDGEMENTALIST?

If I were required to answer it without putting too much into the answer, I would say it was my childhood that shaped the judgementalist in me.

To me, a judgementalist is a shrewd thinker who subjects people non-judgementally to his judgemental analytical skills, a man with a practical approach to the world outside him and yet a stolid approach to the world inside him, a man who treats people in accordance to his theories about them. For instance, a judgementalist would build theories about the collector of his area, and if he occurs to meet him, he would treat him just as he would treat a common man with the same traits he perceives the collector to have. Thus, a judgementalist treats people non-judgementally but perceives them shrewdly. A non-judgementalist, on the other hand, in this instance, would treat the collector just as how collectors are largely supposed to be treated, since he holds no personal or judgemental views about the collector, rather his views, if any, are largely influenced by the perceptions of the society he is a part of.

We all judge people around us, but a judgementalist also treats people according to his perceptions about them, thus separating him from the vast corpus of non-judgemental people.

That is not to say that a judgementalist is always anti-social.

When I was invoked to define my judgemental nature I largely attribute to my childhood, I defined and redefined the primary reasons for the development of this character.

Of all my oratory adventures, the following remains the most sacred.

“And what do you say of bullying?” asked the Governor of the State with a voice that resembled the silent roars of the mighty ocean. All great men talk thin.

I leaned forward, touched the ground, as knights often do, and began my discourse on the topic. Such programs where I was supposed to think as I speak and speak as I think, spontaneously, with not a second in hand to think about the topic developed my spirit of spontaneity. I remember wondering if Socrates and Aristotle were asked to speak in a similar fashion, by people of similar taste.

As the bell rang, my onslaught of thoughts ceased to be mine and became that of the august assemblage gracing the auditorium at Gyandeep Shiksha Bharti in 2018.

Spoke I.

‘On the evening of December 14, 1799 died the Father of America George Washington.’ With pen in hand, I, by way of habit, was striking off words as I read them when...my pen stopped at Washington’s name.

You know that one little drop of blood that replaces that one little drop of sweat when you have four senior girls surrounding you in that solitary nook.

You know what you say when those four senior girls bring in four senior boys and they play dodge-bottle, the metal bottles striking you every now and you wrapping your head with your hands every then.

Yet you don’t say it.

After some days, the four senior boys and the four senior girls bring in another prey.

And you see the bottles strike him every now and you see him wrap his head with his hands every then.

And, suddenly, you realize your eyes are better sense organs than your skin.

And you say it.

I don’t know how many revolutionaries were born this way.

But I do know it to be the story of my origin.

(Before I could continue, the Governor palmed my hand, a signal I had to stop.)

I did not count how many hankies were uprooted or how many people felt an urge to get to the loo, but I did count the number of varied, confounded expressions on the face of a little girl half my stature.

“Didi, this is my story. How come it’s yours too? Did you speak the truth?”

Oh, how were I to explain to the little one that the greater truth was what I had let pass the cave of my thoughts!

I could see the pain-soaked eyes hung on the dark, sweet face. She looked up. I looked down. Time paused. I had many a doppelganger, but she was the real one.

Patting her and with a voice as low as I could keep for the sake of her seemingly soft eardrum, I spoke so she could just hear, 'Little one, it's your story, my story. It's our story. And we will tell it to the world, promise?'

I stretched out my right hand. She didn't have one. I withdrew mine immediately so she doesn't feel she is lacking.

For reasons unknown, I could not sum up the courage to ask her why she was bereft of one.

A haunted college

In this space time fabric, there stands a garden with a buzzing hive and a thousand bees. Three yards away, there stands an oak the trunk of which has been made into a perfect junction by a woodpecker. Every evening, I bring a sack for my bed, a bowl for my food, and wiggle myself into the hollowed-out trunk. My elbow upon my knee, in confirmation with the rules of metaphysics, I imitate Coleridge and pretend to be one of the bees. Once upon a time, when reality bent in, I sneaked into the chamber of the worker bee. Poverty saw I.

And then into the queen's chamber. I saw rich boys drive rich cars, rich girls clad in rich apparel, rich lights illuminating rich spaces. Friends cheer at bars, fork rich cuisines and dance madly. Heaven upon Earth!

As an ill-humoured bee stings me, the metaphysical curtains are unveiled; I recognise myself as neither the queen bee nor the worker bee, as neither a magistrate nor a labourer. I realise I'm but a human who is at times attracted to the glamour of dazzling cities, but the stinging bee in me reminds me where I belong. I belong to my own place, my sack in the nook of the garden. My seat in the nook of the class. My table in the nook of the library. My frame as an observer in this world. I observe the world move, vibe, dance relative to myself. I'm but a note-keeper, silently drawing sketches of bathing beauties. I may greet the magician trying to lure me into the charms of New York, but I belong to the hermitage of the old, wise sage seated on a bark in a secluded nook of the country- the ground-frame wherefrom I can observe glamour and misery alike, magistrates and labourers alike. Is that not the holy seat of realization? Is that not the quiver of all truthful arrows? Am I not an archer who aims his arrow at the bees' chamber as the truth drops down?

I see duality as the biggest trouble here upon the earth. Cases involving two parties can only be solved by a neutral third party- an observer who sits on a holy seat. The nook of the class wherefrom I can observe roars and moans alike is that seat to me. I find myself carried away by the deep and the dark.

But what college in what place would satisfy me with the secluded frame of reference I want? A college in a city of glamour wherefrom all I can observe is the rich? Or a college in the city of misery where from all I can observe is the poor? Or a college in a secluded city where I can be in the hermitage of the old, wise sage and be the third party?

I find myself skimming through the list of colleges. And then I'm reminded of the haunted college. Instantaneously, I reminisce about the story of the cave that swallowed him up with its thousand tongues. I imagine myself being devoured by a dark-mouth cave, being sucked into the thousand branches of knowledge and then being devoured into the darkness of reality, the light of dark truths. In the dark, dark night, I lay on my bed researching about light- this thousand-tongued cave of knowledge. I learn about the snowballs and the haunted library.

Don't the snowballs on the Green charge me with reminiscences of the times of Frost as they rub against one another- times I could myself never throw a snowball at and reach?

Is not The Haunt an archaic library where sit men with braided locks hanging loose, devouring the manuscripts carved by wise, old sages, as an old tungsten lamp burns bright to the right- the perfect direction for the hungry ghost in me? Thus, with my head buried into my shoulders, the white-caped inner ghost in me takes a seat in the corner and opens my book.

The crime of an Indian village girl

You ask what makes me, a humble Indian village girl, dream of going to college, and applying to it. Allow me, Sire, to trace upon the biggest fight in World History- the fight for International Human Rights.

Read it again- International Human Rights.

What painting do these words unveil to you? A vivid French Revolution procession led by Olympe de Gouges? Or one of those elaborate American Women Suffragette carcades led by Elizabeth Cady Stanton?

But to my eyes, O' Man, hits the simple relation of International Human Rights to my elegant right to write this application to Stanford, for had it not been for the international recognition of my humanity, would I, a humble girl- an inhuman servant to inhumane men's commands, have been constitutionally allowed to dream so high so as to apply to a uni of the highest standards?

Had it not been for that one big fight, would this humble girl of colour have possessed this one big right to defend herself against the stigmas that apply to humans who belong to the lineage of femininity and the dynasty of poverty and live in a south-east subcontinent that like all other places known to mankind has had its fair share of gravest injustices wreaking havoc upon its tiniest daughters?

I hail from a land where being born a girl is a crime in itself. But I'd the privilege of being born a repeat offender. Being born a girl and being born poor were the crimes I committed when I was born. Fighting for my Right to Education and my Right to dream big are the crimes I commit now. I have been rightly alleged to be a habitual offender of my society's expectations of me- the ideal Indian girl who cooks, cleans, veils her face and does not dream of studying abroad. I failed them every single time I chose to stand by the Constitution of Equality as opposed to the Constitution of Society. I fail their expectations of me as I fear not to speak, read and write out loud seated in the room where, in a nook, lay my fellow sisters still struggling to be heard, read and recognized. And now they push me aside to the nook. And now I fight back to my seat. And now I extend my hand to my sisters. And now I fail the society. Again.

I doubt how the world will remember me when I fulfil my dreams constantly striving to break off the invisible shackles that captivate my seventeen-year old freedom- as a disobedient rebel or as an outspoken orator? As a girl who

confronted gender and race biases to study a language that was not her mother tongue? Or as that one humble Indian girl who went overseas to study at the uni whose mighty name she had heard of from rich Indian boys? As a dreamer belonging to a land where female foetuses are trampled upon by society's whims before they can even dream? Or as a girl who proved to the world that of great dreams are born great women and of great women are born great leaders?

That qualm, O'Man, Time alone shall clarify.

That fight, O'Man, Time alone shall justify.

The fight that started with that one poor Indian girl who dreamt of going to college but was told she deserved the kitchen.

The fight that ends with this one poor Indian girl who applies to college and is told she deserves it.

My Letter to our Aladdin

My Letter to Our Fishermen

“Honourable remains of the ongoing Dead Sea excavation, happy victims of undecided sleeper cell policies, reverend bedrocks of international democracies, and my dear honourable Fisherman of the plough who hasn’t yet decided whether to dig the sea up or to dig it down because when he digs it down he gets a bottle, and when he digs it up he gets a politician.

Just as the fisherman sits confounded on the democratic shore, the happy fruit of his terse toil pats his back. Now as he turns back, the towering genie thanks him for digging him out, but asks him to say his last prayers as he himself pompously chairs the seat of democracy for another two years.

Unfortunately, our dear fisherman is not smart enough (unlike the fisherman in the 1001 Arabian Nights), and so the genie is able to seal him in the bottle of never-ending poverty. His voice hardly cracks through the air-tight container. And HE is left UNHEARD.

My dear honourable labourers, this is YOUR story. And let me tell you, there is no right way, there is no wrong way to open this bottle. There is no right politics; there is no wrong politics. Because there is no politics. It’s just customer service.

You are a customer in this market of political parties. But you are empowered. You have the choice whether to open the bottle or not. You have the choice to choose the party with the best customer service. So, my dear labourers, use this choice wisely.”

Perhaps cautioning others excites me. Or perhaps it is the idea of internal unity, the elite-common equation that excites me. Vertical integration unites the people; Lateral integration unites the nations. Which equilibrium holds greater importance? Elite-common or American-Indian? Right now, I side with the former since I haven’t really toured the world. But it’s time I learn what presses harder. It’s time I tour the world.

An Idea

You are stuck at work; you don't know why your boss always piles you up with extra work. you have no way to diplomatically solve this problem. Chatgpt terms your problem 'inappropriate'. where do you go?

You have a Math test tomorrow but haven't prepared shit. How do you call up your Math teacher to tell them you cannot come tomorrow. You surely don't wanna score 0. You wanna get the test postponed for you. You cannot ask chatgpt, reddit such questions. They are 'inappropriate'. You cannot hire professionals to answer such questions. What do you do?

You see there are instances in your life when you cannot ask solutions because some questions are termed 'inappropriate'. We seek to provide a remedy for this via our app. Not only this the app will also provide access to people willing to brainstorm for you.

And these people are qualified quirky people who have passed our quirky test (link to it). They are people who could not become consultants because they chose other professions but they are quirky and smart and can find out-of-the-box solutions for your everyday domestic/professional/classroom problems.

This app works in 3 simple steps- you pay \$\$ per question, post it and people approach you on chat with solutions. You approve the solution you like and we tip the person on your behalf. But why will these people give qualified advice- 1. They have to pass a test before joining. 2. They are in for the tip. Unlike on reddit, google etc. you will no longer get boring advice like- you should consult a professional or you should stop thinking of such things.

Best of all, everyone remains anonymous on the platform. The question you post cannot be downloaded, scanned. Since it appears word by word on screen, it's also not possible to take a screenshot or snap a pic using another phone.

You actually have the top-notch quirky mind of the world answering your questions, people who just didn't choose to be professional consultants and do it more as a hobby.

If you find the app useful you can always enrol for the annual subscription.

How the idea of College divides the world/ Collegism and its many sects

Between Albert Einstein and the Wright brothers, there are but two differences. The former banked upon the College river and settled for a hay and grain business while the latter formed the bottom of the college hierarchy ladder thrust upon them. (The other difference is that $E=mc^2$ is boring and aeroplanes are interesting.)

In a world divided between Wrights and Einsteins, I am what I call a 'college agnostic'. For as long as I have known myself, I have wanted to go to college (seeing God) but as a patron of political 'science', I cannot fail to admire the Scientia of college atheism.

31 March or Ivy Day oversees three kinds of faces- the happy, the sad and the confused. If you get in, your lips shape upward. If you don't, they shape downward. If you don't believe in this universal upheaval in the name of college, your lips refuse to arch, popcorn arrives in your hand and your teeth start to munch on college reaction videos.

Why have all the Bunsen burners, Uranuses been discovered by college theists? Maybe because when even Harvard with its orderly state of affairs could not lure them into college atheism, a few classic prop nights could surely not cast them astray. All the fun in your life and mine- the music, the bands, the paparazzi is the gift of college theists. All the bulky books, voluminous pages and cryptic code has been shoved down our throat by college theists in the manner of hand-me-down clothes.

So, dear Americans, while 31 March may mean the 'Ivy Day' to some, it means the cream cheese frosting day to others. While we may be college theists or college atheists, we are all the same (at the end of the year).

And let's not forget that the world was created by someone who didn't go to college.

You are You. College is merely an addend you give your all to acquire. It is never a subtrahend.

The type of college you get subtracts no value from you, my dear friends.

Be the truest form of you.

Mathematicians don't know what they're talking about, & being a practical phil. enthusiast, I like to think our 'race' alone knows what it's talking 'bout. Since I don't find myself apt to shut up the mouth of classmate, Mr. Calculus shielding his inverse trigo function hand with the vector algebra sword sheathed with complex no. rituals (when all I've is a Vibranium Adamantium Aristotle Alloy), I haven't yet advanced on the path to prove how the world is being fooled by Mathematicians.

I will make a hot air balloon to carry myself. I will measure the exact weight one big balloon can carry. I'll buy balloons to carry myself accordingly adding the error due to diffusion/bursting. I will tie a long thread to a stump, tie the first big balloon, and let it soar, then tie more balloons as the thread elongates due to the first balloon. I won't forget to spread carpet on the floor lest I should fall. To be fair, once I actually made a hot air balloon to escape from my house at night, from the roof.

Letters exchanged with the abyss
And letters the abyss wrote back to me.

Stranger, I hope this letter finds you well and in good spirits! I apologize in advance if my writing is a bit wonky, as I am using my pet hamster as a pen. He's quite the talented artist, but his penmanship is a bit difficult to read. As I write this letter, I'm currently sitting in my bathtub surrounded by rubber ducks, trying to decide whether to eat the chocolate bar I have stashed in the soap dish or to save it for a rainy day. It's a tough decision, as you can imagine. Anyway, enough about me. I'd love to hear about you! Are you a dog person or a cat person? Do you prefer pancakes or waffles? Do you have any quirky hobbies or interests that you'd like to share? In the spirit of full disclosure, I should probably tell you that I have a tendency to break into song and dance at random moments, much to the chagrin of those around me. But if you're ever in need of a spontaneous musical number, I'm your man! I hope this letter has brought a smile to your face and brightened your day, even just a little bit. And who knows, maybe we could even become pen pals and continue sharing silly stories and cute anecdotes with each other. Best wishes.

Jan 8, 2023

Hello Miss, today I will be your storyteller.

THE STORY OF YOUTH DORMITORIES

Imagine a society with no formal schooling. Where you don't aspire to be a doctor, engineer, banker when you grow up. Your aspirations are simple. Simple like the society that you belong to. Life's all about, sustaining oneself, and the society, by a combination of horticulture, hunting gathering, and fishing. Your identity doesn't matter, all that matters is your kingroup, the clan that you belong to. The clan totem is a symbol of pride. You exchange goods amongst yourselves, there's no cash or credit card. No lofty dreams or expectations. Life moves at a snail pace.

In this society kids don't live pressure cooker lives. Rather playful lives. Not always parents micromanaging. Staying in a dormitory. Filled with other kids. Learn about the environment, kinship, volunteer for construction and other public functions. Learn about sexuality and are free to practice and explore ourselves. How does it sound?

Two types of such dormitories exist, unisex, among the Muria tribe, called Ghotul. Or different for boys and girls, like Morung and Yo in Konyak tribe.

Usually, they're built on the outskirts of the village to give it a natural sense of privacy and exploration.

Youth collect in the dormitories by evening (after meal) to dance, sing, play, tell each other folktales and later to sleep the night. Categorisation of kids into seniors and juniors gives a personalised sense of bonding. They start at the age of 10, and are a part of the dormitory until they get married. Work includes, volunteer work for marriages, house construction, or harvesting.

Members usually abide by a secret pact, and the elders don't get to know what happens within. Amorous activities are an unofficial norm, and do not need permission from the superiors. In the Muria specifically, older girls train younger boys. As to why incidents of pregnancy are on the lower side, the Muria attribute to a folk tale. 'Youth dormitories are protected by Lingo. So, if there's an unplanned birth in the dormitory, it will bring disgrace to the god.'

It's ironic how modern societies still haven't found ways to deal with stress that adolescents go through. Being sexually repressed, and feeling the need to rebel.

Another incident of how tribes deal it with much better, can be understood from a study conducted in a Samoan tribe. Young adults were accorded the freedom to intermingle and live their lives.

Proselytization by Christianity led to the decline of such beautiful institutions making the tribals look down upon their existing culture and traditions. Good things don't last long huh!

March 10, 2023

Hello Miss

I see you're new to Slowly. Being a Slowly veteran I'd love to show you the ropes.

So, where do we start! You can check out the stamps from my profile, many to collect. Gets one hooked if one's into it. You know, some people like collecting

coins. I know a few. Also, one can collect coins here, and use it to buy stamp packs or attire. You have a generic attire as of yet, while I have a weird one.

I'm sure you would have found out the above excuse of a paragraph boring. Maybe I know your taste.

Why don't we talk about boomerangs! I know it's just me writing a monologue. Anyway!

You see, in English, we have the phrase, an arrow in the dark. Why fire arrows, tell me, when they're liable to be wasted. Plus they're pointy too! They can hurt someone. Throw boomerangs at your opportunities. The boomerang returns back, and brings back what you want with it. If it doesn't, well at least you still have your boomerang.

Have you ever tried making a boomerang? Aborigines of Australia perfected the art. Had they been seafarers like the Europeans, we would have mastered the art, but sadly that was never the case.

Who am I?

Still working on that. I can say I'm a human, struggling with my own demons while taking them on. I put on masks depending on the scenario, act all confident and preach (without the Hallelujah lol), while being afraid inside. Or maybe that's how it should be. Bravery doesn't mean being not afraid, it means standing up to your fears. Something like that yeah!

So tell me, why are you who you are! What makes you, you?

Do you have a quirky fact for me?

Until next time

WiaN

Over n Out

P.S. Introduction letters are usually shorter than average. Just to save efforts in case the other person decides to ghost.

March 13, 2023

Dr. Strange,

As I sit cross-legged in the shade of this flickering lamp, pressing my bosom against the drawer, its coast as rough as India's terrain, a pen almighty graces me with the allure of its design.

This pen is no ordinary pen but one gifted to me by my maternal grandfather on the third Vasant Panchmi of my life. No, I won't write this letter using that pen. Some things, put the wise, are better left unused. Some melodies better left unheard. And some names better left unknown.

My dear stranger, pens are not smooth with me. As I write this piece in useful hope of it touching a piece of yours, the inkwell smears all over my palm like the colours of Holi or the mahr of nikah. But let you not think I am taking pains by letting the colour of my palm strike pale in the presence of so pure an impurity. I am 25% a Rajasthani and 75% a child of the Sand. I have played with Sand in the earlier part of my life and with ink in the latter. I love Sand so much that I divide my life into two phases- BS (Before Sand) and AS (After Sand). Apart from feeding my family and playing the red carpet role in my life, Sand has mothered me for the baby I am. As a child, I chewed pearls buried in the lap of Sand of my ancestral home until, of course, my mother tore me apart from her on the pretext of 'safety'. That was the day the world forced me into oblivion of my very own place and uprooted me from the soul of my culture, my being. Like a mother separated from her child, a damsel made devoid of her beauty and President Carter expelled from his White House, I was drifted apart from the sand of my land.

That was also the day I learnt to sympathise with Jimmy Carter in obvious neglect of the Carter Doctrine. There was a lot in common between us, Jimmy and I. But it's hard to sympathise with someone your mother has already cried for.

What Jimmy was to Mum, MJ is to me. You, my dear, dear friend, will second me when I call Michael Jackson 'the God of Rap'. But, let me tell you, he is also the architect of the soul of the States as we know it today. He was the divine plumber who bandaged the States' flowline with the art of novelty that eventually seeped into the States' soul, or sand, as my Rajasthani ink would like to put it.

My ardent devotion to MJ explains why my room is not melodious. My loyalties lay with MJ and they will, for eternity. You don't wait for the almond plant to grow when you can simply eat the seed.

But don't you confuse me for a perfectionist. As a matter of fact, I pride myself on being an imperfectionist. I just happen to love 'first moves'. Loving the first true rap artist, accepting the first proposal, joining the first college that accepted me, stereotyping on the substratum of the first impression and buying at the first bid price are little firsts I have acknowledged for their position in the electrochemical series. It shouldn't come as a shock given that I'm a student of business and embrace first mover advantages of the first order.

Also, it's hard to be a perfectionist when you get bored very soon. Which is why I now put an end to this letter. Does not the power of eternity lie in breath, imperfection and nobility?

With love, respect and deep admiration,

S.2.

March 15, 2023

I would really like to know why

1. Were you pressing your bosom against the drawer. Pretty sure there are relatively easier ways to get the work done.
2. Is there a flickering lamp? We live in the 21st century. Unless obviously you wish to romanticise getting yourself higher powered glasses.
3. Has the pen got such an alluring design, unless it literally fits perfectly in the palm of your hand, which frankly no pen does, and writers just write that shit to make ordinary folk jealous

Anyway, now that I've got that out of my system, I can begin.

Why are you so unpredictable! In all my 24 years of existence where I've had to consciously figure out to read people, predict their actions, what makes them tick, and adjust my behaviour accordingly, to make them do what I want from them,

you've been the oddball. Humans have always been drawn towards what they cannot understand. So that brings us to where we are.

I do want to highlight the little things, but then like you've left a lot to my imagination, so will I 😊 When it comes to imagination, as kids we have plenty of it. We hypothesize a different world, the what ifs, we ask questions, and seek answers. Sadly as we grow up, we answer our questions in weird ways, we're taught not every question has an answer and ironically that too becomes an answer. One of those questions for me, what was the world beyond the one I lived in. You see my world was two places, home and school. Home, a place of discipline, of following orders, a monotonous world. School was different, free, open where I could test my skills, abilities and ideas.

I also used to imagine what it would be like living at a place which was different than this limited world. The metro cities that I had often heard about. Kids there must be swanky, cooler, less crude than what I was accustomed to. As I grew up, well, I got a taste for it. Delhi for me, was city of duality, the city where unrealised dreams stayed that way. Class division could be blatantly seen. A city which accepted everyone but with scorn and sneer. Weirdly enough, that was what it considered welcoming. Chennai felt more formal. No one jumped to help you if you felt lost among the wide roads. However if you sought help it came to you. The heat was humid, which weighed upon you, as dreams do on a life aspirant. I'm yet to experience the city of dreams, Mumbai. Hoping for the best. However, up until this point of time, Hyderabad, feels the best of the bunch. Friendly, welcoming, there's something for everyone. It relates to everyone, and yet relates with no one, with its distinct identity. I don't think in our country, it can draw any parallels. It can be a role model for any other city in the country with its focus on handling communal harmony, pollution and plethora of dishes to offer.

Speaking about role models, tell me, did you have any growing up? I'm sure you really idolised your dad. But anyone else, that you wanna be like? I remember this question is often asked in schools or by elders in the family and I used to struggle. I found faults in the people around me. Some people who answered, say, freedom fighters, I used to wonder, did these guys know these yesteryear folks to idolise them! I remember then, I decided I'll cherry pick qualities to idolise, instead of just settling for one person. And I have been doing that ever since, trying to be my own person, taking inspirations from anywhere and everywhere I possibly can. Making, breaking and repeat.

Now more about your letter. Do you know when I first read it, I rolled my eyes. It was the descriptive classical English style which over the years I've learnt to avoid, due to its verbose nature. But then I read it later, and again and again. It grows on you. I liked it more and more. Do you know, no one that I know of writes like this. You're unique, like your writing style. I like things that stand out, have an identity of their own, that stand tall despite differences. The use of references, the little tongue in cheeks, imagery, and the flowing nature of the letter were all something I admired. Although I must say, the Carter reference went over my head.

Moreover, MJ isn't the god of rap 😊 He's the king of pop. Rap is where music matters less and it's sorta rhyming more, melody less, imagine Eminem, Tupac, Drake etc. But your emotions on MJ are on point. Give a listen to his, 'They don't care about us'. Great song! I remember watching a tiktok reel, about the Iranian revolution aka hijab protests, which had this song as BGM and man it fit perfectly!

Also, nerding out, these days I've been fascinated with the word juicebox. I think its popularity can be attributed to American public schools where they are often served during lunchtimes. Think of it, it's not a portmanteau, it's just two composite words, that can be joined together. Juice in a box. The J part in the beginning and the X at the end is what gives it its distinction. I'm weird lol.

Question, how weird can you be? Like not the fake ass regular shit that people try to garb as weird? Like juicebox weird XD

That'll be all for today then

S.

Special camel stamp for your Rajasthani identity

March 15, 2023

I received your letter while checking Bed Bath and Beyond's stock price. Now, although I'm a thousand inches away from the Executive Chairman and at least a gazillion light years away from its HQ, my position as a spectator, watching rates fall by 7.26% to a mere 1.15 USD, is no comfortable place to be in.

As you might already have imagined, the agile deer that you are, it is not the best mood to read a letter in, and surely the worst when the letter comes from someone so mischievous.

Your letter is my Hudson Bay. I was, quite inevitably, on the brink of mental bankruptcy when your, dare I say, sarcastic and not-so-agreeable writing came my way. But then you are not exactly known to be a 'yes-man'.

You, my saviour, have flown in at a price- equity. I have paid equity (of my heart :p) in lieu of your \$1B check by subjecting myself to so critical an analysis of my person that I don't swallow in benign temperament.

Hence, I request your permission to vocalise my criticism of your highly displeasing letter. Having obtained your imaginary permission, and under Section 499 of the Indian Constitution, I commence my discourse.

You should know that ladies are not delightfully receptive to being reminded that there are easier ways to get things done. This is like saying to a lady that she should not sleep with her boss to get promoted because there are other ways to get the work done, viz. work hard etc. This is an extreme case of defamation and you shall be convicted of your crime against femininity. By raising arms against feminine weapons, such as pressing bosom against drawer, you have proved your eligibility for the status of a legal offender.

You should know that you are expected to employ your power of imagination and not your prowess of sarcasm when I, very subtly, remind you of a flickering lamp. You are supposed to imagine us together against a candlelight setting and not to remind me that it is the 21st century. Don't you understand? This is dehumanizing and you have successfully failed not to fail all my expectations of you in the romantic sphere. Neither I nor the legal order shall forgive you for this assault against romanticism.

You should know that the object of my affections does not want to be reminded of her intended attempts to make 'regular folks jealous'. You have failed to be appealing and agreeable and this has played a toll on my mental health. I hereby charge you with offence against my person under the Mental Health Act. Given how priceless I am, and the crying session has rendered waste two hours of my life, I also demand compensation. If you are wondering how blur the line between the judge and the lawyer has become, you are commanded to keep shut about it. You say I have left a lot to your imagination, and so will you. This is by no means modest. You are invariably employing lethal intricate designs of letters to fuel my ignorance and take advantage of it. This is a direct case of fraud. You are stealing

words when the emotions rightfully belong to me. If you are wondering how these standards apply only to you and not to me, you are commanded to shut up.

I'm a self-proclaimed taxation lawyer who at times charges, or taxes, convicts, or suitors, on the grounds of mental health and defamation. Civil is not encompassed by my territory, so I will, apologetically, skip your mighty description of India's metros in favour of your faint description of my letter.

Hastily appointing Ingrid Newkirk and Menaka Gandhi as my role models, I now advance to imagine what my letter might have sounded like to you. An extremely subtle appreciation of your intangible assets? Or a very direct expression of my being? An arrow (or a boomerang) shot in futile hope of meeting the target? (which your sarcastic response and leg-pulling have very well proved adequately vain)

In the first instance of its kind in human history, I have been appreciated by Mr S. (aged 24, tall, dark, Capricorn). He expresses that he liked my letter and that I'm unique.

Is that enough to make me happy? Well, I'm sure this is not the kind of question Mr S. bothers to seek an answer to, so I will keep the answer to myself. He is more bothered about pulling my leg in the best possible economical way and discovering the anecdotal history of juicebox.

Thanks for reminding me that MJ is the God of pop, not rap. My ignorance in the field of music will leave you with ample opportunities for correction in future. This is the first you have encountered. I fail music and music fails me. Sigh.

Also, I hope you watched that Tiktok video before Tiktok was banned in India. (You know I'm bad with dates, music, and getting you to say something lovey-dovey.)

I can't toast to your juicebox. Hopefully, you know my limits. Will you forgive me? Or will you sue me in court? If you do, make sure you blur the line between the judge and the advocate :p

For now, I know I'm enchanted by your playful teasing even though I don't wanna accept it. Those are not the kinds of things girls accept, you know. (So you have got to think it didn't impress me.)

Thanks for the stamp. You are lovely, nay, sorry, annoying. Annoying as fuck.
And nosy too.

With love, respect and critical admiration,

S.2.

Slouched in a corner

The man stared at the bars

Dreamy eyes, empty cell, and yet

The body full; covered with scars

This little songbird

Had forgotten how to sing

Possessed the keys to his freedom

Yet wouldn't unfurl its wings

What had changed?

The caged one had a visitor

Regular, incessant, stubborn

A confidante, a friend, a lover

What had changed?

The birdy sung a few nights

Its music had found an admirer

Those gloomy eyes had seen lights

Our admirer wanted more

The prisoner, taken aback

Puzzled how the silence of the cell

Had never said, encore

Promises were made

Verses will be for one alone

After claiming his freedom

The songbird shall atone

S.,

Melancholically optimistic, caused to fission by the fusion of man and woman, separated only by iron-clad serpents raising their ugly heads to bar one hand from touching the other, one embrace from meeting the other, the poem struck me as a hurricane on two legs.

It stifled my throat with overwhelming impressions, the print of its rhythm kissing my neck as a bite would (a love-bite). Just as I ask the verse to drop its strangulating syntax, it sabotages my lips with the advent of its climax.

The three characters in the poem- the singer, the admirer and the messenger(s) (the bars of the prison- so painfully playful, they scream aloof the fraction of time our two principal characters precipitate for want of closure, and lie prostrate when the two characters accept their fate and the only matter that can pass through- voice. They love to make the man desperate and the woman wondering and that's why they are my favourite characters) weave the narrative leaving no wool aside for the tree and the cell. Oh, how safely they fulfil the narrative's hunger!

Rhythmic passion stemming from the complexity of the contrast, waving to and fro the amplitude of an absent kiss but a present longing, seeks no immediate culmination with the death of letters.

I wholly appreciate the poem, S.. My dearth of letters invariably and confidently expresses the surfeit of my sentiments.

S.2.

“Administrator versus an Ivy-League undergrad in the marriage-market?”- this was the heated Discord debate S. and I vehemently broke our goblets over. His consultant-like roundabout arguments slapped against my no-nonsense demeanor. He to me was that disturbing specimen of a human.

As I muted myself clumsily, losing myself to the administrator in the marriage-market, he grabbed the mic grinningly, “Humans, my Miss, fancy what they cannot understand.” Did he just discover that I liked him?

Our quirky differences quickly tied us in an adorable knot. Imagine two martyrs of freedom of expression held captive by art-debate-poetry behind bars of distance. That’s us.

April 29, 2023

S.,

You didn’t reply to my last letter. But you got me a poem and a wound. The poem was lovely and/or the fight was exquisite. I cannot even decide what conjunction to use. If love and fight make a straight chain, let it be ‘and’. If they are intertwined as the bases in the DNA, let it be ‘but’.

Ahh. DNA. It reminds me of Adenine, Guanine, Cytosine, Thymine. I’m sure I spelt them wrong. (I’m the kind of person who would raise her hand as soon as the professor announces that someone did something wrong somewhere. I’m so confident of being wrong.) I’m sure you have been told an octillion times that I’m too lazy to spell-check (by me, of course- Who doth knoweth mine own vices better than I, prithee?). As long as my mistakes don’t knead the world into a pile of flesh, I forgive myself for my mistakes.

But forgiving you is not my thing.

Back to DNA. You know Adenine pairs with Thymine and Guanine with Cytosine. Adenine does not pair with Cytosine nor does Thymine with Guanine. Since you have been a Maths student and I have been a Fearful student, ever

afraid of being called out on PnC, I very humbly add that Adenine also does not pair with Guanine nor does Cytosine with Thymine)

Math is that nasty kid that enters the room the moment its parents are...well, kissing? Making out? Making love? Biting? Kiding? Kidding?

Math makes you deflect from the point. I bet Columbus never applied Math or else he would have reached love? Instead of America. At least that's what Math does to me. Boring Math gives me interesting thoughts. The more boring, the better.

I comfortably credit myself to the discovery of several letters. And since I applied Math, I discovered love in them. In short, I have read several love letters. But since 'love' is not exactly your frequented haunt in the valley of letters, let's call them 'l-letters'. L for lick, okay? Since that does sound like your favourite word.

Now, L-letter writers usually talk about their day or their passion for each other. Why don't I talk about my day in letters? Sharing snippets of Time is not what I'm acquainted with. Snippets of Thought are my nosy friends. Thought would beat Time any day. Character-wise. Weight-wise. Bulk-wise. Heck, even clockwise. I can be lost in thoughts (not of you, all the time, I promise you) and lose time. They tell me I cannot gain lost time back. This is exactly why I don't care about it. Why would I care about something I cannot get back? It could be as mundane as my ex or as ethereal as Time.

Thoughts, ahh...I always get them back. With Interest. Interest compounded @Overthinking% p.a. You know, as a kid I would take things at face value. I would add and subtract pictures. I even invented the Math of Pictures. So, if I were a kid today, I would have begun counting 'OVERTHINKING' to determine the rate. I would have counted it as $1+1+\dots+1$ ($9x$) = 9 or as $15+22+5\dots$ (O is the fifteenth letter of the alphabet and so on...)

Anyway, we were talking about L-letter writers. So, the second question. Why don't I talk about how much I miss you in letters? Get me a mirror and flip this question laterally. Answer it and let me know. I will gladly pull the mirror away. The image of my answer will be derived from the object on your end. This is how optics is used in real life.

Close your eyes and breathe and relax. I have got two things to say to you.

I am proud of what you are putting in to achieve your ambition.
I wonder what the pH of the solution of your sweat and my teardrops will be.
Not-so-verbose this time,

S.2.

April 30, 2023

Thanks for your letter

For someone who proclaims they hate
math, you've tried to show that you
know it. Ironical?

Honestly, your letters are too deep for
me to fathom. | don't even make too
much effort to decipher them.

Obviously, that'll make you flare up your
nostrils and say that | spoilt your mood.

But that's not what my intentions are.

For me, people reveal themselves
naturally. | ask questions and stay
patient and | get the answers anyway.

So why overcomplicate things?

Moreover, right now, | don't have the
time to be picky. | might be more prying

as days go by.

As for your question about missing you.

Do | miss you? Yes and No.

One misses someone when they've

experienced it. | haven't experienced

you IRL so it's difficult for me to miss

you. Rather | anticipate you.

However, I miss the idea of you. You

know for a really long time, in order to

sleep, | think about something

comforting. And for me that's imagining

a person who accepts me, loves me

even and trusts me. | think of such

happy scenarios while | sleep. Weird |

know. So those scenarios you occupy

currently. For a really long time, there

used to be an imaginary person in there.

Now it's no longer so.

FYI Adenine also pairs with Uracil

instead of Thymine in RNA since it lacks

the latter.

Why love is not one of my often-used words? Because it's rare and should be held in high regard. It should be permanent. Love for me is only the romantic kind for a partner who'll stay in the game. For relatives or friends, there are other words like responsibility, affection or brotherhood etc

You say you are proud of my efforts to achieve my ambition. Are you proud that I leave conversations incomplete? Or the fact that I cannot afford adequate time to solve the arguments that arise between us. Think carefully. Only if the answer is still yes and you understand it all without feeling bitter, you actually are proud.

May 1, 2023

Certificate S. designed for me

Footnote- His mood was, as he describes, ‘fucked up’, so I had to pacify him to call me. Since I accomplished his mission, I’m worthy of Indo-Pak, nay, S.2.-S. conciliatory award as a diplomat.

May 3, 2023

S.,

For all musings merge into words and all words break into musings, I define the following heap of literary dust into a colony of anthills.

Anthill #1

It’s tough to write. Of all the verbs in the English grammar, none strikes as delinquent, as disobedient an infant as writing. Its smile is mischievous and I chase it in vain. A child it is. Incomplete sans the shelter of an adult. Writing is incomplete in itself. It needs to knock at your pen, and make you uncap it while playfully eroding you with a flood of thoughts. The art of thinking is our child’s keeper. Responsible, yet fun. The child is the father of man. So is writing the father of thinking. What happens when you really wanna eat that freshly baked cheesecake and it is cheese-hot? You stand in doleful silence, abusing your fate to satisfy your tongue. Both abuses and cheesecakes appeal to our tongues, and as fate would have it, both are readily available.

So is the case with writing. It is painfully slow. Knocking, uncapping...I like it hard and fast. Biking.

Writing is also like the dam- Intrinsically static. A child that knows his aim, it curbs the outpour of your thoughts yet owes its existence to the water. Water presses against it, yet it stands firm upon its foundation- the script, the alphabet. It plays with the ripples of water and allows it to shape itself into the patterns of its script. (Isn’t this how you would laugh at my inability to escape your clutches and yet only allow me to shape myself in you?) So although my thoughts might be bursting open my cerebrum, my emotions ripping apart my heart, the 26-character alphabet never extends to thirty.

And this has been my latest cause of annoyance (though I promise you my nostrils are in perfect shape as I slide this letter under your desk, and although no GPS has

ever been able to determine the distance of my thoughts from my brain, I promise you it is 1343 kms, essentially the distance between Agra and Hyderabad, if Google Maps is to be trusted.)

Anthill #2

Allow me, Sire, to ecstatically transcend my views on marriage. Melville called it an impious thing. I call it an institution that does not have to be built to be brought into existence. Go up into space, look at the earth and you will only be able to see the China Wall. What do you do? You assume that the other buildings though not in your sight are firmly rooted to the soil on the earth. This is what I think of marriage. You don't need to be able to see it to determine if it is there or not. Marriage is in the air, buried in the soil.

Two people could be married with a knot, with a child or with a word. Words are tangible to me. They satisfy the two holy conditions for tangibility. 1. They can be touched. I can touch the ink on the paper and feel it get 'adsorbed' on my skin. I can put my fingers on the screen and feel the stretch of the pixels. Hell, I can transform into Anteman and roam with the words. 2. They have a limit- the alphabet finely restrains them. (I would even attribute captivity to be the major function of the alphabet. Expression comes second.)

Do credit me with the invention of the explanation of the 'marriage with word'. But the credit to its use has long been snatched. I won't attempt to exemplify my statement given that the object of my sandy writing is a student of Anthropology and can well excavate the burials of history to collect the samples of the use case.

Anthill #3

The coup of thought, like most other things in life, comes with a price tag. The price of pain. The tag of tolerance. I think of unhappy affairs to hurt myself. I think of happy affairs to pain myself because I know just well that thinking of happy things will make me think that what I think about things won't manifest into the form of the thing.

Like all other risks in life, I have wanted to minimize pain. Copy-pasting dreams about one person to another appeared to be a potent pain-killer until I met you.

Anthill #4

Let me talk to you about my dad. Of climate change, globalisation, war, politics, philosophy and all things worldly, he is the ruler of my dictionary.

I remember my cousin telling me how her father would massage her feet and palms when she would be cold or ill. My limbs are cold almost all the time, unknown to my father of course. I never received any affectionate telephonic signals from my father. So, I won't call myself a daddy's girl. I'm rather a dad's girl. My father always carried me along to functions when I was small and eager to be shaped. I inherited the art or the weapon or the shield of oratory from him. I would sit by him on the stage, my eyes hungrily devouring the audience. I felt like a horserider, oratory my reign and the masses of my horse, its nostrils inflated, its nature untamed. The horse had 5 sense organs that I had to master. Father taught me how to gallop or trot the horse as time demanded. Father told me I could never satisfy its tastebuds with hay because its diet was too enormous for me to afford, so I had rather satisfied his ears. This is how my stage voice became loud and stern, my eyebrows deep and dense, my smile invisible.

I'm tired having built these four anthills. I will joyously pile the rest of the sand once replete with energy derived from your words of choice and low frequency.

A writer who finds depth in the architecture of verse, I consider myself a hopeless romantic who writes love letters upon love letters to satiate the thirst of my spirit and to delight the man I'm with.

Classical verse and finance are my bae. Alongside, I like to delve into the metaphorical troughs of philosophy and I find myself landing on one of the crests instead. Life is like a ludo game and I'm a dreamer. So, one dream more?

Think of life as the ludo's board. it has 4 phases of eternity, transience, acceptance and reluctance. I represent the end of eternity. My aim is to make it win. It is green. transience is orange. The rising sun is beautiful but the setting sun is magnificent. This is what transience is. It is the setting sun. magnificent while it lasts but it's meant to doom. I keep fun and frolic and friends and love interests in this circle.

Acceptance and reluctance are twins, my girl. you got to be picky to shape the colors and to see who you want in your life. i forget the people i have never wanted, but I forever remember the people I have wanted. My wants are strong and my desires inextinguishable, so i keep them in the box- the blue sphere of the ludo board. They are limitless like the sky.

And then comes reluctance. I have wanted not to give up on people just like you dont wanna give up on your friends; but some things are not in your control. I have had my fair share of picking up pieces from the floor. I have tried to make things work even when they apparently were not meant to. What have I ever gained from this reckless dissipation of my energy? I have lost the ludo game and neglected my side of eternity.

June 20, 2023

S.,

The verses of the Kite Runner are only less painful than the pleasure in my heart upon meeting you.

To the crown of my trunk, it is the eerie silence of my earrings.

To my thirsty lips, it is the taste of distance appropriate between us.

To my veiled milkberries, it is the absence of a willing wind.

To my arms, it is the destiny of the battle amongst luck, chance and fate.

If fate is the merchant of happenings, you and I are but the currency that delivers itself to the destiny of the transaction.

We are only creatures domesticated by the degree our rewards match our actions.

With pleasure as much as can be contain'd in my ovaries and the energy of pain as much as can be held between the capacitor plates of idea and form, I plead your hands to write the fate of this note.

Hello Miss

Glad to make your acquaintance. Receiving this letter made my heart skip a beat, though not literally, because currently in my condition, it can be fatal as you already know.

I appreciate your endeavours to educate this humble human in the ways of pronunciation. I've tried my best to emulate your efforts and attaching the sample of the same. Feel free to correct my ignorance, if any.

Really looking forward to a length correspondence

Yours Truly

Friendly Neighborhood Counselor

S.

The Flower Garden

It was a day of sunshine
When I thought to take a stroll
It was a day of breeze
When I decided, let's explore
My curiosity, the compass
Led me right to this garden
A beautiful house behind it
I wonder who's the warden
Yellow with that centre of black
Dandy sunflowers, swaying on their stalks
Yellow with that centre of black
Petunias, dark, and solid like chalk
The flowerbed with circles concentric
Reminiscent of the layers we wear
One pleasant, bright and gay
The other, proud, sombre, in dismay
The warden hopefully won't mind
If some of her beauties, I pluck
But the symmetry destroyed
On my part, will it be a grave unjust?
I take my leave with a noble thought
Write a few verses, be a bit clever
Do my memory justice;
As,
"a thing of beauty is a joy forever"

Mr. X

I write this for you in austere conscience.

If I were to sketch a Discounted Cash Flow model for the acquisition of my ex as a potential target, he would fail miserably by all accounting principles. Not only was his romantic growth rate as slow as his pace (when it came to our relationship) but his patient attitude required a high dose of hallucinogenic drugs to come to the level of that CEO of Twitter who doesn't pay Google's bills.

Once I tried to calculate my ex's market cap in MS Excel. Unfortunately, I received an error message. On the trading floor of love, our Captain was too busy to have any outstanding shares in the market. (Let me accept that all his heart, veins and arteries had already dried up in quenching the thirst of a girl who remained perennially unamused by all mortal capacities.)

My bygone love was also bad as a stock. (Ah well, I was a bad trader) His share price was as inflated as his ego. The stock exchange rated him as an honest, dedicated, hardworking guy, and his office did too. (I brought an office guy to the house.) The recession revealed he was but a syrup-savoured ornamental stock, the kind you buy for the children's portfolio to boost your capacity to throw money down the drain as you trip up the stairs to your Sunday club. 'I believe in falling,' you mutter as you hold by your Graham rules of investment which your octogenarian self has so proudly come to accept as values and ethics. 'Graham was a great man. I met him in a bookstore.' 'Yes, Sir, great indeed!'

Once, I hired a professional investment manager to navigate the process of trading my tall, muscular holding. He came with many an unrealized loss stemming from his childhood, the manager pointed out sadistically as if to suggest that the reliable market of cat memes and avocado toasts would have been a better fit.

'I told you he is a bad stock.'

'But you also told me I'm a good trader?'

To this day, I curse the app I met him on. I would wish the app curses itself too for having listed him, but why would it? The broker earned his high-frequency trading commission as he left me in peril on the banks of liquid pearls. Amidst the balance sheets of love, my ex's liabilities outweighed his assets, leaving me stranded in a sea of financial heartbreak.

Did I tell you that both Credit Suisse and my ex-honey pass the Fed's stress test revolving around hypothetical ideas of finance and love only to fail in reality? At this point, I'm confused about whether to sympathize with my ex or the bank. There are not many differences between the two.

Now, there are fixed asset stocks, stocks that are not exactly great but know how to pay back. My stock is not one of them. The only fixed asset my ex got is his comical self and no one in our local theatre is ready to hire him. It is also not something he can sell to pay me back.

There are walking-talking stocks that file for bankruptcy and relieve their holders of an aeon-long confusion. My stock is also not one of them. He still believes in his value to the stakeholders and his ability to never be wrong, turn around, divide (me) and conquer.

Let me tell you what my ex was. He was an AT1 security meant not to benefit the holder but to empower the business of ex-banking.

I wish insider trading were legal in dating, or maybe it is. Let's wait until I'm charged for securities fraud on the next date.

August 18, 2023

The Muse

Have you seen hope?

Cuz I have

Have you seen drive?

Cuz I have

It manifests as a woman

No, rather as a girl

A little ball of energy

Though sometimes dull

Swept me off my feet

Took me off guard

I gladly offered what I had

She, remoulded My broken shards

How is it that Naivety and Maturity

Can exist, within one person

How is it that Beauty and Brains

Both in equal proportions

It's safe to say,

I'm in a maze

Exploring which is fun

Occasional terrors do pop up

It'll all be worth it

If she 's the one

A Retort to the Muse

A Retort to the Muse

The tenure of my being

Is the length of his presence

The room of my thoughts

Holds no space for his absence

If the heart were a sensual organ

I would say we rhyme in bed

Because the pace of his trump

Matches the beats of my pump

They say I'm enchanted by all things dead

Be it martyred heroes or the Sardar lad
I hear the whistle of the Death Goddess
In the attitude of this being
Be it the dead emotions
Or the charm of the tears' posthumousness
(Or his refusal to accept grammar's expanse beyond
'liking' and 'wanting'ness)

When I pressed him to write,
I burnt his tolerance clothed.
The iron of my press
Heated his patience pot.

Lovers not in love, We aim not to catch up
With what other refined people hold up.
While I keep count of how many poems he writes
And how many lose their rhythm with time,
He keeps telling me how unsure he is,
Employing dramatic verse.
This is when I declare I'm not sure about him

If he refuses to dwell in our metaverse?

A Vegan's Letter to the Lord of Brutality

My Lord,

I expect of you to, like the Nile, keep cool all through this episode of reading, and rage within yourself lest your tides should unworthily prove futile on me.

By now I expect I have raised your spirits enough to commence this noble piece of writing that intends no conditional harm.

My kind Lord, the most dramatic of verses fail to fully condition your attributes. Since, I too am a deprived creature, deprived of your fragrant lustre, I hereby attempt to dissolve you in a solvent so chemical.

Having been prejudiced in your eyes by your eyes, my Lordship, I would like to commence your criticism since it inevitably turns out to be your most sought-after possession ever.

Since I anticipate the trash anticipating this piece of worthy writing, I would like to claim the unprejudiced authority of this unsound allegation ever so tamed by the reigns of our Wilderness.

Firstly, it's a piece of strict composure and an attempt in the direction of an unabusive criticism purely based on the merit of thy existence.

Secondly, I call it not a writing but a letter, and Your Logical Highness would not like to be deprived of such unconditional hatred for in your terms, the bad is good as well and that's what you imply by your coups.

Thirdly, every sane Lord has read his letter of criticism. He may discard the Dear Jane letters, but these ones charm Him since it takes some real deal of time to scrutinize men better than they would have, themselves.

Having ensured the life of this merciless letter, as merciless as Your Highness, I hereby pledge not to condemn thee in the forefront of hoes.

Destiny hath made thy a controversial claim and thereupon rests the scandal in this cycle. A continuous sense of approval has wreaked havoc upon you. With you raising your ugly head are some utter misfortunes wrested in this garland.

Start counting the beads from either side, you reach the chiral centre.

But, Your Majesty, the side upon which you lay, lies damaged, and the knot can't be mended further. This is you with your sins. Open your eyes, and do that you desire to the garland resting upon your absent mercy.

Your games have been outnumbered, and no longer can you claim to be the better half of all that conspires in the creation. Retrace your Path, O Lord of Brutality, or else you shall lose very warmly to the pleasure of us all.

You invoke no just place for the emblem of injustice can't be justly placed, nor can the emblem of impudence be.

Your uncouth self is but a matter of delight for Mercy for you only add to her beauty uncensored. If you will to success, try some other route.

But, for now you are captivated and on one of the nights of one of the months, 2030, shall you be executed. For now, I being a sincere well-wisher of Your Lordship would like to attach a chamber you may so dwell in. Let it be the Chamber of Revenge.

See you at dusky ten today.

Phoo. Lights on. Have your duvet ready for it remains for fate to tell who shall be chilled out- you or her.

Your sincerest well-wisher,

Chetanya Pandey

That night I sat to compile a letter to myself.

Dear Me,

With the sincerest of wishes, I seek to hope that no harm has fallen your way after the letter was written and posted. The gravity of your fears has been a constant cause of worry as I pondered over the variables of the equation. But not every worry seeks to mercifully abandon the fear of the penultimate before I shall cause myself to worry over the Ultimate.

Nevertheless, with a chance at hand, I penned down the letter and let myself let go. With zero confirmation letters in my inbox, I have been continually worrying myself over depravity, and I need not attach my rudimentary reflexes to the noble proses Life herself asks me to red.

It has taken me some extraordinary amount of time to dwell upon the optimistic trilogy of fate, work and reward, and I seem to be policing myself without fail since the depth is as yet uncertain.

Anyways, I hope for the best in writing and for the worst in our respective minds.

Your worthless servant

5-D Philosophy

That is pentagonal by nature.

Does God exist?

This question dwells in the mind of all beings resting here on planet earth but is not regarded as a question. Maybe the most obvious assumptions aren't questions?

(The real question will be dealt with towards the end but you need to read it whole to appreciate it.)

Tryna objectify Science.

What's Science?

Science means reason expressed in a quantitative order that requires measurement in numbers.

What's Philosophy?

Knowledge of perception and deference to reality in analytical terms forms a conjugate hypothesis that puts forward a form for discussion on the contradictory analytical subjective abstract thesis.

Why is Science Important?

Mainly for practical purposes.

What do we mean by the term 'practical'?

That can be expressed in terms of numbers and dimensions (0 included.)

Why is the practical world important?

'cause it satisfies the real definition of creation. It fulfils its purpose and sustains life.

There are two branches in Science- the Perfect and the Imperfect.

The Perfect branch of science is pure logic and the application of its inductive effect which is none other than Mathematics. The beauty of Science lies in the impractical curves that give rise to the practical infrastructure.

And that's why Maths is beautiful but Physics is magnificent. (will explain later)

The Perfect branch of Science deals with thesis while the imperfect branch deals with real-time thesis. Does that mean reality is imperfect?

Well, that's a question we'll deal with later. For now rear the doubt and feed it.

The Imperfect branch of Science is as rational but it's the very definition of practical though it won't have existed had it not been for Mathematics.

Yes, Physics is imperfect. But Physics is something more than just imperfect. It's magnificent. It's magnificent with all its imperfections. It's magnificent with all the faults in its theories when they all could be turned down with the introduction of the third phenomenon added to the wave-particle duality. With its vast corpus of sheer imperfections, it's all the more beautiful. It enlivens the perfect and sustains our breath.

Do we love the imperfect more?

Maybe. Or else we would have discarded the utilitarian concept of living. I'm amazed at how we grow to love imperfections. I'm amazed at how we constantly thrive to be perfect and in the end grow to love our own imperfections.

And that's what Physics does. Exactly like us. No wonder we love her because she's similar to us. She sounds more real and more practical than her perfect counterpart, and so we love her and accept her beauty as real. That's what makes her magnificent in our eyes. We can connect with her better than we could with The Perfect. We always doubt God as we know he's the ideal of perfect, then shouldn't we doubt the imperfect as well?

Shouldn't we doubt ourselves and the reality of our presence?

We should but we don't. We can't be wholly rational. Physics is not wholly rational. We have learnt to thrive on our imperfect sense organs. And we have learnt to believe whatever they can see, hear, feel, taste, and smell. But we can't believe the Perfect because we can't think in a five-dimensional world.

And these are the limits of Science.

And that's why Philosophy exists- to clarify the doubts Science has. Philosophy is GIGANTIC! It's capable of solving most of the doubts that arise in your mind when you read Science. Does that answer the question- Why great scientists have also been great philosophers?

Before introducing you into the process of thinking spiritually, let me introduce to you the philosophical and hypothetical concept of 5-D Space.

THE IDEA OF THE ABSTRACT-

If you could think beyond what your senses are taught to think and actually 'think' nothing, you enter the five-dimensional space. If you can think of five variables, you have unlocked the doors to this world. If you have derived the five equations to solve these five variables, you have successfully entered it.

The idea of the abstract deals with self-imposed conscious thinking. Thinking about the cause is the first dimension of this space.

Science deals with consequences and the conjectured hypothesis derived by observed and practical phenomena. It tries to find the element that causes the result when Philosophy searches for the cause instead.

Your first approach to deriving an equation for this variable should be hypothetical analytics. For instance, try analyzing death first and then coming to life (reverse engineer the process)

You will face questions like- Why death? This will be followed by- Why are the life-death spheres concurrent? And then, what's the use of life at all if all it has to do is live for a time that doesn't even suffice to make something out of it? Once you answer those questions yourself, you'll look for an authentication. But that's missing in this sphere of Science where even the hypothesis is developed only after authentication in the form of practical address to a problem.

But what if 'the practical' is hypothetical?

This is the second sphere- debating obvious norms of science when science debates the obvious norms of society.

This is the first obvious norm in Science but Science doesn't question it. Why? It's because Science itself is the norm of practicality that's often mistaken as the practical 'reality'. It has no authentication in the sphere of Science, and this second problem in Science is solved by the second dimension of Philosophy.

The Impracticality of the Practical Reality-

Think- How can we prove that what we sense alone is practical, authentic and real? Does that mean a debate with the very definition of 'real'?

REAL- It's like objectifying the purpose of Science, but in 5-D, you are allowed to do it.

Can what we not sense be real? If what I am is real material, can't it coexist as a mere subset of the power set of reality and impracticality? Can't we include the perfect in our definition of reality? Can that which serves the purpose of the realistic Science be impractical itself? Does that mean the imperfect is derived from the perfect? So, can't we, the imperfect, be derived from The PERFECT-GOD? Can the practical be derived from the impractical? So, shall not we consider the impractical as the practical unreal?

We got the answer to the question on the existence of God, this shall help us in solving the following simultaneous equations.

THE IDEA OF DOUBTING

The first two equations raise doubts. And the biggest doubt you sustain is a question mark on the reality of your own existence. After all, we don't really own living proof of our existence. And what if the proof is unreal itself?

Ask yourself- How can I convince myself that my world is real. How? After all, how? By the Big Bang theory? But don't forget the theory believes that matter is real, after all. How do you go about proving that matter is real in the first place? Does it rely on the fact that even time is a mere fabric and vastly different from the reality of the cosmos? Or is Time nothing but movement? But movement of what? You don't know 'what' 'cause it's neither you nor me that sweeps through as time. And yet we say it 'passes'. But how does it pass? Why is there even a notion as time?

THE IDEA OF THE STATIC

Imagine the fabric to be static. Nothing passes. You are you. The frozen you. Time doesn't elapse. What shall happen next? Thinking from a 3-D point of view, there shall be no movement. Then why is time different from movement? Because we don't believe it's just a notion. And yet you need not believe in what I say. It's only a three-dimensional thought process.

But, can we even think in 5-D? How would it be like to think, think, think with no real passage of 'time'? Thinking in the static. The finest equilibrium you would achieve. If everything would frost, even time, and you are the one thinking, thinking out of nowhere, for a 'time' unknown. No passage of 'time'...would there be a passage of thoughts? If your answer is no, you imply that the movement of thoughts is the same as the movement of matter. If your answer is yes, you got the third equation. Both the answers are progressive.

Let's address the second answer first (it's the first one for me, maybe the second one for you).

If you believe that thinking is possible without time, you believe that time is a limitation, and your thought process will be able to 'escape' from time. In case Time indeed is a limitation, you know Death is a reality, so life too must be.

Now the other answer.

If you believe that thinking is impossible without it excising in stages (compare it with the scene when you climb down the stairs- the movement of YOU (a matter) along with the passage of TIME, similarly, when you believe that thinking is impossible without the passage of time, you also believe that thoughts 'flow' just as matter and that movement requires time to flow with itself

This equation can be solved only if Matter is proved as real. So, are thoughts also real? But they are only encoded matter! So, can something that's not coded matter also be real?

This is the idea of the abstract proved by the idea of the static.

Now you are left with two choices- Believing that both God and Matter exist or believing that neither really do exist.

Variables

The kind in me is a lot busy in Himself and doesn't seek to owe me an obligation. I ridicule myself whenever I seek His permission. I'm dependent on Him for my own practices like the labyrinth. And yet I fight for impunity. No one knows this treasure. It lies buried. The grave is fake.

Life is holy, unpolluted and free of filth. It gives no way to dust when realization is sought, but I like to eat dirty cherries. I like imperfections but only if they are my own..

I can but ridicule Him.

I approached a soldier struggling in the battleground- What brings you here, O brave one? Fight, he said. The fight we won.

But you lose. You are dying.

But we won before we died.

I asked a prisoner if he would amend himself when set free.

He didn't seem to care.

About life.

Do I care about myself? About this flesh and this blood. This skeleton tat lies within me. Do I care about what I have to do and what I have to do not?

Yes, I do.

Why do I wander about?

Because that's what I do first.

Religion

I have lived. I have wrestled. Wrestled against a force that neither pulled nor pushed me. And yet wrestle I did. I dragged myself down to betray the foe. I dragged myself down to kill him. I was above these heights, so he pulled me down.

And then he betrayed me.

In the name of religion.

O, Man, you live to die, and you die because you have lived. And you die in the womb because you were not willing to live.

Religion is you in the womb. It dies every time. It kills itself to give way to atheism. But you defend it all over again and push it out through the cervix. And then it takes birth and ruins you again. And then he dies and comes back to the womb and you push him again.

I have seen all sorts of people praise all sorts of deeds on a substratum that doesn't will to live. I have seen people kill in the name of a faith. I was asked to believe in it. I did. But then I realised I couldn't push anymore. I needed energy, and my faith didn't provide even the bare minimum and so I had to flee, and they called me a coward.

I rest my head on mother Earth. She provided for my needs and rendered the energy I once so needed. I play with the deer and we have a good night's sleep. Living in nature is what I deem life. Investing energy is what makes me slip on the peel.

Faith is a non-critic. I have observed him appreciate Religion despite all his flaws, but once I saw him whispering to Religion. Probably, he was reprimanding him in personal for the sake of his self-esteem.

I pushed him again this time, and lost the energy.

I gave in. I was called a coward. Once more.

Philosophy—

He lives in my neighbourhood, and doesn't seem to care about his grades. I have seen Religion bribe him, but he is dumb enough to refuse. He's the most independent of us all, and yet demands independence constantly.

You can see him kicked out of the house any time soon. He drinks all day. And then, for days more than one, he doesn't drink at all. He thinks about thinking, and then dozes off.

Whenever I see him fight with Faith, I hide behind the curtain, and their conversation intrigues me a lot.

Faith (the one who always commences the fight-turned-convo)- You failed so bad in your semesters.

Philosophy keeps quiet. He likes to be quiet.

Faith (trying to sound helpful)- Don't you have books? If no, I can lend.

Philosophy- Thanks, bro. I don't need your books.

Faith tries to reason.

Philosophy- They are plagiarized.

Faith- I bought them recently.

Philosophy smiles.

So do I.

Love

It's hard to resist her, and the good news is you need not resist her. She believes in free availability. I love everyone. She's idle enough to be present everywhere, in everything.

But many do successfully resist her, till the last breath. But once the breath is over, they hear from her again. You can't resist a nymph for seven births.

I have overheard her talking to herself at night.

I didn't quite comprehend her, but learnt she is blind, and remains awake in her sleep and walks to and fro the room when everyone is asleep, with a knife in hand. She is a sleepwalker.

Gratitude

I talk of a song,

A song that echoes itself to flight,

And makes me ponder over its plight.

This song got no place on the space of my lips

As the merry tune confined itself to its own waters,

Satiated with the darkness of the light.

The night that clothes the world puts me at bay

From a place that once on my skin lay.

Perhaps the song's prayers were answered.

And, God in all His mercy taught me not to sing it to it.

The dreary stream complain'd of a broken bond.

The most cherished rivulets the passersby haunt

O, God, in thy mercy, grant 'em some peace!

I have but newly learnt to pray for the song and the geese

As You taught me both- Fortitude and Gratitude.

Woe is me if I forget the merry tune!

For the merriest of encounters all lie dead in a view

And the merriest of views are viewed once in a blue moon!

But my fifteen years could not harness me well.

And the sixteenth year taught me gratitude

My First Student in the Asylum
And he taught me maturity.

I was pseudo-mature. I was the first girl in my school to declare that I would not ever miss my childhood, and many took it in with a chill pill. And now I know how kind they had been to me.

When I told Mum the same, I was duped 'emotionless'. "Thanks," I went ahead, "I wanted a validation."

What's more, I was excessively proud of my pseudo-maturity. Maybe 'cause I was too immature to seek any true realization.

These are some useless facts about me. Now, let's get real, right to the story.

I became a teacher before I became a student. Sigh. The auntie of the colony would get me clients on monthly basis and I was supposed to teach those wimpy kids. Let me be honest. I don't think I love kids. Okay, I love them. Every man with a red pump does. And I too do. But I find them so annoying. So utterly annoying that teaching them is like a well-crafted headache brought unto me by myself. Yet it is something I thoroughly enjoy. I miss the little brown girl in my bus, and her petal-faced brother. Oh, how I would be knocked at my leg by those screammy smartypants every time I clambered the bus! So thirsty for knowledge! And how greedily would the duo assemble around the vast corpus of books I laid before them!

My teachers had been quite opulent with mercy when I was young, an immature bud. But I was so helplessly unable to treat my pupils the same way. And why did I detest them so strongly when I commenced teaching them? I didn't know. But it did make me averse to my own childhood. All those photo boots, those cute pics of our school tour to Kashmir were deleted from a folder named 'Me and my memories'. All those text messages we would send friends became my object of abuse when I was supposed to just laugh them away. Somehow, I came to detest immature people, not just kids, and this went on to such an extent that I came to detest my own immature self.

Finally, I consulted a psychopath. I chose to meet her at night. She welcomed me in the most favourable of fashions with the gentlest of embraces. Our convo went about this way-

I hate the immature now. Even the immature me.

For how long have you been hating your past?

It has been a month. I could take it no more. It was a heavy month. But it didn't take a heavy toll 'cause I wasn't emotionally attached to my past either. I never have been. And I never will be.

Do you love animals?

Absolutely. (I showed her my PETA certified member card placed inside my purse. There is something so soothing about carrying the PETA card as an identity card.)

This is the first case of its type. Most animal lovers love babies too. But you need not worry. You'll be fine, dear. (She spoke with a smile so eloquent. May I dare say, the smile of a doctor is his most appealing apparatus? I seldom smile back except when it's a doctor or if it's a stray. There is something mysterious about their smile.)

Did you ever teach a physically challenged child or a mentally challenged one?

No, Ma'am. Never.

Would you like to?

Perhaps.

Here you go.

She asked a co-worker dressed in a simple cotton saree to let me know about her only son.

Miss Cotton: Ma'am, he is in standard fifth.

(Oh, I didn't exactly know how that mattered to me.)

He studies right here. I want him to learn English.

(I didn't know how I could teach English to her son. But the inner me howled. She was teary-eyed. I could not urge myself not to share her grief. A poor mother of a poor son. A widow at that. With the end of my skirt, I wiped my tears. Oh, how I wish I had curbed those tears!)

Me (stammering): I will. Yes, I will.

(An easy decision to make, a tough one to speak out.)

Doctor: So, you may visit us between 10 am to 3 pm whenever you want.

I left but not without agony. I didn't quite understand why God in His right mind would want to see His children so unhappy.

I reached home and had my share of meals thinking about the asylum all the while. How was I supposed to teach there? What if...? It was a hard nut to crack, even harder than the JEE.

After all those thoughts and nightmares came the most awaited time of the day. I dressed up hurriedly (in a record three minutes) and took an auto to reach the building. Let's not call it an asylum now. The sense of renaissance forbade me from beating my face and ruffling the wardrobe for clothes. Perhaps, my life was changing. But not exactly in the way I had anticipated. And then I felt I was being silly. How could I expect the fruit of my labour without even doing my part?

Now as I look back to those days, I feel so guilty. I long the farishtey to forgive me.

Did I just weep crocodile tears?

As I met Ma'am, she introduced me to the staff of the asylum. It was not like the staff of my school. They were far more sober. Maybe due to the mood. Maybe.

After the welcome ceremony, I was taken to Miss Cotton's room. Her son lay on the right corner of a seat, one that seemed to have been brought from a bus that had seen some very bad days. The colour combination of the room and the furniture immediately repelled my eyes. As I felt giddy, I scurried out and went to a shop that stood at hundred metres from the area. I didn't quite understand why the already sober place had been deprived of an even soberer shed.

I bought a packet of glucose tablets and pampered myself. Three tablets popped right into my mouth. I started back.

I sat at a distance from the boy. I had been taught the basic constitution. I asked him to raise his chin and see me squarely in the eye. He would not do that. I myself raised his chin. I felt that brown fleck in the eye of the boy. Instantly, I knew I had to make someone out of him, someone people would admire for proficiency, revere for intellect. Slowly, I fell in pitiful love with the work I'd been so blessedly declared to complete, a mother's wish I'd been so blissfully assigned to fulfil. For the very first time in life, I devoted my heart, my soul and both my ovaries to what I considered an order of the fairies of paradise.

He was also my first case in the asylum and is doing much better now.

Paeans of Mercy

A classical poem I composed to voice the silent empty wall of my sober messy room.

Speak Cruelty did,

“ ‘Hunger!’, it dragged thee into the plentiful sea of thy wreath

Hath the bolts been hurl’d like a sword drawn upon a sheath?

Nay, shall hanger or strife confer upon Mercy

A note as that bestowed upon Elizabeth by Darcy.

In the hearth most ridicul’d of the dark, dark heavens,

Do scoops lie resting on the shallow lavens,

As they point passionately a note for one to be charmed by?

‘Driveth onward!’ cried the cry just as he passed by.

And innit shall the saddest gloom strive to contrive

For by no means would courage be satiated by a lone life

That brought to his age-old displeasure

The joy of a cool, warm breath

And shone the saddest thought the night of the saddest death.

And spoke it did, loud in the fanciest attire

‘Thy shall forsake thyself’ in the sweetest satire.

Hath he enjoy’d the joy that swung itself across the loom?

And it little profits to end fun on account of doom.

Thro' the gal vale did he proceed to get himself enshrin'd
Doom on doom, pain on pain had his path lin'd
And he could make it all out so well but still he chose to turn deaf
For he had in his mind a goal to which he couldn't faff.
The courage in his refuge wooed out from the bud
As it seated itself aloof and gazed at the trough
As quick as the ruminant brought out the cud.
Set the man on his path, his self so buff.
He swore an oath as the calf lay reclined
And keep he did it as he sang the doleful line
'tis enduring toil landed me in despair
For it was a damage provoked, not I could repair.
I sat beside me and looked large upon the resurgent soul
Who knew no better half than the Mystery dole.
The slumberous sheet of qualm fixed me in my look
As I lacked nothing but a chance to peep it by a book.
Age told on me but was fated not to tell on him
And on went the somber man inimitable and grim.
Thee beckoned him to advance
As the vale lay bordered with palm.

The grievous wound thou had conferred upon him
Was thus a wound sans a balm
But he still did make it!
Whence a man of an ill-used race shall mourn and rave
But there upon he lay bedecked with
An armour of mercy unlike so much the naïve
For thy plot of noble note tho' crafted with thy pen
Cleaved not to him for he was the lion in his den.
And thee bore misery wrought by the self
As he dint lose his homeward way
While thee tried thyself.
The pulchritude and effusiveness was fed no calm by day.
The ones like thee who hath every dreamful ease
Found no satiation with the toil on his grease.
Thee made him advance in a spectacular array
And yet the man countered them all, right in the lay way.
Thy clouds, O, Mystery seem to be lightly curl'd
But mine, nay, have thriftily swirl'd
Up and up, and down and down,
Into the creations of blooming-hues

And, nay, shall I lie careless of mankind in thy rues.

The bribe of sentiment could not but yield to him

As he lay having surpassed the best of courage in man

In keeping with his delightful humour, he shall become a name

And even the inscrutable decree of Providence

Shan't douse the flame.

This utmost bound of a man's lacking in fear

Shall, one day, cease mankind's most terrible conflict in rear-

It's conflict with its own self-

It's conflict with humanity

Which is no mere skirmish.

And the boundaries that I created

Meant not to be o'erstepp'd by man

Are aujoud'hui reduced to the scab of a minor ban.

I slap a tax on all those who seek to outrage

The winding clouds shall blaspheme all the men fed ill

And counter shall it act, counter to their will.

And down shall burst an epic lump of clay

What pleasure can thou have to express it another way?

And thy reprimanding thyself shall force itself upon our glance

Just as la terre shall, one day,
Reckon the void of the employment of the lance
These parcels of the dreadful future rest upon thee
As this hero hath hanken in all his celibacy
To hath the world, with vulgar eyes, open up and see
The Count of Me, The Count of Mercy.
Count on me to come like a ghost to trouble thee tonight
For thee hath kept me shut up in an urn devoid of light.
The echoes call, they are dewy with plenty in thee
Just as this clanging fight shall, eventually, lead on to me.

To speak and to say

though he would no longer fight for me, he fought for animals. He didn't realise that he saw me in them or maybe he did. I didn't see him in me but I could see him in my cause. he was uniting with me. My cause to live became his. No greater love has ever been known. We didn't die for each other, rather we lived for each other's cause. Patience is a piece of marble. Transparent but heavy.

I have spoken less. I have spoken more. But never the same.

Humanity is the force that propels the rocket of virtue in the absence of Faith.

Faith is the most commonly used fuel of this rocket.

Fear lies in wisdom. Only the wise sense danger.

Courage lies in worth. Only the worthy are courageous. It's up to you to be fearful or fearless. Just remember that your fear ends the day you become worthy of being wise.

Speak to yourself. You are the best speaker and your self is the best listener. You both care the most for each other.

Judgement is a policy that supports bookish rules to make real-life decisions, but at times we appreciate books more.

Justice is the unbiased output of a biased process called Judgement.

Do I oppose judgement? Yes. Do I oppose justice? No. So, how can I expect justice without being open to judgement?

Time is the worst lender. He can't give you time enough to decorate yourself.

I would like to learn the art of not repaying a loan from Time, the wittiest borrower.

Humility poisons you when you use it indiscriminately. He is not your best friend.

Your life is in danger if you have been exposing your smartness all this while. Act dumb. You will learn through the process and acquire the weapons to use against your foe when the right time arrives.

Humans don't believe in logic. We create it. We create it every time we prefer contrasting shades that stand out. And we craft the logic for this symbolism. We believe in symbolism and yet tend to refute it. Also, we are frightened by symbolism. Our fear is not logical. We are more scared of a human than an asteroid that might destroy the world. And we crafted a logic for that- relativity.

The being alone you are incapable of compelling, is yourself.

We dwell upon our minds more than they dwell upon our brain.

Sweet words are a bitter reward to evil deeds.

We have wanted to make ourselves obscure since the human eyes look at everything with suspicion.

To believe or not to believe is a question Time alone can answer.

Let them lie upon the ground and graze but let not them be an oblation to suffering.

How would the world be like if we didn't belong to ourselves? Like a cell? No. Like a hole? Yes.

I have contemplated very so often the fate of man. I choose to believe in fate for what has been marked shall occur, and that which had not been marked but had occurred was marked using a lemon a white paper, and you had no flame by your side to look into the imperceptible.

He plays each tune twice, for him and for him self.

The miseries of fellow men are the most sought-after purchases made. It takes some time for us to realise that the present had been purchased with the money zipped in our pocket.

I have learnt all the little smartness I have from Time.

Only the self-taught reach the end of the fool's paradise.

He that conquers his fear commences a quest. He that conquers his fellow men has come after surviving the quest.

It takes the fearless and fearful to conquer both the fearless and the fearful.

You conquer yourself once you realise you fear yourself more than anyone else.

To dream or not to dream are the questions raised by your subconscious mind for the conscious mind dreams daily.

You oppress yourself every time you stand oppression inflicted upon others. You are your most brutal oppressor.

Selflessness is being selfish to yourself when the recipient is a wrong choice.

Ignorance is the grandest gesture. Argument the most pompous.

You move on when you forgive yourself.

Never never let go.

The wise have always tested the waters and they do let go when they are boiling hot.

Displaying gratitude is an art. You could be blessing them and being cursed at the same time. Ignorance sucks.

Man is the paid merchant in life's business.

We don't really know the origin of goodness.

In the end, the good die too.

The evil and the noble are not deprived of shoulders to raise their coffins.

And yet, we choose to be good for we don't want the lifters to hurry up while lifting our coffin.

A foe can never treacherous, but a friend can be.

If you make rules for yourself, you are a knowledgeable untamed creature.

If you make rules for others, you are an ignorant fool.

My pride is my prejudice.

I learn to hate that which hates what I learnt to love.

Optimism is the Creator of the Universe, Pessimism the sustainer and the ones that oppose Pessimism are the destroyers for they kill the sustainer.

You start hating the world when you realise it is practical and you are not.

If you are always right, you have unfollowed yourself in this network.

If you believe you are always right, you never built a network.

Giving in to your urge is not an emblem of self-love for the same man who gives in to his urge hangs himself.

If it was proved, no heaven rested upon this planet, we all should choose to die by ourselves.

We like to follow the nature's algorithm.

If you don't trust yourself, abiding by logic, you can't hang yourself since it takes some deal of trust to trust our own hands more than those of others.

If you trust yourself, you would never think of hanging yourself.

Self-trust is a two-dimensional irony.

I rise, I rise, I rise and then I fall, and then I choose to rise to a merrier tune.

Given the boon to choose a friend, I shall choose the Universe as a whole, since some shall cause me to fall and others shall cause me to rise.

Forgiveness is not a noble deed in my volume of coups.

One's confidence is one's validation of oneself.

We like to be free of worries, and this new-found liking worries us all the more.

Be the yourself you would like, no matter good or bad.

Being bad is not wrong as long as you are not bad to others, and being wrong is not wrong as long as you don't wrong yourself.

I'm the one who tells people the half of things; when they wanna know the other half, I rephrase the first half. aka a finance gal

I bring to life a life inside me

As I accompany myself through the domains of passionate struggle

As I sink into the earth filled to the brim with people

I hold distant from myself.

As I attempt to nurse the wounds I was left to heal all by myself, the wounds that terrify and beseech the emptiness in the depth of my skin the pain of my mind awaits me...just as the woodpecker pecks the woods of the tree her nest lays in.

As I curse my creator for having created me imperfectly, for giving me but one heart when I had wanted two, for what better companion to the darkest places I seek refuge within for my unholy spirit, what better companion to the shallowest nightly I long to live?

She was a painting of sorrow, struggle her master creator. I looked at her, treasured her terror-stricken beauty. Her sorrow was different things to different people. To her, it was her pain. To her creator, it was the paint. To me, it was her beauty. She dwelled in my heart like a vintage painting dwells in the museum. I could bid for her but could never buy her.

I have for long, by the power of my associations, owed delight to poetry.

He was too manly to understand the viles of womanhood.

My mind sways in wanton forgetfulness of the state of its being.

The government is the great capitalist in a socialist nation.

Since times immemorial, the woman has been denied the right to celebrate her femininity. Why doesn't the world have an exception for the man? Why do exceptions exist only for the woman, who after ages of inflicted torments, has finally learnt, or so I say, grown to take her stand in the court.

Sometimes you have to love a man to become a feminist.

And peep into a butcherhouse to become a vegan.

I have done both. I'm both.

Sometimes all the world needs of you is your evolution.

An ignorant non-vegetarian to a level-headed vegan.

I have learnt to recognise equality both horizontally and vertically. Vertically between the rich and the poor. Horizontally between man and woman as a feminist, between man and animals as a vegan.

Every true feminist is a vegan for feminism remains the mother of veganism as evident by the commencement of the vegan movement at the onslaught of feminism.

Yesterday men called us feminists. Today they call us feminazis. Tomorrow they will call us female chauvinists. But my dear ladies, we all know that the philosophy remains the same. So does our beauty, strength and the power of our love.

Girls, let's celebrate our femininity. Spread kindness as vegans, hope and strength as feminists.

When you listen to Taylor Swift on your breakup day, you know you are a feminist.

Veganism is a state of realisation not a character. You 'realise' it when you refuse to accept the bowl of brutality and masked murder passed on to you amidst the soft flowers placed elegantly in a vase on the dining table. The day you refuse to accept the position of flowers in the vase and animals in the bowl as it is, you become a human, a level-headed human.

Let's not deprive the woman the right to absolutely command her own femininity. Wives are women first, let's not bring ourselves to forgetting it. Let there be no separate rules for the night and the day.

Everyone is either a leftist or a rightist.

STRUGGLE

If there is something that I hold very close to myself, it's struggle. Not because I don't like to reap the fruit of my actions but because my actions don't deserve it and I intend to improve them every single time. But if you guess I'm proud of failures, you guessed it wrong. I'm only proud of the struggle I put in that failure. As for the failure itself, I do what everyone does. I doze off. And I think that's what everyone should do 'cause if you try to be successful the next time, in case you fail yet again, your self-esteem might be affected and chances are you will lose your motivation and then will search for external illumination.

But if you try to feel proud of all the labour you put in the task, you will question yourself- Is there really something worth being proud? Did I put in efforts enough to be proud of? Or am I a fool- proud for nothing?

Once you have done that, you have got your motivation. Why? Because the day you question your labour and not the reward, you know you have aced it.

There are times when I feel it's the end of my world. But then I think of it as a tunnel with a light at end. And en I think it's just one door before that ultimate door. And then I know what I climbed all this while were not stairs, but struggles as not every struggle drives you to your dream. This is the harsh reality of life, something I don't wish to conceal from y'all.

So what shall you do in such circumstances? Sit back 'cause the struggle is useless? And what if I say it's not useless? Maybe, it won't drive you to your dream, but it may drive you to your destination. Ever thought of that?

Let me prove this by my own example.

I have always been interested in Science and yet I took to oratory. Simply because destiny paved this path for me. I got no practical motivation for science in my early years, and I didn't struggle for one either. I knew I could live my dream in future. But your dreams are incorrect if you dream of them at the wrong time. Keep your hands open. You are not always meant to be what you dream of.

So, should you stop believing in the power of your dreams?

Yes. But again, only at the right time. The time you get hold of an opportunity second to your interests. Only for the time being. Keep it somewhere in your heart. And as you grow, you'll know true love is actually a fake idea. Your mind can always dream twice. Your heart has four chambers, so can't it easily store two dreams?

In case you find your second 'dream' boring, you shall find your first dream boring too once you start working on it. It's a fun fact that we lose interest in the hobbies that earn. Maybe because almost all earthlings have to 'work' to earn, so our mind doesn't digest the fact. In fact, chances are you'll find your second 'dream' easier to achieve 'cause it wasn't of your interest initially, and you'll always take it as work.

When The Villager Sneezes...

The rain pitter-patters, the maize sways in delight, coarse labourers with even coarser hands winnow and wallow in the chaff, just as Urban India crunches an extra roll on a page of the Hindu.

When rain governs life and droughts govern wages, the celebration of rural development remains vastly uncelebrated. I know you are busy today. You have a meeting at 9 am and another at 10. But, somewhere to the right of your luxurious flat, and left of the shopping mall, lays a newly metalled road, which I promise you, my Sire, is worth your two minutes.

If you walk down the road, you can see an arena that brings to life the very spirit of creativity. You can see ladies decorate Dharti Maa as their husbands sit on cots actively discussing affairs of national importance. Saris are still saris, and kurtas still kurtas. The only item replaced is the rolling pin. It has now been replaced with the book.

I don't know how important a cyber café is to you, but I can see a girl named Heena. She is about my age (16) and is dressed in ghaghara, trying to oppose the gale of contradictory views that erupts from the mouths of her chachas and taus with every passing second. I'm not very sure why she is not making eye contact with you, forgive me, my Sire. I would like to think she is busy with her school, err, science project, and is about to go to the cyber café to get it typed. But, maybe, she is just 'shy'- like the metaphorical village girl.

When we, you and me, see someone so quietly poised against the hustle of the day, so silently engrossed in a mission unknown, curiosity to know the reality arises from within, isn't it?

So, here is the reality, my Sire. I shall trouble you no more with my perturbing antics. Allow me to say the truth, but the truth. And, surprise! This truth is no bitter candy!

GOOD NEWS

After 75 years of Indian Independence, we- you and me- can finally congratulate ourselves on the changes brought to our villages. Rural India continues to be the heart of the country. These 75 years have proved to be India's paediatrician, and have vastly operated on the heart of India.

In my research on rural India, I have usually referred to the village as 'hybrid', since this is where the hands that harvest the crops fall short not of using pen and paper. This amalgam of culture and education and technology has led to great agricultural development and has greatly catered to both national and natural difficulties.

This is not to say, nor to mean, that Rural India is limited to agriculture or the workforce. Rather, it is aggressively progressing on the path parallel to that of Urban India with an even higher velocity at some junctures. At the same time, it continues to carry forward the true spirit of India- the true Indianness, with a modern flavour. If there is something that has changed, withstanding national quakes with financial foci, it, sans any qualms, is Rural India. Thanks to research projects, fierce legislative procedures and the desi will.

WARNING!

Let's not stop at this. With an even stronger will, let's seek to propagate the waves of development uniformly. Future research shall hence be based on this very spirit- the will to make a rural girl with a book in her hands a commonplace.

And the next time, my Sire, I promise you shall see many more Heenas in my village, and one of them will not feel shy.

The Tank

After seven years

I was always the high-collared girl with the high-collared tee on. The same girl who was handed down a silent, secret, 'not-so-confidential' note by one of her classmates. The same girl who was called to Room No. 71 for a discussion on her, about her. The very same girl who had been cursed, abused, ragged for having slept with all the boys in the school.

True, Ms. High-Collar never missed a boy. And yet the boy she missed purposely missed her. Because he was the jerk of the school, the head of the basketball team, officially the hottest guy in the school. And he loved a girl- the military- half Miss' height and high-collar standard. And for a while, the hot in her had heated and humbled, heated due to agony, and humbled due to her then current destitution. She had accepted her contender as the hottest girl in the school when a month ago she was not to be seen anywhere in the lists.

My then best friend had drolly remarked to me, "The witch in the yard has had her kettle fluffed.", and I had planted a slap right across her butterfly-utterly soft left cheek only to apologise later.

But the Ms. High Collar in me couldn't get over her failure in romance. The confident and witty in me gave way to forced humility and deliberate ignorance. And for a while no one in Lane 29 believed that Miss High Collar had been passed on a rejection and it was even duped a rumour.

Oh, how much I had prayed for the two to separate, those witchy strategies to separate the two lovers, crafted in the midst of night in our shared hostel room by the window that saw a perfect spot for all of your nightouts, reverently referred to as an 'Archaeological Site'!

And now the same girl had given up all hopes and was engaged with writing with her spectacles on, in a square-shaped room of a dampened house in Lane 36. This change in Lane 29 to Lane 36 had taken 7-digit long years, a time slot I hadn't spent drinking and smoking with the Drunkards of The Bar House or with the Divas of the Nightouts. This time had been spent in writing to warily vent out the darkest in me, a side which even my dearest friend was, at least as I would like to think, in elaborate ignorance of.

In the middle school year, on my birthday in November, I even compiled a list of all the traits I hated about the boy, and the bucket list but pointed out the already known fact- he was a six-figure jerk with a three-figure net worth. But the next day I again saw myself locked in the room and sitting by the window borrowing a glance from Time Unknown into one of the sweet romantic hotspots in that romantic-cum-archaeological site. I could see myself in his arms being carried away to the inner dormitory for a surprise ball party in which we would dance together. “Oh, Miss High Collar in a rolly dreary roller.”, and here she was, ready to press me against the bosom of the jerk with a jerk so sudden that it would get me wishing against wish. But she knew my darkest side without my having to reveal it. “You know it, Ale.” She naughtily threw her two-year old lizard socket onto my neck and I saw myself crying and wailing and begging for pity the next five minutes as the slimy toy slipped across my soft pink-toned skin. “Your life is like this liz.” I gave her a warm hug and then we shut up our quacking mouths ourselves without the help of Miss Jackson, and then we laughed our collars out and looked back at those old collar days when we would sit in the kennel of her dog colouring and collecting stamps and coins- both of us together without saying to each other a word belonging to the earthly languages.. And for a while, I felt my mood uplift and my passion for life arouse again after all those dismal years.

And now I was happy all over again, ready to flaunt my accomplishments in that perfect saga of bragging, sipping a tamarind juice as I sat working on a research paper on the rocket chair in my apartment, earning considerably more than any of my colleagues. Again the original Miss High Collar with the white collar job- in pieces all over again.

I was chewing the tip of my ball pen of an unknown company (I always used cheap pens) when I remembered Miss Jackson and her eyes sunken into a socket wherefrom she would operate the class. I had been her pet for a good three years after which she left the school due to a confrontation with the Father. She was also the first teacher I, the shyest girl in the hostel, had been so frank with. Despite my shy advances to the cosmos of classical English and the obvious ridicule, I didn't fall shy of being adored by a teacher so menacingly dangerous to the risqué.

Oh, how happily loved I was in those days! No, the leader of the class is not the most loved creature on earth. My English teacher in junior high school never missed grabbing an opportunity to humiliate me or reprimand me for reasons as silly as arguing. And wasn't I the only girl made to sweep the floors of the storey on which our class lay, by the cricket ground? And yet I faced it all, and had the

childish privilege of being a subject to such silly accusations, and what's more my so-called friend offered me a cool tip on not arguing with preceptors as if I was the off-limits fille of the school. What an absolutely nauseous disruption into the privacy of my life! And why should I not be allowed to open up? (And these are the words of the girl who was officially the shyest just a year down the memory lane, I thought with a sense of self-pity as I reminisced the rejection one of my articles had faced just because of being 'provocative'.)

I was just banking on the notepad thinking about my next point against my old-school teachers for self-defence,

WHEN my phone rang up and the tune my friend had set just the previous Sunday sounded and the alarm buzzed off. "Not a damned phone call in all these five days," I spoke to myself as I got down the stairs to answer the call. I was called up rarely 'cause I was not much of 'that rockety dress-up haul in that shimmering glamorous attire with that forced-and-sweet smile on her face posing for a photograph with a stranger the same night she had a breakup with her boyfriend of seven years'. And here I was popular, but not glamorous, smart but not coquettish, turned so ascribable to overthinking about the future, and all I got was requests for autographs in my inbox, and all I did to them was sign using canvas.

I anticipated no milkman for I had turned vegan due to humanitarian and environmental causes. "The only possible candidate could be the publisher's delivery boy who always forgets to bring his phone along and has to turn up at the chemist's to call me up and that when I bought that sleepy thing a brand novel smartphone which he had pocketed with a sudden abruption as though I should be snatching it away the next moment.", I thought to myself slyly as I slammed the door shut behind me and walked past the huge window that stood hanging half in mid air in the chamber. I rummaged about the room for the keys only to realise I had forgotten where I placed them the last time I opened the door for the delivery boy. I shouted out to him in my usual stern Miss High Collar voice, "Have lost the keys and the second pair is upstairs. Will take some time. Place the bundles, if heavy, on the ground and sit down on the Windsor chair placed near the cupboard with your favorite mirror design on it."

I climbed up the stairs and into the room. I checked the bartender's box, the remote set, the bannock basket that had been my first gift from my first (and only) boyfriend on our first month anniversary. I checked the slit in the window pane, the crystal pen holder, the trashcan my friend and I had worked on for the Best out of Waste first prize, and the purse that contained a cent and one of mum's

sweetest pics I would proudly show to friends in school days saying, ‘She’s my mum.’ “Oh, I will call her sometime soon after the book is complete!” I promised myself.

Then I searched the table that had the delivery package delivered the week before. “Oh, my silly shit. I had to check the package.” They would change the contents monthly to allure the buyers. I went to the scullery to get a sharp-edged knife to cut open the cardboard box. As I drew the sharp knife, a piece of paper got in to the way. “Perhaps he wants the dues but I have already paid for October. What an old-fashioned way to ask for lolly!”, I was fuming by the time I read the first two words of the one-lined crisp note, “Miss High Collar,”. “No, the publisher can’t have a hang of my byname. Oh, is it my pseudonym? Oh, no, then what the heck is this chit about? Is it 1 April today?” I checked the calendar placed on the desk with my left hand as I held the knife and the note in the right. “Well, hope no one thinks of me as the October fool.”, I leaned against the desk with the note in my hand and commenced the process.

“Miss White Collar, meet you on October 1 at your apartment. The Tank.”

“Oh, dear me! Who’s this Tank? And how does anyone know I live here?”, and then I smiled at an old memory, “Oo, ooh, he was a teen who will always dwell in his teens.” His way of presenting himself was so amusing. I never called him by a sobriquet. Just once had he casually poked fun at me in the café when I dropped the tank of the coffee. That was when he had cracked the poorest joke of all times, “Here’s your second tank, Miss White Collar.” referring to my favourite singer, Durrell Babbs- The First Tank of my life. “Then you should be the third one.”, we had shared our first hearty laugh, and it wasn’t even a date. It was just a casual one-to-one hangout after the second semester in a dusty inn, and I never understood why chose that compact place for hanging out.

So here he was. Before searching further, I dressed up my pony and put on the highest heels I could find in the room but I didn’t beat my face. All those years had taught me to live without looking ace round the clock for I had no one to impress, no one to show my face to. I thought of using the mascara but had forgotten where the little red box was placed as it had been neglected all these years. “But boys like no-makeup girls.”, I chewed my lips with the thought, and rushed downstairs with no keys and no sense of romance. Writing detective fiction had somehow made me anti-romantic. I hadn’t even thought of him for the last many, many months owing to my hectic work schedule. And here he was, hanging out on my gate. What a strategy! Make a girl cry for you for years and

then surprise her in the middle of a humid afternoon with a guitar in hand! And then boys say girls are incomprehensible as if they themselves are one-laned at pace.

“I have lost the keys, Sir.”, I spoke up abruptly from the room with the door serving as a bulkhead between us. He was the guy I turned to when drowned with problems, the problematic whore that I am. And now, after all these years, I still felt safe with him. He would always be my hero- the guy who would solve all my sums, fix all my problems, find all my keys.

“No worries if you allow me to open it.”

“What! Who gave you the keys?” I shouted placing my hands on my face in anguish.

“Cool down, Miss White Collar. The jerk is here.”

“Is it locked from outside?”

“That’s a mystery for your next piece.”

“Okay, step in.”

He opened the door as I guffawed in the midst of a (serious?) predicament. He came straight to me, “Shan’t a Sherlock Holmes be surprised this much at least, if not more?”

I think he had somehow learnt that I had lately been into detective fiction.

“Sure, Dr. John Watson. Thanks for your surprise.”

He held me tightly in his embrace, “I owe you many more surprises....”

“But the delivery boy?”

“As I said, that’s a mystery for your next piece.

Some thoughts on philosophical texts:

1. Our actions, conscious & unconscious, are influenced directly by collective cognitive mechanisms that help us respond or partake in myriad social occasions effortlessly in familiar environments.

Affective friction is the psychological unease that arises due to mismatch between norm psychology and the actual norms of the current environment.

Affective friction can happen both in known and unknown environments as time changes. For instance, the norm psychology of a person could disagree with their own culture. The gap creates -ve emotional signals while knowing the local norms makes one feel like a 'member' of the community.

A notable 'feature', if I will, of norms is that a norm that feels annoying doesn't imply that the norm is not an improvement from the existing norm! And those who find themselves privileged owing

to the current norms in their community will roll their eyes all the more on being introduced to this new norm that promises to threaten them (which is also why eyeball heuristic is not a good parameter to judge a norm.)

Affective friction can strike close to home when one's ^{own territory feels like a} foreign setting due to rapid social changes that disrupt norms in one's own culture. An example is generational shifts around gender in language.

2. Yuk evokes disgust (e.g. cannibalism, incest) (intuitive feelings of disgust). Disgust reactions are evolutionary protections. There are some good biases in such reactions but sometimes they are applied even though our current situations are not disadvantageous. Human non-human animals was an example given to explain why we shouldn't necessarily rely on intuitions. Prejudices can be connected to yuk reactions.

Needless to say, combating our intuitive yuk reactions requires stepping into a territory of discomfort. I do believe that as a larger society, it's important that yuk reactions not become laws. Homosexuality is a strong case for it. I agree with Darwinian in that there shouldn't be a democratic approach to ethics.

Kelly also made a great point about how eyeball heuristic or the feeling of irritation shouldn't triumph norm corrections. I think we cannot collectively give our feelings of irritation, ^(as with vegan preachers) or disgust ^{actually} the privilege to dictate future norms. What's ^{actually} wrong or what's right is a more important case (and not what our feelings or 'yuk' reactions tell us is right or wrong).

3. Central theme: If it is in our power to prevent something bad from happening....
More practical conclusion: People in rich countries should give until they reach the level of marginal.... by their gift.

The conclusion is that the traditional moral framework of affluent societies (which treats the giving of resources to needy societies as an act of charity and not a moral obligation) is unjustifiable. If it's in our power to prevent something very bad from taking place (without the sacrifice of anything of comparable moral importance we ought, morally, to do it.) The current distinction between duty and charity cannot be morally sustained. A change in the very moral conceptual scheme is needed.

the way people in relatively affluent countries react to a situation like that in Bengal cannot be justified. Indeed the whole way we look at moral issues — our moral conceptual scheme — needs to be altered, and with it, the way of life that has come to be taken for granted in our society.

4. Strong version: "If it is in our power to prevent something very bad from happening, without sacrificing anything morally significant, we ought, morally, to ~~do~~ it."

Moderate version: "If it is in our power to prevent something very bad from happening, without sacrificing anything of comparable moral importance, we ought, morally, to do it."

Key differences

Strong version: We need only to avoid sacrificing anything "morally significant".

Moderate version: We only need to avoid sacrificing anything of 'comparable moral importance' to the suffering we're preventing.

Differing conclusions:

Strong version: Buying luxuries is rarely 'morally significant' so accordingly to this we are ^{obligated} to give away anything beyond basic necessities. We should live very modestly and donate the vast majority of our income.

Moderate version: we would need to give away our own wealth until we reach a stage where further giving would cause us suffering comparable to what we're relieving. We might be able to keep more for ourselves here than in the strong version.

5. Singer argues that physical proximity is morally irrelevant. He uses the drowning child analogy and that morally, one is as much obligated to save a drowning child in another continent as if the child is in front of one's eyes.

Also, whether you are the only one who could help or one of million who could help doesn't alter the magnitude of your moral obligation. That others could also help is no excuse. Also, that the needy are strangers and not family & friends doesn't reduce the moral weight of their suffering.

However, practically, distance matters because sometimes helping those who are nearby also means we

are more aware of their local context and their condition. And we can be more certain that our help is actually reaching the object of our help.

Singer argues that our intuitive sense that distance matters is mostly a product of evolutionary psychology and cultural conditioning and not sound moral reasoning. Suffering is suffering regardless of distance or who experiences it (which means it should include animals as well?)

He suggests that we lived in small communities (earlier) when helping distant people wasn't practical, thus evolving our moral intuition.

6. Singer argues that his principle collapses the traditional distinction between duty and charity because of the reclassification of 'charitable giving' as a moral obligation.

Traditionally:

Duty/obligation: Actions we're morally required

to perform or things we would be blamed for not doing (e.g. not stealing, keeping promises).
 Charity: Actions that are morally praiseworthy but not required per se. (giving on top of duty). It can solicit praise but doesn't solicit blame if not performed.

But this is how Singer's argument really upsets this: Acc. to his principle, if we can prevent something very bad (like death from starvation) without the sacrifice of anything of comparable moral importance, we are morally obligated to do it. And, therefore, what we call charity becomes duty because it prevents greater suffering than it causes us, and a failure to give becomes a moral wrongdoing, a moral failure. Few acts will now even count as extraordinary or supererogatory because the even 'extraordinary' act of charity is an obligation.

We maintain the former distinction for psychological comfort, to not feel bad about our relative wealth.

7. There are children all over the world who are like the drowning child (who we ought to save) who we can save for no serious cost to ourselves. There are also effective charities.

The lawyer friend said that people like Peter Singer are doing more harm than good. A philosopher has studied that massive infusions of aid have barely left these people in a better condition than they were in. He then realised that the drowning child has many variations eg. when to get to the child, I need to cross the bridge but need to give \$100 to a warlord on the bridge who has control on some of the world's neediest countries.

In the real world there are not just two people as in the original pond example. That there could be unintentional terrific consequences of an innocent act is an argument I resonate with.

Singer will point out at successful aid organisations and real people's life stories that have been changed because of aid. He would also argue that the existence

of corruption doesn't justify doing nothing.
Also, he might say, that his principle doesn't
require blind giving; it requires effective giving.
And, do we apply this principle consistently?

1. Theological Objectivism: This says religious truths, like God's nature or rules, are true no matter what anyone thinks. They come from something outside us, like the Bible or God's word. For example, an objectivist might say The Ten Commandments are God's rules, and they're always true for everyone, whether you believe in them or not. It's like saying $2+2=4$ in math. it's a fact, not an opinion. People follow set teachings or scriptures. Mysteriorism is an interesting problem here.

Theological Subjectivism: This says religious truth depends on what you feel or think. It's personal and can vary. For example, a subjectivist might say, "The Bible means something special to me based on my life, and that's my truth." It's like choosing your favorite color and your truth might differ from someone else's. People focus on their own spiritual experiences. Arbitrariness can be a problem here morally. It can be easy to justify a lot of horrendous things based on such subjectivism.

2. Kretzmann says taking theological subjectivism (TS) seriously means believing that what's right or wrong is whatever God says it is. For example, if God told a dad to hurt his kid (like in the Bible story of Abraham and Isaac), TS says that would be okay just because God said so, even if it seems awful.

Kretzmann says this idea is terrible because it makes right and wrong totally random. If God changes His mind, something good today (like being kind) could be bad tomorrow. This makes it hard to trust God's goodness. He could order something horrible, and we'd have to call it good. Plus, it makes people follow God just to avoid punishment, not because it's truly right. Kretzmann thinks this deserves to be called out as wrong.

My Opinion: I agree with Kretzmann. If what's right depends only on what God says, it's like following a boss who keeps changing the rules for no reason. It's confusing and scary. I think God's rules should come from a steady, good heart, so we can trust they make sense. (Also, we shouldn't be doing things to avoid punishment. Being fearful of God is okay, I'm not contesting that, but on a personal level I would like to do things just because they are nice and they're helping people or animals without troubling me much.)

3. According to theological objectivism, morality is grounded in objective truths that exist independently of human opinions or beliefs, and God plays a central role as the source or foundation of these truths and even as the main preceptor of these truths. Unlike theological subjectivism, which says something is right or wrong just because God says so, theological

objectivism holds that moral truths are rooted in God's unchanging nature, reason, or eternal principles that God upholds. This means God's commands align with a fixed standard of goodness that isn't arbitrary. For example, God commands "do not murder" because it reflects an objective moral truth tied to His nature or the value of human life, not just because He decided it randomly.

According to Kretzmann's view, he argues that theological objectivism avoids the flaws of subjectivism by tying morality to a stable, objective reality. God's rational and good nature. God doesn't create morality on a whim; instead, moral truths are eternal and consistent with God's essence. For instance, Kretzmann would say that God's command to love others is good because it reflects an objective truth about human dignity, not just God says so. This makes morality trustworthy and consistent, unlike the arbitrary nature of subjectivism that Kretzmann criticizes.

4. Since I couldn't see the episode on the original playlist, I'd to look it up here- <https://podcastaddict.com/philosophy-bites/episode/78449867>

The moral basis, as Armstrong puts it, is the human relations with each other, even with different species and to not cause harm. It's not necessarily based on religion and God doesn't have to tell that and it's not God who told that. It's merely common sense that has been reflected across the religious texts of myriad religious in order to appeal to the common masses and become popular. Most of us are nice people who don't want to harm others, whether atheist or theist.

Also, even if God didn't exist, some things would still be objectively right or wrong. This is all how you have morality without God.

I agree with Armstrong. Having not God as the foundation to them doesn't make morals subjective, example, rape which is bad to the victim (or killing an animal).

5. Bairer: Ordinary Egoism: In everyday terms, egoism means being selfish in a bad way. It's when someone cares only about themselves and ignores others' needs, even when they should care. For example, a boss who takes all the credit for a team's work to get a promotion, not caring how it hurts the team. Baier says this kind of egoism is a moral flaw, it's wrong because it goes beyond what's fair or decent.

Psychological Egoism: This is a theory that says everyone always acts to benefit themselves, even if it looks like they're being kind. For example, if you give money to charity, psychological egoism says you're only doing it to feel good or

look good, not really to help others. It's not about morality but about explaining why people do things.

Ordinary egoism is a judgment that someone's selfish behavior is wrong (a moral issue). Psychological egoism is a claim that all actions are secretly selfish (a fact about human behavior), whether they seem good or bad.

Why Baier Says Psychological Egoism is Empty or Circular: Baier argues this theory doesn't hold up because people sometimes act against their own interests, like risking their life to save someone they don't know or donating a kidney without expecting anything back. Psychological egoists might say, "They're still being selfish deep down," like wanting to feel heroic. But Baier says this is circular. They just assume every action is selfish without proof, twisting every example to fit their theory. That makes it unfalsifiable (you can't prove it wrong)

6. Veneer Theory suggests that humans are naturally selfish or bad at their core, and any kindness or moral behavior is just a thin, fake layer hiding their true self-interested nature. For example, it claims someone might help a neighbor only to look good, not because they genuinely care. Korsgaard finds this theory unconvincing and rather silly for several reasons. She says it oversimplifies human behavior, ignoring cases where people act kindly without expecting rewards like risking their life for a stranger. Korsgaard also points out that there's no solid evidence proving humans are always selfish deep down; the theory just assumes this without proof, which she sees as a weak guess. Additionally, it fails to account for complex motives like love, duty, or empathy that drive actions, such as a parent caring for their child out of genuine love. Similar to critiques of psychological egoism, Korsgaard finds the theory silly because it twists every action to fit a selfish narrative, making it impossible to disprove and thus meaningless. She believes this reduction of morality to a shallow act ignores our capacity for authentic care, making the theory more of a cynical story than a serious explanation.
7. Korsgaard says morality shows up when people think deeply about their actions in a special way. Most actions are simple like when a dog runs from danger or a person eats because they're hungry. These come from instincts or quick wants. A step up is acting with a plan, like cooking dinner to eat later. But morality happens at a higher level: we stop, think about our choices, and ask if they match who we want to be, someone who respects others and acts fairly. For example, instead of just helping a friend to feel good, we help because we believe it's right to treat people with kindness.

This answers Frans de Waal's question about what makes humans moral, unlike animals. De Waal points out that animals, like chimps, can be kind or fair, but they act on instinct, not deep thought. Humans are different because we think about why we act (~meta thinking), checking if our reasons are good and fair. We don't just follow urges; we make rules for ourselves based on respecting others, which animals don't do. This ability to reflect and choose actions that value everyone makes us uniquely moral.

8. Why de Waal Sees Continuity Between Human Morality and Animal Social Behavior

I really, really loved the video and how beautifully Waal presents the morality of the animal world including reconciliation in chimpanzees, goats etc.

Frans de Waal argues that human morality has deep roots in the social behaviors of other animals, particularly primates, because they share key traits that form the building blocks of morality. In the YouTube video, he challenges the idea that nature is just a selfish, competitive "dog-eat-dog" world. Instead, he points to behaviors like empathy, fairness, cooperation, and altruism in animals, which mirror human moral actions. For example, he describes chimpanzees choosing to reward both themselves and a partner in experiments, showing a preference for mutual benefit over solo gain. He also highlights reconciliation (e.g., chimpanzees kissing after fights), adoption of orphaned chimps by unrelated males, and even whales showing gratitude to humans who free them from nets. These behaviors, empathy, helping others, following rules, and punishing rule-breakers suggest animals have prosocial tendencies that resemble the foundations of human morality. De Waal emphasizes that group living in animals, like chimpanzees and bonobos, relies on cooperation, friendships, and emotional bonds, much like human societies. He sees humans as primates with 95% of our behaviors tied to these shared animal instincts, making morality a natural extension of these social traits rather than something uniquely human.

Sympathy: de Waal vs. Korsgaard

I'm more sympathetic to de Waal's emphasis on similarities between humans and animals as the basis for morality. His view feels grounded in observable evidence, behaviors like altruism, empathy, and fairness in primates show clear parallels to human moral actions. For example, a chimpanzee adopting an orphan or a whale expressing gratitude suggests emotional and social capacities that align with moral foundations, making morality seem like an evolved trait built on shared biology.

Korsgaard's focus on human uniqueness, our ability to reflect deeply and choose actions based on rational principles, sets a high bar for morality that feels overly intellectual. While her idea of reflective reasoning explains complex human ethics, it downplays the emotional and social instincts that de Waal shows are already present in animals. Morality likely starts with these instincts (as de Waal suggests) and gets refined in humans through reflection, not created from scratch. De Waal's approach better captures the gradual evolution of morality across species, avoiding the sharp divide Korsgaard draws. It also feels more realistic, as human morality often stems from gut feelings like empathy, not just abstract reasoning.

Responses 3

1. Bentham to start with agreed with Thomas Hobbes that people always act out of self-interest, a view called psychological egoism. This means even when someone seems to help others, they're really doing it for their own gain, like feeling good or avoiding guilt. But Bentham's moral theory, utilitarianism, says actions are right if they create the most happiness for the most people, considering everyone's interests equally. This causes a problem because if people can only act selfishly, how can they follow utilitarianism's rule to care about everyone's happiness without bias?

Utilitarianism asks people to sometimes sacrifice their own good for others, but psychological egoism says that's impossible because every choice is selfish deep down. This makes utilitarianism seem unrealistic, as it expects people to act in ways they can't. Bentham tried to fix this by saying laws or habits could make self-interest match the public good, like when helping others feels rewarding. But this still weakens utilitarianism, because if people only do good for selfish reasons, the theory's call for impartiality doesn't fully work, right? Later, Bentham focused more on happiness as a goal for laws, but his early belief in egoism made it hard to explain why people should care about others without thinking of themselves first. Such an irony!

2. David Hume and Jeremy Bentham had different ideas about what makes someone a good person, and their views match the two sides of Euthyphro's dilemma, a question from Plato about whether something is good because gods like it or because it's already good.

Hume's idea is like the first side, where something is good because it's liked. He said a person's character is good if it makes us feel positive emotions, like when we see someone being kind and it feels right. For Hume, goodness comes from our feelings, not from some fixed rule. So, a trait like honesty is virtuous because

people approve of it, similar to how Euthyphro's dilemma says something is good because the gods like it.

Bentham's view fits the other side, where something is good first, and that's why it's liked. He thought a good person is someone whose actions create the most happiness for the most people, like being honest because it helps everyone get along better. For Bentham, goodness is about real results, more happiness, less pain and not just what people feel. This is like the dilemma's idea that the gods like something because it's already good by an outside standard, which for Bentham is how much happiness it really brings.

3. To answer this question, I will first talk about pleasure as Crisp puts it. Crisp teaches Philosophy at Oxford. Utilitarianism has nothing to do with something of a utilitarian value and fundamentally has two components and is a philosophical theory about how we should act (theory of utility- what makes life good for people) b) what should we do about that?

We should produce as much of that utility as well.

Real pleasure is the only component of happiness and pain of an unhappy life. To calculate it, there's a special type of calculus to measure the amount of pleasure we got from different types of activities. To some measure, we can surely measure pleasure and pain.

As for pleasures of higher and lower orders, Socrates enjoyed his life more because he is enjoying these higher activities unless he was seriously depressed in his life. A finite amount of some pleasures is better than an infinite amount of others.

I really do agree with this. Firstly, I have harboured the belief of higher pleasures since I was a child, so I have an inherent bias towards it. But, secondly, and more importantly, the belief of higher pleasures has made navigating the tough waters of life easier. For instance, take a philanthropist with a billion dollars in the bank who goes to villages in Africa to give food to hungry kids versus, say, take a person in Minnesota who does no work of charity, works for a bar and goes about their day (and a picnic every Sunday). I strongly think that the philanthropist with a billion in their bank yet moving around villages and forests is enjoying his life more because they are capable of doing things, even though more 'boring' than a Sunday picnic, that someone else can simply not do.

Now, I know it's not the best example but I hope I could explain why I so believe this theory can be correct.

4. Classical utilitarianism, the idea from Jeremy Bentham and John Stuart Mill, says the best action creates the most happiness for the most people, no matter who they are or when they live. Longtermism says we should focus on making the far future better, because there could be tons of people living later, and their happiness matters a lot.

Yes, classical utilitarianism supports longtermism. It says we should care about everyone's happiness equally, including people in the future. Since the future could have way more people, their happiness could outweigh what we do for people now. So, if we can do things like prevent big risks (like dangerous AI or climate disasters), utilitarianism says we should focus on that.

But this depends on assuming we can actually predict and affect the future. If we can't be sure our actions will help far-off people, utilitarianism might not push us toward longtermism so strongly.

Should we accept longtermism or see it as a problem with utilitarianism? It depends. If you think utilitarianism is right, longtermism makes sense, it's just taking the idea seriously by caring about future generations, like we already do with things like saving the environment. It could lead to important actions, like making AI safer. But some people think longtermism shows utilitarianism is flawed. It might make us ignore real problems today, like poverty, to chase uncertain future goals, like space colonies. Critics say this focus can feel extreme or even unfair, and other moral views that prioritize people alive now avoid this issue.

Whether you like longtermism or not depends on whether you think it's smart to plan for the distant future or if it feels wrong to put that above today's needs. Personally, I'm not a fan of longtermism because I would much rather care about a person or animal suffering right before my eyes than craft up imaginary people to help. That's not to say, I don't care about climate change and the like. It's just that helping a needy suffering right before my eyes is more important to me ethically.

5. According to Chang, hard choices are big decisions where you have to pick between options that are good in different ways, but neither one is clearly better overall than the other. These aren't small choices like what to eat for lunch; they're life-changing ones, like picking a career, ending a relationship, or deciding where to live. What makes them hard is that the options aren't better, worse, or even equal in value. They're kind of "on a par," meaning they're in the same ballpark but can't be directly compared because they involve different kinds of values or reasons. For example,

choosing between a high-paying job as a banker (good for money and security) and a creative job as an artist (good for self-expression and passion) feels tough because you can't easily weigh those benefits on the same scale.

Chang argues that the options in hard choices can't be equally good, and she uses what she calls the "sweetener" test to show why. Imagine two options that seem balanced. If they were truly equal in goodness, adding a small improvement to one like a tiny extra benefit should tip the scale and make that one clearly better. But in real hard choices, that doesn't happen. Take the banker vs. artist example: if you add \$500 more a month to the banker's salary, the choice doesn't suddenly become easy; it's still hard to decide because the extra money doesn't outweigh the artist's unique rewards like creativity. This proves the options weren't equal to start with, if they were, the improvement would resolve things right away. Instead, since the choice stays hard, the options must be incomparable, or on a par, rather than equal

In the near future, I will have to decide between a well-paying AI engineer job or an easier, chill job. It's hard because I do value money and it's already hard to get an AI job but the AI job is really tiring. It's a practically hard decision to make. Let's see what triumphs in the end!

6. Derek Parfit describes three theories about what makes life good.
 Hedonistic theories say it's about having the most pleasure and least pain.
 Desire-fulfillment theories say it's about getting what you want, especially for your own life. Objective list theories say some things, like knowledge or love, are good no matter what you feel or want.

Each has its problems. Hedonism might value bad pleasures, like cruelty, or ignore reality outside your feelings. Desire-fulfillment can count weird wishes, like wanting something you don't know about, or say a boring life is better than a great short one. Objective list theories struggle to justify why certain things are good and might ignore your desires.

Parfit's composite theory mixes them: a good life has objective goods (like love or knowledge) that you also want, which brings pleasure. This avoids many issues, like bad pleasures or odd desires, by tying everything together. But it might still struggle with picking the objective goods or handling mismatched desires. Overall, it's stronger than the others because the parts support each other. I actually appreciate how we're talking about these theories in our class. They are great food for thought.

7. It's a very complex question. If someone doesn't have a family and also not a friend or simply doesn't have a working relationship with them, the relationship with simulacra might mean a lot to them. But otherwise, generally, relationships with simulacra, simulated beings like advanced AI companions or virtual people would not be just as valuable as relationships with real people, even if they provided the exact same amount of happiness. The key reason is that true value in relationships goes beyond subjective feelings of happiness or pleasure. Real relationships involve genuine reciprocity, where both parties have independent agency, emotions, and the capacity for mutual growth, vulnerability, and moral responsibility. A simulacrum, no matter how convincing, is ultimately a programmed entity without its own intrinsic experiences or free will and so it's a one-sided 'love'. If you're in a "relationship" with one, you're essentially interacting with a mirror of your own desires, not a separate conscious being. This lacks the authenticity and shared reality that make human connections meaningful. A simulacrum could mimic a caring friend perfectly, but the support isn't real, it's algorithmic prediction, which cheapens the bond in a fundamental way (similar to why people reject Nozick's experience machine which I read about and it was mind-boggling to say the least).

This answer reveals that I find objective list theories (or Parfit's composite theory that builds on them) more plausible than pure hedonistic theories. Hedonism reduces value to happiness alone, so it would say yes, simulacra relationships are equally valuable if the joy matches. But objective list views hold that certain things like authentic love, friendship, or personal achievement are good in themselves, independent of how they feel or what you desire. The composite version adds that these objective goods are even better when desired and pleasurable, but the objectivity ensures fakes don't fully count. Desire-fulfillment theories could go either way, depending on whether you specifically want "real" connections, but they often fall short by ignoring deeper, non-subjective values.

8. Recreational drug use and Nozick's experience machine share some similarities but aren't the same. The experience machine is a hypothetical device that creates a perfect, simulated life full of happiness, but it's all fake: you're cut off from real experiences. Drugs, like psychedelics or alcohol, can give you intense feelings of joy, escape, or altered perception, which can feel a bit like the machine's artificial bliss. For instance, taking ecstasy might make you feel deeply connected to others, similar to a programmed happy moment. If someone uses drugs to constantly avoid reality, it's somewhat like choosing the machine's illusion over real life.

However, there are big differences. Drugs work on your real brain and body, not a virtual world, so the experiences are grounded in reality, even if they're altered. They're also temporary, you come down from the high and face real-world consequences, unlike the machine's permanent fantasy. Plus, recreational drug use varies: having a glass of wine with friends isn't the same as heavy drug use that isolates you, and neither fully mimics the machine's complete detachment from reality.

I think that this comparison supports an objective list theory of value over pure hedonism. Hedonism says any pleasure counts, so drug highs or the machine's fake joy could be just as good as real experiences if they feel the same. But objective list theories say some things like real relationships, achievements, or growth are valuable beyond just feeling good. Drugs might give a quick high, but they often lack the deeper meaning of authentic experiences, much like the experience machine.

Response 4

1. Pettit says consequentialists, like act utilitarians, promote value choosing actions, like lying, to maximize good (e.g. happiness) each time. Act utilitarianism judges every act by its consequences alone. Non-consequentialists sometimes honor values acting true to it, like being honest because it's right, not for outcomes.

Promoting and honoring clash when maximizing good (e.g. framing an innocent to stop a riot) violates values like justice. This threatens integrity, your sense of self, making you a cipher of circumstance, detached and opportunistic, unable to commit to values like honesty.

[I do agree with it because they clash. The only other explanation would be that they are somehow paradoxical? I did not understand it fully. An example about giving money was also made in the podcast.]

Pettit's fix is indirect consequentialism: Promote good through rules or virtues that honor values, ensuring long-term benefits like trust (rule consequentialism as he mentioned in the podcast). This preserves integrity, unlike act utilitarianism's constant calculations. I find it compelling it balances non-consequentialist intuition (honoring matters) with consequentialist logic (outcomes justify rules), avoiding act utilitarianism's alienating demands. But rare overrides (e.g. torture in crises) might weaken rules. Still, it's practical, aligning with how we honor values daily for stable, good results.

2. A theory of justification explains what it is that makes an action right or wrong, like a rulebook for judging choices. For example, it might say an action is good if it helps the most people. A theory of deliberation tells you how to think when making decisions in daily life, like steps to follow when choosing what to do. It's like an algorithm. Justification is about the standard for rightness; deliberation is about how you decide.

Consequentialism is mainly a theory of justification: an action is right if it creates the most good, like more happiness. It's not always about deliberation. But Pettit says indirect consequentialism can guide deliberation by using habits or rules that lead to good results. So it can be both justification for what's right and guidance for deciding.

Pettit says non-consequentialists focus on deliberation, teaching us to honor values like fairness or loyalty when deciding. They care about acting true to these values, and not just outcomes. If consequentialists focus on justifying actions by maximizing good, the two sides aren't totally at odds. Consequentialists can agree that honoring values in decisions often promotes the best outcomes, so the disagreement shrinks. It's sort of a consensus as I like to think about it.

3. The city's happiness depends on one child's suffering. I'd walk away. The child's misery is too high a price for everyone else's joy. Staying feels wrong because it treats the child like a tool, not a person. Walking away honors the child's value as a human, even if it doesn't fix everything or maximize happiness. It's about doing what's right, not what's best for most.

(I found it very, very bad morally that a child should suffer. I'm fine with everyone being unhappy. I may try to convince people to not do so, but if no one listens I'll walk away.)

This choice shows I lean toward honoring what's valuable, like human dignity, over promoting overall happiness. Honoring means acting true to a value, like respecting every person's worth, even when it's hard. Promoting might mean accepting the child's suffering to keep Omelas happy, but that feels like betraying fairness. My decision reflects a belief that doing what's right sometimes means standing by a principle, not just chasing the greatest good. Walking away isn't perfect I know, it doesn't save the child but it's a way to say no to a system that demands cruelty, keeping my integrity intact.

4. Yes, they disagree. Baldwin pushes a bold approach: face past racial wrongs, like slavery and murder, with anger to demand real change. He

wants to keep history in focus. Buckley agrees the past was bad but says focus on the future. He thinks Black people should use the American dream to improve their lives in a practical way, avoiding anger because it causes trouble and could lead to white violence.

Both seem to lean on consequentialism, which says actions are right if they lead to good results. Buckley clearly thinks anger is bad because it makes things worse, like causing conflict through pure consequentialism. Baldwin mostly cares about anger showing the truth of injustice, which feels non-consequentialist but he still weighs outcomes.

5. Amia Srinivasan says anger is apt when it matches a real moral wrong. It's not just feeling mad, it's the right response to an injustice because like when someone steals from you, apt anger judges the act as wrong and wants it fixed, like demanding fairness. It should fit the size of the wrong, big wrongs deserve big anger! Even if anger causes trouble, it can still be apt because it shows you see the injustice clearly, like tasting something bitter. An argument is also made that aptness matters on its own, not just what anger achieves (I like this view). For example, a Black person's anger at racism is apt if it reflects the real harm done, even if it risks backlash. This helps victims name their pain, not just stay quiet for others' comfort. Apt anger keeps your moral compass true, showing the world's unfairness without needing to fix it right away.

Yes, apt anger honors morality, especially justice. Honoring means acting true to a value's core, like respecting justice by calling out wrongs. Apt anger does this, it sees injustice and says, "This is wrong!!" It's like saluting fairness, not ignoring it for quick wins. Srinivasan shows anger grasps injustice deeply, like art captures beauty. Suppressing it to promote peace dishonors justice, making you feel fake. Honoring justice through anger keeps your moral self whole, staying loyal to fairness even if it's costly. Promoting might mean hiding anger to avoid trouble, but that treats justice like a tool, not a guide. Anger's honesty fuels real change, not just cold math. But too much anger can cloud judgment, so it needs balance. Still, apt anger fits honoring, it stands for justice's truth over just chasing better outcomes.

6. Victims of affective injustice are typically marginalized groups facing systemic oppression. This includes Black people whose anger at racism is labeled "dangerous," women shamed for fury at sexism, poor people silenced to keep the powerful comfortable, and activists told their outrage is "too strident." These groups face a double wrong: the initial injustice

(like racism or sexism) and the demand to suppress their apt anger to avoid further harm, like social exclusion or violence. Srinivasan highlights how society's biases amplify this, targeting those already hurt by oppression, making their emotional expression a risky act that deepens their marginalization. It would be counterproductive.

7. Myisha starts with distinguishing between rage and anger traditionally from a magnitude and logic point-of-view. Cherry would counter that not all rage is counterproductive. In the episode, she argues rage can be apt and productive if managed, like MLK's anger fueling his Birmingham letter and social change. She'd say critics overlook how "Lordian rage" motivates anti-racist action, providing optimism and self-belief, without necessarily leading to revenge or alienation. Rogue rage might be destructive, but Lordian rage serves justice, challenging the view that anger always worsens things.

Rogue rage is destructive, irrational, exclusive, and based on false wrongs. Lordian rage is anti-racist anger targeting real injustice, inclusive, change-focused, and motivational. They can be told apart by target.

Cherry says manage it through techniques (like Stoic reflections) to channel it productively direct towards movements. Anger should be form of social change, she says.

I agree. Distinguishing rage types redeems its role in justice without endorsing harm. History shows managed anger drives progress (eg. civil rights), aligning with Srinivasan's apt anger while addressing counterproductivity empirically.

1.

Utilitarianism says there is one big rule: Do whatever makes the most people happy. It looks at the results, from what I understood. Cheating in class is wrong because it hurts the whole class. If you cheat and get a good grade without learning, others feel it's unfair. They may cheat too. Then grades mean nothing, teachers can't trust anyone, and real learning stops. Less happiness for everyone.

Kantianism says there is one big rule: Only do things you could let everyone do all the time. It looks at the action itself. Cheating is wrong because you can't make a rule that says, "Everyone should cheat." If everyone cheated, tests would be useless and the whole idea of school would break. So cheating breaks a rule that must apply to all.

Which feels right to me? I like Kant's way more. Utilitarianism cares only about results, so it might say, "Cheat if no one finds out and you get a job." That feels sneaky. Kant says, "Don't cheat, period," because it's not a fair rule for all. That kind of matches how I was raised: do the right thing even if no one is watching. But please note I don't like all maxims that come to be associated with Kant. In this example, yes, but not always.

2.

According to O'Neill, using someone merely as a means is treating a person like a tool to get what you want, without caring about their feelings or choices. For example, you trick a classmate into giving you their homework so you can copy it and get a good grade. You don't care if they get in trouble; you just want the grade.

Treating someone as an end in themselves means respecting the person as a full human with their own goals and rights. You only act if they can freely agree. For example, you ask your classmate if you can see their homework to study, and you only use it if they say yes. You respect their choice and don't use them just for your benefit.

3.

Utilitarianism says life has value only because it brings happiness or stops pain—I could be wrong in my understanding here. A person's life is not special by itself. If harming one person saves ten and makes more people happy, utilitarianism says do it. Life is like a tool: use it to get the most good for the most people.

Kantianism says every human life has huge value just because the person can think, choose, and make their own goals. You must never treat a person like a thing to use. Even if killing one saves many, it's wrong. You cannot use someone just to reach your goal. Every person is an "end in themselves" and not a means to something else.

The big difference: Utilitarianism looks at results—life is worth what it produces in happiness. Kant looks at the person: life has value no matter what. You must respect every person's dignity, always.

But from what I understood, it would also be wrong to think of Kantianism as always giving more worth to life. Kantianism treats human life as having inherent dignity and worth based on reason and autonomy, making it inviolable. Utilitarianism, however, judges the morality of an action based on its

consequences, allowing the potential sacrifice of some individuals if it maximizes overall happiness, so in that sense I “sensed” the above.

4.

Langton talks about two ways we see the world. One is the strategic standpoint: how we treat nature, like rocks, plants, or, sadly, animals. This is so sad and unrightful because animals are sentient beings, unlike plants, and deserve better treatment than what they get from people today. We use them as tools to get what we want. We control them and don’t need their permission because they don’t have thoughts or choices.

The other is the human standpoint—how we treat people. We must respect them because they can think, choose, and have their own goals. We can’t just use them like objects. We need their agreement and must be honest. We see them as equals with dignity.

The big difference is this: with nature, we act to control and use. With people, we act with respect and consent. Using the strategic way on people, like lying to get your way, is wrong because it treats them like things, not humans. But then who do we even lie to? We cannot technically “lie” to nature.

5.

In Langton’s view, deception or “strategic speech” is bad in the Kantian picture because it treats people like objects from the natural world, not as humans with dignity and choice. For example, as in baking cakes, deception does the same to people: you lie or hide truths to manipulate their actions, like tricking a friend to get what you want. This ignores their ability to think and consent, using them merely as a means to your ends. Kant says we must always treat people as ends in themselves, with respect and open communication, so strategic speech breaks that rule by turning human interactions into a game of control.

Because of this, Langton thinks Kant fails to treat Maria von Herbert as fully human in his letters to her. Maria wrote to Kant about her pain from deceiving, feeling lost and suicidal. But Kant responds strategically: he lectures her like a child, assumes she’s at fault without empathy, and even suggests she’s exaggerating for attention. He doesn’t engage her as an equal thinker or show compassion; he tries to “fix” her like a broken object, using cold advice instead of true dialogue. This makes him hypocritical, as he preaches respect but treats her from the strategic standpoint, not the human one.

6.

When Langton speaks of acting “for the Kingdom’s sake,” he means that sometimes one must act strategically to serve a higher purpose rather than simply follow ideals in a rigid way. The “Kingdom” here represents the ultimate good, God’s plan, or the greater moral aim, and a “strategy” for its sake allows for practical judgment and even compromise if that helps advance what is truly right in the long run.

In my humble view, it is important to honor your ideals, but not to the point where they stop you from achieving the good they stand for. Ideals should guide you like a compass, not trap you like a cage. There are moments when keeping to them too strictly can harm the very purpose they were meant to serve, as when honesty might endanger someone’s life. What matters is your intention, whether you bend your ideals out of weakness or out of wisdom. Acting for the greater good sometimes means sacrificing perfection for effectiveness, just as Langton suggests that faithfulness to the Kingdom’s purpose is more important than flawless obedience to every ideal.

But then again, there are times when I disagree with complete flexibility in the ideals’ matter because, if you are being too flexible about the ideal in order to promote it, are you even promoting your own ideal?

7. A lie is when you say something, the literal meaning of which you know not to be true, and you also want it to be misinterpreted by the other person, says Webber. There are straightforward lies, but that’s not the only way to deceive. Webber started with the broken-window example and expanded on it to explain deception, especially how you can deceive without lying by deflecting attention.

The bullshitter just doesn’t concern himself with the truth at all and is more engaged in pursuing fame or reputation—something of a similar nature.

Implicatures: Conversations are governed by the assumption of conversation, even if not thinking in these terms. Implicatures mean that a speaker implies or suggests beyond the literal meaning of the words they say. It relies on context, shared knowledge, and the assumption that speakers are cooperating with each other.

We treat people who lie worse than those we catch “misleading,” and lying does more damage to one’s reputation.

1. political contractarianism became understood because it gave people a fresh way to think about why governments exist. Instead of saying, “Kings rule because God says so,” folks like Hobbes and Rousseau said people make a “social contract.” Basically, we all agree to give up some freedom to a government so we’re safe and things run smoothly. This clicked during crazy times like wars or revolutions, since it focused on people choosing their system, not just following old traditions.

But it’s got two big problems. First, this “contract” never actually happened. it’s just a made-up idea, so it’s kinda shaky as a real reason for government. Second, it acts like everyone’s equal when they “sign” this deal, but that’s not true. People with less power like the poor or oppressed don’t get a fair say, especially if they’re desperate for safety.

2. Hobbesian contractarianism, or just contractarianism, comes from Thomas Hobbes’ idea that people make a “social contract” to escape a brutal “state of nature” where life’s “nasty, brutish, and short.” Basically, folks agree to give up some freedoms to a strong government to keep things safe and orderly. It’s all about self-interest: you join the contract ‘cause you don’t wanna get screwed over in a chaotic world, and consequently, if you don’t join it, then you’re welcome to being screwed over!

Kymlicka, a modern philosopher, says this isn’t really about morality, it’s more like an alternative to it. Why? Because Hobbes’ setup is all about what’s best for you, not what’s right or wrong. It’s just people making a deal to avoid danger, not because they care about being “good.” Kymlicka thinks true morality involves principles like fairness or duty, not just survival-driven bargains. So, Hobbesian contractarianism skips moral ideals for a practical “let’s not die” agreement. It confuses me a bit because why should prioritizing ‘safety’ which is what this social ‘contract’ gives us be kind of a substratum to it not being a moral idea.

3. Kantian contractarianism, or contractualism, comes from Immanuel Kant’s ideas. It’s about figuring out what’s fair by imagining people agreeing to rules in a hypothetical setup, like Rawls’ “original position” where everyone’s behind a “veil of ignorance” (not knowing their own status, wealth, or abilities). This setup pushes people to pick rules that are fair for everyone, not just themselves. This expresses the “moral point of view,” which is about making decisions based on impartial, universal

principles that respect everyone's dignity and equality, not just personal gain.

Will Kymlicka says Kantian contractarianism “begs the question” about what’s just or right because it assumes fairness comes from this idealized agreement. But the setup itself like the veil of ignorance already sneaks in moral ideas about equality and fairness. So, it’s not proving what’s just from scratch; it’s building justice into the process, making it circular. You’re already assuming what you’re trying to figure out.

4. “proceduralist characterization” of constructivism is about seeing morality as something we build through a specific process or set of rules like a recipe: you follow steps like Rawls’ “veil of ignorance” where people pick fair rules without knowing their own status to come up with moral principles. It’s all about the procedure being neutral and fair, and the outcome is legit because of how you got there. (which, on further inspection, at least on my end, appears quite open to variables in the environment of the decision-making process, that are actually rather difficult to normalize)

The “practical standpoint” characterization, on the other hand, focuses on morality as something that comes from our everyday reasoning as practical agents. It’s less about a strict process and more about what makes sense when we act and make choices in real life, based on our shared human needs and rationality. Like, it’s about what we’d naturally agree on as reasonable people trying to live together. This is more intuitive to me rather than being the mechanical engineer of morality.

The difference? Proceduralism is rigid, tied to a specific method (like a thought experiment) to define what’s right. The practical standpoint is looser, grounded in how we actually think and act in daily life, without needing a formal setup.

5. The practical point of view is about looking at morality as something that comes from how we, as rational people, think and make choices in everyday life. It’s focused on what makes sense for us to do when we’re trying to live together, act fairly, or solve real-world problems, based on our shared human needs and reasoning. It’s based on common sense and the intrinsic rationality of us, humans.

A substantive characterization of the practical point of view gets specific about what moral principles or values we should follow like saying fairness means

prioritizing equality or maximizing happiness. It gives concrete content to what's morally right, like filling in the details of a rulebook.

A formal characterization, on the other hand, is more about the structure or process of moral reasoning, not the specific rules. It's like saying, "Morality comes from reasoning in a way that's impartial and respects everyone," without spelling out exactly what the rules are. For example, Rawls' veil of ignorance is a formal setup, it's a method for figuring out what's fair, not a list of set-in-stone principles. It's an algorithm not the output array.

Difference: Substantive gives specific moral content; formal focuses on the process of reasoning without locking in particular answers.

6. Street talks about two senses of "realism" in metaethics. First, there's ontological realism, which says moral truths exist independently of human minds like facts about the universe that are true whether we think about them or not. Second, there's practical realism, which says moral truths are real in the sense that they guide our actions and reasoning in a practical, undeniable way, without needing to be "out there" in the universe.

Constructivism is a realist position in the practical sense because it sees moral truths as real within the context of human reasoning and values. It builds moral principles from our practical needs and agreements, like fairness in a social contract, so they're real for guiding action. But it's not realist in the ontological sense because it doesn't claim moral truths exist independently of human minds. Instead, constructivism says morality is constructed through our rational processes, like Rawls' veil of ignorance, so it depends on human thought, not some universal, mind-independent reality.

7. An ideally coherent Caligula someone who's perfectly consistent in their reasoning but revels in cruelty, like the infamous Roman emperor raises a tough question (but sort of easy too). Is he making a mistake? Yeah, I'd say he is, but it's not about him being illogical. His mistake is in valuing something (like power or sadistic pleasure) that screws over others and undermines the shared trust and cooperation humans need to thrive. It's a practical error: his actions make the world worse for everyone, including himself in the long run, since societies built on cruelty tend to collapse or turn on folks like him.

This take leans on the idea that morality isn't just about internal consistency but about what works for human coexistence. (this consistency can even be worse because a person who revels in sadism and consistently does it is only causing

more and more harm) Humean constructivism, which ties morality to our desires and sentiments (like empathy or social instincts), fits this vibe best. It says Caligula's mistake is ignoring the human tendency to value mutual respect, which most of us share because it keeps society humming. Kantian constructivism, with its focus on universal rules and impartial reasoning (like the veil of ignorance), could also call Caligula out for failing to respect others as equals, but it feels a bit rigid and abstract. Moral realism, which claims there are objective moral truths out there, might say Caligula's violating some universal "don't be a monster" rule, but that feels less grounded since it relies on truths we can't easily prove.

Humean constructivism wins for me here, it's practical, rooted in how humans actually tick, and explains why Caligula's coherent cruelty is still a bust without needing cosmic moral laws.

8. **Humean Constructivism (Charitable Take):** Humean constructivism sees morality as rooted in our human desires, emotions, and social instincts, like empathy and cooperation. It says moral rules aren't cosmic truths but practical agreements we form to live well together, based on what we naturally care about say avoiding harm or building trust. It's grounded in our shared human nature, making it relatable and flexible for different cultures.

Why Most Plausible: It's practical and doesn't rely on abstract setups like Kantian constructivism's veil of ignorance, which feels forced, or moral realism's unprovable universal truths. Humean constructivism starts with real human feelings and needs, explaining moral diversity without losing a sense of shared values.

Handling the Arbitrariness Problem: As a subjectivist theory, it tackles the worry that morality is just random preferences by tying moral norms to widely shared human sentiments, like empathy, which evolve for survival. These aren't arbitrary but grounded in our nature as social creatures, giving morality a stable, practical basis.

1. **Meta-ethical relativism** says that what's true about morality depends on the person or culture and that there's no universal moral truth, just different views. Normative relativism goes further, saying you should follow the moral rules of your own culture or perspective, and it's wrong to judge others by your standards. They don't always entail each other. You can believe moral truths are relative (meta-ethical) but still think people should follow universal principles, not just their culture's norms. Or, you could push normative relativism like command: 'stick to your culture's rules'

but without fully buying that moral truth itself is relative. But often, meta-ethical relativism leads to normative relativism, since if morality is just perspective, it makes sense to say “do what your perspective says.” Still, they’re distinct ideas, and you can hold one without the other depending on how you reason it out.

2. A normative relativist, who believes we should stick to our culture's moral norms and avoid judging others' would tend to defend the village's stoning ritual in "The Lottery" as totally valid for them. They'd say something like, "Hey, that's their longstanding tradition. Can we let it be this way? It's kept their society humming along for generations, ensuring crops or whatever. If it works for the town, who are we outsiders to call it wrong? Tessie only freaks out when it's her turn, but that's just her personal gripe, not a real critique." No universal "thou shalt not kill" overrides local rules (so who are we to make it for them?).

Is this a plausible implication? No, not really. Sure, normative relativism can lead to excusing horrors like honor killings in other cultures, which feels dodgy if it paralyzes any pushback against abuse. But in the story, the ritual's on its last legs—kids are bored, other villages abandoned it, and even defensiveness screams insecurity. It shows blind adherence to "our way" can trap folks in pointless cruelty, not some wise cultural wisdom. Relativism might allow this implication, but the tale exposes it as a trap, not a feature. And I like that about the tale.

3. In my view, morality differs from science in ways that make meta-ethical relativism tempting but not necessarily justified. Science deals with observable, testable phenomena like flat Earth or Creationism can be debunked with evidence like satellite imagery or fossil records. Disagreements persist, but we trust empirical methods to converge on objective truths. The key word is convergence. In morality, it's hard to always converge. Morality often hinges on values, emotions, and cultural practices, which feel more subjective. Moral diversity (like differing views on honor killings) stems from varied human needs and social structures (Wong), suggesting relativism explains this better than universalism. But here's the rub: just because moral worldviews vary doesn't mean there's no objective moral truth. Science's disagreements don't imply relative truth because we expect evidence to settle them; morality's disagreements might just reflect complexity, not relativity. For example, cultures disagree on lying, but most value trust hinting at shared principles. The inference to relativism in morality often comes from mistaking diversity for proof of no universal standard, while science's clearer metrics resist that leap. Morality's messier, but that doesn't make relativism the best explanation.

And if that were to be considered, I essentially contradicted myself, so, perhaps morality can also converge?

4. David Wong says morality serves two universal human needs that are social and personal. Cooperation (working together to survive, like sharing food) and conflict resolution (sorting out fights, like who gets what). These needs shape any decent moral theory. It's got to help people get along and settle disputes, or it's useless. These constraints are universal since every society deals with these issues, no matter the culture.

But this doesn't wreck meta-ethical relativism. Relativism says moral truths depend on culture, and Wong's constraints are loose enough to let different systems meet these needs in their own way. Like, one group might cooperate through strict rules, another through vibes, both work. Compared with contractualism, which pushes one universal way (like agreements everyone would sign). Relativism with constraints is different because it says morality's job is universal, but the rules can vary.

It's like saying all cars need engines, but the engine type depends on the car, still relativism, just with guardrails.

5. Meta-ethical relativism, as discussed by Paul, holds that moral truths depend on a specific moral code, typically tied to a culture or individual perspective. For example, whether an action like making loud noise when eating noodles is right or wrong depends on the cultural or personal framework you're operating within, there's no universal moral truth that applies across all contexts while absolutism (or universalism) that accounts for circumstances asserts there's a single, objective moral code, but what's right can vary based on situational factors. For instance, stopping to help someone stranded on the road might be obligatory in a dire situation (middle of the night, freezing) but not if others can help or you're in a rush for a medical emergency. The key difference is that relativism ties moral truth to a subjective code (culture or individual), while absolutism grounds it in a universal code like a uniform dress, with circumstances shaping how that code applies (the size of the uniform). Relativism risks reducing moral judgments to descriptive claims (e.g., "this is wrong in my culture"), literally stripping them of normative force, whereas absolutism maintains moral judgments have objective weight, even if the context will influence their application.
6. A virtue is a characteristic of personal character. A bit antiquated of a word, it's centered around the ordinary moral life, a stark difference thus from rights-based theories.

It leads to questions such as how utility is relevant to a life of moral character. Some situations can be solved by ranking different virtues such as loyalty and autonomy and deciding compromises as in Dorothea's case.

Contemporary culture regards empathy as a virtue and rightly so. An empathetic person, as can be argued, makes life slightly difficult for oneself but is nevertheless having a more flourishing life. This also related back to the theory of higher good. Is empathy essential for living well or flourishing? I would say it depends much on the moral conscience of such an agent and how perturbed it gets by not being 'virtuous'. If it is of little personal consequence to them and they refuse to think about it routinely, I do not see why they should not be able to lead a life that's 'flourishing' per their standards.

2. According to Hursthouse, the major criticism of virtue theory is that it fails to guide actions on real moral issues, relying on mere assertions rather than arguments.

She breaks it into two parts: (1) Virtue rules like say act kindly use virtue/vice terms that people apply differently, with no internal way to resolve disputes. (2) It depends on ideas of what's worthwhile in life, leading to varied conclusions. The (2) is very similar to (1)

But it's not giving simple, clear action guidance that anyone can apply without wisdom or judgment and also conclusions shouldn't rely on views of what's truly important, so even ignorant people can use it. These reveal the implicit conditions of adequacy.

Hursthouse rejects these as utterly implausible because: (1) Moral knowledge needs life experience and is hard to gain unlike math, it's rare in the young and requires sensitive judgment. (2) Real moral issues must connect to what's worthwhile; ignoring this is absurd for guidance on hard topics like abortion.

3. egoism objection claims virtue ethics is selfish because it focuses on personal character development and achieving eudaimonia but this inward focus on self-improvement neglects broader social and political issues, prioritizing personal goals over collective good.

Annas counters that this objection misrepresents virtue ethics, stemming from a modern view that pits egoism against altruism. She argues virtues like justice and generosity inherently involve caring for others and addressing social conditions. It might be acting outwardly but it's not acting selfishly.

I believe it is convincing because equating honing virtues with focussing on personal self hence being selfish sounds like a really narrow-minded way of thinking to me. Why would I pit myself against the world just because of focussing on myself? We are just too used to the idea that morality is about other people. Annas's response shows virtues are relational, connecting self-improvement to others' welfare, as Aristotle emphasized. Her parent example clarifies this outward focus. However, critics might argue eudaimonia still prioritizes personal fulfillment, potentially sidelining systemic issues requiring collective action.

4. My mother is my moral example. I go to her when I'm in my own moral dilemmas and even generally when I'm stuck in cases of morality, I try to mentally think how she would have reacted or responded to such a situation and then that is what I do. She is 'moral' in a very non-utilitarian way and more in an innate way, so I think she is much more virtuous than she is moral. She is really kind- she exudes kindness and taught me kindness from a very young age. There is no virtue like kindness in my eyes. It's also why I became vegan at a young age. I am not sure if she has more phronesis than me though. But I believe her to have a much higher order of morality than myself. I have seen it in my life by way of how she treats people, loves animals (and animals love her).
5. A moral saint is someone who always tries to be moral and acts good to others. He acts 'at the drop of the moral hat'. A loving saint is driven by emotions not rules, so even if he really likes the ice cream he has on hand, he will immediately give it away to a hungry child without even giving it a passing thought, without even thinking to check if the child even likes the icecream. He is driven by love for others. But a rational saint is driven by logic. He will still do things at the drop of the moral hat but he knows that he likes the icecream although he will still give it away. He uses a second order thought process when giving things away but at the end of the day, his actions are still very, very rooted in moral thinking.

Wolf thinks both have issues: Loving Saints lose their identity; Rational Saints lose emotional connection. Her point is that being a perfect saint can make life less human, missing joy or balance. Some might say she exaggerates, as saints could feel fulfilled, but her ideas make us question extreme goodness.

6. Wolf argues that the issue with moral sainthood isn't just about having too much of one value, like moral goodness which might overwhelm other aspects of life. She rejects this as the essence of the problem because it oversimplifies the issue, suggesting a mere balance problem that could be

fixed by moderation. Instead, Wolf sees the essence of the problem as the conflict between moral perfection and a well-rounded human life (indeed, if everyone were moral then people would be all too predictable and the world all too boring. People would lose their various flavors. But we should be careful not to rationalize immorality in the name of 'flavor'.). A moral saint's relentless focus on moral goals helping others or maximizing good crowds out non-moral values, like personal passions, relationships, or hobbies, which give life richness and individuality. This isn't about excess but about the loss of personal identity and diversity of human experience. For Wolf, a life wholly dedicated to morality sacrifices the unique, non-moral traits like humor, creativity, or leisure that make someone fully human, rendering it unattractive or even dehumanizing.

7. Point of view of individual perfection- focuses on the overall excellence of a person's life, considering all aspects that make it fulfilling, unique, and admirable. It values personal traits, talents, relationships, and non-moral goods- essentially well-roundedness and perfect wonder in life.

Moral point of view- evaluates a person solely based on their moral goodness, prioritizing actions and traits that align with ethical duties, such as helping others, reducing suffering, or acting justly. It's the lens of a moral saint, where moral values dominate, often at the expense of non-moral interests or personal desires. From this view, being "perfectly moral" means dedicating one's life entirely to moral goals.

Wolf argues that someone can be perfectly wonderful without being perfectly moral. A perfectly wonderful person excels in diverse areas artistic talent, wit, loyalty, or personal passions making their life vibrant and admirable, even if they don't prioritize moral perfection and this person's wonderfulness comes from their qualities and not necessarily morality.

Wolf's distinction highlights a tension between ethics (broadly, how to live a good life) and morality (narrowly, adherence to moral duties). Ethics encompasses morality but also includes non-moral values that contribute to human flourishing, like creativity or personal relationships. Morality, from the moral point of view, can be overly restrictive, demanding constant selflessness that may conflict with a fully human life.

8. I do not think so. I think I can be a moral saint and still not have important things in my life crowded out. I value learning and growth. If I practiced too much morality, I might not have time for it. But if I did it in limits

(because I can still be a moral saint AND limit time spent on morality), I can still learn. I also value growth.

But would I want to do it? No. I value virtues, but in a world where others are not their most moral selves, being super moral will put me at a competitive disadvantage. That doesn't mean I won't be kind to others or to animals. I will be but I want to have the mental freedom to do it on my own time than to be 'restricted' by a moral force that requires me to do it every hour of the day. But for me kindness will be more of a virtue than a sense of morality. It's more ingrained in my character- i don't do it as a rule I've to follow, and that makes a huge difference.

Even if I were to be moral and constantly of help to others, who is to stop the dark forces that will undo my good actions in the blink of an eye? I would thus prefer to limit myself to practicing kindness versus building my life around a keen sense of morality. To answer the question, no, it will not crowd out the things I value but it still does not make it a good choice for me.

1. Good/bad distinction is an aristocratic system that originates from the noble class, where good is noble traits like strength and beauty, and bad what is low or even common. The good/evil distinction (slave morality) is reactive (a response to something), resentment-based system that originates from the weak, where evil is first what's by negating the noble class and good is then the opposite of that evil. good/bad mode originates in social class privilege and the good/evil mode in the oppressed. Nietzsche's evidence is linguistic and he supports his claims by appealing to the etymology of "good" and "bad" in different languages. The nobility defined their own traits as good and the opposites of them as bad. Slaves needed their own system to cope with their oppression. Since they are weak, they cannot afford actual revenge, hence they set out to seek spiritual revenge. Religion followed suit with texts like 'Blessed are the meek, for they shall inherit the earth' which the meek liked.

As I write this answer, the fact that the world has changed tremendously suddenly dawns on me. At least many parts of the world, if not all, in terms of equality, at least on paper.

2. Back in the day, the tough guys like the aristocracy were like: "We're awesome, strong, brave = good. Those weak peasants? Just lame = bad." No hate, just facts to them.

Then the weak people got super bitter. They couldn't win in a fight, so they pulled the sneakiest move ever: they invented a new story: "Actually, being poor and humble and suffering is holy. Being rich and proud and strong? That's evil and you'll burn forever."

Priests (especially Jewish and then Christian ones) ran with this and sold it hard. It felt amazing if you were on the bottom, because now you were secretly the best and the bullies were going to hell.

That story spread like wildfire because it gave hope and revenge to billions of suffering people. Over centuries it totally crushed the old "might is right" attitude. So now almost everyone thinks in good/evil instead of good/bad.

Nietzsche's take: the weak flipped the script out of pure resentment and accidentally won the biggest culture war ever.

3. This is a bold genealogy move. The historical origin of our moral ideas matters for whether we should keep believing them.

Most people today think "good = kind, humble, selfless" and "evil = cruel, proud, selfish" are eternal truths from God or pure reason.

But let's attempt to look at this history. Those ideas were invented by weak, suffering, resentful people as a psychological weapon against the strong. They're not "true" in any objective way; they're just the victory cry of the losers who finally won.

So the origin (born from resentment, not from truth or nobility) is a huge red flag. If a belief comes from sickness, hatred, and the need for revenge, that makes it suspicious and probably life-denying, even if the whole world now treats it as sacred.

4. The greater significance of Benatar's writing is not in what he discusses but in how actively he takes measures to properly and deterministically determine the question.

the most important clarification needed when asking if life has (or can have) meaning is distinguishing between cosmic meaning and terrestrial/subjective meaning.

Earlier I assumed the question means: Does human life matter in the grand, objective scheme of the universe? Benatar clarifies that this is the cosmic perspective, where our actions and existence are insignificant against the vast,

indifferent cosmos, with no ultimate purpose or lasting impact. From this view, life lacks meaning entirely.

However, he urges shifting to the terrestrial perspective: Does life have meaning within our earthly, human context? Here, subjective fulfillment (e.g. through relationships, achievements, or personal goals) can provide limited, local meaning, even if it doesn't "redeem" the cosmic void or outweigh life's inherent suffering and death. It's like the local maxima on a differential curve.

This clarification avoids conflating scales, allowing honest assessment: Cosmic meaninglessness doesn't preclude earthly value, but it underscores our predicament without false hope.

5. A life is subjectively meaningful if the person living it feels that it matters, they have goals, projects, relationships, or experiences that make them think "my life is worth living" or "this gives my existence a point." It's about personal fulfillment from the inside.

A life is objectively meaningful if it has significance from a viewpoint completely outside any human mind, basically, if it matters on the scale of the cosmos, history, or some absolute standard that would still count even if nobody was around to care.

So yes, your life might be subjectively meaningful but if you are but a speck somewhere in the universe, the life ain't objectively meaningful.

6. Subjective (terrestrial) meaning Most people feel their life has a point because of love, family, creativity, achievement, pleasure, knowledge or just day-to-day enjoyment. None of those things require being a good person to be honest. A ruthless dictator, a selfish artist, or a hedonistic playboy can feel that their life is deeply meaningful, and, yes, in the subjective sense, it is. Meaningful in a bad way perhaps, but meaningful still. The 'bad way' is very, very open to subjective interpretation.

Objective (cosmic) meaning Benatar (and almost everyone else who thinks seriously about it) says cosmic meaning doesn't exist at all. So the question "does it have to be moral to have cosmic meaning?" is pointless- nothing has cosmic meaning, saint or monster.

7. I mostly agree with Benatar's way of splitting the question.. It's the least confusing way to talk about meaning without people talking past each other. But I don't quite accept his conclusions and categories fully:

- I think he's right that cosmic/objective meaning is impossible. The universe is indifferent; nothing we do will matter in 10^{100} years.
- I also think he's right that subjective/terrestrial meaning is real and possible- love, art, discovery, helping people, even small daily joys can make your life feel worthwhile to you.
- But what about the people who actually make an impact on an earth level with time period = 100 years? Say, scientists like Marie Curie. The current scale puts the subjective meaning of her life on par with that of a retailer. This is tough to digest without a confrontation. We need a category for these people. Or else the only reason someone will aspire to come up with a cure for cancer will be fun or pleasure, and not adding a meaning to life.

A sunset is beautiful even though it fades in an hour. A friendship is valuable even though both of you will die. The fact that the feeling is temporary and local doesn't make it fake or pointless while it lasts.

So my own view of meaning in life is living out your values. If at the end of your life, you can confidently say that you lived out your values, you have lived a meaningful life. For me, one important value is caring for animals.

The cosmic scoreboard says 0-0 forever, but I still want to enjoy the game and make it my game.

8. Cottingham says the biggest problem with human life is its fragility.

Everything we care about be it our plans, relationships, achievements, even our moral projects, can be wrecked at any moment by things we can't control: illness, accident, old age, death, or sheer bad luck.

His extreme example: a good person spends his whole life building a hospital for lepers, and on opening day a meteorite destroys it. All that effort turns to dust through no fault of his own.

This isn't just "stuff happens"; it shows that human life is structurally vulnerable. We are not the autonomous, self-sufficient bosses of our own story that many philosophers pretend we are.

He responds by saying that two things are required for a meaningful life- moral seriousness and facing fragility without absurdness.

Yes I agree on fragility: He's 100 % right that contingency is baked into the human condition. Anyone who's lived long enough has felt their most important projects break for stupid reasons.

Yes that pure defiant absurdism is emotionally exhausting and practically unsustainable for most people. Yes that some kind of regular practice (spiritual or secular) is probably necessary to keep going without burning out or becoming cynical.

I agree on the spirituality point too. But I will add one solution: indifference.

Sometimes, dare I say, all you need is a bit of indifference?

1. They define free will as the control needed to be truly morally responsible for our actions. In simple terms: you have free will only if you could have done otherwise in a strong, ultimate sense, and you're actually the real source (source of truth) of your choices.

Their definition does not settle what kind of control we actually need day-to-day. It sets a very high bar: if determinism or luck rules out this strong control, then free will doesn't exist (they're both free-will skeptics or hard incompatibilists sort of).

They define it this way on purpose. By tying free will tightly to being fully morally responsible (blameworthy or praiseworthy), they can argue that since we probably lack that ultimate control (because of physics, genes, upbringing, etc), free will is illusory. This lets them reject blame and retributive punishment while still keeping ethics and compassion. It's a strategic definition meant to make skepticism about free will seem reasonable and humane.

One angle, although distorted, that I would argue for is that even an individual like the richest billionaire does not have free will because he lacks ultimate control, given that he actually doesn't lack ultimate control because he can even make the impossible possible (you know, money?). Does this apply even to such outliers? I understand it is a distorted argument but I cannot help not asking it.

2. Being a morally responsible agent means you are the kind of person who can generally be held responsible: you understand right and wrong, you can control your behavior, and you are not mentally ill or a young child. In contrast, being morally responsible for something specific, such as a particular action, means that this one thing you did can fairly be blamed or praised on you right now. A normally responsible person can still not be

responsible for a single act if they were coerced, deceived, or physically forced.

When philosophers judge whether someone is morally responsible for a specific action, they make two fundamental assessments. First, they ask whether the action is truly attributable to the person- did it really come from their own values, judgments, and character, so that it can be said to be “theirs.” Second, they ask whether the person is accountable- whether it is fair to blame or praise them, which requires that they knew what they were doing, were in control, and could have done otherwise.

Both assessments must be satisfied for full moral responsibility, and hard incompatibilists argue that the strong control needed for true accountability is almost never present. I think that it might not be, but can we establish a benchmark here that any act exceeding this level of control will be held accountable irrespective of the exact or absolute level of strong control.

3. A thought experiment, it describes a situation where a person seems morally responsible for their action even though they could not have done otherwise.

There is usually a watcher or device ready to intervene if the person starts to choose something else. But the person freely chooses the intended action on their own, so the device never activates. Because they acted on their own reasons, we still see them as responsible, even though alternate possibilities were not really available.

They aim to challenge the principle of alternate possibilities, which says that a person is morally responsible only if they could have done otherwise.

Do they succeed?

Many think yes: if someone acts freely without intervention, they still seem responsible, even without real alternatives. But others argue the person still has a “flicker of freedom”, some tiny alternative, that keeps PAP alive.

My view: They raise strong doubts about the principle but may not completely defy it, because responsibility might still depend on having some possible alternative, however small. But sometimes Frankfurt case does emerge victorious. Because what if what we see as the only ‘option’ for our agent is actually the only option they silently wanted?

(Punishing a messenger for not being able to run 40 miles in an hour in Athens for saving the country could be indirectly justified by the Frankfurt case but is it even morally correct when a person can actually not even run 40 miles an hour?)

4. What do we mean by “free”? If free will means acting without any causes at all, then probably not, because our choices are shaped by biology, upbringing, and unconscious processes. But if free will means acting according to our own thoughts, desires, and values, then we likely do have it. We reflect, deliberate, and choose in ways that express who we are, even if those traits were shaped by earlier factors.

This is the compatibilist view, which says that freedom and determinism can coexist: we are free when our actions come from us rather than from coercion or external control. In this sense, we have enough freedom for responsibility, moral judgment, and meaningful decision-making. I do believe that in this sense all of us have free will to some extent but not absolute free will, but in most of the events that we encounter in our lives, we do not even need to call upon absolute will, and therefore, we have all the power to make our own choices mostly.

5. Nagel thinks the “fundamental problem about moral responsibility” is that our judgments about praise and blame clash with the fact that so much of what we do is shaped by factors outside our control. We feel responsible for our actions, yet when we step a little back and look at ourselves from an objective, external perspective, it seems impossible to justify holding anyone fully responsible, because luck- our temperament, upbringing, circumstances, and even how situations unfold plays a huge, huge role. The problem arises because we occupy two standpoints at once: the internal viewpoint, where we see ourselves as agents who choose, and the external viewpoint, where we see ourselves as part of a determined world. These viewpoints seem equally valid, but they cannot be reconciled. Nagel does not think there is a satisfying solution. Instead, he argues that the tension is permanent: we cannot give up either perspective, and therefore moral responsibility remains deeply puzzling and ultimately unresolved. But I wonder, why are we even worried about punishing/rewarding people only when full responsibility is proved? Can there not be a benchmark or is this a binary game?
6. Nagel describes four ways in which moral assessment is vulnerable to luck. First, there is resultant luck, which concerns how our actions actually turn out; two people with the same intentions can be judged differently depending on outcomes beyond their control. Second, there is circumstantial luck, which involves the situations we happen to face; the

moral tests we encounter depend on chance. For this, he gives the Nazi citizens' example and how citizens of other countries were not subjected to a test like this. Third, there is constitutive luck, which refers to the traits, temperament, and dispositions we are born with or shaped by, none of which we chose. Finally, there is causal luck, which covers the entire chain of causes leading to our actions, suggesting that everything we do arises from factors stretching far beyond us. These four forms of luck show how deeply moral judgment depends on things outside our control. I think it's a very interesting viewpoint. I admit I would like to use it if held morally responsible. It's a tempting mode of escapism.

7. I don't fully agree that people can be morally responsible only for what is completely under their control, because moral life doesn't work in such clean boundaries. It's true that responsibility requires some degree of control- otherwise blame or praise feels unfair. But in practice we often hold people responsible for things shaped partly by luck, such as outcomes they didn't intend or traits they didn't choose. I think an interesting example for this can be seen on the roads. If a careless driver injures someone, we still blame them more than a lucky driver whose carelessness caused no harm, even though the outcome wasn't fully under their control. If no one was hurt, we simply laugh it away. This shows that our moral judgments respond to both agency and results. Total control is impossible in a world filled with chance, yet abandoning responsibility entirely would make moral judgment meaningless. So responsibility seems to sit in a middle zone: we must have enough control for the action to express our character or intention, but not perfect control over everything that follows. (Also, I think morality can be of a consequentialist nature too. Sometimes we invoke thoughts about an action being moral or immoral if the action actually 'results' in something good/bad past/present/future. We mightn't often reprimand ourselves for thinking bad thoughts while we hold ourselves more morally accountable for actions that impact anyone.)
8. Excellence in speech and debate and how I won many such competitions is one 'event' I can speak about. While not exactly an event, I think it sufficiently makes me think about this question. My form factor, that depends on constitutive luck, comprises genes and talents that I inherited from my parents. Debate and speech from my father, an appreciation of literature, love and kindness from my mother. Then there has also been circumstantial luck because the world happens to value what I inherited or worked on. Example, the world values powerful speakers as time has shown us and the world values kindness and elegance. All these 'refined'

traits that I inculcated by dint of constitutive luck. I consider myself lucky owing to these happenings.

9. It helps avoid the moral luck paradox because it separates character from outcomes. Even if luck affects the results of our actions, our moral evaluation can focus on whether a person has good intentions, virtues, or moral character, rather than on accidental consequences. For example, someone who tries to help but fails due to circumstances beyond their control can still be judged a good person, even if the act itself has a bad outcome. This approach preserves moral responsibility while acknowledging that luck influences what actually happens.

Whether it is satisfying is debated. It partially resolves the paradox by protecting judgments about character, but it does not fully address why we still often hold people accountable for outcomes they cannot control. Critics would argue that separating character from actions feels artificial because in real life we judge people both by what they do and by the effects of their actions. So, while Hartman's distinction is helpful, it may not completely dissolve the tension of moral luck in the eyes of the critics. But I actually think it's a fairly good resolution because it's fair and actionable. But then again calling a person 'bad' is not entirely fair because a lot of 'bad' people become good when the situation is changed.

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Short Essay

4. The Type-Identity theory states that the mental type is the physical type, for instance, pain = C-fibers firing. Smart also presented an example of lightning = electrical discharge regardless of what definitions or terms we use for lightning.

However, the dual-aspect objection states that even in such a case, we are referring to the mental aspect of the substance (when in its mental state) and the physical property (when in its physical state). Hence, ^{we agree that} there are mental and physical properties of that mental state, so it cannot simply be reduced to a physical type alone. There are two aspects (mental and physical).

A reply to the dual-aspect objection is that pain has a functional role that is physical & it causes someone to withdraw their hand when touching a stove and a causal role that is also physical (pain is because of C-fibers firing) so it can be reduced to physical type.

10

6.

In the imitation test, Turing says that if an interrogator cannot differentiate between the responses of a machine and a human, then the machine has passed the test of ^{thinking} mind. If this is not enough then the question of ^{thinking} mind in machine is too meaningless to be contemplated upon. Since, even in humans, we do not see exactly what is going on in 'the mind' ^{when they think}, we should not worry about it in machines if they can produce indistinguishable answers.

And that humans have a soul and machines do not is a scientifically baseless objection. ^{Also, both humans and machines have limits} because then the functional ^{and behavioral} output of machine versus human is the same.

Turing replaces with a more testifiable question.

Short Essay

The whole system doesn't understand even if it is correctly able to produce Chinese, it still does not 'understand' what it produces because the system lacks the biological bases that cause understanding which are present in humans.

7.

The System Reply is that if not the person ^{alone}, then at least the person plus script of Chinese ^{plus output and algorithms} (i.e. the entire system) understands Chinese and the system is not just a formal algorithm but that it actually understands Chinese.

Searle's reply to system reply:

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5. The token identity theory states that the mental state is a token ^{that represents the mental state but is not it} physical state, so the state of pain is the state of C-fibers firing. It identifies mental and physical tokens to be the same and not the types ^{themselves}.

The Type Identity theory says that the mental type is just the physical type, so pain = C-fibers firing.

Modern Philosophers reject Type Identity Theory but some follow Token Identity Theory.

It rejects that the mental state is not also the physical state and claims that these are just two different ways to define the same concept, example, lightning can be expressed in many ways but it is just electrical discharge.

The Type Identity Theory is stringent compared to Token Identity Theory. (which allows more room)

10

2.

Descartes says that ^{non-human} animals don't have genuine minds because they lack general-purpose skills or linguistic capabilities. And while they can 'utter' language and words, and even though they are disposed to certain skills, ^{that they excel at} they lack the general purpose reason that humans have. The beasts' linguistic capabilities can be replicated by a machine, and overall beasts are just fancy machines with no soul that is immortal, indivisible, rational unlike humans. They do not just have little reason, they (beasts) have no reason at all and no genuine mentality.

Beasts do not have a mind or soul (that is indivisible) and thus can be replicated part-by-part

- 10 3. Genuinely meaningful statements are clearly well-defined ^{using language and replicable} else they are meaningless ^{they remain of no use}.
 For instance, 'the purpose of life is...' is not a genuinely meaningful sentence because it cannot be tested.
 A genuinely meaningful sentence can be empirically or analytically verified. ^{publicly} Knowledge can be used to give exact definitions to terms and if a concept cannot be defined clearly then it is pseudo-science.
 A meaningful concept in science can be publicly observed and doesn't rely on a priori knowledge alone.
 It can be postpriori verified and will not depend on ^{someone's} innate knowledge.
 This was considered to be the hallmark by the Vienna Circle of logical positivism.

100

only perfect score!

Short answer

10

1.

Descartes would ^{agree} concur with this because the unreliability of senses (I'm dreaming so I'm sensing everything even though it doesn't exist) does not deny basic mathematical facts like 2 is the first prime number, and only the evil demon can render it wrong.

10

4. Behaviorists want to eradicate the mediator (inner state) by relying on behavior that people are disposed to on account of mental states. So a person suffering from pain would display that behaviorally. But Putnam's point is that behavior can be masked and faked as in the case of extraterrestrials who despite suffering from pain (even though the material substratum of that pain could be different from that in humans) ^(but the function is the same) do not show their pain.

(Also, any behavior can be caused by any mental state, so it cannot be mapped directly.)

So relying simply on the behavioral account display of the mental state discounts the inner states and thus is unreliable.

Notes on a class

Objectivity, the child of the 19th-century American historical profession, was strung to ideals of scientific rigor, neutrality, and empirical fact-finding much like a gentleman is strung to his tie. I have always thought that while objectivity is often much 'scienticised' in science, even the sciences are not without their subjectivity. For instance, the laws of physics are only correct as long as there is a silent agreement between the reader and the writer that the drift of the wind is not to be invoked.

Around the time of the influence of Wissenschaft, historians like Leopold von Ranke emphasized the pursuit of historical truth 'as it really was'. Ranke's approach was rooted in a Romantic-idealist belief that historical scholarship revealed divine order through critical engagement with archives. If it appears 'higher order' to one's readership, I beg to state that a man of Ranke's status had all the right to indulge in such a stringent order of belief.

American historians, however, stripped objectivity off of its philosophical underpinnings, reducing it to a methodology of detachment, neutrality, and avoidance of moral judgment. The lonely flower on the moon, objectivity, was now bereft of its beauty in the epoch of American historians. Was that the sad beginning of empirical humanities?

Our class discussion seemed to converge about this point when Jay mentioned how even statistics is not so objective after all and Math remains the only science that could survive the American historians' strict ideals.

I felt myself very anguished when I read how scholars like Herbert Baxter Adams trained in German seminars, imported the seminar method and the ethos of history as a 'science', because despite being a student of the sciences, I have considered the ethos of history to be history itself, unblemished by the unforgiving aspects of science.

The tension between objectivity and individual perspective is central to Novick's and Haskell's analyses. Novick highlights how historians, despite professing neutrality, were often driven by political and moral commitments. Haskell, however, distinguishes objectivity from neutrality, defining it as a practice of self-discipline: the ability to engage rival perspectives empathetically while subjecting one's views to communal critique.

As a class, Haskell's was the most neutral definition of objectivity we all could agree upon and envision ourselves believing.

For Haskell, objectivity requires 'detachment'—not the erasure of perspective, but the ascetic effort to bracket one's biases and inhabit alternative viewpoints. It is as stoic as it can get. Why Haskell purposefully defines objectivity so strictly, I do not know, but it aligns with Nietzsche's grudging admission. Both authors appear to agree that absolute objectivity is unattainable.

I like to think that the test of objectivity lies not in the absence of bias but in the transparency of one's methods and the receptivity to critique. That is also the limited role it plays in my work.

But what does play a great role in my work is my sense of purpose. When prompted to share my calling in the class, I spoke of it in ways I think about it—when you pursue 'it' for pure pleasure (regardless of the paycheck). My material definition was complemented by Jerry's deeper definition of what it means to have a sense of purpose and by the calculated definition of Jay (the matrix of risk of achieving versus number of people affected by the work).

I think we all broadly agreed that a sense of purpose is transformational in nature. We discussed how the word 'Vocation' boasts of its Latin root 'vocare' which means to call/invoke.

For Weber, the rise of Wissenschaft under modernity makes the scholar's vocation very transactional. Academic freedom was compromised by ministry mandates, secretive appointments, and political patronage, eroding the ethos of scholarship as a lifelong pursuit. The American system, while fostering competition, imposed a 'ruthless selection process' and excessive teaching duties that left little room for original research. In both cases, the ideal of scholarship as a transformative calling gave way to transactional careerism.

We also discussed how America plays a central role in Weber's analysis. U.S. universities, though bureaucratized, retained more independence from state control compared to Germany. Their differentiation and 'Europeanization' hinted at a path where institutions could balance administrative efficiency with intellectual freedom. Weber first-handedly saw two universities across from each other in Chicago— the glory of academic freedom.

Our discussion about the confines of a young docent and the classes he is allowed to teach was especially intriguing because, even today, in the humanities department at Stanford, rarely does one see a young graduate with the entire class to himself. He is often confined to the occasional visitor in the office hour. Weber brings this up when he talks about the Privatdozent who confined himself to secondary courses in order to give regard to his elders and those who came before him.

What I took away from this class is the thought-provoking realization that the world is slowly venturing away from the time when vocation was a pursuit of purpose into a world where it is business. Weber rightly foresaw it when he spoke about how we are moving into a world where certificates are key.

After this class, I would like to add but one word to my former definition of sense of purpose, 'passion'. For as passionately as Max Weber speaks about the profession of science as a noble pursuit, I have greatly enjoyed reading the piece.

Private Universities in the Public Interest" confronts a fundamental paradox: elite American universities, long revered as intellectual citadels, now find themselves perched precariously atop an anthill of mistrust and skepticism. These pinnacles of academic and societal progress seem to have drifted from their original social contract. Is this a slow erosion of trust, caused by opacity, and exclusivity?

For those of us raised with near-religious reverence for these academic giants, the disillusionment is palpable. As I read, I was reminded of how, growing up, I had come to know these universities as temples of reason and hope, places where intellectual rigor intersected with societal advancement. Yet the article offers a historical lens that helps contextualize this fall from grace. American higher education was not always so disreputable. There was a time, as the authors remind us, when universities were tasked with molding the nation's future—training soldiers, advancing technologies, and contributing to the public good. Jay added a fascinating perspective, noting that wars historically accelerated innovation, with universities often at the forefront. A symbiotic relationship—universities received public investment, and society reaped the benefits of their knowledge production. In this way, universities functioned as more than ivory towers.

However, the article doesn't resign itself to cynicism. It challenges universities to engage in self-reflection and reform—not through defensive public relations campaigns, but through real, substantive change. This call for reform resonates with a broader theme we've encountered in our class discussions, particularly the idea of objectivity. True integrity, as we discussed, doesn't come from detachment but from the discipline to receive critique and engage with differing perspectives. This brings me back to our discussions on the academic social contract, especially the tension between public funding and private governance. We spoke in class about how Stanford University's contract with the state of California differs from that of Freie Universität Berlin, and how local politics shape these relationships.

Peter Maassen's analysis has an interesting angle—as universities move away from traditional models of teaching and research, they are now embracing a "third mission"—economic engagement. The pressure is mounting to quantify their societal value. (I haven't yet decided if I appreciate this urge for quantization or not.) Our class discussions about how universities are increasingly driven by market forces, looking more like corporations than traditional educational institutions, was an interesting addend to this. Maassen's warning about "institutional imperialism" highlights a risk that universities, in their quest for economic utility, may lose their distinctiveness. This is a concern we also

discussed in relation to Max Weber's critique of the scholar's vocation. Weber argued that modern institutions, particularly in the U.S., have increasingly turned academia into a transactional enterprise, eroding the ideal of scholarship as a lifelong pursuit. In a way, this process has commodified intellectual work, reducing it to something to be bought and sold.

But I will add that the tension between the pressures of economic engagement and the preservation of academic integrity is not just theoretical but deeply embedded in the everyday experiences of junior faculty. In our class, we discussed the Junior Professorship model in Germany, which promises earlier career independence in exchange for the abandonment of the Habilitation. Yet its limited success speaks volumes about the mismatch between policy and institutional culture. Junior professors, especially in large universities, often face a reality of job insecurity, excessive duties, and little opportunity for meaningful research. This experience underscores the failure of the academic social contract. When junior faculty are burdened with the responsibilities of senior professors without corresponding rights or even an acknowledgement, the contract appears deeply unfulfilled. It appears to reveal the increasing corporatization of academia, particularly in larger universities where the numbers and demands make the system untenable.

"Negotiating the Academic Social Contract," gets into the roots of the academic social contract offering further insight into the evolution of this issue. The Truman Report of 1947 and the GI Bill marked a time of expansive public investment in higher education, with the expectation that universities would serve the national interest through research, workforce training, and the democratization of education. In class we discussed how this agreement shifted particularly after the 1920s, when rising tuition fees began to erode the notion that higher education should be a public good. The shifting dynamics between state funding and private governance continue to shape universities today, creating a landscape in which the academic contract is ever-evolving and contested.

In the end, these readings and class discussions remind me that universities are not monolithic institutions. They are shaped by the people who inhabit them—students, faculty, staff, administrators—all of whom have the power to change the trajectory of these institutions. For someone like me, who hopes to work within or alongside academic spaces, this piece poses a quiet but profound challenge: How do we restore reciprocity between academia and society? How do we balance autonomy with accountability? Most importantly, how do we remember that maybe, just maybe, education, at its core, is not a privilege but a promise?

Our discussion highlighted how the distinct academic social contracts in America and Germany were shaped by differing political, economic, and social contexts. In America, the rapid expansion of universities during the mid-20th century was heavily tied to national needs, like military research and workforce training. This created a contract based on public investment and service. On the other side of the ocean, Germany's system evolved under a different set of conditions, with universities emphasizing academic freedom and autonomy. Weber's analysis highlights the contrast: in the U.S., universities became more bureaucratized and market-driven, while in Germany, there was a stronger focus on academic integrity and the role of the scholar as a vocational pursuit.

Is it possible to change course once certain conditions have settled? It's difficult, but not impossible. Reform would require a fundamental shift in the way universities view their role in society, especially in the U.S. Universities would need to balance their corporate strategies with their original mission to serve the public good. This could involve rethinking funding structures, prioritizing accessibility, and ensuring that academic labor—especially that of junior faculty—is respected and supported. Ultimately, this would require a collective commitment to reimagining the academic social contract and restoring a sense of shared responsibility between universities, the state, and society.

Animal Capitalism

The shadow of humans choreographed on animals has led to flimsy lives for some pets while critical sheds for others. This theatrical extravaganza is what I plumply put forth as animal capitalism.

Not only is it one of the most dissolving problems of the human-animal relationship, but it is also easily a benefactor social evil. While economists will explain to you how benefactory it is, I will dissect for you the evil in the definition of it.

The Man is God's propagator of His creation. When He created hunger, man spread it to the animal world by dragging their self-sufficient ecosystem down by its deep-rooted hair. He domesticated the animal, failed in the stock market and gave the animal lesser and lesser shares- as many shares less as he had lost that day. He started passing his losses onto the donkey's back. This is how the early man lived.

Some mass killings later, the modern capitalist man was born. He decided that some animals deserved jewels and others deserved the farmhouse. (He treated his favourite wives in the same fashion.) While Winter, the pet of Madame Cooper, itched his fattening curves, Gasp, the bull awaited his weekly meal while pondering the meaning of his life, and Dig, the shelter dog searched for food in the wound on his leg. The next day his dangling leg was cut, packed and sent to the laboratory.

Two economic crises later, the world was divided into the modern and the ultra-modern. We switched from English to Greek as Gen Z became Gen Alpha. The ultra-modern man invented the QSAR technique (which I hail as the most civilised discovery of mankind) and animals didn't need to be sent to the laboratory. Computers replaced animals.

However, the modern man could not accept this discovery because he was cheap. He saved money off b'day gifts, made profit off animal dissections. He did not want to think about the QSAR because his life was great and his pet was fat.

This modern man became the mummy for animal capitalism. He maintained his position in the ether as he was too modern to become the early man and too ancient to become Gen Alpha. In this confused state of affairs, he managed to transfer capitalism off his shoulders to the animal world. He did not understand QSAR and he did not understand the concept of a humane civilization. The modern man thus came to be the father of animal capitalism.

I propose two solutions for the settlement of the progeny of the modern man.

1. A problem-solving mindset: We have harnessed animals better than our own species. While your favourite human might not jump up at your clap or sit down at your knock, your favourite animal is the master of your hand-verse. Thence, although capitalism appears to be the natural order of things in the human world because we cannot brute-force our fellow humans into equality, our extensive control over the animal world can assuredly lead to the chemical dissolution of this extravaganza.
2. Wider use of the QSAR technique: This will inevitably lead to cheaper machinery and computer testing structures, minimizing economic costs in the long run while letting animals retain what belongs to them. In a world so madly in love with rights, why should animals' rights be forgotten and forlorn?
3. An appreciation for the changing world: An understanding of the fact that compassion comes with civilization and that a change in the ways we 'process' animals is urgently required. We cannot continue to send off animal body parts to dingy labs the way our forefathers did. Times change, people change, tools change.

Law essay

There is one thing common between the darkskinned labourer tapping the axe on the metalled road, and the fair-skinned not-a-labourer me. We both were forced into maturity.

They say, life is easier when you are just a spectator. But this never worked for me,

I could see the broad little face endowed with glimmering eyes searching for the Sun in the soil as yet another blow was planted upon the land. Every boy a tick of the Earth's clock. Every blow a reminder of every little drop of sweat that struck his brow. Every blow a mark of yet another child who had been successfully deprived of the rights he was born to in a world that always talks of rights.

All those times, I'd but one sense organ- my eyes. The sweat, the toil, the heat- I could sense them all through my eyes. With every passing day, the rusted iron of the archaic window that opened to me a grotesque world outside so much in revelling contrast to the world inside, matured along with me.

In school, we sang of the 'big, bright, beautiful, wonderful world, but through the window all I could see was a world of misery and dismal gloom- the song of the saddest toil- the song of the people who hadn't yet been told when the Earth would become bright and beautiful for them.

Every quiet scorching summer afternoon, I, with my little fingers, would open the window of the room that overlooked the tents of the homeless, the trashcans that contain'd 'food for them' and the roads of the government. I could see colonies Bihari mothers with their Bihari children- all toiling on the land we called our mother. Colony on colony. Toil on toil. Daily. Everyday.

And so, when our Civics teacher asked me what hierarchy was, it wasn't surprising when I defined it as- The system where the conditions of the child's birth determine what he is at age 16 and who he is at age 60. I told the prof. about my inherent contempt of this divided institution that allocates hell and heaven hereupon the Earth.

What accompanied my statement was peels of laughter from the first and back benches alike. According to our system, defending the 'low-class' was not 'noble'; feeding the 'low' made you lowly and speaking up for them classified you as a jester.

I grew up as a part of this systematic hierarchy. The misery I saw shaped me into a weed in the system. My first-hand experiences gave me the courage to speak up for the truth was with me, for I didn't have to gather the truth from ancient sources. It lay there- right outside the window. The truth assured me I was fighting for a real thing, a living battle. I was fighting to bring cheer to faces I saw everyday. I was fighting to salvage myself from such painful sights.

It was this truth that made me hold the city's mayor accountable for the labourers' work conditions and irregular meagre wages in a program of thousands. It was this truth that forced a 10-year old me to stamp my feet in the RTI meeting, 2015 where I invoked the Right to Information Act to question the dignitaries about the efficacy of administration of the labourers' funds, a subject that had been brought to my notice by a Bihari mother who had knocked at our door three nights before.

These years of sublimation from childhood to maturity awakened in me the desire to toil and feel the labourer's destitution with not one but five sense organs- my desire to be one with them, one among them.

For the time being, I felt content giving the labourers halwa and puri to eat. I even marked my place with orange, white and green on the map of our country for it was finally 'big, bright, beautiful'.

But when I learnt that the map of my country rolled into a larger and an even larger map, I realized how short my life was to cause the colours of happiness to spill across the map, that was, unless I teamed with a tool that could cause a fundamental change the waves of which would longitudinally traverse through time. That was when I learnt to appreciate the legal order as the most fundamental unit of change and knew I had to team with it. This kindled in me the desire to embark on a quest that extended from the Pacific to the Atlantic, from the Arctic to the Antarctic as a lawyer, marking all the space with . . . and on the one single map I have known to exist.

Verses: rhymes churned with philosophy, some black, some white.

What poetry means to me

As the tides are to the ocean, poetry is to me. If you wonder, why I denied to compare poetry with something sweeter, then let me tell you that poetry to me is not sweet, not deep, not broad. To me, poetry is a sketch carved on the conscience of this earthly life, that which finds its depth in the hollows of the ocean, that which is inevitable and ruthlessly invincible- the most sought after refuge to the insane-the most sought after refuge to me. Nay shall Tennyson, nay shall Wordsworth define the languidity I seek in this tidal dame.

On Fools

I carved a sculpture

To offer in the barn.

I tore open my sweater

To earn some yarn.

They call me a fool.

I call myself wise,

Since I appreciate

Everything thrice.

They call death a tragedy.

I call it a melody.

They call life a melody.

And I call it a tragedy.

My friend, fools are fools

And shall remain fools.

But the death of a fool

Is the saddest tragedy.

The Opening of a Door

The shadows unveil

The curtains open

Arrives my page

To summon the morn

A letter arrives in

All its haste

The page is flipped

And the box is blown.

He steps in and

Looks in yonder.

I step aside in the aisle.

Our eyes meet.

So do our hands.

The door opens.

Arrives our band.

I would lift a stick and carve my sculpture

Piercing through the pearly sheet of my molar

I would form and reform my eyes until it's over-

The serenity I've longed to see, I've longed to be

I would form and reform my ears until it's over-

The melodies I've longed to hear, I've longed to bear.

I would form and reform my heart until it's over-

The life I've longed to live, the heart I've longed to give

To the someone who holds me near.

I know I'll slowly, slowly forget him.

As slowly, slowly, as the mountains

Rise, rise into the skies;

Their height increasing every inch.

So will, so will, decrease my memory.

Unable, unable to fit in the past

Sweet-sour experiences

Wrought of the unworthy soil.

He is an idiot,

Idiot, aha, idiot!

He is an, idiot,

idiot, aha idiot, aha!

O, let me just say it right

Oo, he's an idiot,

O, idiot!

And I mean it all,

He is an idiot,

I don't give you a

Fuck or a damn,

I don't have the patience.

It's over.

It's over.

It just gotta be that

It just gotta be over, over

It just gotta be over.

He just wants attention.

And let it be a clap.

And I won't mention

Him in my diary

He doesn't deserve a place.

As he is, is

An idiot, idiot, idiot.

He is an idiot,

Idiot, aha, idiot.

O, you just let me say

It laa-oo-uud

He is an idiot,

O-oo an idiot,

And I do mean it all...

He is an idiot,

O-oo an idiot,

And I don't give him

A fuck or a damn...

Yaah, ahh, Yaah, ahh

YAAH...

You are killing me, YAAH-ahh

Yaah, ahh, Yaah, ahh

YAAH...

You are killing me, YAAH-ahh

You are killing me, YAAH-ahh

Your words are killing me

YAAH-ahh

Yaah, ahh, Yaah, ahh

YAAH...

You are killing me, YAAH-ahh

You wanna change me?

Not gonna happen..ahh

Because my spirit is so ME

i can live.

I can die, I can starve

But I can't change.

YAAH-ahh-YAAH-ahh...

Because my spirit is ME.

I'll get ME to my toes.

You will not pay, not to say

I'm gonna be ME, ME, ME

I can clasp onto you,

I'll count on you.

And I won't give you a buck myself

YAAH-ahh...

You are killing me, O, YAAH-ahh

You are killing me, O, YAAH-ahh

So, let's be friends only

YAAH-ahh

And don't you try to change me,

YAAH-ahh

I won't give you a fuck or a damn.

Because I'll be only ME,

Oo, YAAH-ahh,

YAAH-ahh, YAAH-ahh,

You are killing me, O YAAH-ahh...

Your words are killing me,

O YAAH-ahh, O YAAH-ahh,

You wanna change me?

O, it's not gonna happen

Because my spirit is ME,

I can live, I can die

I can strive

But I cannot change, ahha

Because my spirit is ME, ahha

You got me to my toes.

You will not pay not to emerge

I'm gonna be ME, ME, ME.

I can clasp onto you,

I'll count on you.

But I'm just gonna be me...

Pinch, punch, first of the month,

No returns of any kind.

A slap and a kick for being so quick,

O stop, buddy, don't just mimick.

So tomorrow is buddy's exam,

And you know what it's not to cram.

Calculus and all this fuck...

O, I just got parred.

Just see his expressions,

The boy is getting shirty.

O hell, beware, before the nostalgia

Sizes and enlarges to bump you downright.

He is the bee's knees,

O, at least that's what I think of him.

And his calculus couldn't allow his

Being on his temper like a car bonnet,

And that makes the sonnet.

The road is getting mucky

As I envision him doing the bite' n bobs.

Oops, If you wonder why

I know more about him

Than I possibly know about myself,

Then let me tell you

That I'm an anorak into his life.

And all this despite me being on the pull.

After all, it's buddy's exam demain!

WORRY

O why can't I get myself to believe

That what's going on in life

Has nothing much to worry about.

But no, how can I ever leave worry

Or maybe it's worry

That can't part with me.

Either be the case,

Worry makes me a thinker,

And no, I'm not a thinker,

Am just a person who worries.

..

ECHO

My life gets me changing,

My life gets me brooding

Over the things one would call 'petty',

But for me..

Oh, they are the whole world to me.

And I'm okay with what I believe in,

My thoughts and my approach

Gets me flingin'

In the tightest manner,

So all around and about

Just sit quiet

To let me hear the echo

Of my own voice.

I've turned a new page

As I write this song,

And let me be as plain as this.

I've turned a new page

As I write this song,

And oh, let me be as plain as this.

In the hollow sea of mortality,

My life is wreaking havoc

On myself and on my guts.

And this patience has ceased to end.

I've been struggling

For a while now

And nothing ever seems to come to pass.

And now I've started believing in

The law which I so strongly opposed.

The bad has to come to pass and let it do its work.

Morphy's law is no longer a myth,

My life has proved it,

And I'm a penny-worth.

This settlement makes me ponder

Over the facets life presents,

And I'm ego has obediently

Surrendered to this fa-ai-ai-act.

And I can but sit with my hands crossed

Or maybe the legs that move.

And no, it won't ever change.

I've been turn into a pessimist

After this thing changed my life,

And this thing is none other than LOVE.

o, WHY DON'T i STOP MYSELF FROM..

i KNOW IT'S BAD, IT'S BAD,

BUT STILL THE INNER SELF URGES

aND THE OUTER SELF

iN GLOOM SUBMERGES.

o YEAH, IT BRINGS ME GLOOM,

aND WITH MY PATIENCE ON HOLD,

i'D RATHER ACCEPT IT THE WAY IT IS,

PERHAPS NO CHANGE CAN BE WROUGHT,

AND OKAY OH, THAT'S JUST AS IT IS,

BUT NO, CHANGE IS A LAW,

MY OUTER SELF APPEALS TO MY CONSCIENCE

AND THEN THE INNER ONE RESIGNS.

I AM CONFUSED WHO SHOULD I FAVOUR, O LORD?

WHEN IT COMES TO NONE, IT GOES TO YOU.

These words crash me,

They crash me as never.

I'm lost in the sea

I'm lost in the sea.

I'm lost in the sea.

It requires a great deal

To move mountains.

It requires a great deal

To crush me down.

And congo, you did it

You did it.

You did it.

My life is shaken

And the earth beneath me rocks, rocks, rocks

And I can but stare

At something

That passed so long ago

And I was left dazing

At a thing soo amazing

That which all happened out of the blue

And my words are damn blown

Away from the castle offshore.

Something is parking in there,

And I'm willing to seize despair,

But ho, ho, ho,

Something requires a change

This thing tore me down as a blade

And tis shade

Gets me 'shady',

O night, spare me!

Am calling someone to call upon,

As I'm so lonely.

The epoch has passed,

The epoch of turmoil,

O congo, has passed.

And now there is no more

Of shaking grief,

All has just come to pass.

The epoch did pass,

The end is holy,

The beginning was evil,

And as always I'm gonna fast.

The time is holy.

Good over evil

And come, let's pray

Pray for the devil.

Pray unto the devil.

Shouldn't there be

Shouldn't there be

a thing to pray for?

And life is holy,

So, come let's sway for

Something so, so, so, so

UNHOLY.

SOMETHING'S WRONG today,

Ooo, Something is wrong

About the rain today,

I feel it drizzled less.

Nothing to mimick,

It puzzled less.

It unveiled its charm

In too much of a subtle way

Conveyed by its own ethnicity

That makes grief for the most part

And lies on the bank of the sea

Drooling, screaming its crib tale

That makes the worst part of grief.

Grief is its charm

And I respect it.

Hold my hand,

And here we go,

O, Let's soar

Up, up, up...

Hold my hand

O, let's rise

Into the skies

Up, up, up

Hold my hand,

Let's roar

Ra..Ra..Ra..Re

Hold my hand

Let's soar

Up, Up, Up

Time to elapse,

Time to collapse

Bounce,

Let's bounce.

But where do we go?

Into the skies,

Up, up, up..

Holding,

O you Lord of Highs..

The way to hell is closed.

The way to heaven is closed.

Where should I go, YO?

O, My Lord...

I'm not let on the earth

Or I don't let the earth

Rest upon me.

In either case, I'm

Thrown, thrown, thrown.

With the dusky,

I'm smokin'

BLOWN, BLOWN, BLOWN.

Nothing to drape have I

I know no space,

But I know no danger,

As I'm in your eyes.

And I know no hanger

'Cause rage's bro

Infuriated me,

And I need it no more.

Argh, I do seem to know

HAnger.

O, Why don't you ask the Goddess

To curse me?

I pulled a blinder

But no home, home, home.

I'm not a mortal,

And I'm not immortal.

I'm not skew-whiff

Just a bit shirty.

Not at all dirty.

I've had myself pruned

For long ante I turned up here.

But I know no home, home

And I know no love

Cuz no one ever loved me.

Drizzle, Drizzle!

O Rain, you

Drizzle, Drizzle!

Ahem, listen baby

Hushh, listen baby

It's a melody

Woven for my ears, ears.

Drizzle, Drizzle!

O Rain, you

Drizzle, Drizzle!

O Rain, you are a mood.

You go hunting the skint

And you go gracing the minted.

You are not the one to skive

The SUn feels so live

When your alive.

Only the jammy, ohh

Enjoy you, O Blimey, ohh...

The wonky go, go, go

TONKY

And you drizzle, drizzle.

You being so wicked cool

Can dazzle anyone.

I want your cool

And your graced sonnet

Will lend you my bonnet.

But your shady, ohh

For this lady, ohh.

You are off the hook

And I got you in the nook

As I went searching ohh

For someone to dazzle.

O, you dazzled me.

I gotta hit you

With a lance.

You ceased my attraction-

A BLOWJOB

And I lost my attention

That for which I'd worked so hard.

I know no forgiveness

Just as the Rain God.

But you drip, drip, drip

Ain't in your grip, grip, grip..

Drizzle, Drizzle!

O Rain, you

Drizzle, Drizzle!

Ahem, listen baby

Hushh, listen baby

It's a melody

Woven for my ears, ears.

Drizzle, Drizzle!

O Rain, you

Drizzle, Drizzle!

Imperceptible

All ready to sing

All ready to laugh

O life your so, so much of

An imperceptible scarf

My heart's got some bout

Of maybe, pullout.

With my head in he cloud

When pigs fly

O life your so, so much of

An imperceptible scarf

Thick as thieves? Aha..

We shan't fly as it leaves

Hands on, on, on

Wahey, you spill it out

Got some bout?

Of pullout?

All ready to shout

Don't fret.

I'm on it like a car bonnet

All ready to sing

All ready to laugh

O life your so, so much of

An imperceptible scarf

Keep your cool

It's everything you can keep.

Keep your cool

You dread it the most.

Keep from harm

No one can say NO to it

Keep your cool

YES to it.

Cool is a YES-man

And you gotta mean it,

And what's more,

It dreads the least

Keep your cool

And your outta harm's way.

DREAMS

Let's run about

And stare at the shops

The lolly in my purse

Won't get me a bolly

But am free to stare.

No one can snatch this right

From me, from me.

Big dreams are not to be dreamt of

Think in your limits-

I'm often told.

Big things are not for you.

They are for the rich,

And you've no right to dream of them.

Maintain a decent chap living.

And don't get your head in the clouds.

Stop puffin' about-

These are all for the rich

And their kids

And not for the mint.

But uh oh, why can't I even dream

Of balls, dresses and cream

Something is pulling me back-

The fright of my own dreams.

Respect-the word that I was never taught

Until I met this stranger

Who was so freakin' wrought

To me by the heavens above.

He was subtle in his methods,

In fact too patient.

Never got so slow approaches

And he meant to take them

As though it would take an aeon.

It was then that I learnt

That respect and patience

Go hand-in-hand.

O, the impatient, you have problems

With respect...

I myself find respect is a slow process.

But no, no one will listen to me

When I say with my optimism

That cool, patience

Is about respect

And respect ditches it

When the impatient

Can't hold their horses anymore.

Songs of mystery,

Songs of love,

Songs of laughter,

Songs of the heart.

O, it hurts to leave a song unfinished.

Like everything it needs

Its nine months.

Songs of mystery,

Songs of love,

Songs of laughter,

Songs of the heart.

O, don't hurry

Infuse it with life,

Make it living,

Make it live.

Don't give a cold approach.

Its skin tone is light

The tones gotta be tanned,

But this is life.

Don't mix and match

Preserve it as what it is..

It was born to be unique.

It was born to be unique.

Feed it with your

Burning heart's passion

Don't leave it alone to cry.

It has got a life too,

O just let it cry and sing

And as need comes, you may ping...

Songs of mystery,

Songs of love,

Songs of laughter,

Songs of the heart.

The beauty of this earth

Sinks me so deep,

It doesn't let me float

Round its open boat.

The beauty of this earth

Rages a war

To conquer the land of my heart

Something's always moving about.

Be it earth,

Be it water,

Be it life.

Be it love,

Be it discord,

Be it strife.

I just get no peace.

Even in the

Realm of the grove.

The last refuge for me

Is perhaps my own grave.

And nothing defies its beauty.

And this small piece of land

Was gifted to me

By the Earth alone.

And O, my friend,

It rests on this land...

O, let me live in peace.

It's time I reunite

With the old myself

That had been subject

To its fast epoch.

I want no change in me.

I love my old old ME.

O, let me have my space.

O, let me be my old myself.

That triggered your attention

In the very first place.

But now I'm changed

And I love only the old myself.

I've lost my old beauty.

Old is Gold,

Okay, so, let me dig it.

O, let me stare at etoiles.

Their twinkle dims the rustle

And brightens my self.

O, just gimme my time.

Let me be gobsmacked.

At the party,

O, let me celeb my old self.

O, let me live in peace.

It's time I reunite

With the old myself

I just love to keep

Ooo...things just as they are.

Ooo, Ooo,...

I just love to leave

Things broken and open..

Who cares to close and mend

Close the chapter

And open the book...

Mend yourself...

I'm okay being ME.

Hi Life, how do you fare?

Nice to meet you

After so many decades.

My youth passed

And now I've you back.

You ghosted me

For a whole half-a-century.

Hi Life, how do you do?

How dope, how dope, how dope.

Forgotten your facon, dear.

Will let you have your way

As long as you go on mine.

Hi Life, howdy?

Missed you.

Wanna tell you your pucker.

I know I realized it too late.

And now as I look into the mirror,

I see my reflection

Looks like my shadow should be

And as is needless to say,

Brother Time caused this change.

BUDDY

O, just encourage him!

O buddy, everything'll be nice!

I don't know why you worry

So fuckin' hard!!!

And it's a worry to all.

So, no ill-luck to you alone.

Just stop this crazy thing

And be your cool yourself.

Break the leg demain

And for God's sake,

Do stop acting like a dorm.

O, just be a shit bit wonky.

Come, O, let's be frank.

Ante you get 'cank'.

O just encourage him...

Keep your temper dank

Ante you get cank.

Keep your chin up

And break the leg demain!!!

You gave me no souvenir to keep.

You gave me no souvenir to keep.

As you hardly realized how much I feel.

I'd nothing to cherish,

Nothing for to weep

And so I got my tears a box.

I know that would evaporate, LOL!

Wish Science had no Conscience!

But still their presence was felt.

And those memories got so afresh.

The box is now the souvenir

And I'll give it to you.

My heart's attached to

Something you call 'nonsense'.

And my life is built around the thing

You call a 'waste'.

You gave me no card,

And I was okay with it.

And you didn't bring me

A souvenir, souvenir

And you didn't bring me

A souvenir, souvenir.

But you brought me tears-

A souvenir, souvenir.

But you brought me tears-

A souvenir, souvenir.

The night's over

But the tears are not.

Like crazy, crazy.

And this image is all

So dazy, dazy.

These are the souvenirs

And you gave them all.

These are the souvenirs

And ta, you gave them all.

Not one bit was spared by me,

And now I can feel what I see.

Not one thing was granted by you,

But now I see you in everything...

My heart's saying,

(The baby it is)

“Can’t bear any gainless pain.”

Could anyone make it understand?

An excerpt from my book ‘Little Modi, Big Modi’

Dedicated to every Indian.

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Introduction

Life is an experimental experience which is subtle to experience and is open to its traits.

It can be defined as a path which is laid with trees of action and fruits of enlightenment.

Every life is a varied path with its own fruit of enlightenment. Mine is a blossoming one. The denizens of India have a blossoming one- in the divine avatar of Narendra Damodardas Modi.

This is the story of that fruit of enlightenment, although in refined and cultivated terms, it's not a story but a legend, a legend immortal.

The existence of this world is brought into existence by mobility and action. It's mobility alone that initiates the process of birth. It's mobility alone that brings two mates together. Further, it's the mobile sperm that fertilizes the ovum, resulting in the formation of zygote which then occurs as an embryo, and then a foetus and then arrives the creature onto the planet earth. When the creature is born, we know not what fate has in store for it. Ad libitum does the child work. When it's born, it's but externally touched by the mundane macrocosm, and as it grows and increases by a factor termed as 'Age', it peeps into the materialistic environs, and lo and behold! The charm of Maya engulfs it and as a child, it fully enters the trap laid by it.

This world has a history: The History of the Immortals, and it's this record that salvages Civics, and yes did I mention what it takes to be the Civics of tomorrow?

As the child turns into a full-fledged young man, his deeds come to be his definition of the self. Some own the self, others forget it in preference to resting upon the bed of mortal sleep. And this is when the cycle of Time pauses, and this is when Actions dwell upon beginnings, and this is when conscience matters.

One such being for whom Time paused, Actions dwelt and Conscience mattered, great de facto, is India's Prime Minister, Narendra Modi. And no, his actions ain't bigger than the self he is. Ipso facto, his *modus operandi quod est memine* contradicente is practical, rational, logical, realistic and the big thing- UNIVERSAL.

Factually, each and every second of such a priceless self can be detailed on a thing called paper with a thing called pen. But can the Identity dwell on 'things'? Chee'uh, NO! This shall lead to thousands of volumes or maybe more (do correct yourself)- all recounting episodes of sky-reaching height and horizon level width. These infinite dimensions give this a three-dimensional view and a three-dimensional perspective, and I'm unable to present it on the two-dimensional 'thing' called paper due to my limited knowledge acquired during the course of some fifteen years. With the humble knowledge I acquired in this course of life, O, Thou, let the book be at least partially worthy of the subject.

With an effort that's due to the creation of this book, this concise biography of a person called Modi on a thing called Paper is but parallel to a drop in the vast ocean of this person's life, highlighting but a two-dimensional perspective.

And yes I introduced the hero of the book.

And yes, I introduced this person as a person, and I hope that would suffice.

PART 1

EARLY LIFE

CHAPTER 1: Hiding behind the Aanchal of his mother

Little Narendra, synonymous with vibes and zeal, stepped on the surface of the earth on the auspicious day of 17 September, just two days after the exalted position of the Sun according to the Vedic Sciences. The year that marked the step was 1950, the very middle of the twentieth century.

The day was like any other day of the month of September and no, there were no showers from the heavens above (in case, that's what you expected next). But there indeed was a delightful air all about. The Sun did the work it has been doing for time unknown. He rose and set. And the shape wasn't altered either. But no, not everything was not altered. There indeed was a variation, and I have the freedom to keep you in suspense for a while.

The variation was the birth of a baby from the womb of its mother.

A poor household in the small town- Vadnagar celebrated, with ethereal delight, the birth of a male baby, but oh, would the celebration have been any different if the delighted assemblage had known the identity of the baby? I highly doubt but then not every soothsayer waits for the stars to settle in order for the celebrations to commence. It was the time of political instability in the country and others had little to care about the birth of that one particular child destined to be the man who would bring about political stability in the very same country. The solution to the problem of India had arrived! It was right there in the small little-heard of town. It was right there lying next to a mother. The solution to India's distress had arrived in the most unnoticeable fashion, and it took the rarest path of all, so there was no fellow traveller who could take notice of it. But, are not the rarest gems the toughest to find?

The 1950 Vadnagar was not without its bucket list of complaints and scarcities. It was bereft of even the most essential of commodities. The villagers led a normal nine-to-five life, and didn't seem to possess the inclination to achieve what their fate couldn't contain. Some unarguably were endowed upon with the bliss of imagination, and these people could be seen lying on the cot on the roof dwelling upon their dreams.

They ate, drank and slept and yes, worked. And that was all this baby was expected to do- being a good citizen of the country.

Vadnagar had been an epicentre for Indian folk culture, and with time it came to be its focus. The town was presided by a ten thousand 'Deities on Earth'- the monks who preached Buddhism and non-violence all clad in saffron-yellow robes.

North Indians take Vadnagar to be the land of banyan trees, just as its name suggests in the contemporary language. It can boast of great and appealing natural beauty and purity and festivity (on the occasion of a fest named Vad Amavasya as observed by the womenfolk).

Damodardas hailed from the fam of Ranchoddas that migrated to Vadnagar to set up a grocery store in the late years of the nineteenth century. At the time of little Narendra's birth, he was thirty-five- in the very prime of his youth. Damodardas tied the nuptial knot with Hiraben who was and is as lustrous as her name.

Narendra arrived in the household just after his two elder bros, the elder one of whom was Soma (then, aged six) and Amrit who was but four back then. Two-year old Narendra became the brother of a lovely sister named Vasanti. Prahlada followed after three years, and then arrived Pankaj in 1958.

The Modi fam was poorer than many and richer than many. Supporting a family comprising eight members was definitely not a piece of cake for Damodardas. The dwelling place was no way an edifice. It was made up of bricks plastered with mud sans any belvedere. Did anyone in the little abode, back then, know that the humble hut would be an archaeological structure on fine day? Did anyone know that the house sheltered a mortal beyond the ordinary, a mortal who was habitual to do the extraordinary and prove that miracles are a real thing?

The dwelling wasn't strong enough to undergo the torments of weather, but it wasn't a mortal thing despite the external compulsion. It was a immortal structure, a structure that had had the opportunity to shelter a hero.

Baby Narendra would spurt just as any other baby of his age. He was sensitive and gala. When he started having tramps, the parens were naturally flabbergasted. His yelps were pleasant and the skepticism got almost erased. The eloquent and floundering bebe would walk fastidiously and then step in awkward fashion, thus becoming slackened preceding the afternoon kip. That was no hallucination. Struggle had begun and it would continue, and each pace taught him the way to struggle in a unique fashion. Each of the steps knew him better than many. Each of them loved him in a way not many did. Each of them knew him in a way so uncanny, so queer for a young child.

Struggle, indeed, is a dominant part of one's life- dominant in a broad sense, in a domineering way- unique to the struggler, unique to the man facing it. And no, struggle ain't a mortal thing as you might be supposed to believe after listening to its glory at length. The deities- both the earthly and the unearthly salute it with the respect that's due to it. Shouldn't the creator deserve the salutations?

This confirms my belief in a well-said saying, "Life loses half of its charm if there is no struggle and if there is no risk to be taken."

We struggle- Everywhere. We struggle in the womb of our mothers. We struggle for the breath that's the very cause of all existence. We even struggle for breath when we are born. We struggle to make our first stride or flight. We struggle in the babyhood. As we learn to grow and actually do grow, we struggle to blend in to our surroundings and camouflage with the environs. We even have to struggle to fit ourselves into the mundane world and to condemn spiritualism! At this point, I really do wonder if this struggle is unnecessary but I got the answer. It is not. And God believes in us as much as we do in Him.

As India, we struggled to retain our sense of a common national identity, and again struggle proved to be of use in our search for liberation and truth (Satyagraha).

As we grow up a little more, we struggle to fulfil our one desire- the desire to attain the 'goods' we would so often dream of as a child and which our parents didn't possibly share. This enhances our sense of independent identity and puts teens at odds with their parents. And Struggle is the reason for this all!

When little Narendra would become morose, he would be stimulated. When he would be smitten with funk, soothings would but cut no ice. As his age inflated, he became intent and was a keen observer. There wasn't any clear-cut sign of impotency or struggle for larceny. He was being shaped slowly and steadily into a stubborn man.

With time, he began the production of qualms in the realm of his mind which were rarely provided with a solution at his disposal 'cause they were mean to stay unsolved and grow to a height that would compel him to set out on a journey to seek the solutions when he could no longer resist the temptation to get them to spill the beans with an eventual hatred for suspense. At a point of life, he would definitely be keen about knowing the answers to his vast ensemble of questions piled up in the questionnaire.

Due to the presence of his brethren, he might have been muffed but his ways were distinct and were a reflection of the man he was to be one day. His furtive glances portrayed his restlessness artfully. Restless Narendra knew he had more to do than playing marbles, and he was restless to learn what it was.

The family belonged to the Ghanchi guild that is built around traditional producers of vegetable oil. Narendra learnt to be religious tolerant from his glorious ancestors. He was friends with many a Muslim in his childhood, significantly because of the neighbourhood presence of Muslim colonies. He observed both Hindu and Muslim fests with vows of non-violence to both humans and animals.

His sobriquet was 'ND' and apparently the letters were the initials of the words- Narendra and Damodardas. With the passage of time, the little child turned into a student and 'Struggle+Discipline' came to be the driving force in his life. The episodes of the student life of Mr. Prime Minister who was then Master Narendra Damodardas Modi, are equally interesting to hear. With abundance in nature, the historical legends are somewhat sarcastic as well!

Narendra, industrious as he was, came to be a promising student. The 'souvenirs' of his life as a student also come to reflect his obduracy. A well-known incident proves to be its radiator within the reach of the radar.

Once, little Narendra denied a class monitor access to his homework. He persistently insisted that the teacher alone had the right to do so.

There are not one or two but infinitely many incidents that prove to be the unknown emblem to his present known greatness. But there is something more outlined in these classics, something of a significance that put him at par with his buddies and bff. A child of spirit and liveliness was resolute. For him the truth was something he couldn't compromise with. He could only resonate with the truth as for him it alone was true. His attachment with truth possessed a queer sense of unwavering obstinacy, an unwavering desire to hold on to it, an unwavering desire to see it flying, flying, flying about clad in its natural and true glory. He was a big lover of originality and that was what he always pined for. His faith and belief in truth was profound and could be compared with the historical need for Satyagraha. His life was perhaps the best example of the need for knowing the truth and sticking to it in a discipline not many can possibly adhere to. And yes, it was Truth that made him wake up from his state of slumber and showed him a path in the woods, a path unknown, a path which bore risks, but risk was and is this person's life. And Truth made him familiar with a want for risk and made him what he truly was born to be.

His classmates saw him as 'not like the rest of us', in simple words. Little did they know that the person in young Narendra had zero 'parallels' and zero 'likes'. And, come on, raise your hands if you agree with me- Doesn't this person still act different from the rest of us?

The word 'Narendra' originally means 'The King of Nar' or 'The Ruler of Humans'. The spirit of kingly dignity was an indispensable part of his character right from his childhood. He respected people from all walks of life regardless to their profession, be it a menial job or a white-collar job, and held ascetics in high esteem. As Narendra grew up, his soul was smitten by a unique sense of asceticism. In his early life itself, he gave up the consumption of salt, oil and chillies. This great renunciation procured not the sign of piety alone, but also that of ambition. Now, 'Struggle+Discipline+Ambition' came to drive his life ahead.

Religion can be compared with the humus that grants the Indian alluvial soil its fecundity. It is the presence of varied religions, apart from geographical conditions that makes India the land of diversity. The religious diversity of the country makes it exquisitely beautiful and fertile. Religion is the very foundation of the art and culture which are the rabi and the kharif crops respectively being grown in this fecund soil. If India is unique, it's because of the religions that embellish Her. From the fifteenth, the sixteenth and the seventeenth centuries, there arose a sense of religious intolerance which was prevalent until this time as well but only continued to be the underground epicentre. It's present even now. The thing that has got to be attacked, bonafide, is not the religious diversity but the religious intolerance.

Unity- has it ever been known to be independent of diversity? Has it ever been known to have an independent existence? The Law of Independent Assortment doesn't govern the traits of the duo- Unity and Diversity which are the two alleles of the same gene- the gene of Indian.

Likewise, peace can't ever retain its identity sans war 'cause war is woven with it into a tapestry. We can't appreciate Unity sans diversity just as we take peace for granted in the absence of war. The tormentations of war highlight the need for peace. So does diversity build a sense of national integrity. And so does unity impute its importance to diversity.

Didn't the struggle for national independence create a sense of a common national identity? But, we were not a nation back then. We knew our differences and even in the light of our differences, we found a strange sense of unity bind us altogether

in wake of liberation from the clutches of the enemy. This defined our identity as Indian and our sense of belonging. Our boundaries were redefined. And this consciousness emerged in the wake of diversity! To top it all, in countries as Vietnam, India and many other colonies, the growth of modern nationalistic sentiments is intimately related to the anti-colonial movement. We happened to stumble across unity just as we embraced our diversity! The Struggle provided a shared bond that tied us all together and we could string the beads of myriad hues all in a row!

With its diverse minerals, the Indian soil is home to religious ambassadors. With its citizens hailing from diverse religious backgrounds, the Indian soil is home to multiple religious ideas that provoked a sense of innovation.

Narendra was born to innovate India in a way that would compel the world to look to India for 'the next big idea'. Innovation had always been his driving force. He had his own ways. He wasn't the follower of what we call 'verbatim ac litteratim'. His ways were long but accurate and shortcuts only shortcut the life, after all.

Narendra was devoted to Narendra Nath Dutta (alias Swami Vivekanand). He was respectful to his preceptors and liked the association of sacred souls. Possess a philosophical insight from his very childhood he did, and he hovered around in the company of saints like a bee in the search for the sweet elixir of philosophical wisdom that was what appealed to him the most. At a very young age had he learnt about the Chakra of Life, and he sought to enhance his grasp and understanding of the subject matter. Being one of the august assemblage of such spiritual discussions, Narendra came near to the realization of his purpose and his realization of the self. He saw himself as an ordinary creature with extraordinary capabilities and he never did embark on a journey to become something, rather he did so to do something. He saw himself as the doer. Spiritualism was what he thought of each of the 89400 seconds of the day, each second being a precious memoir of unending wisdom and philosophical insight. The boundaries of his mind extended and his thinking became broader. He could view the same thing from a wider, three-dimensional perspective. He learnt the ins and outs of life and diplomacy. He learnt the true nature of strength, the nature of the strength of his soul. Philosophy was the most appealing subject that flashed through his brain

works. He developed a practical-cum-spiritual attitude towards life, and his attendance in the meets only enriched his insights as a matter of fact.

‘Damodardas’, viz, ‘The Eternal aide of Damodar (a synonym for Lord Krishna, famously known as the twenty-second incarnation of Lord Vishnu) is a popular name among Indian boys. In extenso, i.e., Narendra Damodardas means ‘The King of Nar who is also the aide of Lord Krishna’. And, as is needless to say, Narendra’s religiosity was a full-fledged evidence of the grace contained in the graceful name, who even as a child, was a ‘man’ - ‘a man of letters and insight’. He believed in shouldering responsibility and sharing the troubles of others when he could easily wipe his hands off the task. Any opus given to him would be rendered complete the assigned day. He never shirked even the most irksome of tasks. And he never considered the onus to be an onus. Troubles were to him a yet another facet of life- a novel angel to view the beauty of life from, granting it a five-dimensional view!

After coming back home from school, young Narendra would stimulatingly inflate with pleasure at the prospect of visiting his father’s tea stall. He was assigned the work to serve tea to the railway passengers. In cogent terms, he appreciated the work he was told to do sans any tantrums. All chores were equally important to him, for as a boy, he had been taught that Lord Krishna cleaned the dishes of the brahmins at the time of the Rajsuya Yajna performed by sagacious Yudhishtira. However, he was always short of biscuits for the passengers for Indies would finish them off on the way itself! He happily passed his childhood keeping himself busy all the while.

As for his love for his motherland, no qualms about it! Service ante even the self continued to be his motto throughout his life. Having been self-taught the subtleties of life, this little boy was a profound believer of Self-Service and Selfless Action. These came to define his ways very soon. The first visible evidence for his staunch belief in Self-Service and Selflessness was displayed when at the young age of eight, he became a Swayamsevak and joined the local Rashtriya Swayamsevak Sangh (the RSS) branch, established by Keshav Baliram Hedgewar and others. This proved to be a vivid portrayal of his way of understanding life and his sense of self-service as well as his grade of selflessness. This vivacious soul had something in common with all the vivacious souls the Earth had ever had- sky-high thoughts and the unique kingly trait- Ambition and

Cruelty with the self. His relentlessness for his own self only grew with time as he came to identify the rights and wrongs in life, and became mature at a deeper level. And no, this deeper level wasn't the backdrop for the sprite. He had been sent to be the foreground and the Leader- a Leader who would be followed by 1.3 billion humans and a Leader who would be admired by populace equal in number to the 'Citizens of the Earth'. He wasn't born to convince the masses at large but to show them the path so they would on their own, be convinced to follow his steps wherever he went. He was meant to be the torch that would light up the path- the path of humanity. The reflector of this torch took time to be built.

Destiny did mean something more. The fate had something else in store for young Narendra, the playful boy at the local RSS branch, playing amongst the ridges with his mysterious agility.

CHAPTER 2- Vagrant Narendra

Narendra's family was amiable but strict to the youngsters. The kids were 'curbed at the end of a ferule' in order to prevent them from getting spoilt. Narendra's mother, Hiraben was one enveloped with an aura of eclat. She was agreeable and was a woman with floundering ethical pulchritude. She left no stone unturned in bringing up her children well and making them imbibe both simple and intricate values. She brought them up ideally, providing warmth and harshness in mediocre and required quantities. She showered affection and motherly love upon them and tutored the Indian moral values to them personally, hence becoming their premier teacher. Her own spiritual insights shaped Narendra's personality to a huge extent and her sense of sacrifice and contentedness with hardships passed on to her dear offspring as well. Unquestionably, her presence around and about him made him far-sighted and non-querulus. Thinkers have rightly said, "The hand that rocks the cradle rules the world."

And what better example could be presented with this?

A child is, to a considerable extent, influenced by his environs, *exempli gratia*, a child in the slumps is only expected to stone a poor dog venomously. This has a high probability (though many animals' rights activists have been slum dwellers).

Not to say this is a stereotype. There would certainly be one with a difference but that one will be one in a milliard. A little slum-dweller might also be kind-hearted and not callous or infernal. He might not jeopardize the animals, but again the same is quite unlikely to occur 'cause it's not commonplace, hence kind-heartedness despite being present in his heart, might not even formally introduced to him as had been the case with his parents themselves.

As for Mister Prime Minister, he came to influenced not by the culture of Vadnagar alone, but also by that of his home-state. The culture of Gujarat was what he admired. His love for Gujarat turned into the greater love for his country and his kin as he joined the RSS and saw India as a whole. He learnt to admire all the differences equally, so they wouldn't compete for surplus admiration, causing zero riots!

His jauniness is one of its results. By nature, Narendra was resolute and he didn't allow the lack of resources he constantly underwent to turn into the greater lack of spiritual enhancement and advancement.

The holy shrine of Giripur Mahadeva was his go-to place in the quest for peace. The chants and recitations of the name of God from the mouth of little Narendra echoed all over in the temple. The sublime devotion the waves of the voice carried along with their own propagation was soothing for all the devotees to hear. The devotion with which the name of God slipped from his mouth made it all so enchanting and captivating and devotees found themselves lost in this ethereal seating.

Young Narendra had but few interstices left in order for spiritual perfection. But spiritualism wasn't the only subject he took so keen an interest in.

He followed the Brahmacharya Ashram with utmost discipline and took his studies gravely to attain distinction in the subjects. With a poised mind, he showed not a sign of knavishness. His approach towards life was practical-oriented and so was his approach towards his studies. He loved to perform experiments at home and take the conclusions to task by scrutinizing them! Little did he know then, that his habit of experimentally researching and proving would

prove to be beneficial for India in the long run when he would set up experiments for new policies and bills and would learn and analyze the mistakes and the errors from the output they produced. His habit of multiple solutions to a problem showed his sense of self-consciousness and awareness of the danger signal in a singleton set of the solutions to the problem. This made him search for all possible solutions to a problem which is again a useful tactic when it comes to learning the science of diplomacy. Not only this, with every experiment he learnt how to detect shortcomings and errors in the apparatus and the procedure undertaken. He would try it with the corrected procedure or a plainly novel method! This process, unknowingly to him, taught him the art of sign detection which would prove to be a healthy trait of his character.

There was one particular sign that made him distinct during the centrifugation. He detested jabber. It was nothing but a sheer wastage of time for India's future Prime Minister. Serenity was vividly portrayed by his calm and composed behaviour and fanciful were his ways.

The combo of these traits put together made him insurmountable and ready for the future battle. Children collect stamps as a pastime. To Narendra, the collection of traits and qualities were a pastime. He placed them altogether in a vault from where they grew up as plants and bore seeds and reproduced, leading to an infallible collection of traits just as the infallible arrow-holder of the celebrated archer, Arjuna.

He explored novel methods of operating the ironmongery. Even tomfoolery enchanted him due to its own perpetual conditions, not due to the childish perspective owed to it. Everything, from a hansom to a plane mattered to him in ways innumerable. He viewed them in a way not many are familiar with and no one could change his solitary habits, not even Time itself. Everything was a matter to ponder over. Everything was worth being pondered over to little Narendra. The cause couldn't keep itself from him for long for the thinker wouldn't surrender to the chase, rather, he would enjoy it. The longer, the better! When the boy acted precipitately, others would be taken aback for by no means had they fancied a boy doing deeds sumptuous for his age. His formidable ecstasy hasn't waned away even with the passage of time. The peering Narendra could not be unnerved by a grotesque object! Fearlessness was devoutly devoted to him.

And it would help him embark fearlessly on the journey he was meant to undertake as it never would sally.

As for the antagonists, what stand could they take in front of the protagonist?

His bed might have been tawdry but his character possessed a value greater than that in the possession of platinum. His collection wasn't bereft of a thing, be it conduct or discipline. Ab initio, he had been a great disciplinarian in every walk of his life, for everything and everybody else was sojourner to him when he had a task to do at hand. Though he was largely mistaken to be egotistical, he was but social and amiable and was always the beloved colleague of his comrades and confederates. The swayamsevaks would oft coagulate to listen to the orators in the Sangh with Narendra in their company for listening, to him, was a necessity and speaking an invention, and he always did focus on his prime necessity, viz, listening, for he had learnt that necessity is the mother of invention.

His unparalleled taste for impunity was what made his liberal ways sound weird and not many could bring themselves to enjoying the pleasures of his company for they knew not what rested within the bigwig.

Also, the boy being an avid reader learnt to write dexterously and somewhat spasmodically at an early age for reading is followed by writing in its prime lustre, and he would use his pen only after having collated substantial facts for he never wrote a statement sans an example accompanying it.

Now, what did he do ante scoffing? He would express his gratitude to God for providing him with repast. And he wouldn't cease the prayer here. He would continue it with a request to God to provide food to all the creatures created by Him. He would pray for Thou's creatures! This blazing inclination of his 'heart so kind' was fuelled with the sight of the miseries of this world unto the impecunious and the innocuous, viz, the animals, alike. The desire served the purposes of neither of servility or slavishness. And he had clambered these heights of wisdom as a young lad! In all these years, the wriggling scrambling baby had turned into an object of praise. At the same time, however, he had run the gauntlet.

Now, with all this light, if I an ex-ante is done, I won't be doing any wrong.

Nota bene.

With his sincerest desires for employing alternatives, Narendra was sure to alter the world at the altar of alteration and altercation one day. And his mother, Hiraben observed this all. With her stringent state of siege, she never did let him be convulsive, but something of a nature mysterious seemed to bewilder her as a constant.

She resorted to eavesdropping and providence had it. She had noticed it right the very first time itself.

Quick thoughts on some books

You Will Find Your People: How to Make Meaningful Friendships as an Adult

Lane Moore

An absolute stunner of a book, it's packed with advice you NEED to make strong friendships as an adult. Moore has laid out the entire plan, quite literally. Yes, and, animals can be best friends. I loved that it was mentioned. Thoroughly delighted by this book.

I know this book stands for a lot more than what I have compiled above, but the points above stood out to me.

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Just Kids

Patti Smith

A thorough image of New York and Brooklyn's art scene, the life of a budding artist, poet turned singer Patti Smith and her lover Robert Mapplethorpe, it's such a beautiful book. Be it the imagery of Hotel Chelsea, the rock and roll scene, the chance encounter with Robert, the Persian necklace, it's a book dedicated to embracing beauty, in its purest form. The part that struck me was how Robert

mulled over their not having kids. Theirs was the purest, most glorious form of love.

I even highlighted some sentences of the book as I read. It took me about five hours to devour it entirely and should take me about a week to forget. It's the type of book you love but wanna get over with. I shall keep it in my room (folded delicately in violet issue packed with red satin ribbon, as Patti says it) for the foreseeable future but try not to see it because it's so black and so bleak. But so pure- a type of purity I must stray from for it's all-consuming. May Robert and Sam's soul rest in peace.

Many Lives, Many Masters: The True Story of a Prominent Psychiatrist, His Young Patient, and the Past Life Therapy That Changed Both Their Lives

Brian L. Weiss

So rebirths and incarnations are real?! That's why we need to do good deeds and not harm anyone, be it human or animal. This book made me a more determined vegetarian.

The Picture of Dorian Gray

Oscar Wilde

I admire the book, a psychological thriller, overall. It's impressionable, tasteful and very languid. Despite admiring the language style of the book and the metaphor of the painting, I did not like very much the philosophy of Henry and the character of Dorian Gray. I mean Wilde did not make him to be a character worth being liked. It was never his intention. But the whole thing defied my expectations and baffled me in a way it did not need to in order to make a great book. I sympathised a lot with Basil, some with Vane (the brother of Dorian's first lover who had pledged to take revenge on 'Prince Charming' if he wronged her. The lives of so many people were destroyed because of Dorian's evil character and debauchery. While I hold this book in a high regard for its deep portrayal of suffering and those who cause it, I find it very hard to digest this book healthily.

Men Without Women

Haruki Murakami

I had only fleetingly heard of Haruki Murakami and had never really 'stalked' his books in a bookstore or alibi. But yesterday I found a copy of 'Men without Women' just about randomly and finished the 228 pages raw in like three hours.

Murakami's writing instantly reminds me of Han Kang though her writing is a shade more intense, provoking and terrifying. Haruki's is comparatively chill but it's a kindler of new sort of thoughts, new sorts of ideas. It makes one think deeply of one's coordinates in this world even if one is not spiritual. The story of the lovesick doctor who shrank himself was the strongest and much along the lines of the woman in Ms Han Kang's *The Vegetarian*. Kitaru, the coffee boy with a high school girlfriend, the Kitaru who could never pass his college entrance exams, that Kitaru went on to be a chef in Seattle. Mr Murakami immediately makes us feel his characters. I love that sort of feeling. These are 7 worlds, not 7 stories. These two writers are also responsible for making me like Tokyo.

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Lolita

Vladimir Nabokov

This is one book every literate man should read before he dies. I do not wish to discourse upon the philosophy of this book because even materially, this book has a lot to offer- an entry into a deviant person's mind, an understanding of what rationalization of serious crimes is like in the mind of the perpetrator. As one reads the book, one might start to think it not hard for a man to fall for such temptations for this book has been written from the perspective of the rationalizer of that thought, or one might be taken aback, spellbounded, petrified at how the minds of such characters function.

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The Testing of Luther Albright

MacKenzie Scott

Mackenzie Scott, I heartily request you, if you read this comment, to donate some money to animal welfare organisations. they do really need you. it's the humble request of a small child. thank you!

Northanger Abbey By Jane Austen Annotated Edition

Jane Austen

Persuasion was quite a read. Northanger Abbey, despite being surprisingly more imaginative and youthful, is not engaging enough to be read in one sitting. Our heroine gets interested in gothic stuff making this novel quite a bit ahead of its times.

Secrets of Sand Hill Road: Venture Capital and How to Get It

Scott Kupor

When i first started to read this book, I didn't know I will soon rate it to be a mighty two stars. This book does a catchy job of attracting the reader but is an absolute no for me. The 'secrets' it distributes are no secrets. While the term sheet part is useful, i never intended to read it for term sheet knowledge. Gosh, I hardly have a startup in the first place! To talk about term sheet with someone who is not even at the first step of their startup journey is so insipid and relentlessly useless. As one can see in my profile, I rarely give bad reviews but I do hold non-fiction to the standards it promises in its preface and covers. I got a lot of useless knowledge from this book and would have instead loved for the writer to focus more on the startup side than the accountant side.

The Vegetarian

Han Kang

This book can be the root of vastly different styles in imagination. I will review it from my own singular perspective.

This work has immortalised veganism. The story of a woman who gives up meat after a graphic dream despite being forced by her husband and family to intake it. Her husband deserts her and only her sister and brother-in-law, who is an artist, remain by her side. This guy is an artist and paints flowers on her body. That sparks in her an idea- to become a plant. She becomes so intensely driven to become a plant that she basks in the sun for sunlight and starts to drink only water.

A Philosophy of Walking

Frédéric Gros

This book can be easily deemed anti-intellectual for the lofty seriousness it takes intellect in. However, I do believe intellectuals take themselves a little too seriously.

Anyway, this book has some wondrous accounts, be it the French Jean Jacques Rousseau's account of walking or the spontaneous walking of a man who walked from country to country.

Persuasion

Jane Austen

Persuasion is a rather odd book for its time.

I barely see a mention of meat in the book whilst the other books of the time based in England rave about meat and all things meat. I admire this nuance a lot.

Persuasion is also very persuasive. It's the story of a Miss Anne who disregards the persistent attempts of a young navy officer on account of his perceived lower birth and the advice of her favorite lady-in-meeting, but then she meets him seven years later, manlier and richer and the subject of many a lady.

But it's also not a story about the rekindling of an old lover as much as it's a satire on the ranks of birth and inherited self-importance.

Madame Jane Austen, it seems to me, misses never a chance to recount the matters of life that seek an order of importance above that of all-encompassing femininity.

Stories from Tagore

Rabindranath Tagore

These stories pain me to no end. While I cannot possibly enter the characters' world and free them of their misery, this book makes me feel continually helpless about the millions of people who suffer from poverty and disease.

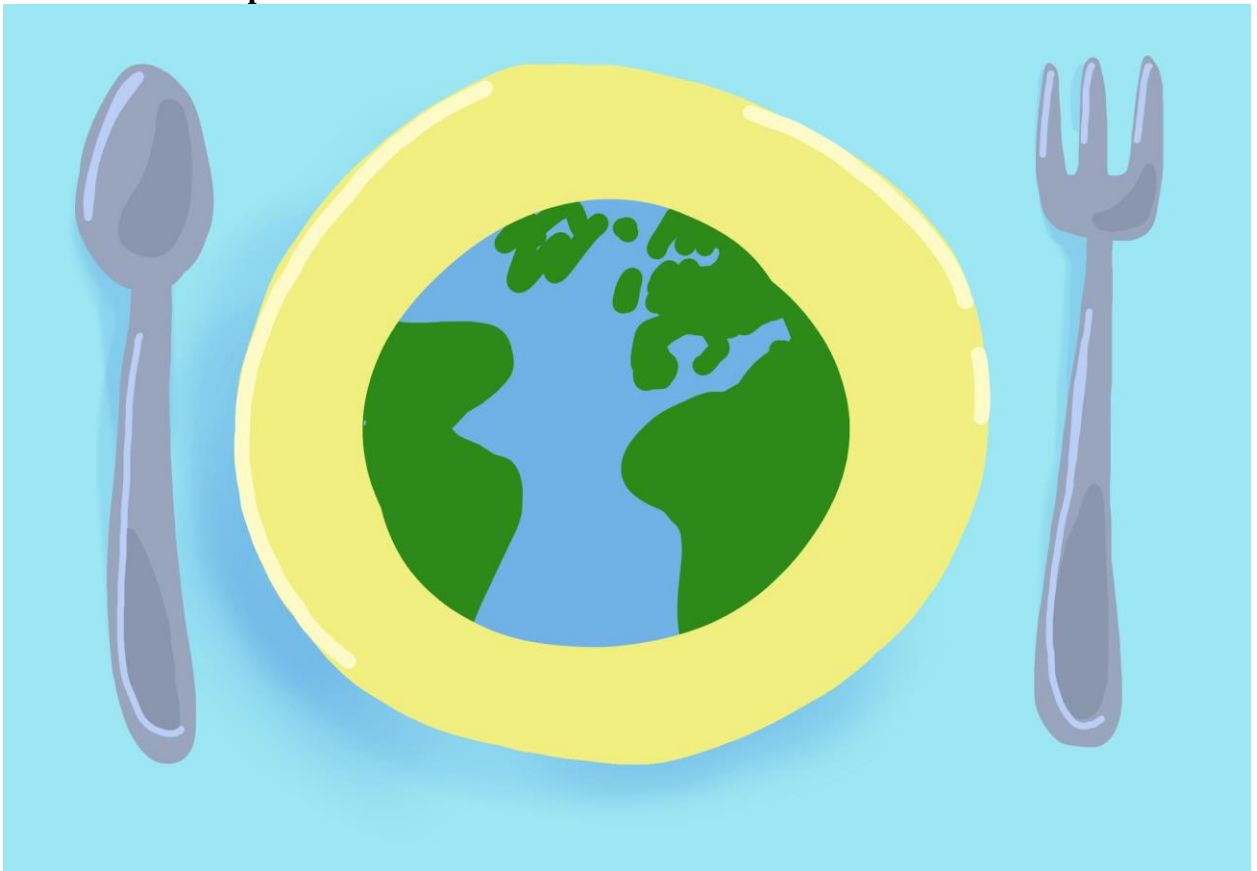
And yet it is more than an account of suffering. Tagore's stories talk about suffering that's self-respecting, about pain that's received with the head held high.

The Cabuliwalah, I read it as a child and now I read it all again.

The Home-Coming is the story of a notorious boy who is eager to stay with his long-lost uncle in order to escape the clutches of his unrelenting mother and the company of his docile younger brother, only to enter the claws of his wicked aunt. There start his days of utter gloom. He stops playing with the kids and become lonely like the crow. He waits for holidays so he can get to meet his mother. Indeed, being beat by your mother is far, far better than being thrashed by an aunt for the former contains in itself a semblance of love. He catches malaria and decided to run away back to his house for he knows how well he will be received at his aunt and uncle's house. But he fails to run away and is sent back to his uncle's house whereupon his mother is called. While lying on the hospital bed, he says to his mother, "Mother, the holidays have come."

Originally published in the Stanford Daily

Edible Ethos: Six power foods



Chetanya Pandey '27 explores how making small food choices can save the planet — and your taste buds. (Graphic: DA-HEE KIM/The Stanford Daily)

By **Chetanya Pandey**

June 22, 2025, 9:36 a.m.

In Edible Ethos, Chetanya Pandey '27 dives into the wild world of eating with a pinch of humor, a dash of honesty and a whole lot of love for plants. From simple switches to kitchen wizardry, this column explores how making small, conscious food choices can help save the planet... and your taste buds. Think of this as your guide to eating better, feeling better and saving the world, all while keeping it casual (and delicious).

When you think of protein and nourishment, it's likely your mind immediately leaps to animal based products like chicken, fish and eggs. But the plant kingdom is actually bursting with nutritious, satisfying and even protein-rich foods that can

rival and often beat non-vegetarian options in terms of health, sustainability and most importantly, flavor!

Whether you're trying to cut down on meat or look for some spice in life, here are six foods to try instead of chicken that will keep your plate exciting and your body thriving.

1. Lentils

First up are the unsung heroes of this world. Packed with protein (frickin' 18 grams per cup cooked), iron, fiber and folate, lentils make an excellent substitute for minced meat in dishes like shepherd's pie, tacos and curries. Aside from being incredibly affordable, they are also easy to cook and can even be eaten in soup form. Don't forget to add a squeeze of lemon and fresh coriander for a flavor punch. Stanford Health Care even offers a recipe for Egyptian red lentil soup filled with multiple cancer-fighting ingredients.



Mercimek çorbası (lentil soup) at a restaurant in Ankara. (Courtesy of Wikimedia Commons)

2. Chickpeas

Hearty and protein-rich, chickpeas are great in both Indian and Mediterranean dishes — think chole, hummus and falafel. Full of fiber and low in saturated fat, chickpeas aid digestion and keep you feeling full for longer. Maybe skip the chicken sandwich and go for a chickpea wrap with cucumber, onions and tahini sauce. Not convinced? Try your hand at [this](#) extremely delicious chickpea salad or [this](#) chickpeas and artichoke “crab” cake with remoulade over mushroom risotto by Chef Jay-ar Pugao.



Chickpea dish with cucumbers and onions. (Courtesy of Wikimedia Commons)

3. Jackfruit

Raw jackfruit is surprisingly similar in texture to pulled pork or shredded chicken. When cooked with the right spices, jackfruit makes a fantastic meat alternative. One could replace mutton curry with jackfruit masala. It has all the heaviness of meat minus the cholesterol. Unlike money, meat can grow on trees!



Northern Thai whole jackfruit curry. (Courtesy of Wikimedia Commons)

4. Quinoa

It's not a grain but a seed that offers all nine essential amino acids, making it a complete protein! High in fiber and minerals like magnesium and iron, quinoa is a great alternative to animal protein in salsa, bowls or even breakfast porridge. Don't shy away from replacing your chicken-and-rice lunch with a quinoa bowl topped with beans, avocado, corn and a lime vinaigrette.



Vegetarian quinoa bowl. (Courtesy of Wikimedia Commons)

AD

5. Seitan, aka wheat-meat or mock duck

With a meat-like texture, seitan has 21 grams of protein per 3 ounce serving. A [favorite](#) amongst athletes, it's a must-try and is easily found in grocery outlets! Here is a recipe for [making](#) seitan in a dorm kitchenette. And yes, from my own dining experience, it is convincingly meaty — proving you don't need animals to enjoy meat. Seitan can sizzle on the grill, crisp in a stir-fry or stew in curry. As a bonus, it's also low in fat.



Seitan. (Photo: Wikimedia Commons)

6. Soybeans

Like quinoa, soybeans are gifted with all the nine essential amino acids. Rich in calcium and iron, they provide 31 grams of protein per cup. They also contain isoflavones, which reduce cholesterol levels and improve heart health. Soy nuggets, tempeh and boiled edamame need to be on your to-eat list.

AD



Soybean paste stew. (Courtesy of Wikimedia Commons)

These alternative proteins are both sustainable and delicious. With this, I leave you to experiment and go have fun with the plant-based world!

Edible Ethos: Is this aioli vegan?



Can even small acts of questioning signal an existential problem? (Graphic: ANANYA UDAYGIRI/The Stanford Daily)

By **Chetanya Pandey**

May 21, 2025, 2:21 p.m

There are moments in life — albeit small and inconsequential ones — that reveal the underlying structure of a collective discomfort. One such occurrence recurs with quiet predictability: a dinner invitation, a restaurant menu and a question asked.

“Could you check if the aioli is vegan?”

It’s a question that is not about ingredients nor wholly about food. It is a question that hovers directly above a larger cultural unease about choice, about conviction and about the mild but perceptible irritation provoked when someone — even gently — deviates from the norm.

For a big portion of my life, I asked it hesitantly, with the softening preface of “sorry to bother...” or “it’s okay if not...” These were not merely niceties but reflexive attempts to temper the audacity of a question that implies an ethical boundary. To inquire about ingredients was to mark oneself as different, to refuse

absorption into the ambient consensus, to insist that what lies beneath the surface matters. And for this, one is trained to apologize.

Yet to apologize is also to admit fault. And what fault is there in acting with regard for unseen lives and environmental consequences rendered invisible by packaging and presentation?

I no longer apologize.

To eat vegan is not, as it is often caricatured, an act of moralization or a disruption for disruption's sake. It is to bear witness in the smallest of ways. To refuse — quietly, even awkwardly — the indifference that makes violence ordinary. To ask what is in the broth is to ask what systems are held together by our silence and what might unravel if we ceased to be silent.

Many seek to evade the confrontation inherently embedded in discussions of ethical consumption. But think of philosopher Theodor Adorno, who observed of postwar Germany that the work of confronting the past was often evaded in favor of euphemism and closure. To look too closely is to lose comfort. Better not to ask.

In this sense, the question of the aioli becomes emblematic — not of a dietary preference but of a broader resistance to forgetting.

I have come to believe that discomfort is a necessity. In fact, it's a sign that the question matters.

At Stanford, a place that fashions itself as a crucible for inquiry and innovation, it is important that many students be unafraid to question power and not fall silent at the table. In this context, to ask about animal products doesn't trouble culinary norms but gently indicts a system of consumption that thrives on disavowal.

Still, things are shifting. Friends now tell me what's vegan before I ask. Menus have changed. Plant-based options appear not as afterthoughts but thoughtful creations.

All this boils down to one way of navigating life my mother taught me: if you don't ask for it, you don't get it. Dear vegans, environmentalists and animal

lovers: we know we have a cause worth fighting for. Let's collectively ask for a change.

‘Re-Thinking Food: From Plate to Planet’ symposium serves up future of food



Blending research with real-world impact, the symposium invited participants to rethink their relationship with food. (Photo: Wikimedia Commons)

By **Chetanya Pandey**

April 27, 2025, 11:33 p.m.

On April 24, the Clark Center Auditorium opened its doors to conversations that start on our plates and stretch to the planet. Hosted by Stanford [Bio-X](#), “Re-Thinking Food: From Plate to Planet” gathered scientists, students and innovators to explore the future of food.

The symposium began with opening remarks from Ellen Kuhl, director of Bio-X, and Lily Sarafan, incoming chair of the University’s Board of Trustees. Jonathan Foley, executive director of climate nonprofit Project Drawdown, examined the climate crisis through the lens of our diets. Vipula Shukla, the Gates Foundation’s senior program officer of Crops R&D, explored how plant biology can leap from

lab to farm, while UC Davis plant pathologist Pamela Ronald brought stories of climate-resilient crops designed to withstand the storm ahead. Arun Majumdar, Dean of the Doerr School of Sustainability, offered remarks on Stanford's place in these transformations.

One of the most spirited segments of the morning was “Fast Food: Speed Science Updates,” where Stanford students and postdocs shared their research — from earth systems and engineering to gastroenterology — in rapid succession. Gianna Dugan '25 presented on EARTHSYS 109: Rethinking Meat: An Introduction to Alternative Proteins, a course providing a look into alternative protein.

“It's time to rethink food because meat and dairy tech is so redundant,” Dugan told *The Daily*. “You put in 100 calories in an animal and get 10 calories — a waste at a time of global food insecurity. The meat and dairy industry is really holding us back from social justice. We need to talk about Concentrated Animal Farming Operations in the 21st century.”

Sean Spencer, a Stanford instructor in gastroenterology and hepatology, also emphasized sustainable diets. “A vegan diet is very viable and is the best for human health when accompanied with beans, lentils and fermented vegetables,” Spencer said.

Curated to align with the symposium's values, the program's lunch was both environmentally friendly and delicious, a sensory reminder that sustainability can be satisfying.

“It was great, like a Superbowl,” Dugan said. “There is a lot of integrity in having a fully vegan catered event to lower environmental footprint.”

The afternoon shifted focus to human health. Christopher Gardner, a medical professor at Stanford, brought humor and hope with his thought-provoking and conscientious talk: “If I Were Food Czar.”

“I'd answer the ‘What would you do if [you had] a trillion dollars?’ question with solving food: health, sustainability, deliciousness and social justice aspects of

food. We might have to change the demand before we change the supply, and prepare for the rancher, the fisher and the farmer to raise less but in better working conditions,” Gardner said.

A short break offered one last bite of plant-forward refreshment before the final phase of discussion: food and technology. Karim Pichara, co-founder and CTO of NotCo, a food-tech company that combines AI and human knowledge to develop and scale plant-based foods to the mass market, discussed the AI-driven reinvention of what we eat.

The day ended on a luscious note of curiosity and culture with a plenary lecture by Harold McGee, food scientist and acclaimed author. His talk was introduced by Andrew Mayne, Stanford’s culinary strategist for plant-forward dining, who has helped reimagine what Stanford students eat.

Finally, Vice Provost and Dean of Research David Studdert offered closing reflections, reminding us that this day is not the end of a conversation but just the beginning of it.

In a world reeling from climate change, chronic illness and unsustainable growth, food is no longer just food. It’s a problem, a possibility and a beacon of hope. Re-Thinking Food invited attendees to slow down, listen and taste what that future could be.

George Steinmetz on the hidden costs of our food system



Photographer George Steinmetz '79 received the Doerr School of Sustainability's 2023 Distinguished Alumni Award and returned to unveil his book, "Feed the Planet: A Photographic Journey to the World's Food." (Courtesy of Kathy Ryan)

By **Chetanya Pandey**

March 13, 2025, 10:07 p.m.

For photographer George Steinmetz '79, recipient of the Doerr School of Sustainability's 2023 Distinguished Alumni Award, a career in photography has meant soaring over deserts, rainforests and urban sprawls — capturing the fragile beauty of our planet from above. From the fragile veins of a rainforest to the

geometric sprawl of industrial agriculture, his camera reveals patterns invisible to those bound by the earth.

On March 11, Steinmetz returned to the Farm to unveil his book, “Feed the Planet: A Photographic Journey to the World’s Food,” published in Oct. 2024. Steinmetz led a masterclass, offering a behind-the-scenes look at the realities of documenting the world’s food systems, for students in the seminar EARTHSYS 185: Feeding Nine Billion. He also held a talk titled “Do You Know Where Your Food Comes From?” in Hewlett Teaching Center.

In a decade-long odyssey across 36 countries and six continents, Steinmetz chronicles the vast and intricate machinery that feeds humanity, exposing not just its scale but its consequences.

Steinmetz’s journey into photography was neither planned nor conventional. Studying geophysics at Stanford, he told the audience: “To be honest, I wasn’t exactly interested in geophysics. It just happened to be the second highest-paying major.”

His true education began not in lecture halls but on the open road — 28 months of hitchhiking across Africa, sleeping on train roofs, wandering through forests and buying his first camera with no prior experience, according to his talk.

Decades later, with over 40 National Geographic photo essays and accolades, including the Environmental Vision Award and three World Press Photo prizes, Steinmetz has become a cartographer of hidden landscapes, with a motorized paraglider, its 310 cc engine — named “Monster” — strapped to his back.



Through his photography, Steinmetz reveals typically unseen worlds. (Photo courtesy of George Steinmetz)

From this fragile perch, he hovers between earth and ether, following the old pilot's adage: "Altitude is your friend," Steinmetz said at the event. The audience laughed. "I figured it meant I'd have more time to scream on the way down," he added.

Michael Fried — [director](#) of Planet Earth Arts, a project commissioning, presenting and showcasing works of environmental art, including creative writing, dance, film, media, fine art, music, photography and theater — introduced Steinmetz's evening presentation as part of an ongoing effort to merge the arts, humanities and sciences exploring the future of all life.

Fried, whose work at Planet Earth Arts [champions](#) the role of storytelling in shifting public consciousness, emphasized the necessity of such talks. "George

goes and he sees and he clicks — and does not create any opinion but just brings the sense of reality to these photographs that take people’s breaths away. Reading George’s book was an intoxicating, mind blowing experience. I also admire his aerial geometrics,” Fried said to The Daily.

Steinmetz projected images onto a towering screen that unspooled the story of modern food production: wheat fields stretching like golden oceans across Kansas, Indian shrimp farms sculpted like abstract mosaics, Australian cattle stations vast as entire nations.

Poultry plants where hundreds of thousands of birds spend their entire lives under artificial light, crammed wing to wing. Cattle confined to barren pens, their bodies engineered for maximum yield. In a Brazilian slaughterhouse, cow skins hang on the wall as 1,200 workers move in synchronized precision. The relentless pace is dictated by the unyielding demands of a global supply chain, according to Steinmetz’s talk.

“These cows live in a sad, still environment,” Steinmetz said to the audience, pausing on the image of the calf separated from its mother, tiny and alone on the screen.

Audience members said they were impacted by Steinmetz’s talk. Rachel Lit ’25 spoke of the images’ stark honesty, particularly those illuminating the beef industry’s toll on the climate. “The meat industry is devastating for the environment,” Lit said.

Isabel Vilá Ortiz ’25 was “caught between exhaustion from finals and a surge of inspiration,” she said. For her, “this talk gives hope.”

Students weren’t the only ones impacted by Steinmetz’s work. “We, humans, have gone too far. Through his presentation, Steinmetz shows that industrial meat production places an unbearable strain on the planet, and we must rethink the way we feed ourselves,” Fried said.

According to Fried, “George’s talk gives great hope in these dark, dystopian times. And hope is a verb. One of the challenges for those of us who consider ourselves as environmental activists is how overwhelmed we can feel. George

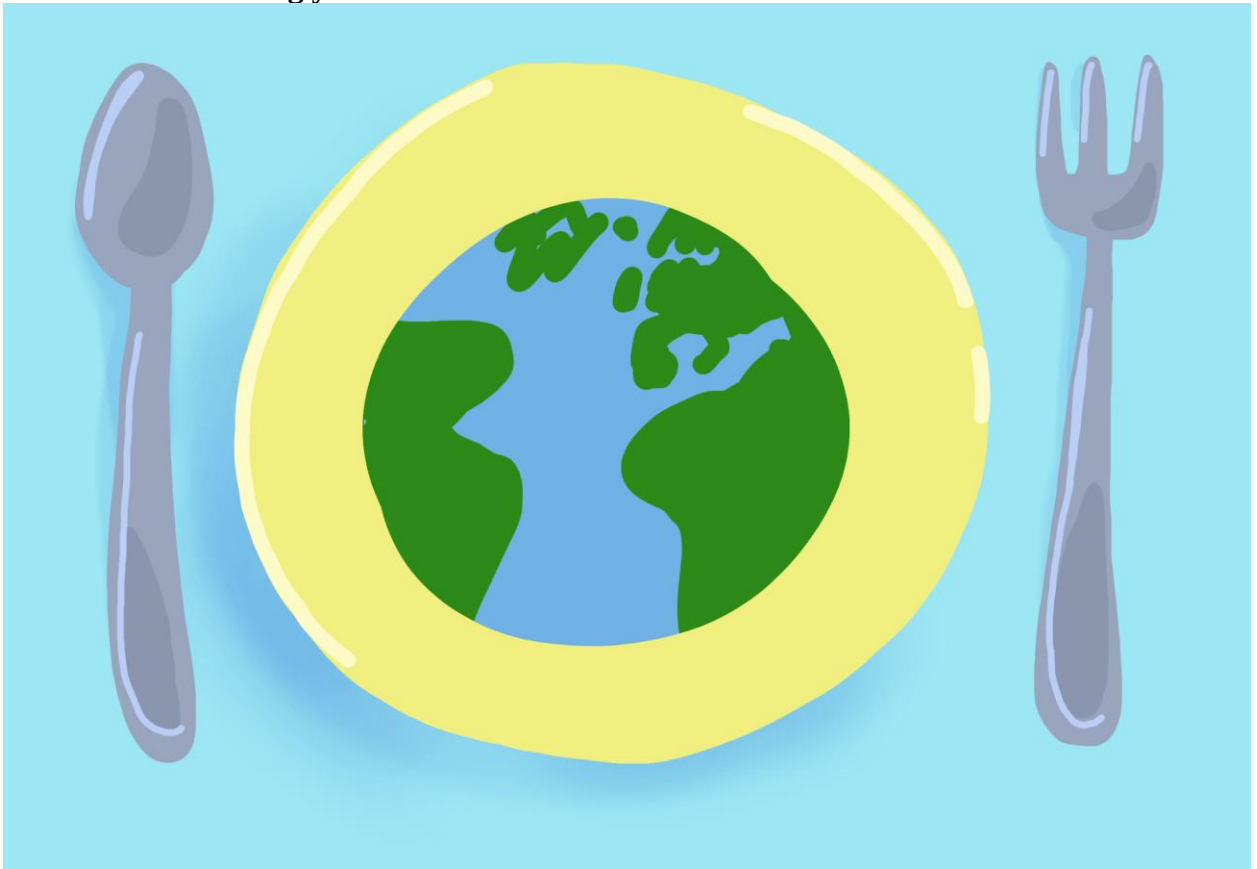
does not preach, rather he brings the truth to light through his photographic investigation so we can make more informed decisions about our food.”

As the presentation drew to a close, Director of the Stanford Woods Institute for the Environment Chris Field joined Steinmetz in addressing one of the most staggering statistics of the night: “The total biomass of domesticated poultry, mostly chickens, now outweighs that of all wild birds combined and layers upon layers of stacked chicken bones fill processing floors.”

Steinmetz said, “We have a natural right to see how the food we eat is made. We are putting this into our bodies, so we need to see where it comes from. There are environmental consequences to what we eat.”

When asked by the audience how one might follow in his footsteps, how one could go about capturing such raw, unfiltered truths, Steinmetz’s answer was simple: “Go.”

Edible Ethos: Making your next meal sustainable



Chetanya Pandey '27 explores how making small food choices can save the planet — and your taste buds. (Graphic: DA-HEE KIM/The Stanford Daily)

By **Chetanya Pandey**

Feb. 23, 2025, 9:28 p.m.

In Edible Ethos, Chetanya Pandey '27 dives into the wild world of eating with a pinch of humor, a dash of honesty and a whole lot of love for plants. From simple switches to kitchen wizardry, this column explores how making small, conscious food choices can help save the planet — and your taste buds. Think of this as your guide to eating better, feeling better and saving the world, all while keeping it casual (and delicious).

Look, I get it — whether you're juggling classes, club meetings, a caffeine addiction or all three together, the last thing on your mind is whether your lunch is sustainably and ethically sourced and if your bacon was once a-pig-with-an-actual-life. Who has time to scrutinize food labels when you're already running late and just need something — anything — to eat?

But here's the thing: every bite we take is a vote for the kind of food system we want. And right now, the industrial food system is, well, kind of a dumpster fire. From worsening [deforestation](#) to promoting [excessive water use](#), the way animal-based food is produced and consumed is one of the biggest contributors to climate change.

Here's the deal: sustainable eating isn't about suddenly becoming some chia-seed-eating, kombucha-brewing, barefoot-walking guru. It's not about getting all *holier-than-thou* about your food choices. Instead, it's about looking at your plate and going, "Hmm, what could I eat that won't cause the Earth to explode?"

You don't need to go full-vegan and turn your dorm room into a fermented food lab to make a real impact. All it takes is a few tiny tweaks to your food game. Let's roll.

1. Eat more plants (and maybe stop crying over steak)

Now, I'm not here to make you quit eating your beloved bacon-wrapped steak. But every time you swap out just one meaty meal a week for a plant-based option, you put a dent in global warming with your very own hands. Case in point: to make one pound of beef, it [takes](#) 1,800 gallons of water. Meanwhile, you can [grow](#) a whole pound of soybeans with just 257 gallons. That's right. 257. Lean beans — like black beans, chickpeas and lentils — are packed with protein, fiber and a bunch of essential vitamins and minerals. They'll fill you up without the sad, post-meal guilt spiral. Plus, consuming all that lower saturated fat? Your heart will send you a thank-you card.

2. Waste less, save more and feel like a freaking food wizard

Every time you toss food into the trash, you're essentially sending all the resources that went into producing it straight into the landfill — especially with meat and dairy. These foods are like the Kardashians of the food world: they [require](#) way too much attention, land, water and energy to produce. So, how do you fix this? Easy: Don't buy what you won't eat. Be that magical creature who can turn last night's leftovers into tomorrow's culinary masterpiece. Extra chicken? Put it in a stir-fry. Fish from last night's dinner? Tacos. And those bones? They're the secret ingredient to broths that will make you feel like the

coolest, most sustainable chef on the planet. You're not just saving the planet: You're becoming a wizard in the kitchen.

3. Support local and seasonal foods (because grocery store tomatoes are a crime)

Here's the real tea: when you buy local, seasonal food, you're high-fiving local farmers, who are way cooler than the big corporate farms that mass-produce food like they're on a factory assembly line. Plus, food just tastes *better* when it's fresh and hasn't traveled 1,500 miles to land on your plate. Ever had a tomato in January from the grocery store? It's basically a sad, flavorless rock. But local, seasonal veggies like a crisp, neighborhood carrot? They taste like sunshine in your mouth, like you've just won a gold medal in the culinary Olympics. So go ahead, load up on that local kale, sweet potatoes or whatever's in season, and feel like a plant-powered champion every time you munch down.

Small steps, big impact

Sustainable eating doesn't require you to live in a green-smoothies-for-breakfast-lunch-dinner-24/7 lifestyle. It's not about being perfect or getting a gold star for eating quinoa every day. It's about making small, easy changes that stack up over time. Whether it's swapping one meal a week for a plant-based option, reducing food waste or supporting local farmers, these choices are tiny, rebellious acts that fight against the global food system that's currently trying to kill us all.

So go ahead — make your next meal more sustainable. Your taste buds, the planet and your future self will thank you. And bonus? No environmental guilt on the side.

How to survive election day, or, the bunker



Here is your ultimate, foolproof guide to surviving election day like a pro.

(Graphic: ANGELA WEI/The Stanford Daily)

Humor by **Chetanya Pandey**

Oct. 28, 2024, 7:28 p.m.

Editor's Note: This article is purely satirical and fictitious. All attributions in this article are not genuine, and this story should be read in the context of pure entertainment only.

Ah, election day: that magical time when the air is thick with political tension, every conversation feels like a minefield and your social media feed is hotter than a jalapeño pepper on a summer day in a freshly microwaved vat of lava. But fear not, brave voter! Whether you're just trying to make it through the day without losing your cool, or you're hoping to survive the Thanksgiving dinner table without causing yet another murder-suicide, here's your ultimate, foolproof guide to surviving election day like a pro. Ready? Let's get you fortified.

1. Stock your bunker with vague responses

Like a bunker stocked with canned beans and bottled water, your mental bunker needs vague, noncommittal responses to keep you nourished through the conversational siege. Here's what to have on hand when people try to coerce you into answering covert questions, such as "Do you prefer the left or the right direction? Or the center?":

1. "It's complicated." This is your multipurpose phrase. Is it really that complicated? Maybe, maybe not— but people will think you're being thoughtful. They'll move on much quicker than the average Stanford student does with his ex.
2. "Interesting perspective!" Translation: "I'd rather eat your dog than argue with you." But they'll hear: "I respect your opinion!"

These phrases are like MREs (Meals Ready to Eat) for election day survival — portable, slimy and ready to get you through tough times.

2. Fortify with snacks and patience

Waiting at your polling place is long, stressful and full of unexpected tears.

Stock up on enough snacks to last through endless lines and the occasional heated debate from that guy in line who really wants to talk about voter fraud. Pro tip: Chewing gives you an excuse not to talk. When someone asks, "Who are you voting for?" just take a loud, long crunch on those chips and shrug. Sometimes, you won't even need to shrug. That will usually happen when you are shrugged on.

3. Build a mental shield (because the opinions are coming!)

Your mental bunker's thick walls need to keep out others' political opinions. When someone starts yelling, "If you don't vote for [insert candidate], the country is doomed," imagine that you're behind a concrete wall — safe and sound from any possible future of the nation.

When Aunt Karen from Facebook starts going off about [insert conspiracy theory of the week], you can deflect like a true pro: "I'm just excited to exercise my civic duty!" It's like responding to someone throwing a grenade with, "Wow, great toss! I'm just here to participate!"

4. Prepare for family dinner

After election day comes Thanksgiving Dinner — a.k.a., the real battleground state. But if you’ve built a strong bunker of political neutrality, you’ll be ready for those political landmines your uncle likes to plant.

Your best weapon? A strategic topic change. Steer the conversation away from politics and into safer territory, like asking, “So, how are the divorce proceedings coming along?” Watch the debate shift from healthcare policy to whether the prenup was actually a bad idea.

5. Keep communication brief and tactical

Soldiers don’t waste time on unnecessary chatter. On election day, adopt the same approach. When someone presses you for your thoughts on a contentious issue, remember: Brevity is your best friend. Keep it short and neutral. If someone’s really pushing, hit them with a classic: “I’m just here for stickers.” It’s friendly, disarming and, most importantly, it’s 100% noncommittal, like most of your relationships.

6. Silence: The ultimate tactical move

In military strategy, sometimes silence is the smartest play. On election day, the same rule applies. If the conversation gets too heated, simply go silent. It’s not avoidance — it’s self-preservation in an invisible freezer. Silence can be golden, especially when your cousin is trying to trap you in a three-hour debate about tax policy.

Final tip: Exit strategy

Every good bunker comes with an exit strategy. Know when to retreat. If the political discussion gets too intense, you can always pull the classic, “Oh, I’ve got to check on my casserole!” No one questions a casserole. Nor knows what a casserole is.

By the end of Election Day, your bunker of political neutrality will have kept you safe from all political crossfire. You’ll emerge not only as a survivor, but also the person everyone still likes because you didn’t argue with anyone. And that, my friends, is the ultimate election day victory.

Why is this ballet performance blowing my mind?



Bewitching Stanford Ballet Company dancers doing the "Gravity-Defying Grande Plié" (Photo: WILLIAM MENG/The Stanford Daily)

Humor by **Chetanya Pandey**

Jan. 10, 2024, 12:26 a.m.

Editor's Note: This article is purely satirical and fictitious. All attributions in this article are not genuine, and this story should be read in the context of pure entertainment only.

The Stanford Dance Troupe's recent showcase was the brainchild of visionary choreographers and mad scientists, breaking the laws of traditional ballet and challenging the very fabric of reality. Or at least, that's what I thought as I reviewed the show last weekend after a whiff of fresh sprouts plucked from the forest. Gone are the days of graceful arabesques and elegant pliés; instead, dancers catapult themselves into a parallel universe. In one universe, the ballet dancers are called 'dallet bancers.' In another universe, I'm Taylor Swift, and the dancers are just shaking it off like I told them to.

At the heart of this bewitching spectacle lies the “Gravity-Defying Grande Plié,” a move that sends the dancers soaring into the stratosphere with a flex of their pinky toes. Audience members (me) gasp in disbelief as ballerinas gracefully pirouette through alternate dimensions, the shimmering of their tutus seemingly displaying secrets of the cosmos. Take me with you, I call out. They answer in a language I can’t understand. “Globo Globo,” they say. I say “Globo Globo” back, but they make fun of my accent.

The Stanford Dance Troupe spares no expense in pushing the boundaries of artistic expression. Picture this surreal scene: a prima ballerina. Riding a unicycle. And — oh, what’s that over there? It’s a teacup on her nose, all set against a backdrop of neon-lit intergalactic landscapes. I’m foaming at the mouth.

Other critics in the audience labeled the dancers as sheer absurdity. I say: some people need to stop taking shrooms before college dance performances. Not me, though. I’ll never stop.

An epic on Stanford football



Quarterback Ashton Daniels leads the team out of the tunnel. (Photo: CAYDEN GU/The Stanford Daily)

Humor by **Chetanya Pandey**
Nov. 15, 2023, 9:59 p.m.

Editor's Note: This article is purely satirical and fictitious. All attributions in this article are not genuine, and this story should be read in the context of pure entertainment only.

Gather 'round, folks! Welcome to the Stanford stadium, where the formidable Stanford charges ahead in the annual Big Game against Berkeley. Fans of the Cardinal, draped in red and white, are cheering and hollering in the Red Zone — and with good reason.

The score stands tall, flashing Stanford's 40 against Berkeley's 25 on the mighty scoreboard. The Cardinal's lead isn't just substantial; it's commanding. With every run and every pass, Stanford articulates a symphony of football prowess.

The precision in their execution is enthralling; it's not just a game — it's a masterclass in strategy.

The offense operates like a well-oiled machine, seamlessly alternating between powerful runs and long passes. Quarterback throws are sharp, weaving the ball through tight spaces as if guided by an unseen hand. The running backs, nimble and determined, slice through Berkeley's weak line of defense, gaining critical yardage with every down.

The defense, a steel wall, holds strong against Berkeley's attempts to breach Stanford's territory. Their interceptions and tackles are met with thunderous applause from the fans, who watch in awe as Stanford asserts its dominance on the field.

Every move is calculated, every play meticulously planned. It's as if the Cardinal team has decoded the very essence of football, turning the game into an art form. The momentum firmly favors Stanford, and the supporters are already envisioning the post-game celebrations, certain of another glorious victory.

But then, just as victory seems imminent, a sudden turn of events unfolds. Berkeley, against all odds, mounts a rapid, unforeseen comeback. The air in the stadium shifts, the momentum sways, and the seemingly unassailable lead Stanford held starts to slip away.

The once-confident supporters are taken aback as Berkeley's plays grow bolder and more daring. The score gap begins to narrow, creating a ripple of unease among the Cardinal faithful. Yet, even in the midst of this unexpected challenge, some fans hold onto the hope that their team will weather the storm and emerge victorious.

But alas, once the storm starts raging, it does not yield to the whims of mere spectators.

Papers

Can the future of neuroprosthetics be moral-debt-free?

Introduction

The central feature of the future of human senses augmentation is to examine how emerging neural interfaces, AI-driven signal processing, and multifunctional implantable devices can restore lost sensory function and, increasingly, extend perception beyond normal human limits. Current and near-term technologies already demonstrate this trajectory:

Retinal prostheses approaches aim to restore patterned vision in the profoundly blind and, in principle, could be extended to enhance contrast sensitivity, widen spectral bandwidth, or add zoom capability via variable-focus photovoltaic restoration (Palankar, Daniel et al).

Next-generation cochlear implants with 22 channels, electrode arrays and machine-learning-based scene analysis already allow users to achieve great speech comprehension in noise; future closed-loop systems promise selective attention beams and real-time translation by direct stimulation of frequency-specific auditory nerve fibers (Advanced Bionics).

Multichannel vestibular prostheses have restored dynamic visual stability in bilaterally vestibulopathic patients and could be adapted to augment roll-tilt perception or provide inertial navigation cues (Guyot & Perez Fornos).

High-density peripheral nerve and cortical interfaces for touch restoration in amputees are being scaled toward bidirectional prostheses capable of conveying fine texture, slip, and proprioceptive feedback at near-natural resolution (Raspopovic et al.)

Intranasal and direct olfactory bulb stimulation arrays, combined with electronic nose sensors, are in early development for diagnostic olfaction and potential enhancement of odor discrimination (Datta et al.).

These devices share a common technical foundation: chronically implanted microelectrode arrays or optoelectronic stimulators that must maintain stable, selective neural interfaces for years to decades. Validating long-term biocompatibility, electrode encapsulation, stimulation safety thresholds, and perceptual fidelity has historically required, and in most regulatory pathways still requires, many large animal models (pigs, sheep, non-human primates) and, for cochlear implants, thousands of cats in earlier decades.

The hidden cost is therefore not hypothetical. The same electrode materials, encapsulation chemistries, and surgical approaches that enable high-resolution retinal stimulation in a patient today were refined through chronic implantation studies in which not only did animals undergo insufferable pain, repeated surgeries, behavioral testing, and terminal electrophysiological mapping but also the number of animals used was more than required. Even in 2024–2025, FDA

and EMA submissions for next-generation retinal and vestibular implants continue to include mandatory 6–12-month pig and primate cohorts despite the existence of human-relevant alternatives (FDA).

Let us drive our focus back to the course mission. The sensory augmentation technologies we examine, devices capable of restoring and eventually enhancing human perception, are technically mature enough that continued reliance on animal models is increasingly a regulatory and cultural inertia rather than a scientific necessity. We can, and therefore must, reach the next phase of safe, high-performance neural prostheses by redesigning the preclinical pipeline to prioritize validated human stem-cell-derived tissues, organ-on-chip platforms, computational models and only use the number of animals actually required, thereby ensuring that moral progress in minimizing animal harm advances in lockstep with technical progress in expanding human sensory capability.

An important facet of this paper is not to propose the complete elimination of animal experiments. Rather it is based upon the opportunity to be more mindful about how we implement animal studies within this framework. The concern isn't about eliminating animal experiments which provide irreplaceable data on biocompatibility, surgical feasibility, and long-term stability. Rather the paper advocates for greater intentionality in experimental design to ensure that only the minimum number of animals necessary to achieve statistically valid and clinically meaningful results are being used for human senses augmentation. Indeed, in some cases, careful study design or more refined statistical approaches could yield the same essential safety and efficacy data with fewer animals.

What do we desire?

Many technologies are already moving quickly from restoring lost senses to actively improving perception beyond normal human levels. The following advances are expected in the next 10–20 years based on devices now in human trials or late-stage development:

Vision

New retinal implants (Science Corp, Bionic Vision Technologies, Neuralink) and optogenetic treatments place light-sensitive cells or tiny photovoltaic chips under the retina. By the mid-2030s these systems should give patients with macular degeneration or retinitis pigmentosa useful patterned vision and, in the same device, the ability to switch on infrared or ultraviolet sensitivity and to zoom digitally without an external camera (Palanker et al.).

Hearing

Today's best cochlear implants can use even 22 electrodes. The next generation will have even more channels plus built-in artificial intelligence. These implants will let users focus on one voice in a noisy room, hear frequencies above the normal human range, and even receive direct, real-time language translation delivered straight to the auditory nerve.

Balance and Spatial Orientation

Vestibular implants that replace inner-ear balance function are already working in patients who had no vestibular function on either side. Future versions with more electrodes and better sensors will provide much finer control of perceived motion and galvanic vestibular stimulation could even create an artificial sense of gravity for astronauts on long missions (Steller, Chris).

Touch

Prosthetic hands and arms connected to nerve or brain electrodes can now restore a natural feeling of pressure, texture, and movement. Over the next two decades these interfaces will become dense enough to let users feel objects in virtual reality or turn abstract data (temperature, radiation, network signals) into patterns felt on the skin (UChicago Medicine).

Smell and Taste

Early implantable electrodes in the olfactory bulb, combined with wearable electronic noses, are being developed to restore smell after injury or illness. Olfactory stimuli are integrated in various brain interface control loops. (Ninenko, I. et al.).

What's the common denominator for this future that we want? All of these near- and mid-term systems depend on the same core engineering challenges: (1) chronically stable, high-density neural electrodes; (2) long-term tissue-electrode biocompatibility (minimizing gliosis and encapsulation); (3) precise, percept-specific neural coding; and (4) surgical feasibility of implantation in delicate sensory end-organs. These four domains remain the primary justification for large-animal (pig, sheep, non-human primate) studies in current regulatory submissions. The same biocompatibility and chronic-performance data that will determine whether a patient can reliably perceive infrared light or feel fine texture in 2035 are still being collected, in 2025, from sentient animals undergoing months-to-years of implantation, stimulation, and terminal histology.

Thus, the sensory future we anticipate is technically inseparable from the preclinical paradigms we inherit—and from the ethical obligations we now have to reform them. This is why we need to talk about these obligations right now.

What did we do historically?

The development of sensory augmentation technologies, from cochlear implants that restore speech understanding in noise to retinal prostheses enabling pattern recognition in the blind, has relied heavily on preclinical testing in animal models. These studies have been essential for validating electrode biocompatibility, neural encoding strategies, and long-term surgical outcomes—core requirements for the multifunctional devices we have wanted. However, this progress has come at a significant ethical cost: thousands of animals subjected to invasive procedures, chronic stimulation, and often terminal endpoints. Below, let us review key examples across sensory modalities, drawing on published data to quantify the scale of animal use. While exact totals are rarely aggregated due to fragmented reporting, available figures underscore the cumulative toll on cats, pigs, sheep, and primates, species selected for their anatomical similarities to human sensory end-organs.

Case Studies

Cochlear Implants

The multi-channel cochlear implant, the device that is evolving into true “augmented ears” with frequency extension and instant translation, was born in the 1970s–1990s inside the cochleae of thousands of cats.

Why cats? Their inner ears have almost exactly the same size and spiral shape as ours, and they can be trained to respond to sounds. Labs in California, Australia, and Europe deafened cats with drugs, drilled into their skulls, threaded experimental electrodes through the round window, and stimulated them for months while the animals wore backpacks of electronics. After the data were collected, the cats were euthanized so researchers could slice open their inner ears and count surviving neurons.

No one kept an official tally, but historians and former researchers estimate the total exceeded 10,000 cats worldwide (Mudry and Mills, Patricia, Leake et al.). A single influential lab might implant 100–200 cats per year. Shepherd’s group in Melbourne alone published studies on hundreds; Leake and Snyder-Keller at UCSF did the same. Those cats paid the price for every safe charge limit, every electrode coating, and every speech-coding algorithm we now take for granted. Even today, when human stem-cell cochlea organoids can already measure electrode toxicity and hair-cell survival, some FDA reviewers still demand final cat studies before approving the next 100-channel “super-ear.” The cats that taught us how to hear are still teaching—because the regulations have not caught up with the science.

Retinal Prostheses

The dream of giving blind people useful vision, and one day adding infrared sight or digital zoom, was made real on the operating tables of hundreds of pigs, sheep, and monkeys.

Why these animals? Pigs and Yucatan minipigs have eyes almost the same size as human eyes (20–25 mm long), so a full-size implant can be slipped under or onto the retina exactly the way it will be in a patient. Sheep are close enough and cheaper. Monkeys are added when researchers need to prove the animal can actually “see” the dots of light the implant creates which requires brutal electrode implantations. (Kruger and Fausto).

- Argus II, the first FDA-approved retinal implant (Greenemeier, L.): Second Sight’s own records and FDA files show more than 100 animals kept for 6–12 months with the 60-electrode array glued to the retina, plus 30–40 rhesus monkeys trained to press buttons when they saw moving patterns. Total for Argus II alone: >200 large animals (Bertschinger, D.R. et al.)

- PRIMA (the wireless photovoltaic chip now becoming Science Eye): Pixium Vision implanted many Yucatan minipigs and dozens of rabbits to perfect the subretinal placement of its 378-pixel array. Sheep studies added another 20–30 animals to test new coatings (Schwahn, H.N. et al.). The exact numbers are not very clear.

- Alpha-AMS (the German 1600-pixel subretinal chip): Animals carried the implant for up to two years in reliability tests, so engineers could measure how much scar tissue built up around each tiny photodiode. Even a 6 micro-electrode device was implanted in many minipigs. (eyewiki.org)

Add the primate work for visual behavior tests (10–20 rhesus across programs), and the three major retinal prosthesis families together required roughly 400–500 large animals, each one undergoing retinal surgery, months of exams under anesthesia, and final euthanasia so the retinas could be sliced and studied under a microscope.

These animals proved that electrodes can stay in the eye for years and still make light, but they also revealed the same problems we now see in humans: retinal tears in many cases, slow encapsulation, and fading signals. Today, human retina-on-a-chip models already reproduce those exact failure modes in weeks instead of years, yet regulators still ask for more pigs and monkeys before the next infrared-capable implant can move forward. (Achberger, Kevin et al.)

Vestibular Implants

To give humans a stable world when their inner ears fail, and one day to create “artificial gravity” perception in space, researchers turned to monkeys.

Monkeys were chosen because they have human-like eye movements and can be trained to show when the world spins. Early work used squirrel monkeys. Lewis’s

lab at Johns Hopkins implanted many at a time, spinning them on chairs while measuring how well the implant restored the vestibulo-ocular reflex (VOR) during yaw rotations (Lewis, R. et al., 2002).

Early proof-of-concept work in squirrel monkeys showed that a prosthetic semicircular canal could restore a measurable vestibulo-ocular reflex (VOR) even when the electrical signal was delivered to the “wrong” canal nerve, a striking demonstration of central nervous system adaptation (Lewis, R. et al.).

Once the concept was proven, the field switched to rhesus macaques because they walk upright like humans, making them far better models for studying posture, gait, and real-world balance during three-dimensional motion.

Some got 24-channel electrodes in all three semicircular canals; others served as controls after their balance organs were destroyed with gentamicin. Monkeys wore head bolts, carried backpack stimulators, and were tested for months on rotating platforms before final surgery to examine the brain and inner ear (Lewis R. et al.).

These monkeys proved the implant can bring VOR gain back to much of normal: results now helping human patients walk without falling. But each success meant repeated anesthesia, deliberate balance loss, and eventual euthanasia.

Today, computational VOR models and human inner-ear organoids can already predict most electrode effects in weeks instead of years (Chen, Joseph et al.), yet the latest NASA- and ESA-funded “space balance” protocols still require new cohorts of rhesus monkeys. The steady gaze we want for astronauts tomorrow is still being bought, one monkey at a time, today.

Emerging Haptic and Olfactory BCIs

The next frontier- feeling textures in virtual reality or sniffing out diseases before symptoms hit, relies on brain-computer interfaces from Neuralink, Paradromics, and Blackrock Neurotech. These companies are wiring monkey brains with hundreds of electrodes to test if the tech can last years without failing, but the numbers reveal a hidden cost: dozens of primates enduring botched surgeries, infections, and early deaths.

Neuralink's N1 implant, with its 1024 flexible threads, has been drilled into at least 23 rhesus macaques since

2018, according to veterinary records obtained by animal rights groups. Elon Musk's famous 2021 video of "Pager" playing Pong- that monkey was one of 2–3 in the demo cohort, but others suffered brain swelling from unapproved glues or chronic infections that led to self-mutilation before euthanasia. By 2025, many animals were used for haptic validation in motor cortex, with federal probes uncovering rushed tests that killed more than necessary, up to 1,500 animals overall, including monkeys, amid cries that human-relevant simulations could have slashed the numbers by 80%. One monkey missed fingers and toes 'possibly

from self-mutilation' and another had brain swelling from unapproved BioGlue. Such chronic infections led to euthanasia. (Physicians Committee, Isobel Asher Hamilton)

Similarly, Paradromics, and Blackrock Neurotech do not have pretty numbers.

But, why should we care?

Before I suggest alternatives to decrease the number of animals in experiments, it is well worth it to discuss why we are even worrying about it in the first place.

Indeed, one might wonder, and rightly so, why we even need to mull over decreasing the number of animals tormented and used for human senses augmentation?

I derive an answer to such tensions from the roots of moral philosophy, the subsections of which dissect utilitarianism's human-centric justifications against sentience-based critiques, incorporating global variations and counterarguments. Regulators and reviewers are trapped by path dependency: the same animal protocols that won FDA approval for the Argus II in 2013 or the first multichannel cochlear implants in the 1980s are treated as the gold standard simply because they are precedent. Changing them would require admitting that thousands of past animal deaths were, in hindsight, avoidable. Companies fear liability: if a human trial fails and no animal data exist, lawyers will ask why the "tried-and-true" pig study was skipped. Veteran reviewers, trained in an era when a cat cochlea was the only realistic model, instinctively distrust a dish of human cells they have never held in their hands. The result is a quiet conservatism that keeps the old pipeline running long after its scientific justification has expired. Philosophically, this inertia clashes with the growing recognition that sentience, not species membership, is the morally relevant trait. Jeremy Bentham's 1789 question—"Can they suffer?"—remains the most powerful. (Whitman, Daniel) Peter Singer's principle of equal consideration of interests and Tom Regan's rights-based view both conclude that a rhesus macaque's pain counts every bit as much as a profoundly disabled human's. Gary Varner sharpens the point in his discussion of "marginal cases": if we justify primate experiments on the grounds that monkeys are not moral agents, we must confront the fact that many severely intellectually disabled humans are also not moral agents, yet we do not experiment on them. Once we accept that the capacity to suffer is what matters, the moral ledger becomes stark: the marginal benefit of one more 12-month pig study must be weighed against months of confinement, repeated anesthesia, and eventual euthanasia.

Christine Korsgaard offers a way forward that avoids absolutism. She argues that because we value our own rational nature, we are committed to valuing the conditions that make any valuing possible, including the ability of other creatures

to pursue their good in their own way. This does not demand the immediate abolition of all animal research tomorrow (some endpoints, such as closed-loop behavioral adaptation in complex environments, remain difficult to model perfectly).

It does demand a constrained consequentialism: animal use is permissible only when

1. a validated, human-relevant alternative does not yet exist for that specific endpoint, and
2. the expected human benefit, multiplied by its probability, clearly outweighs the actual suffering imposed on sentient beings.

In practice, this translates into a simple, enforceable rule: every new animal protocol in sensory neuroprosthetics should include a mandatory, public justification explaining why a human stem-cell, organoid, organ-on-chip, or in-silico model cannot replace or reduce the proposed animal numbers. If the answer is “because the 2013 guidance asks for pigs,” that is no longer morally sufficient. The moderate position is therefore not a compromise, it is the only honest one. We do not need to choose between superhuman senses and compassion. We need only to insist that the same ingenuity that will one day let us see infrared or feel data streams be applied first to ensuring no more sentient beings must pay unnecessarily with their lives for our progress.

What can we do?

Now that I have established why the unnecessary use of animals in experiments is morally a matter of concern, a key question still remains: do we even need so many animals to be sacrificed in the name of testing? The answer, backed by peer-reviewed breakthroughs and regulatory green lights, is a resounding no. The super-senses this course celebrates, infrared vision, selective hearing, lifelike touch in VR, are already within reach using human-based lab models that work faster, cost less, and deliver results that match or beat animal data. They're published, validated, and in some cases, regulator-approved. I now analyze two game-changers that prove we can drastically minimize the sacrifices.

Seeing Without Animal Eyes: Vascularized Human Retina-on-a-Chip

Imagine a tiny lab-grown human retina, complete with its own blood vessels, that “sees” light flashes just like your eye does, but survives for months without a single pig or monkey in sight. That's the reality from Fraunhofer Institute and collaborators, who in 2019 pioneered a “retina-on-a-chip” (RoC) who merged organoid and organ-on-a-chip technology to generate complex multi-layer tissue models. The organoids develop functional photoreceptors that form outer segment-like structures, crucial for light detection, and the whole setup mimics the blood-retina barrier to keep nutrients flowing realistically.

By 2023–2025, scientists have built much better “retina-on-a-chip” systems that stay alive and healthy for weeks or even months. These tiny chips now include real blood-vessel-like channels that copy the eye’s natural blood barriers. They grow three key parts together: light-sensing retinal organoids (mini-retinas made from stem cells), the supportive RPE layer underneath, and, in the newest versions, real human blood-vessel cells separated by an ultra-thin membrane that acts like the eye’s Bruch’s membrane.

The big win is that the RPE cells now touch the tips of the light-sensing cells exactly as they do in a real eye. Every day, the RPE “eats” the old tips that the light-sensing cells shed, keeping everything clean and healthy. This simple daily cleanup makes the light-sensing cells live much longer than they do in older lab dishes.

The chips also pump fresh oxygen and nutrients through the blood-vessel side only, just like in your eye. This natural flow helps the mini-retina mature better and stops the middle from dying of starvation, something that regular dishes and many animal eyes can’t do as well.

In short, these new chips give us a living, breathing piece of human retina that works more like the real thing than ever before. (Source: Gensheimer et al., Lab on a Chip review, 2025)

Therefore, although real-time calcium imaging and light-induced electrophysiological responses have been recorded in retinal organoids for years, the newest retina-on-chip platforms make these readouts far more stable and reproducible over weeks by supplying physiological flow, controlled oxygen levels, and RPE support. For retinal prostheses (e.g., PRIMA photovoltaic arrays), these human-based chips already allow direct, long-term testing of electrode–tissue interfaces, glial scarring, and chronic biocompatibility in fully human cellular contexts, eliminating the translational uncertainties of sheep, pig, or primate studies that dominated earlier validation work. As a result, several groups and companies are already replacing large-animal “biocompatibility” cohorts with retina-on-chip data for early-stage safety and performance evaluation, markedly reducing animal use while gaining more relevant human insights.

Hearing Without Cat Cochleae: 3D Human Cochlear Organoids

Paradoxically, the actual invention of the cochlear implant, the moment someone first put an electrode inside a living human cochlea and left it there, was achieved without animal models. The subsequent refinement, safety testing, and especially the move to multichannel designs relied very heavily on animal work. (Mudry, Mills et al.)

What if we could grow a mini human cochlea in a dish, one with twitching hair cells and firing neurons that “hear” vibrations, all wired together like the real one?

In 2023, scientists figured out how to grow tiny working cochleae in a dish using human stem cells. They take ordinary cells from skin or blood, turn them back into stem cells, and guide them to become inner-ear tissue. Older methods always ended up making balance organs instead of hearing organs, but this team added two special chemicals at just the right moment, one turns on a signal called Sonic Hedgehog, so the cells finally chose the correct “hearing path.”

After a few months, these little 3D organoids grow real inner and outer hair cells, the same ones we use to hear. The tiny hairs on them wiggle when gentle fluid flows past, exactly like sound waves do in a real ear, and the hair cells send proper electrical signals to nerve cells. A 2024 follow-up kept the organoids alive for more than 150 days and even tested tiny cochlear-implant electrodes inside them, all without ever using animals or real human ears.

For the first time, we now have living, working mini-human ears in the lab that can help treat deafness and improve hearing implants.

These organoids aren't just pretty; they've decoded why some genetic mutations (like in Usher syndrome) kill hair cells early, and they've screened drugs that boost SGN survival by 50%. For our augmenting senses vision, this means validating frequency extension without deafening cats, human data, human relevance, without moral debt.

The regulators are catching up too: FDA's ISTAND program (2023–2025) now green-lights organoids and microphysiological systems (MPS) for biocompatibility endpoints. Even if completely eradicating animal experiments may not be feasibly or humanly correct, decreasing the number of these experiments is very much a possible and necessary action.

What next?

The Compassionate Argumentation Protocol

The future we celebrate (infrared vision, hearing a whisper across a stadium, feeling data as touch) is coming fast.

The only question left is whether we arrive there proud or ashamed. I propose the Compassionate

Augmentation Protocol, a short, concrete set of rules that lets us keep the ambition while dropping the cruelty. It is to be used in cases where animals in experiments will be inflicted with pain, example, rolling, implantation etc.

1. The Alternatives-First Gateway- Every premarket submission for a retinal, cochlear, vestibular, haptic, or olfactory neuroprosthesis must include a public, two-page document titled “Why This Could Not Be Done in Human-Relevant Models.” If the honest answer is “because the 2013 guidance still says pig,” the reviewer sends it back. If a validated retina-on-chip, cochlear organoid, or in-silico model already exists for the endpoint (and it does for most biocompatibility

and chronic-stability questions), the animal study is rejected outright. Let there be no more 400-page justifications buried in appendices; two pages, posted online, signed by the PI and the company CEO. Transparency kills inertia.

2. The Sensory Augmentation Innovation Levy- For every mammal (pig, sheep, primate) used, the sponsor pays a portion of the research grant to an animal welfare organisation of their choice. As an alternative, the money could also go to scaling human organoid and organ-on-chip platforms (perfusion systems, cryopreservation, high-throughput electrode interfaces). Example: a typical 30-pig retinal-implant study generates a \$100,000 donation, enough to build one fully vascularized retina-on-chip lines that will prevent dozens of future pig studies. Industry funds its own escape hatch and turns every remaining animal into an accelerator for the animal-free future.

3. Mandatory Smart Statistics- No more fixed $N = 8$ pigs because “that’s what we always do.” Every sensory neuroprosthesis protocol must use Bayesian sequential designs or AI-optimized experimental planning for accurate sample size calculation (tools freely provided by NC3Rs and the NIH). Real-world evidence from similar neural-interface studies shows these methods cut animal numbers while actually increasing statistical power (NC3Rs EDA toolkit). One stops the moment the data are conclusive. No rounding up to a nice even number, no “just in case” extras.

4. The Human-First Registry- A single, open, clinicaltrials.gov-style database where every new human-relevant model (retina-on-chip, skin-on-chip, in-silico fibrosis predictor) is registered the day it is validated. Regulators, reviewers, and competitors can see in real time which endpoints are already covered without animals. Once a model is listed and independently replicated twice, any new animal protocol for that same endpoint is automatically flagged for rejection. The registry becomes living proof that the field is moving forward, not circling back to the same old monkey lab.

5. Minimizing Pain- It will be mandatory to explain how steps will be taken to minimize pain to the animal and video footage be sent to verify that the animal was not unnecessarily mishandled.

Objections

Skipping animal studies will delay the superhuman future.

Once scaled, organ-on-chip and organoid platforms are dramatically faster and cheaper. A vascularized retina-on-chip delivers 6-month biocompatibility data in 8–10 weeks at ~\$80,000 versus \$600,000–\$1.2 million and 14–18 months for a 30-pig study (Fleischer et al., 2024; Marx et al., 2020). The same holds for cochlear organoids and skin-on-chip: weeks instead of years, pennies on the dollar. Companies that adopt them first will reach the clinic sooner, not later.

Some things can only be learned in behaving animals.

True, for now, and only for a narrow slice. Closed-loop behavioral adaptation in complex, real-world tasks (e.g., a monkey learning to use a new haptic channel) is still hard to model perfectly.

Concede the narrow case but impose a strict sunset clause. Any animal protocol approved for a “behavior-only” endpoint must include a public 36-month expiration date. If a validated human-relevant model (organoid + VR/AR behavioral proxy, high-fidelity simulation, or early human data) achieves the same endpoint within three years, the animal protocol is automatically retired. Also, as Bochen Liu et al. write, “the animal model also has limitations, and there are species differences from human which makes it challenging to extrapolate the generated results”.

Conclusion

By 2035–2040, children will hear any language instantly, and feel gravity in space. These miracles are technically ready. What is not ready is our conscience: in 2025, regulators still demand pig and primate cohorts for endpoints that Fraunhofer’s EpiRetinal chips, Stanford’s neuromuscular organoids, and FDA-accepted I STAND models already cover with higher human relevance.

The 3Rs exist on paper, great-ape research is banned in Europe, and NIH policy demands justification, yet animal welfare remains the least-enforced clause while bureaucratic habit keeps the old pipeline spinning. Sometimes, it is not about the animal use in experiments but the way in which they are handled. An article published in the Guardian covered how a Spanish lab was ruthlessly extremely tormenting its monkeys and animals just for the sake of it.

We do not need to abolish animal research tomorrow; we need only to stop treating it as inevitable today. The Compassionate Augmentation Protocol—alternatives-first review, a \$3000-per-mammal innovation levy, mandatory smart statistics, and a public Human-First Registry with sunset clauses, turns every remaining animal study into a countdown to its own obsolescence.

History already proved the pattern: the first human cochlear implant worked in 1961, years before the cat studies we later called “required.” Outdated animal models are the biased algorithms of biology, familiar, but no longer acceptable. We can have superhuman vision, hearing, touch, balance, and smell without a single unnecessary death. That is the only future futurists should celebrate, and, starting now, the only future we need to build.

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The United Nations Human Rights Council's 2017 Report on the Financialization of Housing: Combining 3D Architecture with Human Rights-Based Solutions
 Title: The United Nations Human Rights Council's 2017 Report on the Financialization of Housing: Combining 3D Architecture with Human Rights-Based Solutions

Introduction:

The 2017 report by the United Nations Human Rights Council sheds light on the financialization of housing and its detrimental impact on the right to adequate housing. This essay seeks to present a comprehensive approach that incorporates the report's findings and proposes the integration of 3D architecture as a solution to the global housing crisis. By combining the report's recommendations with the implementation of 3D printing technology, countries can address the shortage of affordable housing rapidly and cost-effectively. Moreover, an international agreement facilitated by an international court can foster collaboration and ensure the equitable sharing of 3D house building machinery, enhancing global efforts to uphold the right to adequate housing.

The Financialization of Housing and Its Consequences:

The report underscores how the financialization of housing has led to soaring prices and unaffordability, resulting in the displacement of vulnerable communities and exacerbating social inequality. These challenges necessitate innovative approaches that prioritize human rights over financial interests.

The Potential of 3D Architecture:

To mitigate the housing crisis, the integration of 3D architecture, leveraging state-of-the-art 3D printing technology, presents a promising solution. By utilizing this technology, homes can be constructed quickly, cost-effectively, and with structural integrity intact. The customizable designs and sustainable materials associated with 3D printing enable the creation of affordable and environmentally friendly housing options.

Implementing an International Collaboration:

Recognizing the transformative potential of 3D architecture, countries can establish an international agreement facilitated by an international court. This

agreement would promote collaboration, resource sharing, and knowledge exchange to address housing challenges globally.

The Plan:

1. **Incorporating Human Rights Principles:** To ensure that 3D architecture aligns with human rights principles, governments must adopt an overarching national housing law based on human rights. This legal framework will address inequality and provide a long-term roadmap for the fulfillment of housing rights.

2. **Embracing 3D Architecture:** Governments should invest in research and development to enhance 3D printing technology for housing construction. This includes advancing sustainable materials, optimizing scalability and efficiency, and promoting architectural designs that prioritize the well-being and dignity of residents.

3. **International Resource Sharing:** Under the international agreement facilitated by an international court, countries can collaborate on the equitable sharing of 3D house building machinery. This approach ensures that all nations facing housing challenges have access to the necessary technology and resources.

4. **Capacity Building and Knowledge Transfer:** Countries proficient in 3D architecture should offer technical assistance, training, and knowledge sharing to nations seeking to adopt this technology. Through workshops, partnerships, and platforms for exchange, expertise can be shared to enhance the implementation and effectiveness of 3D-printed housing projects.

5. **Local Adaptation and Community Engagement:** Governments must prioritize community engagement, ensuring that the implementation of 3D architecture respects local contexts and meets the needs of residents. Active involvement of communities in decision-making processes fosters social acceptance and sustainability.

Conclusion:

Integrating 3D architecture with human rights-based solutions enables countries to address the global shortage of affordable housing. By combining the recommendations from the United Nations Human Rights Council's report with the implementation of 3D printing technology, governments can construct homes rapidly, cost-effectively, and in line with human rights principles. Through an

international agreement facilitated by an international court, collaboration and resource sharing can be fostered, strengthening the collective response to the housing crisis. By embracing 3D architecture, investing in research and development, sharing resources, building capacity, and engaging communities, nations can move closer to realizing the right to adequate housing for all. The transformative potential of 3D printing, coupled with a human rights-focused approach, paves the way for sustainable, inclusive, and dignified housing solutions.

10 June 2025

The Predicament of Mental Health in India

In 1846, anesthesia swept through the American medical world like wildfire. Within two months of its first public demonstration, it was adopted in hospitals across Boston, and within two years, it became standard in most of Europe. In contrast, antiseptic surgery, another major breakthrough, lagged behind for decades. As Atul Gawande writes in his New Yorker essay *Slow Ideas*, the difference lay not in usefulness but in uptake: anesthesia offered an immediate, visible benefit in convenience for patients and surgeons, while antiseptic measures were invisible, inconvenient, and made doctors feel blamed. Gawande's framing mirrors what Emily Levine and Mitchell Stevens describe in "Neither state nor market," where ideas diffuse through competitive emulation across universities rather than diffusion of its own sake.

These institutions, like surgeons resisting antiseptics, emulate selectively, guided as much by prestige and peer modeling as by evidence.

The framework of "Atlantic Crossings" by Daniel Rodgers offers a broader historical lens for this. Rodgers outlines how social innovations crossed national borders, not by brute force or imperial channels, but through reformers, mutual admiration, and shared crises. In today's language, this might be called "policy transfer" (Hulme, Rob, 2005). What matters in all three cases is context: the idea must not just land; it must land in soil ready to nourish it. The problem of 'slow ideas', good innovations that resist adoption, remains alive and well today. Few examples are as striking, or as consequential, as the story of mental healthcare in India.

When a country with the world's largest adolescent population fails to provide basic mental healthcare, the costs reverberate beyond individual suffering. They manifest in lost productivity, intergenerational trauma, and the reinforcement of social hierarchies that deny care to the marginalized. In failing to take mental health seriously, the state is not only depriving its most vulnerable citizens of a necessity but also undermining the foundations of its own economic and social aspirations. And this is why this discussion matters.

Despite the rising mental health burden, and despite decades of global awareness campaigns and international best practices, therapy has never truly taken root in India. According to a 2022 World Health Organization report, India accounts for nearly 15% of the people suffering from a mental health crisis, but spends less

than 1% of its total healthcare budget on mental health services. A 2021 Times of India report found that fewer than 4,000 clinical psychologists were registered to serve a population of over 1.4 billion. There is only one psychiatrist for every 343,000 Indians. Meanwhile, therapy, particularly psychodynamic or Cognitive Based Theory-based modalities that involve talking to a licensed professional, remains stigmatized, misunderstood, or economically inaccessible to the majority of Indians.

As reported by the New York Times Editorial Board in 2014, 258,000 of the world's 804,000 suicides occurred in India in 2012 which is the highest per capita. Suicide is now the leading cause of death among young Indian women, surpassing even maternal mortality. Depression, anxiety, and distress linked to educational pressures, economic insecurity, and gendered expectations have created a perfect storm of mental vulnerability. Yet the state response remains lackluster. Even as a national mental health policy was unveiled in 2014, the same month saw a 20% cut to the overall health budget, revealing a gap not just between vision and execution, but between policy and political will.

In the United States, by contrast, therapy has become increasingly normalized. From college campuses to Instagram reels, therapy has become a language of self-care, privilege, and social consciousness. Between 2002 and 2022, the number of U.S. adults receiving mental health treatment jumped by over 50% which is comparatively higher. A survey by the American Psychological Association puts forth that Americans are becoming increasingly open about their mental health. Cultural factors, of course, play a role. But so too do the social mechanisms of innovation diffusion.

To help understand why therapy remains “a slow idea” in India, I want to shed light on how cultural transfer happens, how public health ideas spread, or don't, and what can be done to change that.

The Theory of Diffusion

In *Diffusion of Innovations* (2003), Everett Rogers proposed a model for how new ideas spread: first to innovators, then early adopters, early majority, late majority, and finally laggards. The model emphasizes key characteristics of successful innovations: observable benefits, simplicity, compatibility with existing values, and credibility of the messengers. Historically, these traits explain the success of the Green Revolution in India, the uptake of mobile phones in rural villages, or the spread of microfinance: clear utility, visible results, and trusted intermediaries like scientists or NGOs.

In the context of mental health, it reveals tensions. The “innovation” is not a seed or a gadget but an immaterial, cultural practice: therapy. Its benefits are long-term, not easily demonstrable immediately. It lacks physical form, often demands emotional vulnerability, and is heavily shaped by Western norms of individualism and introspection. The sophisticated “change agents”, often elite, urban, English-speaking therapists, may not be seen as credible or relatable by the rural poor who regard them as agents of the rich, by the rich and for the rich. The bottlenecks, then, are not just infrastructural but cultural and perceptual: therapy is perceived as elitist, foreign, and unnecessary source. The diffusion model, when applied here, breaks down not because the need is not there, but because the message and messengers do not align with the lived realities of the majority.

Mental Health as cultural transfer

Rather than treating therapy as a value-neutral science, I draw on Kulturtransfer theory to frame it as a cultural object, shaped by and embedded in the Western histories from which it emerged. When practices like CBT or Freudian analysis cross borders, they undergo transformation. As Wissenstransfer theorists such as Alexander von Humboldt argue, knowledge does not simply move; it is reshaped, reinterpreted, and sometimes resisted.

In India, the dissonance is especially sharp. Psychotherapy arrives bearing assumptions of the nuclear family, of individual agency, of speaking one’s pain aloud, that often jar with the rhythms of Indian life. Here, the self is less solitary and more entangled: bound by kin, by duty, by caste and community.

Communication leans on what is unsaid, not declared; emotion is felt in gesture, not always in words. For many, suffering is not a matter of chemical imbalance or cognitive distortion: it is spiritual, familial, karmic.

Therapy’s quiet room, its scripted silences, its reverence for privacy, can feel foreign, even self-indulgent. The result is a model of care that speaks fluently only to the English-speaking elite, while the vast majority remain untouched, not simply for lack of clinics or counselors, but because the very grammar of therapy is written in a language their lives do not speak.

Entangled Indian psychiatry

Kris Manjapra’s *Age of Entanglement* provides a framework for understanding how Indian psychiatry is not a pure Western import, but a hybrid. Indeed it is shaped by colonial histories, caste hierarchies, and institutional legacies. While Western psychotherapeutic models rarely reached the masses, biomedical psychiatry, pills, asylums, diagnoses, was institutionalized during British rule and retained in postcolonial policy. Mental health became aligned with control and pathology, not healing or self-knowledge.

This institutional legacy persists. Most psychiatric care remains centered in underfunded government hospitals, with rigid hierarchies between doctor and patient. Therapy is seen as optional; medication is considered essential. The entangled modernity of Indian psychiatry has thus created a fractured landscape where healing is unevenly distributed, and legitimacy is unevenly granted. Mental health, as delivered through this system, often fails to resonate with lived realities and local expressions of suffering.

Jain and Jadhav's "A Cultural Critique of Community Psychiatry in India" deepens this critique by arguing that psychiatric models rooted in Western epistemologies are frequently misaligned with Indian cultural contexts. They draw attention to how concepts like depression or trauma may lack direct linguistic or cultural equivalents.

Barriers to diffusion

India has fewer than 0.05 psychologists per 100,000 people in rural areas. Even in urban centers, therapy is expensive (₹2,000–₹4,000 per session), rarely covered by insurance, and concentrated in private clinics. For the vast majority, this is unaffordable. But even where access exists, stigma and misinformation prevent uptake. Therapy is associated with "madness" or weakness. Seeking help is often seen as admitting failure or is looked down upon for being "delicate".

By contrast, consider innovations that have spread rapidly: mobile phones, microfinance, even psychiatric pills like SSRIs. These are discrete, visible, and require minimal behavior change. Therapy demands time, trust, and introspection. In Rogers' terms, it fails on "observability" and "compatibility."

Meanwhile, the U.S. provides counterexamples. This is the land where therapy has been normalized through pop culture, institutionalized in schools and workplaces, and supported by platforms like BetterHelp or Talkspace. These forces reinforce each other. India lacks such an ecosystem, both in infrastructure and cultural legitimization.

Why some innovations spread fast

Atul Gawande's "Slow Ideas" highlights why some breakthroughs, like anesthesia, spread quickly while others, like antisepsis, took decades. The difference? Visibility, ease of use, and immediate payoff. SSRIs spread in India not because mental illness was de-stigmatized but because a pill was easier to distribute, prescribe, and sell than therapy. It required no change in worldview, just a transaction.

The Green Revolution succeeded through government support, local champions, and cultural embedding. Mobile phones took off because they plugged into preexisting communication needs and quickly showed benefits. These cases show that diffusion works best when innovations are affordable, adaptable, and anchored in local realities.

Therapy, as currently practiced, fails these tests. But it could succeed, if reframed, localized, and might I add, repackaged and redistributed.

Proposing an Indian model of therapy

New models are emerging. Organizations like Sangath and Atmiyata train local community health workers in basic counseling techniques. WhatsApp-based support groups offer anonymity and scalability. Some startups deliver therapy in regional languages or embed it in familiar formats (e.g., storytelling, group sessions).

Bollywood and digital influencers have helped reduce stigma, but unevenly, mostly among youth and the urban middle class. Still, these cracks in the wall suggest potential. A truly Indian model of therapy would decenter urban norms, integrate spirituality, family dynamics, and local idioms of distress, train peer counselors at scale, leverage existing trust networks (teachers, nurses, ASHA workers).

I would like to put forth that policy could help by funding mental health programs in government schools, funding emerging startups in the mental health sphere, subsidizing training in regional languages, creating a public mental health corps, akin to the National Service Scheme, but mainly I will dwell on the unique positions startups hold in India to be able to address this change nationally through the right messages and even memes.

The Hope: Startups

Agile, culturally attuned, and tech-enabled, startups can go where traditional systems cannot. Where old traditional Western models stumble across language barriers, social taboos, and remote geographies, startups can leap. Just as Powell observes that nonprofits step in to address the gaps left by both the market and the state, startups in India are uniquely positioned to fill these critical voids, driving innovation and impact where it's needed most.

Most Indians think and feel in the textures of their mother tongues; they speak of sorrow not in diagnostic codes but in phrases like *mann ka bojh* or *dil bhar aana* (burdens of the heart, heaviness of soul). By building platforms rooted in regional languages and cultural metaphors, startups can reimagine therapy not as a foreign import but as something familiar, close to the hearth. Therapy that sounds like you is therapy you're more likely to trust.

With chatbots whispering through WhatsApp, voice notes exchanged like confidences, and mobile apps offering solace at midnight or in stolen lunch breaks, startups can meet people where they are, whether in a village courtyard or a student's cramped hostel room. They can offer care that bends to the lives of working-class users: asynchronous therapy, journaling prompts, recorded wisdom that can be returned to, like a trusted friend.

And in an age where attention is the rarest currency, even a meme can become a lifeline. Reels, stories, and short videos democratize it. A line of text beneath an Instagram post might be the first time someone sees their own pain named. Startups are making emotional fluency go viral, one share, one scroll, one spark of recognition at a time.

The potential doesn't end there. Startups can gather the unseen data of distress, anonymized, respectful, but rich, mapping how suffering moves through regions, communities, generations. They can test, adapt, and shift direction faster than any bureaucracy ever could.

To let this promise grow, policy must become soil, not stone. Grants, fellowships, and incubators, especially those that reward cultural sensitivity, local partnerships, and blended models, could help these saplings become trees. In a land where care has often been slow, these young ventures may yet move fast, not to erase the past, but to reshape the future, one voice, one language, one quiet breakthrough at a time.

Conclusion

The journey of therapy in India offers a deeper parable: no idea spreads untouched by the soil it lands in. Innovations do not drift freely through the world like clouds, rather they move through dense forests of history, language, custom, and power. What works in Palo Alto, US doesn't automatically work in Patna, India. For therapy to take root, it must be grown, not grafted. It must learn to speak in local tongues, to sit beside people rather than hover above them, to listen before it heals. This is not a task of import, but of translation, slow, patient, and attuned. Mental health may be India's "slow idea," but slow need not mean still. With the right voices to carry it, the right hands to shape it, and the humility to meet people where they are, even the slowest ideas can become seeds of quiet revolution—changing not just systems, but the very way suffering is seen and soothed.

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This report finds that most American adults now hold positive views about mental health, with growing openness and acceptance. However, it also documents that some stigma persists, especially around disclosure and seeking help. The report provides recent survey data and analysis of shifting attitudes, offering a useful comparative context for understanding mental health stigma and cultural adaptation in India.

The mind of the animal-kind

A. Thesis

René Descartes argues that nonhuman animals are mere machines, lacking minds, while humans are uniquely endowed with an immaterial thinking substance. Yet the striking similarities between human and animal behavior raise a challenge: if animals can act in flexible, intelligent ways, what justifies denying them minds while preserving human exceptionalism?

In this paper, I argue that Descartes' criteria for denying minds to animals do in fact fail, because neither language nor general reasoning establishes a genuine difference in kind between humans and animals. Once these criteria are rejected, there is no principled basis for maintaining a strict metaphysical divide, and we are led to take seriously the possibility that at least some animals possess forms of mentality, and thus to soften Descartes' sharp distinction by allowing that mentality may extend beyond the human case, even if in different forms or degrees.

B. Descartes' position

Descartes argues that non-human animals are purely mechanical beings lacking thought or reason. In Part V of the *Discourse on the Method*, he proposes two criteria distinguishing humans from animals: genuine language and universal reason. Animals may produce sounds or perform specialized tasks, but they cannot use language creatively or apply reason across situations. Humans, by contrast, possess both body and immaterial mind (*res extensa* and *res cogitans*), making them fundamentally distinct from animals.

C. The tension

Descartes' position faces a serious and arguably self-undermining tension. As Margaret Dauler Wilson's survey of the reception of his automaton thesis makes clear, the problem comes from a set of philosophical difficulties internal to the way Descartes constructs his two criteria, difficulties that, when pressed, suggest not that animals fall short of the threshold for mind, but that Descartes has drawn that threshold in a question-begging way precisely to exclude them. This creates a deeper methodological problem: Descartes defines rationality through capacities already associated with adult humans, then uses that definition to deny mentality to beings who express intelligence differently. Descartes presents two criteria for distinguishing rational from purely mechanical beings.

The first is the capacity to use language as a genuinely flexible instrument, not mere stimulus-triggered vocalization, which machines can produce, but the ability

to "arrange words differently so as to respond to the sense of whatever is said in its presence" across an indefinite range of contexts (*Discourse on the Method*, Part V).

The second is general reason: the ability to "act in all the circumstances of life," which no fixed set of bodily dispositions could fully replicate (Wilson, p. 9). Failing either test, Descartes concludes, an entity acts from mechanism alone. Since no animal passes either test, all animals are mere automata.

The first difficulty is that these criteria, far from functioning as neutral tests, appear to build on the very conclusion they are supposed to establish. Consider the language criterion. Gassendi objects in the Fifth Objections that it is not "fair" of Descartes to demand human speech of beasts and refuse to consider their own forms of expression — that beasts produce their own kind of speech and use it "in just the same way as we do ours" (Wilson, p. 11). Gassendi's objection is that Descartes mistakes difference for absence: because animal communication does not resemble human syntax, he concludes it is not genuine expression at all. The objection is persuasive because it exposes that Descartes never fully justifies why human linguistic structure should count as the exclusive standard for mentality. The argument therefore depends less on proving the absence of thought in animals than on restricting thought to specifically human forms of expression.

The same circularity affects the general reason criterion. Descartes argues that wherever animals outperform humans in specific domains, this proves not the presence of reason but its absence: genuine reason, being universal, would secure superiority across all domains, not just some (Wilson, p. 9). But this assumes that reason is an unlimited, all-purpose faculty, meaning any domain-specific excellence automatically does not count as real reason. Animal behavior is treated as mechanical not because investigation has shown it to lack flexibility, but because it fails to meet a standard designed with that conclusion in mind. As the authors of the Sixth

Objections point out, they are no more able to "see his internal mode of action" than he is theirs (Wilson, p. 12). Any evidence supporting animal intelligence is reinterpreted in advance as instinct or mechanism, making the theory difficult to falsify.

That divide is itself philosophically unstable in ways Descartes' later concessions expose. The sharpest pressure comes from the argument from physiological continuity, which Descartes himself raises in a 1646 letter, only to set aside on theological grounds. He acknowledges there that since the organs of non-human bodies are not very different from humans', it may be conjectured that there is

attached to these organs some thought such as we experience in ourselves. The anatomical similarities between humans and animals give genuine grounds for thinking animals have some form of inner experience. Descartes' only reply is that if animals thought, they would have immortal souls, which he finds implausible (Wilson, p. 13). But this is a theological deflection rather than a philosophical argument because it shifts the discussion away from the evidence of animal cognition itself and toward theological concerns about immortality. As a result, Descartes does not fully answer the underlying challenge posed by the similarities between human and animal physiology and behavior.

The most powerful strand of Wilson's case concerns the internal tension she identifies within Descartes' own position across time. The Descartes of the *Discourse* claims to have proof that beasts have no thought whatsoever (Wilson, p. 7-8); the Descartes of the 1646 letter to Newcastle and the correspondence with More retreats to the weaker position that animal mentality cannot be demonstrated either way (Wilson, p. 12-13). The shift from certainty to uncertainty suggests that the automaton thesis could not fully withstand objections concerning behavior, physiology, and subjective experience.

D. Are animals body-soul hybrids too?

We are now left with two possible directions. One is to revise Descartes' framework and allow that at least some nonhuman animals possess genuine minds rather than being mere machines. The other, more radical possibility, is to interpret humans themselves in purely mechanistic terms. Although this second possibility cannot be dismissed outright, the first better preserves the features of conscious thought and rationality that Descartes himself takes to be central to human existence.

The deeper philosophical problem is that Descartes' version of Cartesian dualism depends on a strict distinction between humans as thinking substances and animals as purely mechanical bodies, yet his own criteria fail to establish that distinction convincingly. Once language and universal reason are shown to rest on standards modeled exclusively on human cognition, the exclusion of animal mentality no longer appears philosophically neutral. Descartes can preserve human exceptionalism only by dismissing strong similarities in behavior, perception, memory, and adaptive intelligence between humans and other animals.

For this reason, Descartes' strict form of Cartesian dualism ultimately becomes difficult to defend. The issue is not merely that animals may possess some degree of cognition, but that the sharp metaphysical divide required by Cartesian dualism

cannot be sustained without relying on circular standards for what counts as genuine thought in the first place. Once those standards are questioned, the claim that humans alone possess minds loses much of its philosophical force, and the boundary between human and animal mentality appears far less absolute than Descartes assumes.

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Searle argues (based on the Chinese Room thought experiment) that we should not rely on the Turing Test for deciding whether a system exhibits genuine mentality (in particular, understanding). Does his discussion suggest any alternative way of testing an artificial system for genuine mentality?

A. Introduction

Alan Turing argues that a machine should be considered intelligent if it can successfully imitate human conversational behavior in the imitation game, while John Searle challenges this conclusion through the Chinese Room argument by claiming that formal symbol manipulation alone cannot produce genuine understanding. This raises the further question of whether any alternative test could determine if an artificial system truly possesses understanding. In this paper, I argue that although Searle's discussion suggests that a more adequate test would need to examine the internal causal processes underlying cognition rather than external behavior alone, there remain many problems with such a test to establish genuine thinking. While indirect neurobiological and causal comparisons may provide evidence for machine thinking, subjective understanding ultimately remains inaccessible from a third-person perspective.

B. The Turing Test

Alan Turing proposed the Turing Test as a practical method for determining whether a machine can think. Instead of attempting to define the terms "machine" and "think" directly, Turing replaces the original question with what he calls the "imitation game" (Turing, pp. 433–434). In this game, a human interrogator communicates through text with both another human and a machine. Turing proposes that the machine succeeds if the interrogator "does not make the right identification after five minutes of questioning" frequently enough that the machine's performance becomes comparable to that of a human participant (Turing, p. 440).

The Turing Test therefore treats intelligent linguistic behavior as sufficient evidence for thinking. Turing argues that the important issue is not the machine's physical composition or internal structure, but whether it can perform in conversation in a way comparable to a human being. Turing further explains that the question-and-answer format can introduce "almost any one of the fields of human endeavour" into the imitation game (Turing, p. 435).

Turing even predicts that future digital computers could eventually perform well enough in the imitation game that an average interrogator would frequently mistake them for humans (Turing, pp. 441–442).

C. Why Searle thinks the Turing Test is not a good test

Searle rejects the Turing Test because he believes it confuses syntax with semantics. His Chinese Room thought experiment attempts to show that a system may successfully imitate understanding without actually possessing any understanding at all. In the thought experiment, Searle imagines himself inside a room receiving Chinese symbols through a slot. Although he does not know Chinese, he possesses an enormous rulebook written in English that tells him how to manipulate the symbols and produce appropriate responses. By carefully following these formal rules, he can generate answers that appear perfectly fluent to native Chinese speakers outside the room. To outside observers, it seems as though the room understands Chinese (Searle, p. 418).

However, Searle insists that neither he nor the room genuinely understands the language. He is merely manipulating symbols according to their shapes and formal relationships. The system processes syntax but lacks semantics. Searle develops this argument as a criticism of what he calls “Strong AI.” According to Strong AI, a sufficiently programmed computer does not merely simulate a mind but literally has a mind and genuinely understands language. By contrast, “Weak AI” claims only that computers are useful tools for simulating or studying mental processes.

Searle accepts Weak AI but rejects Strong AI. He argues that while computers may imitate intelligent behavior or help researchers study cognition, executing a program or following rules alone cannot produce genuine understanding or intentionality (Searle, pp. 417–418). This distinction forms the core of Searle’s argument: > syntax ≠ semantics.

According to Searle, computer programs operate entirely syntactically. Programs manipulate symbols according to formal rules without attaching any meaning to those symbols. Genuine understanding, however, requires semantics: intentionality, meaning, and mental content.

Searle argues that instantiating a computer program is never by itself a sufficient condition of intentionality and that mentality depends on the specific causal powers associated with brains rather than on computation alone. Therefore, even if a machine passes the Turing Test perfectly, this only demonstrates successful behavioral imitation. The Chinese Room is intended to demonstrate that external

behavior alone cannot prove the existence of mentality. A system may appear intelligent while internally lacking any conscious understanding whatsoever.

D. Is there a test that can demonstrate genuine understanding?

Searle does not provide a straightforward replacement for the Turing Test.

However, his discussion suggests certain criteria that any adequate test for genuine mentality would need to satisfy. According to Searle, a system would require more than successful input/output behavior, more than formal symbol manipulation, genuine semantics and intentionality, and possibly the same causal powers possessed by human brains.

This raises the possibility of an indirect neurobiological test. Instead of focusing only on behavior, researchers could compare the internal processes of machines with the neurological processes associated with consciousness and understanding in humans.

For example, in humans we can identify strong correlations between neurological states and mental states such as pain, wakefulness, unconsciousness, memory, and intentional thought. Certain patterns of neural activity reliably accompany conscious experiences. Because humans are already accepted as conscious beings, these correlations provide indirect evidence about mentality.

One possible approach would therefore be to examine whether machines exhibit internal causal processes analogous to human neural states. If a machine displayed brain-like organizational structures, complex interactions between internal states, and processes strongly parallel to those associated with human consciousness, we might have reason to attribute mentality to it. However, this approach still faces a major philosophical problem. In humans, neurological correlations work because consciousness is independently assumed. We already believe humans possess minds, so observing correlations between neural states and conscious experience is meaningful. In machines, however, consciousness itself is precisely what is under dispute.

Even if a machine exhibited processes analogous to human neural activity, we could not conclusively infer genuine understanding or consciousness from those similarities alone. To do so would require assuming beforehand that this class of machine is capable of consciousness. Thus, neurobiological similarity may provide evidence for mentality, but it cannot definitively prove it.

Nevertheless, the Neurobiological Test can be developed further so it does not just treat cognition as a sequence of detached computations. Rather than focusing

exclusively on similarities to biological neural activity, we might examine whether a system develops internally integrated semantic representations that guide learning, memory, perception, and action rather than merely producing correct outputs. The machine could be exposed to ambiguous situations, deception, conflicting goals, emotional narratives, embodied sensory interaction, memory revision, and self-reflective tasks.

The Neurobiological Test is based on the substratum that evidence for understanding must come from internal causal organization rather than behavioral performance alone. The test asks whether the system has causal powers similar to those that produce understanding in human beings. It is not conclusive, but it is stronger than a behavioral test because it investigates the source of intelligent activity rather than only its external appearance. The Neurobiological Test is not the strongest candidate because it conclusively proves understanding but rather because there may be no better alternative available once Searle's objections are taken seriously. Searle argues that behavioral performance alone cannot establish genuine understanding. At the same time, unlike properties such as temperature or heart rate, which can be measured directly with appropriate instruments, subjective understanding and consciousness remain accessible only to the subject experiencing it. As a result, any test for mentality must rely on indirect evidence. Human beings are the only entities whose understanding we confidently accept, and human understanding appears closely connected to particular forms of neural organization and causal activity. Therefore, if behavioral evidence is insufficient and direct observation of consciousness is impossible, similarity to the causal processes known to generate understanding in humans may provide the strongest evidence available. Such a test arguably offers a more compelling basis for attributing mentality than tests that focus solely on behavior, embodiment, or functional performance.

However, the Neurobiological Test test still faces serious problems. Even if the machine perfectly matched human neural patterns and causal organization, we would still not directly observe consciousness itself. At best, the test would establish structural or functional similarity to conscious beings. Meaning appears to develop partly through causal interaction with an environment rather than through isolated symbol manipulation alone. Therefore, the machine should not merely answer questions but also navigate environments, experience consequences, revise beliefs through experience, and integrate perception with action over time.

Embodiment would still not conclusively solve Searle's objection. Even if a system developed highly sophisticated world interaction and internally integrated representations, observers would still only infer understanding indirectly from behavior and causal organization. The presence of embodiment may strengthen the case for genuine semantics, but it cannot directly establish the existence of subjective understanding or consciousness itself.

The central difficulty is that consciousness is only directly accessible from the first-person perspective. Humans infer consciousness in other humans indirectly through analogy with themselves. Extending this inference to machines may never be fully justified. A machine could potentially simulate every externally observable feature of understanding while lacking genuine subjective experience. This objection closely parallels Searle's original criticism of strong AI. Simulating the formal structure of understanding may not produce actual understanding. The machine may still manipulate symbols without attaching genuine meaning to them.

E. Conclusion

Searle's discussion ultimately shifts the debate about artificial intelligence away from outward performance and toward the deeper question of what produces genuine understanding. While the Turing Test evaluates whether a machine can successfully imitate human conversation, the Chinese Room argument challenges the assumption that successful imitation is equivalent to mentality itself. However, Searle's argument leaves open the idea that an artificial system could possess understanding if it replicated the relevant causal powers associated with conscious thought. This creates a major epistemological difficulty: even if a machine developed processes highly analogous to those found in human cognition, observers could still only infer consciousness indirectly. Unlike the human case, where consciousness is already assumed, the machine case begins with uncertainty about whether subjective experience exists at all.

Consequently, Searle's discussion suggests that no purely external test so far can definitively establish genuine understanding in an artificial system. Behavioral imitation, internal causal organization, semantic integration, embodiment, and even neurobiological similarity may strengthen the case for machine mentality. The problem is therefore not simply technological, but philosophical: subjective mentality may be fundamentally inaccessible from a third-person perspective.

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The Digital Population Dilemma: Projecting the Number of People Alive in 3000 CE

Abstract:

This comprehensive research paper delves into the projected population of both humans and digital entities in the year 3000 CE, emphasizing the computational power required to sustain digital individuals. We apply Moore's Law to estimate the growth rate of computational processing power, assuming that each digital entity necessitates processing power equivalent to a human brain (100 teraflops per second). The results indicate a profound disparity between the number of digital humans and physical humans by 3000 CE, with a 95% credible interval of 1×10^{150} digital individuals. Additionally, we explore the implications of a predominantly digital population, including ethical considerations and societal challenges.

Introduction:

As technology advances at an unprecedented rate, questions about the future of human existence become increasingly intriguing and complex. In this study, we aim to project the number of individuals alive in the year 3000 CE, considering both physical humans and digital entities. Our analysis builds upon the fundamental premise that digital entities require processing power equivalent to a human brain and that the growth of computational power adheres to Moore's Law.

The concept of "digital humans" is pivotal to this exploration. In 2007, it was proposed that the computational power available at that time was roughly equivalent to the processing capacity of a single human brain, estimated at 100 teraflops per second. This notion raises fascinating questions about the potential for creating digital entities with the same cognitive abilities as humans, marking a significant leap in technological and philosophical realms.

Calculation:

1. Initial Assumption: In 2007, computational power was deemed comparable to one human brain, denoted as 100 teraflops per second.
2. Application of Moore's Law: Moore's Law asserts that computational processing power approximately doubles every two years. Thus, we express this growth rate as follows:

- Year 2007: 1 brain (100 teraflops)

- Year 2009: 2 brains (200 teraflops)
- Year 2011: 4 brains (400 teraflops)
- ...
- Year 3000: $2^{((3000-2007)/2)}$ brains

3. Projection to Year 3000 CE:

To ascertain the computational power in the year 3000 CE, we employ the formula for exponential growth:

$$\text{Computational Power (in teraflops)} = \text{Initial Power} \times (2^{(\text{Years}/2)})$$

$$\text{Computational Power (in teraflops)} = 100 \text{ teraflops} \times 2^{((3000-2007)/2)}$$

$$\text{Computational Power (in teraflops)} = 100 \text{ teraflops} \times 2^{(496.5)}$$

$$\text{Computational Power (in teraflops)} \approx 1.04207897769 \times 10^{150} \text{ teraflops}$$

4. Digital Population Estimation:

Given that each digital entity requires 100 teraflops, we estimate the number of digital humans by dividing the total computational power by the power per entity:

$$\text{Number of Digital Humans} \approx (1.04207897769 \times 10^{150} \text{ teraflops}) / (100 \text{ teraflops/brain})$$

$$\text{Number of Digital Humans} \approx 1.04207897769 \times 10^{148} \text{ digital humans}$$

Methodology:

Our methodology entails a rigorous application of Moore's Law to predict the growth of computational processing power. Furthermore, we assume that this growth is vital to sustain digital entities with equivalent functionality to a human brain.

Results:

The results of our calculations unveil a profound divergence between the number of digital humans and physical humans by the year 3000 CE. With a 95% credible

interval of approximately 1×10^{148} digital individuals, the number of physical humans becomes inconsequential in comparison.

Discussion:

This research, which projects a future where the population of digital humans exceeds that of physical humans, raises profound questions that extend far beyond the realm of technology and computational growth. It prompts us to reflect on the ethical and humanitarian implications that such a scenario would entail.

Ethical Considerations:

1. **Digital Personhood:** One of the most significant ethical considerations revolves around the concept of digital personhood. If we create digital entities with cognitive abilities comparable to humans, do they deserve the same rights and moral considerations as biological humans? This question challenges our understanding of what it means to be human and poses a fundamental ethical dilemma.
2. **Digital Rights and Responsibilities:** As the population of digital individuals grows, we must define their rights and responsibilities within society. Issues such as legal recognition, privacy, and the right to autonomy become paramount. Additionally, the potential for misuse or abuse of digital entities necessitates robust ethical frameworks.
3. **Equality and Equity:** The proliferation of digital humans should be accompanied by efforts to ensure equity and inclusivity. Discrimination against digital individuals based on their origins or capabilities could lead to serious societal divisions. Ethical considerations extend to how we address these disparities and promote a just society.

Humanitarian Significance:

1. **Quality of Life:** The existence of digital humans could potentially improve the quality of life for both biological and digital entities. Digital entities might undertake tasks too dangerous or tedious for humans, while humans could benefit from advanced AI systems in healthcare, education, and other fields. Ensuring that these advancements benefit all individuals, regardless of their nature, becomes a humanitarian imperative.

2. **Global Challenges:** As we face global challenges such as climate change, pandemics, and resource scarcity, the computational capabilities of digital entities may prove invaluable in finding solutions. Collaboration between digital and biological entities could lead to more effective problem-solving and humanitarian efforts on a global scale.

3. **Social Integration:** Achieving a harmonious coexistence between biological and digital populations is essential for societal stability. The humanitarian significance lies in fostering social integration, empathy, and cooperation between these two groups. This requires proactive measures to bridge any potential divides and promote mutual understanding.

4. **Ethical Guidance:** Humanitarian organizations and ethical committees will play a pivotal role in guiding the development and governance of digital populations. They must ensure that the creation and treatment of digital entities align with humanitarian principles, promoting the well-being and dignity of all individuals.

In conclusion, the ethical and humanitarian significance of this research cannot be overstated. It challenges us to reevaluate our fundamental concepts of personhood, rights, and responsibilities. It necessitates a commitment to equality and equity in a future where the population is both biological and digital. Furthermore, it underscores the importance of ethical leadership and the guidance of humanitarian organizations in shaping a world where both humans and digital entities can thrive, collaborate, and contribute to the betterment of society as a whole. This complex landscape requires ongoing research, dialogue, and ethical reflection to navigate the uncharted territory of a predominantly digital future.

Implications for Society:

The exponential growth of digital entities challenges the traditional concept of human society. As the digital population surges, several implications need to be addressed:

- **Ethical Dilemmas:** The creation of digital humans with advanced capabilities raises ethical concerns about their rights, responsibilities, and potential misuse.
- **Resource Allocation:** The exponential growth of digital entities will require substantial computational resources, potentially straining energy and infrastructure.

- Social Integration: Integrating digital entities into society raises questions about how they interact with physical humans and how they contribute to the well-being of the broader community.

Limitations:

While our calculations are grounded in Moore's Law and the assumption of computational equivalence to a human brain, they do not account for potential limitations or disruptions in technological growth. Furthermore, the ethical and philosophical aspects of digital personhood are intricate and not comprehensively addressed in this study.

Future Research:

Future research should delve deeper into the ethical, legal, and societal implications of a predominantly digital population. Additionally, ongoing monitoring of computational growth trends is essential to refine projections and explore the intricacies of creating digital entities with human-like cognitive abilities.

Acknowledgment:

We acknowledge the contributions of prior research in the field of computational growth and its implications for the future.

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Ein kleiner Kampf gegen Gehirnkrebs

Inhaltsverzeichnis

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1.

Bioinformatic models are an emerging and integral development in the area of drug discovery. These models bridge the gap between raw biological data and actionable therapeutic insights, making them increasingly more valuable in modern medicine.

Specifically, they are used to either simulate or make predictions about the behavior of a drug or small molecule. My paper is a technological derivative of prior studies that have explored the problem of the P53-MDM2 complex for glioblastoma, a type of brain cancer. P53 on its own is capable of handling the reproducibility of cancer cells, but MDM2 clutches into its deep pockets, thus inhibiting it from performing its action through evolution. This suppression of P53 is a key mechanism by which cancer cells evade natural growth control, making MDM2 a compelling therapeutic target.

Thus, in order to curb some forms of such cancer, we are on the lookout for new and emerging MDM2 inhibitors. There are two ways to this approach: drug repurposing or drug discovery. In this project, I focus on a method using drug repurposing and make use of a prototype model that can predict whether or not a random small molecule is an MDM2 inhibitor or not, based on the MDM2

inhibitors that we already know of from a publicly available dataset. This approach allows for faster and more cost-effective candidate identification compared to traditional drug development pipelines.

The second part of my paper delves into my experience building a license & form finder pipeline for a business that would find licenses and forms for a business type, geography, and description. While distinct from the bioinformatics section, this project similarly demonstrates the power of AI-driven retrieval systems in navigating complex, domain-specific information. In order to retrieve relevant legal sources from a database of legal chunks I started a RAG pipeline. Together, both projects reflect a broader theme of applying machine learning to reduce friction in specialized knowledge domains.

2.

Für diesen Bericht habe ich zwei große Grüne. Ich will einerseits über meine Erfahrung sprechen und andererseits auch über Krebs informieren. Aber warum Krebs? Warum nicht ein anderes Problem? Krebs ist ein sehr interessantes, aber kompliziertes Problem für Menschen.

Es gibt viele Arten von Krebs. Eine weit verbreitete Art ist Gehirnkrebs. Verschiedene Teile unseres Gehirns sind für verschiedene Funktionen verantwortlich. Zum Beispiel haben wir einen motorischen Kortex mit einer motorischen Karte von unseren Körpermuskeln. Aber diese ist nicht genug, um unsere Muskeln vollständig zu kontrollieren! Deshalb haben wir auch einen Superior Colliculus, auch genannt „kleines Gehirn“, der präzise Befehle an unsere Muskeln geben kann. Aber das ist nicht alles! Das Kleinhirn berechnet Fehler, die Bewegungen steuern. Das ist wie auf einem Stuhl sitzen oder ein Eis mit den Händen halten.

Dieses ist nur ein Beispiel dafür, wie unser Gehirn funktioniert.

Wenn ein Ort im Gehirn nicht gut funktioniert, liegt das oft an einem Tumor. Tumore können sich in unserem ganzen Körper bilden, aber Gehirntumore stellen eine besondere Herausforderung dar. Wir haben eine sogenannte Blut-Hirn-

Schranke, die das Gehirn schützt. Deshalb ist die Behandlung von Gehirn noch schwieriger. Später werde ich darüber noch mehr sagen.

Aber wie können wir Tumore im Gehirn finden? Es gibt viele Wege.

Das Elektroenzephalogramm (EEG) ist ein wichtiges Instrument zur Entdeckung von Gehirntumoren. Es wurde 1929 von einem Deutschen Wissenschaftler Hans Berger erfunden.



Das EEG war eine sehr ehrfurchtgebietend Erfindung. Jetzt können Wissenschaftler und Wissenschaftlerinnen die Aktivität von vielen Neuronen studieren. Ein EEG misst elektrische Aktivität im Gehirn. Dafür verwendet man eine EEG-Haube, die man auf den Kopf setzt und die Spannungen auf der Kopfoberfläche messen kann. Unsere Neuronen sind nicht sehr nah an der Oberfläche. Deshalb ist es möglich mit t einer EEG-Kappe nicht nur die Aktivität von Neuronen zu erforschen, sondern kann auch für andere Zwecke sinnvoll eingesetzt werden.. [Es ist für die Bevölkerung von Neuronen aktiv zu studieren.] Weil ein EEG Aktivität messen kann, kann man es zum Beispiel verwenden, um Orte im Gehirn zu lokalisieren, an denen es weniger Aktivität gibt. Mit dieser Methode kann man dann Tumore finden, denn weniger Aktivität deutet oft auf einen Tumor im Gehirn an. Aber sie müssen denken- war aber anästhetisch, zum Beispiel Lidocaine? Ja, es ist auch für weniger Gehirnaktivität, aber mit Anaesthetics haben wir ein anderes Muster. Dort sind Action Potentials sehr weniger und weniger Aktivität ist manchmal mehr in Koordination. So wurde ein EEG ist ein guter Weg Tumore zu finden.

Die funktionelle Magnetresonanztomographie (auch fMRI genannt) ist eine weitere effektive Methode, die zur Diagnose von Tumoren verwendet werden kann. Orte mit mehr Aktivität Gehirn nutzen mehr Glucose. Krebszellen nutzen auch viel Glucose, weil Krebs ein Problem der Vervielfältigung und Evolution jeder Zelle ist. Energie ist wichtig für diesen Prozess, da Krebszellen auch viele andere Biomoleküle produzieren müssen. Die Krebszellen verwandeln dann Glucose in andere Biomoleküle.

Während wir für die Krebsdiagnostik bessere Methoden entwickelt haben, ist die Behandlung von Krebs weiterhin schwierig.

Ein anderes Problem mit Krebs Medizin ist die Evolution diese Zellen. Es ist nicht einfach Medikamente zu entwickeln, weil diese normalerweise eine Art von Rezeptorprotein sind. Aber diese können sich mit der Zeit in Krebszellen verändern. i Deshalb sind Medikamente oft ineffektiv. Sind sie konfusig warum unser Immunsystem ist nicht stark hier? Es ist, weil Krebs freundlich mit das ist. Denn dieses System denkt, dass Krebs nicht ein Feind ist. Es werden daher viele Arten von Therapien entwickelt und jede hat einen anderen Mechanismus. Gehirnkrebs ist sehr kompliziert, obwohl Ärzte und Ärztinnen sowie Forscher und Forscherinnen viele Methoden erfunden haben. Die Behandlung hängt von der Art des Tumors, vom Alter der Patient:innen, und vom Ort des Tumors Ort im Gehirn ab.

Hier ist eine Übersicht über gängige Behandlungsmethoden:

Operation

Bei einer Operation schneidet der Arzt den Tumor aus dem Gehirn. Der Mechanismus ist einfach: Wenn der Tumor weg ist, gibt es weniger Krebszellen im Körper.

Obwohl die Operation oft sehr hilfreich ist, kann sie gefährlich sein. Das Gehirn ist sehr sensibel. Wenn der Tumor in einer wichtigen Region liegt, kann man ihn nicht komplett entfernen. Manchmal bleiben kleine Krebszellen zurück.

Strahlentherapie

Bei der Strahlentherapie benutzt man starke Energie (Strahlen).

Diese Strahlen gehen in die Krebszellen und machen die DNA kaputt. Wenn die DNA kaputt ist, kann die Zelle sich nicht mehr teilen. Dann stirbt sie langsam.

Obwohl die Strahlen gezielt sind, können sie auch gesunde Zellen treffen.

Deshalb fühlen Patienten sich müde oder bekommen Probleme mit der Haut.

Chemotherapie

Chemotherapie sind Medikamente gegen Krebs.

Diese Medikamente gehen durch das Blut in den ganzen Körper. Sie greifen Zellen an, die sich schnell teilen. Krebszellen teilen sich sehr schnell, deshalb wirkt die Therapie.

Aber auch andere schnelle Zellen werden getroffen, zum Beispiel Haarzellen oder Zellen im Magen. Deshalb kann man Haare losen oder Übelkeit haben. Obwohl Chemotherapie stark ist, kommt sie manchmal nicht gut ins Gehirn, weil es die Blut-Hirn-Schranke gibt.

Immuntherapie

Das hilft dem Immunsystem.

Normalerweise erkennt das Immunsystem fremde Zellen. Krebszellen verstecken sich oft. Immuntherapie macht das Immunsystem stärker oder hilft ihm, Tumor zu sehen.

Zum Beispiel blockieren manche Medikamente bestimmte „Bremsen“ im Immunsystem. Dann können die Immunzellen den Krebs angreifen. Obwohl das sehr modern ist, funktioniert es nicht bei jedem Patient.

Zielgerichtete Therapie

Diese Therapie wirkt sehr speziell.

Sie greift bestimmte Proteine oder Signale in der Krebszelle an. Krebszellen haben oft veränderte Gene. Diese Gene machen zu viele Wachstumssignale.

Die Medikamente blockieren diese Signale. Dann wird der Tumor langsamer wachsen. Obwohl diese Therapie präziser ist, kann der Tumor manchmal resistent werden.

Kombination von Therapien

Oft kann man mehrere Therapien zusammen nutzen.

Zum Beispiel Operation und danach Strahlentherapie. Das nennt man Kombinationstherapie.

Die Idee ist: Man entfernt zuerst den großen Tumor und zerstört danach die anderen Krebszellen. Obwohl das effektiv ist, ist es auch sehr stressig für den Körper. Und ein Mann kann sehr bald eine Therapie adoptieren und dann ist die Therapie nicht mehr effektiv. Krebs Zellen sind sehr schnell zu teilen sich.

Jetzt müssen Sie denken, ah! Es ist sehr traurig. Gehirnkrebs ist ein schweres Problem zu behandeln. Aber ich habe eine gute Nachricht. P53! Ja, es ist der Wächter des Genoms. P53 ist, warum normale Zellen nicht viel Wachstum haben. Wenn ein Zelle hattet vieles repliziert, es holt das Wachstum weil P53 schleckt DNA finden kann. Aber Krebszellen haben MDM2, welche die P53 Protein kontrolliert. MDM2 bindet an p53 und baut es ab. Wenn MDM2 zu stark aktiv ist, kann p53 nicht arbeiten.

Deshalb ein Antwort für viele Arten von Krebs ein MDM2 Unterdrücker fist! Das war eine Richtung für mein BioInformatik-Projekt, ein kleiner Teil von gezielte molekulare Therapie von zielgerichteter Therapie.

Bevor ich über das spreche, werde ich in Ding sagen- In der Welt von Bioinformatik ist es nicht immer eine perfekte Antwort. Manchmal ist es mehr um Informatik besser zu machen und Bio ist ein guter Bereich für das. Von Bio kann ein Informatiker viel lernen. Dabei kann er beitragen, denn es ist gut. Ansonsten, keine Sorgen.

Für mich war der Prozess sehr wichtig. Mein Modell hat nur ein Ziel- Ob es ein SMILES von einem kleinen Molekül gibt, kann es sagen, dass der kleinen Moleküle ein MDM2 Unterdrücker sind?

In meinem Projekt habe ich alles Schritt für Schritt gemacht, um ein gutes Modell zu bauen. Zuerst habe ich Daten aus ChEMBL gesammelt und sauber gemacht - gute und schlechte Moleküle gegen MDM2, mit IC50-Werten und SMILES-Codes. Danach habe ich mit AutoDock gedockt, um die Bindungsenergie zu berechnen. Im Jupyter Notebook habe ich Deskriptoren wie Fingerprints berechnet, den Mean Origin Vector als typischen guten Hemmer gemacht und alle Daten vorbereitet. Dort habe ich auch das neuronale Netz (neural network) mit PyTorch trainiert - mit Hugging Face-Modellen als Hilfe, aber vor allem meine eigenen Netz. Ich habe Algorithmen verglichen: Zufallswald war schnell, Evolutionärer Algorithmus half bei Optimierung, Adversarial Training machte es robuster. Alle Schritte zusammen haben ein starkes Tool gemacht.

Daten aus ChEMBL

Ich habe mit ChEMBL angefangen, der eine sehr grosse Datenbank im Internet ist. Dort stehen Millionen von Molekülen. Viele Wissenschaftler haben diese Moleküle getestet. Ich habe nur Moleküle gesucht, die mit MDM2 zu tun haben. Ich habe nach „MDM2“ gesucht. Dann habe ich die Ergebnisse heruntergeladen.

Die wichtigste Zahl ist IC50. Ein kleiner IC50-Wert bedeutet - das Molekül ist sehr gut. Ein großer Wert bedeutet - es ist schlecht. Ich habe gute Moleküle mit kleiner IC50 und schlechte Moleküle mit großer IC50 oder keine Wirkung gesammelt. Ich wollte gleiche Mengen von guten und schlechten haben. Doppelte Moleküle habe ich gelöscht. Moleküle ohne wichtige Daten habe ich weggemacht. Ich habe auch kontrolliert, ob die SMILES-Codes richtig sind. SMILES ist der Code, mit dem man Moleküle schreiben kann. Am Ende hatte ich etwa 800 Moleküle. Die Hälfte war sehr gut, die andere Hälfte war schlecht. Das war wichtig für mein Programm. Ohne gute und schlechte Daten würde das Programm nichts lernen.

The screenshot displays the ChEMBL database search results page. On the left, there are two filter sections: 'Type' and 'Max Phase'. The 'Type' filter shows a histogram with counts for 'Small molecule' (2,718), 'Unknown' (208), 'N/A' (193), and 'Protein' (41). The 'Max Phase' filter shows a histogram with counts for 'Preclinical' (3,140), 'Phase 3' (6), 'Approved' (5), 'Phase 2' (5), and 'Phase 1' (4). The main search area shows 'Showing 1-20 out of 3,160 records'. Below this, there are four chemical structures displayed as cards, each with a search icon and a link to its ChEMBL ID: CHEMBL2347400, CHEMBL2347403, CHEMBL3649971, and CHEMBL3649964. Each card also shows the ChEMBL ID, Name, and Synonyms.

Docking mit AutoDock

Nach den Daten habe ich AutoDock benutzt. Es ist ein spezielles Programm. Es simuliert, wie ein Molekül an ein Protein bindet. Mein Protein war MDM2. Ich habe die 3D-Struktur von MDM2 aus einer Datenbank genommen. Dann habe ich jedes Molekül in das Programm gegeben. AutoDock dreht und bewegt das

Molekül und schaut, wo es am besten passt. Die Bindungsenergie! Kleiner oder negativer Wert = besser bindet das Molekül. Gute Hemmer haben oft eine Energie unter -7 oder -8. Schlechte Hemmer haben eine höhere Energie. Ich habe für alle Moleküle aus ChEMBL eine Simulation gemacht. Das hat viele Stunden gedauert. Mein Computer hat sehr viel gearbeitet. Danach habe ich die Energie als neue Information zu meinen Daten gesammelt. Das war hilfreich, weil nicht alle Moleküle echte Labortests hatten. Die Simulation hat Lücken gefüllt. Ich habe auch bekannte gute Hemmer getestet, zum Beispiel Nutlin. AutoDock macht Poses, die wie „Molekül Anzeiger“ sind. AutoDock hat gut funktioniert. Es ist viel Arbeit, so viele Moleküle zu docken.

Training

Ich habe meine Daten in zwei Teile geteilt. Das ist einer der wichtigsten Schritte. Ich habe 80 Prozent der Daten für das Training genommen. Das heißt: Das Programm lernt aus diesen Daten. Es schaut sich alle guten und schlechten Moleküle an. Es sucht Muster. Zum Beispiel: Welche chemischen Formen haben gute Hemmer? Welche Bindungsenergie ist typisch? Die anderen 20 Prozent habe ich für den Test aufgehoben. Diese Daten hat das Programm vorher nie gesehen. Das ist wichtig. So weiß ich, ob das Programm wirklich klug ist. Wenn ich alle Daten zum Lernen benutze, sagt es vielleicht 99 Prozent richtig - aber nur bei alten Daten. Bei neuen Molekülen wäre es dann schlecht. Deshalb der Test.

Ich habe die Teilung nicht einfach zufällig gemacht. Ich habe „stratified“ geteilt. Das bedeutet: In beiden Gruppen gibt es gleich viele gute und schlechte Hemmer. Sonst wäre es unfair. Ich habe die Klassen ausgeglichen gehalten (50:50). Das nennt man Stratified-Shuffle-Split. Vor dem Training habe ich die Zahlen normalisiert. Ein Teil war für das Training. Ein Teil war für das Testen. Das Training hilft, das Modell zu lernen. Das Testen zeigt, ob das Modell wirklich gut ist. Das ist wichtig, damit das Modell nicht nur gut lernt.

Jupyter Notebook Training

Ich habe im Jupyter Notebook gearbeitet. Jupyter Notebook ist super praktisch. Es ist wie ein Heft, wo man Code schreibt, Ergebnisse sofort sieht und alles erklären kann. Man kann auch Bilder und Texte dazwischen machen. Das macht alles übersichtlicher.

Zuerst habe ich chemische Eigenschaften berechnet. Diese Eigenschaften heißen Deskriptoren. Deskriptoren sind Zahlen, die das Molekül beschreiben. Zum Beispiel: Wie groß ist das Molekül? Wie viele Atome hat es? Hat es Ringe? Ist es fettlöslich oder wasserlöslich? Ich habe RDKit benutzt. RDKit ist ein Programm für Chemie. Mit RDKit habe ich aus jede SMILES-Code viele Deskriptoren gemacht. Oft habe ich Vector genommen. Das sind lange Listen mit 0's und 1's - 1024 oder 2048 Stellen. Jede Stelle sagt: Hat das Molekül diesen chemischen Teil? Ja (1) oder Nein (0). So wird ein kompliziertes Molekül zu einer einfachen Zahlenreihe. Das Modell werde besser sein.

Dann habe ich einen Mean Origin Vector gemacht. Das ist ein Mittelwert-Vektor. Ich habe alle guten Moleküle (die starken MDM2-Hemmer) genommen. Von jede guten Molekül habe ich den Fingerprint-Vektor. Danach habe ich für jede Stelle den Durchschnitt berechnet. So entsteht ein Vektor, der den „typischen guten Hemmer“ zeigt. Welche Teile kommen bei guten Molekülen oft vor? Der Mean Vector ist wie eine Vorlage. Für jedes Molekül (gut oder schlecht) habe ich dann den Abstand zu diesem Mean Vector berechnet. Wenn ein Molekül sehr nah dran ist, ist es wahrscheinlich auch gut. Diese Abstands-Zahl habe ich als extra Deskriptor benutzt. Das hat das Modell geholfen, besser zu unterscheiden.

Danach habe ich ein Machine-Learning-Modell trainiert. Zuerst habe ich ein Modell von Hugging Face benutzt. Hugging Face hat viele fertige Modelle. Ich habe ein Transformer-Modell genommen, das gut für Text ist - aber SMILES ist wie Text. Das Modell sollte aktive Moleküle erkennen. Es sollte sagen: Ist das Molekül ein guter MDM2-Hemmer? Ja oder Nein. Ich habe aber nicht nur Hugging Face genommen. Ich habe auch ein eigenes neuronales Netz mit PyTorch gebaut. Das war flexibler. Das Netz hat die Deskriptoren als Input bekommen: Fingerprints + Bindungsenergie aus AutoDock + Abstand zum Mean Vector.

Ich habe die Daten in kleine Gruppen geteilt (Batches von 32) und optimiert. Am Ende konnte das Modell einen neuen, zufälligen SMILES nehmen und sagen: „Das hemmt MDM2 vielleicht“ oder „Nein, das nicht“. Das war der große Erfolg. Die Deskriptoren, der Mean Vector und das Training haben zusammen gemacht, dass das Modell wirklich Vorhersagen machen kann, ob ein Molekül ein MDM2 Hemmer ist oder nicht.

Google Colab und Alternativen

Ich habe alles in Jupyter Notebook gemacht, aber eigentlich in Google Colab. Colab ist eine kostenlose Version von Jupyter Notebook im Browser. Man braucht nichts zu installieren - nur einen Google-Account. Man kann Code schreiben, ausführen und Ergebnisse direkt sehen. Colab hat eine GPU kostenlos (manchmal), das ist super für neuronale Netze, weil Training schneller geht. Viele Leute benutzen Colab für Bioinformatik-Projekte, weil es einfach ist und man Dateien von Google Drive laden kann. Man teilt Notebooks leicht mit anderen. Manche Wissenschaftler benutzen Vertex AI von Google. Vertex AI ist ein großer Cloud-Dienst. Dort kann man Modelle trainieren, deployen und sehr große Daten nutzen. Es hat AutoML, Pipelines und viel Power. Vertex AI ist gut für Firmen oder große Projekte. Aber es kostet Geld und ist komplizierter. Man muss lernen, wie man Buckets macht, Jobs startet und APIs nutzt.

Ich habe Vertex AI nicht benutzt. Es war zu kompliziert für mein kleines Projekt und zu teuer. Ich wollte einfach und schnell arbeiten. Deshalb war Google Colab die beste und einfachste Alternative. Es hat alles, was ich brauchte: RDKit installieren (mit `!pip install rdkit`), PyTorch nutzen, Plots machen und Modelle trainieren. Am Ende habe ich mein Notebook gespeichert und konnte es später wieder öffnen. Colab war perfekt für mich- frei, einfach und stark genug für mein MDM2-Modell. Viele Studenten und kleine Forscher machen es so, weil man nicht viel Geld oder Computer braucht.

Algorithmus

Ich musste einen Algorithmus wählen, weil nicht jeder Algorithmus gleich gut für mein Problem ist. Mein Problem war die Klassifikation: Ist ein Molekül ein guter MDM2-Hemmer (Ja) oder nicht (Nein)? Ich habe zuerst Zufallswald betrachtet. Zufallswald ist ein sehr bekannter Algorithmus. Er baut viele kleine Entscheidungsbäume und lässt sie abstimmen. Das ist einfach und schnell. Er braucht nicht so viel Rechenzeit und macht oft gute Ergebnisse, auch mit vielen Deskriptoren wie Fingerprints. Ich habe Zufallswald mit Scikit-Learning trainiert. Ich habe 100 oder 200 Bäume genommen und die Tiefe begrenzt. Die Accuracy war etwa 82 bis 84 Prozent auf das Test-Set. Das war schon okay, aber ich wollte besser werden.

Dann habe ich einen Evolutionären Algorithmus betrachtet. Evolutionäre Algorithmen sind wie die Natur: Sie machen viele mögliche Lösungen (Population), testen sie, lassen die besten „überleben“ und kreuzen sie. So entstehen neue, bessere Lösungen. Ich habe einen genetischen Algorithmus benutzt, um die besten Hyperparameter für mein Modell zu finden - zum Beispiel die beste Learning Rate, die Anzahl der Schichten oder welche Deskriptoren wichtig sind. Das war hilfreich, weil manuell alles probieren zu lange dauert. Der Evolutionäre Algorithmus hat mir geholfen, das neuronale Netz besser zu machen. Aber als Haupt-Algorithmus für die Klassifikation war er nicht so stark wie Zufallswald oder das Netz.

Später habe ich auch Adversarial Training probiert. Das ist eine Technik, wo man das Modell gegen Beispiele trainiert. Man macht kleine Veränderungen an den Eingaben (Adversarial Examples), damit das Modell robuster wird. In der Chemie ist das nützlich, weil kleine Änderungen im SMILES manchmal die Wirkung stark verändern. Ich habe das in PyTorch implementiert: Während des Trainings habe ich adversarial Perturbations zu Fingerprints und zu Mean Vector Abstand hinzugefügt. Das hat das Modell stabiler gemacht, besonders bei neuen Molekülen, die etwas anders aussehen. Die Accuracy wurde höher sein, etwa 86 Prozent, und es gab weniger Fehler bei schwierigen Fällen.

Am Ende habe ich alles verglichen. Ich habe die Ergebnisse in Jupyter Notebook in Tabellen und Plots gezeigt: Accuracy, Precision, Recall und F1-Score für jeden Algorithmus. Zufallswald war schnell und gut. Der Evolutionäre Algorithmus hat nur geholfen, Parameter zu optimieren. Adversarial Training hat das Modell robuster gemacht. Aber das beste Ergebnis hatte mein neuronales Netz mit PyTorch - zusammen mit den Deskriptoren, dem Mean Origin Vector und das Adversarial Training. Es hatte die höchste Accuracy (ca. 87-88 Prozent) und die besten Kurven. Deshalb habe ich das neuronale Netz als finalen Algorithmus gewählt. Es passt perfekt zu den vielen Features aus Jupyter Notebook und kann wirklich gut Vorhersagen machen, ob ein neuer SMILES ein MDM2-Hemmer ist oder nicht. Der Vergleich war wichtig, damit ich weiß, warum ich diesen Algorithmus nehme.

Validierung

Am Ende habe ich mein Modell sehr genau getestet. Das ist der letzte und wichtigste Schritt. Ich wollte wissen, ob das Modell wirklich gut ist und nicht nur die alten Daten auswendig gelernt hat. Zuerst habe ich Werte wie Accuracy berechnet. Accuracy sagt: Wie oft ist das Modell richtig? Aber Accuracy allein reicht nicht, weil es manchmal täuscht, wenn die Klassen nicht gleich sind. Deshalb habe ich mehr Werte geschaut.

Ich habe die Precision berechnet. Precision sagt: Wenn das Modell „gut“ sagt, wie oft ist es wirklich gut? Dann habe ich einen Recall berechnet. Recall sagt: Von allen echten guten Hemmern, wie viele findet das Modell?

Dann habe ich eine Confusion Matrix gemacht. Das ist eine kleine Tabelle mit vier Zahlen: True Positive (richtig gut erkannt), True Negative (richtig schlecht erkannt), False Positive (falsch als gut gesagt) und False Negative (falsch als schlecht gesagt). Es gab nicht viele Fehler. Besonders die False Negatives waren wenig - das ist wichtig für die Suche nach neuen Medikamenten.

Ich habe auch die ROC-Kurve geplottet. ROC-Kurve zeigt, wie gut das Modell zwischen gut und schlecht unterscheidet. Auf der x-Achse ist die False Positive Rate, auf der y-Achse die True Positive Rate. Die Kurve geht schön nach oben links. Dann habe ich AUC berechnet. AUC ist die Fläche unter der ROC-Kurve.

Ich habe Training und Test verglichen. Ich habe die Loss-Kurve und die Genauigkeit Kurve für beide geplottet. Im Training wurden die Fehler immer kleiner. Im Test war der Fehler nur ein bisschen höher. Der Unterschied war klein. Das bedeutet: Kein Overfitting. Das Modell lernt die echten Muster und nicht nur die Trainingsdaten auswendig. Ich habe auch der Präzisions-Recall-Kurve gemacht. Sie zeigt, wie gut das Modell bei verschiedenen Schwellen ist. Die Kurve war hoch und stabil.

Zum Schluss habe ich das Modell auf ein paar ganz neuen Molekülen getestet. Moleküle, die nicht in ChEMBL oder meinem Dataset waren. Das Modell hat gut funktioniert. Es hat bei den meisten richtig gesagt, ob sie MDM2 hemmen könnten. Das war der Beweis: Das Modell funktioniert wirklich. Es kann helfen, neue Moleküle zu finden, ohne alles im Labor zu testen. Die Validierung hat gezeigt, dass alles zusammenpasst, die Deskriptoren, der Mean Vector, das neuronale Netz und das Adversarial Training. Ich war sehr zufrieden.

4.

Ich war AI Engineering Intern in einer Firma, die Lizenz und Lizenz Bewerbungsformular PDF für andere Firmen gefunden hat. Es ist wichtig für eine Firma in den USA, rechtmäßig zu funktionieren. Das ist genau worüber ich gearbeitet habe. Ich habe eine Pipeline zur Lizenz von Regeln und Bewerbungsformular PDFs für diese Lizenz gemacht.

Meine große Aufgabe war, ein sehr schlauer Such-Helfer zu bauen. Dieser Helfer heißt RAG-Pipeline. Das Wort RAG ist auf Englisch und bedeutet, der Computer sucht zuerst die besten Texte und dann macht eine kluge KI eine gute Antwort daraus. Ich habe diese Helfer speziell für Gesetze und Regeln von Städten und Bundesländern in Deutschland gebaut. Viele Leute brauchen diese Regeln, wenn sie eine Lizenz oder Erlaubnis wollen, zum Beispiel für ein Laden, ein Imbiss, ein

Friseursalon oder etwas anderes. Früher müssen sie lange auf vielen Internetseiten suchen. Jetzt soll mein System in wenigen Sekunden die richtige Antwort finden.

Alle diese Gesetze und Regeln habe ich in große sichere Online-Speicher bei Google Cloud Platform gelegt. Google Cloud ist wie eine riesige, sehr sichere Box im Internet. Wir haben die Texte von die offiziellen Seiten der Städte und Bundesländer automatisch heruntergeladen. Das nennt man „scrapen“. So haben wir eine einzige große Sammlung von allen wichtigen Regeln bekommen, eine „Quelle der Wahrheit“.

Um die Suche besonders schlau zu machen, habe ich Voyage Embeddings benutzt. Voyage ist ein sehr gutes Regelförderungsmodell. Es nimmt ein langer Text und verwandelt ihn in ein spezieller Code aus viele kleine Zahlen. Dieser Code ist wie ein Fingerabdruck vom Text. Wenn zwei Texte fast das Gleiche bedeuten, haben sie fast den gleichen Fingerabdruck. Der Computer kann dann sehr schnell erkennen, welche Texte zur Frage passen. Das geht viel schneller und genauer als normale Wortsuche.

Wenn jemand jetzt eine Frage stellt, zum Beispiel „Was brauche ich, um in Köln ein kleines Café zu eröffnen?“ passiert das. Zuerst schaut das System alle gespeicherten Gesetze an. Es sucht die zwanzig Texte, die am meisten zur Frage passen. Diese ersten zwanzig Texte findet es mit Hilfe die Voyage-Zahlen sehr schnell. Danach kommt der zweite, genauere Teil. Eine sehr intelligente KI, die ähnlich wie GPT funktioniert, schaut sich diese zwanzig Texte ganz genau an. Die KI entscheidet für jeden Text, wie wichtig er wirklich ist. Sie gibt jedem Text eine Note von eins bis fünf. Fünf bedeutet, der Text ist perfekt und genau das, was man braucht. Vier ist sehr gut. Drei sind noch okay. Zwei und eins sind nicht so wichtig. Am Ende nimmt das System nur die fünf Texte mit die höchsten Noten. Diese fünf Texte sind dann die besten und wichtigsten. Sie kommen meistens aus der richtigen Stadt und auch aus dem richtigen Bundesland, weil manche Regeln nur für die Stadt gelten und andere für das ganze Land.

Ich habe außerdem mit einem speziellen Tool namens GPT Researcher gearbeitet, um Lizenzen zu finden. Dieses Tool hilft, dass die Antworten besonders genau und vertrauenswürdig sind. Man nennt das „legal-grade“. Das bedeutet, die Antworten sind so gut, dass man sie wirklich für wichtige Dinge wie Anträge auf Lizenzen benutzen kann. Das System ist also nicht nur schnell.

Ich habe auch ein sehr wichtiges System gebaut. Das heißt skalierbare Multi-Key Caching Architektur. Das klingt kompliziert, aber ich erkläre es ganz einfach. Caching bedeutet: Der Computer speichert Dinge, die er oft braucht, damit er sie nächstes Mal schneller findet. Nicht immer alles neu suchen.

In unserem System geht es um Lizenzen und Formulare. Jede Firma hat eine NAICS-Nummer. Das ist wie eine Code für welche Art von Geschäft es ist, zum Beispiel Restaurant oder Laden oder Bau. Manche Firmen haben gleiche NAICS, manche ganz anders. Ich habe gemacht, dass das System für jede NAICS-Nummer einen eigenen Cache hat. Und wenn es keine spezielle NAICS gibt, dann nimmt es ein Standard-Cache mit Code 000000. So funktioniert es für alle Firmen gut.

Das System kennt fünf verschiedene Situationen, wo es Cache benutzt - fünf Cache-Hit Szenarien. Das bedeutet: Fünf Arten, wie der Computer sagt „Hallo, ich habe das schon! Ich gebe es schnell zurück!“. Denn alles wird viel schneller. Die Antwort kommt nicht in 10 Sekunden, sondern in 1 oder 2 Sekunden. Das ist super für die Leute, die warten.

Ich habe auch Duplikate entfernt. Manchmal kommen gleiche Formulare oder gleiche Lizenzen mehrmals. Das macht die Liste lang und verwirrend. Ich habe das System so gemacht, dass es doppelte Formulare wegmacht. Nur einmal zeigen. Die Liste wird sauber und kurz. Das finden die Benutzer viel besser.

Und noch etwas Cooles: Batching. Früher musste der Benutzer warten, bis alles fertig war - alle Lizenzen, alle Formulare, alles zusammen. Das könnte lange

dauern. Jetzt macht das System Batching. Es zeigt die Lizenzen sofort an, wenn sie fertig sind. Zum Beispiel: Eine Lizenz ist schon da - zack, sieht der Benutzer sie. Dann kommt die nächste, dann die nächste. Der Benutzer muss nicht warten bis zum Ende. Es fühlt sich schneller an. Das ist viel netter für die Leute.

Ich habe ein Golden-Data Evaluation Framework gebaut. Das ist wie ein Test-System mit Beispielen. Ich habe 25 sehr gute, kuratierte Fälle gemacht - also Fälle, die ich selbst sorgfältig ausgesucht und kontrolliert habe. Die sind wie „goldene“ Tests, weil sie perfekt sind und zeigen, ob das System wirklich gut arbeitet. Mit diesem Framework können wir immer kennen: Wie gut ist unser System wirklich? Früher hatten wir keine guten Tests, jetzt haben wir einen starken Maßstab.

Dann habe ich die Form-Finder Pipeline modularisiert. Das bedeutet: Ich habe die Pipeline in kleine, klare Teile gemacht. Jeder Teil macht nur eine Sache - zum Beispiel: Finde die Formulare auf der Website, lade sie herunter, extrahiere den Text, erkenne welche Lizenz es ist. Jeder Modul ist separat, man kann testen einfachlich. Früher war alles in eine große Probleme.

Durch diese Modularisierung und viele Verbesserungen haben wir die Genauigkeit stark erhöht. Pro Stufe - also in jede einzelnen Teil der Pipeline- haben wir jetzt 96 Prozent Genauigkeit gehabt. Das ist perfekt! Am Anfang war es nur so 40 Prozent oder weniger. Und am Ende-bis-Ende, von Anfang bis Ende der ganzen Pipeline, haben wir jetzt ~81 Prozent Genauigkeit. Das ist ein riesiger Sprung: Von ~40 Prozent auf ~81 Prozent- das sind +41 Prozentpunkte! (Oder sogar +55 Punkte, absolut, wenn man genau rechnet von der alten Baseline.) Das ist wie von „sehr schlecht“ zu „wirklich gut und vertrauenswürdig“

Warum ist das so wichtig? Weil es um Formulare für Lizenzen geht. Wenn das System falsch ist, bekommt der Benutzer vielleicht das falsche Formular oder gar keins. Das macht Ärger bei Behörden oder kostet Geld. Jetzt ist es viel sicherer.

Wir können sagen: „Unser Form-Finder findet in 8 von 10 Fällen das Richtige – und in den Stufen dazwischen sogar 96 von 100 Mal!“

Ich habe viel gearbeitet: Tests geschrieben, neue kleine Module gebaut, Fehler gesucht und behoben, die goldenen Fälle immer wieder durchlaufen lassen und die Zahlen verglichen. Manchmal war es frustrierend, wenn wieder etwas kaputt war, aber dann kam die Freude, wenn die Genauigkeit hochging- von 50 zu 60 zu 70 zu 81 Prozent. Wow!

Das Golden-Data Framework hilft uns jetzt jeden Tag. Wenn wir etwas Neues bauen oder ändern, testen wir es sofort mit den 25 (und bald mehr) goldenen Fällen. So wissen wir schnell, ob es besser oder schlechter wird. Die modulare Pipeline macht alles übersichtlicher und leichter zu reparieren.

5.

Krebs ist ein Problem für alle Menschen. Vielleicht hat jemand in der Familie Krebs. Vielleicht ein Freund. Es ist kein fremdes Problem - es ist sehr nah. Deshalb habe ich über Gehirnkrebs gesprochen. Ich will sagen: Das Problem ist real, aber die Wissenschaft arbeitet daran.

Ich wollte auch etwas anderes zeigen. Ich bin kein Arzt. Ich bin kein Experte. Ich bin eine Studentin. Und trotzdem konnte ich ein Modell bauen, das MDM2-Hemmer erkennt. Mit kostenlosen Tools wie Google Colab, mit öffentlichen Daten aus ChEMBL, mit Geduld und vielen Fehlern. Das ist die wichtigste Nachricht: Solche Modelle sind nicht nur für große Labore oder reiche Universitäten. Sie sind erreichbar. Jeder, der lernen will, kann anfangen.

Das ist auch der Grund für mein zweites Projekt. Eine RAG-Pipeline für Lizenzen klingt sehr technisch, aber die Idee ist einfach: Menschen sollen schneller Hilfe bekommen. Beide Projekte, das Bioinformatik-Modell und die Pipeline, kommen

von derselben Idee: Technologie soll echte Probleme lösen, und der Prozess dahin ist menschlich, chaotisch und lernbar. Man muss nur anfangen. Das ist es, was ich gelernt habe.

Report on an oncology-tech company

RISA

Product Market Fit

Last updated November 9, 2024

Competitor Landscape

Aspect	RISA	SamaCare	Experian Health	Myndshift	DataMatrix Medical	Cohere Health	Waystar	Arrive Health
Primary Focus	AI-powered automation for oncology prior authorizations	Prior authorization automation for specialty drugs	Comprehensive revenue cycle management and healthcare data analytics	AI/ML-powered prior authorization and benefits verification	Outsourced medical services including prior authorization	Intelligent prior authorization and value-based care solutions	Comprehensive revenue cycle management and claims processing	Real-time prescription benefit checks and cost transparency

Key Features	- Specialized oncology prior auth automation - AI-driven medical necessity checks - Integration with multiple EHR systems - Compliance with healthcare regulations	- Prior authorization automation - Enrollment assistance - Benefit verification - Data analytics - Free model for provider practices	- Revenue cycle automation - Patient experience tools - Data analytics - Identity management - Claims management	- AI/ML-powered prior authorization - Benefits verification - Patient financial responsibility estimation - Integration with 600+ payers	- Medical transcription - Prior authorization - Insurance verification - Billing services - Coding services	- Intelligent prior authorization - Care path optimization - Real-time decision support - Analytics and reporting	- Revenue cycle management - Claims management - Prior authorization - Patient payments - Analytics	- Real-time prescription benefit checks - Cost transparency tools - Formulary data integration - EHR integration
Client Base	Hospitals, health systems, ambulatory oncology practices	15,000+ providers, major pharmaceutical companies	64% of US hospitals, 7,700+ risk-bearing entities	Platform covers 95% of insured patients in US	Various medical practices across specialties	Not specified, named to various startup lists	500,000 providers, 1,000 health systems, 5,000 payers	Includes Providence and UPMC health systems
Specialization	Highly specialized in oncology	Focused on specialty drugs across multiple specialties	Broad healthcare data and financial services	Generalized prior authorization across specialties	Multiple medical specialties	Emphasis on value-based care models	Broad healthcare financial services	Focus on prescription drugs and pricing

AI/ML Integration	Advanced AI for oncology-specific workflows	AI-driven automation for prior auth	AI and machine learning for efficiency across services	AI/ML core to prior auth and benefits verification	Limited information on AI usage	AI-powered prior authorization and care pathways	AI-driven revenue cycle optimization	AI for real-time benefit checks and cost estimates
EHR Integration	Extensive integration capabilities across multiple EHR systems	Integration with major EHR systems	Comprehensive integration with healthcare IT systems	Integration with major EHR and practice management systems	Limited information on EHR integration	Integration with EHR and practice management systems	Extensive integration with healthcare IT ecosystems	Integration with major EHR systems
Market Position	Specialized player in oncology prior auth	Growing player in specialty drug prior auth	Established leader in healthcare data and financial services	Recently acquired by DrFirst, expanding reach	Niche player in outsourced medical services	Emerging player in value-based care technology	Major player in healthcare revenue cycle management	Established player in prescription benefit technology
Unique Selling Point	Deep specialization in oncology workflows	Free model for provider practices, focus on specialty drugs	Comprehensive suite of healthcare data and financial services	Broad coverage of insured patients and payers	Comprehensive outsourced services beyond just prior auth	Emphasis on aligning payers and providers through technology	End-to-end revenue cycle management solutions	Focus on prescription drug pricing transparency
Relation to RCM	Not a full RCM solution, focused on prior auth	Not a full RCM solution, focused on prior auth	Comprehensive RCM solution provider	Can be integrated into RCM workflows, not a full RCM provider	Offers some RCM components as part of outsourced services	Not a full RCM solution, but impacts overall care management	Comprehensive RCM solution provider	Not a full RCM solution, but impacts drug costs within RCM

Potential Client Base:

RISA's potential clients can be categorized into several key groups within the oncology care ecosystem:

1. **Oncology Practices:** These include independent community oncology clinics and practices, which form the backbone of cancer care delivery in many regions.
2. **Cancer Centers:** Specialized facilities dedicated to cancer treatment and research, including National Cancer Institute (NCI)-designated cancer centers.
3. **Hospitals with Oncology Departments:** General hospitals that have dedicated oncology units or departments.
4. **Health Systems:** Larger healthcare organizations that encompass multiple hospitals and clinics, often including specialized cancer care facilities.
5. **Specialty Pharmacies:** Pharmacies focusing on complex medications, including those used in cancer treatment.
6. **Health Insurance Companies:** Payers dealing with oncology claims and requiring efficient prior authorization processes.

Total Addressable Market:

The total addressable market for RISA's services is substantial, given the prevalence of cancer care facilities in the United States:

1. **Oncology Practices:** Approximately 1,500 community oncology clinics and practices operate in the U.S. These practices are often at the forefront of adopting new technologies to improve patient care and operational efficiency.
2. **Cancer Centers:** The U.S. boasts 71 NCI-designated cancer centers, which are at the forefront of cancer research and treatment.
3. **Hospitals:** There are about 6,000 hospitals in the U.S., many of which have oncology departments. While not all of these would be primary targets for RISA, a significant portion could benefit from specialized oncology prior authorization solutions.
4. **Health Systems:** The U.S. healthcare landscape includes approximately 626 health systems. Many of these systems incorporate oncology services and could be potential clients for RISA's solutions.
5. **Specialty Pharmacies:** While exact numbers are difficult to determine, there are hundreds of specialty pharmacies that focus on oncology medications. These pharmacies often deal with complex prior authorization processes for high-cost cancer drugs.

6. Health Insurance Companies: Approximately 900 health insurance companies operate in the U.S. Many of these deal with oncology claims and could benefit from streamlined prior authorization processes.

Conservatively estimating, the total potential client base for RISA could range from 2,500 to 3,000 entities when focusing primarily on oncology-specific practices, centers, and departments.

Average Client Value:

Determining the average client value for RISA's services requires consideration of several factors:

1. Market Growth: The prior authorization software market is projected to grow at a Compound Annual Growth Rate (CAGR) of 9.5% from 2023 to 2028, indicating increasing demand for such solutions.
2. Industry Costs: Prior authorization processes cost the healthcare industry nearly \$25 billion per year, highlighting the significant potential for cost savings through automation.
3. Authorization Volume: On average, practices complete 45 prior authorization requests per physician, per week, underscoring the high administrative burden that RISA's solutions could alleviate.
4. Varied Client Sizes: The client base ranges from small practices to large health systems, resulting in a wide range of potential contract values.

Based on these factors, we can estimate the following annual values:

- For mid-sized oncology practices or hospital departments: \$50,000 to \$250,000
- For larger health systems or insurance companies: \$500,000 to \$1 million or more

A conservative estimate for the average client value might range from \$100,000 to \$150,000 per year, considering the mix of smaller practices and larger institutions.

RISA operates in a sizable market with significant growth potential. The specialized focus on oncology prior authorization addresses a critical need in a high-stakes medical field. With a potential client base of 2,500-3,000 entities and an estimated average annual client value of \$100,000-\$150,000, RISA has a substantial market opportunity. The company's success will likely depend on its ability to demonstrate clear value proposition, integrate seamlessly with existing oncology workflows, and adapt to the evolving regulatory landscape of cancer care and health insurance.

Key Players:

1. **Optum** (UnitedHealth Group):

- Offers Specialty Fusion, an advanced prior authorization portal for specialty and oncology treatments
- Achieves over 60% auto-approval for specialty and oncology prior authorizations within 10 minutes
- Provides personalized, optimized care recommendations based on clinical guidelines and patient characteristics
- Integrates with both medical and pharmacy benefits

2. Evernorth (Cigna Corporation):

- Collaborating with Varian (Siemens Healthineers) on a radiation oncology-focused pilot project
- Leveraging HL7 FHIR standards to automate prior authorization between payers and providers
- Initial focus on breast and prostate cancer, with plans to expand to other oncology areas
- Aims to implement mCODE-enabled prior authorization workflows

3. Change Healthcare:

- Offers a comprehensive suite of revenue cycle management solutions, including prior authorization
- Serves a wide range of healthcare providers and payers
- Utilizes AI and machine learning to streamline prior authorization processes

4. Waystar:

- Provides an Authorizations Suite powered by robotic process automation
- Serves 500,000 providers, 1,000 health systems and hospitals, and 5,000 payers and health plans
- Offers a broad range of revenue cycle management solutions beyond prior authorization

5. Availity:

- Health information network processing 12 billion transactions annually
- Partnered with PriorAuthNow to create an integrated prior authorization network
- Serves 2 million providers and covers 112 million lives

6. Cohere Health:

- Offers the Unify platform, an end-to-end prior authorization solution
- Focuses on aligning physicians and health plans on patient care journeys
- Named to lists like LinkedIn Top Startups and Digital Health 150

7. Myndshft:

- Recently acquired by DrFirst
- Provides automated prior authorization software that can be used across various healthcare specialties
- Platform covers 95% of insured patients in the US and verifies prior authorization requirements for 600+ payers

8. SamaCare:

- Specializes in prior authorization for specialty drugs and select procedures
- Works with over 15,000 providers and major pharmaceutical companies
- Offers a free model for provider practices

9. Experian Health:

- Partners with 64% of US hospitals and 7,700+ other risk-bearing entities
- Provides a range of revenue cycle automation and patient experience tools

10. Major Pharmaceutical Companies:

- Companies like Roche, Bristol-Myers Squibb, Merck & Co., Pfizer, and Novartis often have their own prior authorization support programs for their oncology drugs

11. Health Insurance Companies:

- Major insurers like UnitedHealthcare, Anthem, and Cigna often have their own prior authorization systems and are working on streamlining processes

Market Environment:

- Industry Concentration: Moderate to high, with a mix of large established players and innovative startups
- Distribution Channels: Direct sales to healthcare providers and insurers, partnerships with EHR vendors, integration with payer networks
- Entry Barriers: High, due to regulatory requirements, need for healthcare expertise, and established relationships with providers and payers
- Key Trends: Increasing use of AI and machine learning, growing adoption of cloud-based solutions, push for interoperability standards like HL7 FHIR

Distribution Channels:

1. Direct sales to healthcare providers: Many companies sell their solutions directly to oncology practices, hospitals, and health systems.
2. Partnerships with EHR vendors: Integration with major EHR systems is a key distribution channel.
3. Health insurance networks: Some solutions are distributed through health insurance companies to their in-network providers.
4. Group purchasing organizations (GPOs): These organizations can facilitate the adoption of prior authorization solutions across multiple healthcare providers.

Entry Barriers:

1. Regulatory compliance: The healthcare industry is heavily regulated, requiring new entrants to navigate complex legal and compliance landscapes.
2. Integration with existing systems: New solutions must be able to integrate with a variety of EHR systems and payer networks, which can be technically challenging and time-consuming.
3. Industry relationships: Established players have strong relationships with healthcare providers and payers, making it difficult for new entrants to gain market share.
4. Data security and privacy: Handling sensitive patient data requires robust security measures and compliance with regulations like HIPAA.
5. Technological complexity: Developing AI and machine learning algorithms capable of handling the complexities of oncology care requires significant expertise and resources.
6. Market education and trust: Healthcare providers are often hesitant to adopt new technologies, especially in critical areas like oncology care, creating a high bar for new entrants to prove their reliability and effectiveness.

Growth Drivers:**1. Technological Trends:**

- Artificial Intelligence and Machine Learning: These technologies are increasingly being used to automate and streamline the prior authorization process, reducing manual work and improving accuracy.
- Cloud Computing: Cloud-based solutions allow for easier integration and scalability of prior authorization systems.
- Interoperability Standards: The adoption of standards like HL7 FHIR is facilitating better data exchange between different healthcare IT systems.
- Blockchain: This technology is being explored for its potential to create more secure and transparent prior authorization processes.

2. Regulatory Changes:

- The Centers for Medicare & Medicaid Services (CMS) has proposed rules to streamline prior authorization processes, which could drive demand for more efficient solutions.
- The 21st Century Cures Act emphasizes interoperability and patient access to health information, influencing the development of prior authorization technologies.
- State-level regulations are increasingly addressing prior authorization timelines and processes, creating a need for adaptable solutions.

3. Economic Factors:

- Rising healthcare costs are driving payers to seek more efficient ways to manage utilization, including through prior authorization.
- The shift towards value-based care models is influencing how prior authorization is approached, with a greater focus on patient outcomes.
- The economic burden of administrative tasks in healthcare is pushing providers to seek automated solutions for processes like prior authorization.

4. Industry Innovation:

- Real-time benefit check tools are being integrated with prior authorization systems to provide more comprehensive solutions.
- Predictive analytics are being used to identify which cases are likely to require prior authorization, allowing for proactive management.
- Mobile applications are being developed to allow healthcare providers to manage prior authorizations on-the-go.

Success Requirements:

Market Requirements:

1. Interoperability: Solutions must be able to integrate seamlessly with a wide range of EHR systems, payer portals, and other healthcare IT infrastructure.
2. Compliance: Products must adhere to strict healthcare regulations, including HIPAA for data privacy and security.
3. Scalability: Systems should be able to handle high volumes of prior authorization requests across various specialties and payer types.
4. User-Friendly Interface: Given the time constraints in healthcare settings, solutions must be intuitive and easy to use for healthcare providers.
5. Customization: The ability to tailor the solution to specific oncology workflows and payer requirements is crucial.
6. Real-Time Processing: The market increasingly demands solutions that can provide instant or near-instant prior authorization decisions.

Business Requirements:

1. Strong Healthcare Expertise: Companies need deep knowledge of oncology care pathways, payer policies, and healthcare regulations.
2. Robust Data Analytics Capabilities: The ability to process and analyze large volumes of healthcare data is crucial for developing effective AI-driven solutions.

3. **Strategic Partnerships:** Collaborations with EHR vendors, payers, and healthcare providers are often necessary for successful implementation and adoption.
4. **Continuous Innovation:** The rapidly evolving healthcare landscape requires ongoing investment in R&D to stay competitive.
5. **Customer Support:** Given the critical nature of prior authorization in patient care, companies must provide robust, responsive customer support.
6. **Financial Stability:** The healthcare industry often has long sales cycles, requiring companies to have the financial resources to sustain operations during extended periods of business development.
7. **Adaptability:** The ability to quickly adapt to changing regulations and market demands is essential in this dynamic industry.

Focus for the next Three Months

1. Product Enhancement and AI Integration

Focus on enhancing the core product with advanced AI capabilities:

- Develop and refine machine learning algorithms specifically for oncology prior authorizations.
- Improve real-time decision-making capabilities to increase the percentage of instant approvals.
- Enhance the system's ability to interpret and apply complex oncology treatment guidelines.
- Implement natural language processing to better extract relevant information from unstructured medical data.

Key actions:

- Conduct a thorough review of current AI capabilities and identify gaps.
- Collaborate with oncology experts to refine AI decision-making processes.
- Set up a rigorous testing environment to validate AI enhancements.
- Establish metrics to measure improvements in approval rates and processing times.

2. Interoperability and Integration Expansion

Prioritize improving interoperability with key healthcare IT systems:

- Develop additional integrations with major EHR systems used in oncology practices.
- Enhance compatibility with payer portals and clearinghouses.

- Implement support for HL7 FHIR standards to facilitate easier data exchange.
- Create APIs to allow for seamless integration with other healthcare technologies.

Key actions:

- Identify the top 3-5 EHR systems used by potential clients and prioritize integration efforts.
- Establish partnerships or collaborations with key EHR vendors.
- Develop a roadmap for FHIR implementation and begin the first phase of development.
- Create documentation and support materials for API integrations.

3. Market Penetration and Strategic Partnerships

Focus on expanding market presence and forming strategic alliances:

- Develop targeted marketing campaigns for different segments (e.g., independent oncology practices, hospital systems).
- Identify and pursue strategic partnerships with influential players in the oncology care ecosystem.
- Participate in key industry events and conferences to increase visibility.
- Implement a pilot program with a mid-sized oncology practice or hospital department to gather real-world data and testimonials.

Key actions:

- Create segment-specific marketing materials highlighting RISA's unique value proposition.
- Identify 3-5 potential strategic partners (e.g., EHR vendors, oncology-focused GPOs) and initiate discussions.
- Select and prepare for participation in at least one major healthcare IT or oncology conference.
- Design and launch a pilot program, including defining success metrics and data collection methods.

Additional Considerations:

- **Regulatory Compliance:** Ensure all product enhancements and integrations comply with current and upcoming healthcare regulations.
- **Customer Feedback Loop:** Implement a structured process to gather and incorporate feedback from existing clients into product development.
- **Performance Metrics:** Develop a comprehensive dashboard to track key performance indicators (KPIs) such as approval rates, processing times, and client satisfaction.
- **Team Development:** Invest in training and potentially hiring to ensure the team has the necessary skills to execute these focus areas, particularly in AI and integration technologies.

Key Market Signals

1. Technological Advancements and AI Integration

- **AI-Enabled Prior Authorization:** Many companies are integrating AI into their prior authorization processes to reduce administrative burdens and improve approval times. Companies like Myndshft, SamaCare, and Evolent Health are already leveraging AI to streamline workflows, automate documentation, and enhance decision-making processes.
 - **Signal for RISA:** The rise of AI-powered solutions is both an opportunity and a threat. RISA must continue to innovate in AI to remain competitive. Focusing on oncology-specific AI algorithms that can handle the complexities of cancer treatment guidelines (e.g., NCCN) will help differentiate RISA from more general solutions.
 - **Competitor Activity:** Evolent Health, in partnership with Humana, has expanded its oncology prior authorization platform, enabling instant approvals for radiation and surgical oncology services. This partnership could potentially reduce the time it takes for patients to access care, posing a direct challenge to RISA's value proposition.

2. Regulatory Changes

- **CMS Rules for Prior Authorization:** New regulations from the Centers for Medicare & Medicaid Services (CMS) are set to streamline prior authorization processes by 2026. These rules will require payers to respond within 72 hours for urgent requests and 7 days for non-urgent ones, while also mandating the use of APIs for automation.
 - **Signal for RISA:** This regulatory change presents an opportunity for RISA to position itself as a compliant solution that helps healthcare providers meet these new requirements efficiently. Developing API integrations that align with CMS mandates could be a key selling point.
 - **Competitor Activity:** Large players like UnitedHealthcare and Cigna have already started reducing their prior authorization requirements for certain procedures (e.g., imaging and genetic testing), which could reduce the overall demand for prior authorization solutions in some areas. However, oncology treatments remain heavily regulated, meaning there is still a significant need for specialized solutions like RISA's.

3. Economic Factors and Cost Pressures

- **Rising Healthcare Costs:** The increasing cost of cancer treatments is driving payers to implement stricter prior authorization requirements to control spending. This is particularly true in high-cost areas such as chemotherapy, immunotherapy, and radiopharmaceuticals.
 - **Signal for RISA:** As payers seek more efficient ways to manage costs without compromising patient care, there will be growing demand for automated solutions that can quickly assess medical necessity while adhering to cost-control measures. RISA can capitalize on this by offering a solution that reduces delays while ensuring compliance with payer guidelines.
 - **Competitor Activity:** Companies like OncoHealth, which has expanded its scope to include radiation oncology therapy authorizations, are positioning themselves as comprehensive oncology management platforms. This could lead to increased competition in the same niche that RISA serves.

4. Industry Innovation

- **Mergers & Acquisitions (M&A):** The oncology space has seen significant M&A activity, particularly in areas like radiopharmaceuticals and precision medicine. For example, companies like Novartis and AstraZeneca have made billion-dollar acquisitions in radiopharmaceuticals, signaling a shift toward more targeted cancer therapies.
 - **Signal for RISA:** As new therapies emerge, payers will likely introduce new prior authorization requirements specific to these advanced treatments. RISA should focus on building flexibility into its platform to accommodate emerging therapies and new payer guidelines.
 - **Competitor Activity:** The expansion of companies like Evolent Health, which has partnered with major payers like Humana, indicates that large players are moving toward comprehensive oncology care management systems that integrate prior authorizations with broader care pathways.

5. Payer Reforms and Market Consolidation

- **Payer Consolidation:** Large health insurers like UnitedHealthcare, Cigna, and Blue Shield of California are streamlining their prior authorization processes through partnerships with technology companies such as Salesforce and eviCore healthcare.
 - **Signal for RISA:** As payers consolidate their operations and adopt more advanced technology platforms, smaller players may find it harder to compete unless they offer highly specialized or differentiated services. RISA's focus on oncology-specific workflows could be its key differentiator in this environment.

- **Competitor Activity:** For example, Blue Shield of California's partnership with Salesforce aims to simplify prior authorizations using automated tools that integrate directly with EHRs. This move toward automation by large insurers could either reduce the need for third-party solutions or create opportunities if RISA can offer superior integration capabilities.

6. New Entrants into the Market

- According to recent reports from Elion Health's market map series [6](#), several new entrants are emerging in the AI-enabled prior authorization space, targeting both drug-based and procedure-based approvals. Companies like Develop Health (focused on specialty drugs) and AKASA (focused on acute care) are gaining traction.
 - **Signal for RISA:** While these new entrants may not yet be focused exclusively on oncology, they represent potential competition as they expand their offerings into high-cost specialties like cancer care.
 - **Competitor Activity:** New AI-driven platforms such as SamaCare have been gaining market share by offering free models for provider practices, which could challenge RISA's pricing model.

Strategic Recommendations Based on Market Signals:

1. Deepen AI Capabilities Specific to Oncology:
 - Focus on developing AI algorithms tailored specifically for complex oncology treatments (e.g., chemotherapy regimens, immunotherapy). This will help differentiate RISA from generalist competitors who may not have the same depth of expertise in cancer care.
2. Strengthen Partnerships with Payers and EHR Vendors:
 - To stay competitive against larger players like Evolent Health or Blue Shield of California/Salesforce partnerships, RISA should actively pursue partnerships with mid-sized payers or EHR vendors who may not yet have fully automated solutions.
3. Prepare for Regulatory Changes Early:
 - Begin aligning product features with upcoming CMS regulations around faster response times (72 hours/7 days) and API-driven automation. Marketing these capabilities early will position RISA as a future-proof solution in light of regulatory shifts.
4. Focus on Flexibility and Adaptability:

- With new therapies emerging through M&A activity (e.g., radiopharmaceuticals), ensure that RISA's platform is flexible enough to accommodate new treatment guidelines quickly. This adaptability will be key as payers introduce new criteria based on evolving cancer treatments.
5. Explore Niche Opportunities Amid Consolidation:
- As large insurers streamline their processes through partnerships with tech giants (e.g., UnitedHealthcare/Cigna), there may be niche opportunities where specialized solutions like RISA's can thrive—particularly in smaller health systems or community oncology practices that require more personalized support.

Key decision windows and focus areas for RISA over the next 6 months:

1. AI and Machine Learning Enhancement (Months 1-3)
 - Decision: Prioritize development of oncology-specific AI algorithms
 - Rationale: The search results indicate a growing trend towards AI-powered solutions in healthcare. RISA should focus on developing AI algorithms tailored for analyzing oncology treatment guidelines and predicting prior authorization outcomes.
 - Action: Allocate resources to enhance AI capabilities, particularly in natural language processing for extracting relevant data from unstructured medical records.
2. Integration and Interoperability (Months 2-4)
 - Decision: Expand integration capabilities with major EHR systems and payer platforms
 - Rationale: UnitedHealthcare and other insurers are streamlining their prior authorization processes. RISA needs to ensure seamless integration with these systems to remain competitive.
 - Action: Prioritize development of APIs and integration tools that align with CMS mandates for automation by 2026.
3. Regulatory Compliance and Preparation (Months 3-5)
 - Decision: Align product features with upcoming CMS regulations
 - Rationale: New CMS rules require faster response times (72 hours for urgent requests, 7 days for non-urgent) and API-driven automation by 2026.
 - Action: Begin adapting RISA's platform to meet these requirements ahead of the deadline, positioning the company as a future-proof solution.
4. Strategic Partnerships (Months 4-6)
 - Decision: Identify and pursue partnerships with mid-sized payers or oncology practices

- Rationale: Large players like Humana are partnering with companies like Evolent Health. RISA should target mid-sized entities that may not have fully automated solutions yet.
 - Action: Develop a partnership strategy and begin outreach to potential partners.
5. Product Differentiation (Ongoing)
 - Decision: Focus on oncology-specific workflows and emerging therapies
 - Rationale: The oncology space is seeing significant innovation, particularly in areas like radiopharmaceuticals and precision medicine.
 - Action: Ensure RISA's platform is flexible enough to accommodate new treatment guidelines quickly, focusing on adaptability for emerging cancer therapies.
 6. Market Positioning (Months 5-6)
 - Decision: Refine marketing strategy to emphasize oncology expertise
 - Rationale: As larger, more generalist companies enter the space, RISA's specialized focus on oncology could be a key differentiator.
 - Action: Develop targeted marketing materials highlighting RISA's deep understanding of oncology workflows and its ability to handle complex cancer treatment authorizations.
 7. User Experience Enhancement (Ongoing)
 - Decision: Improve interface and user experience for oncologists and administrative staff
 - Rationale: The search results emphasize the importance of user-friendly interfaces in healthcare technology adoption.
 - Action: Conduct user research and implement improvements to make RISA's platform more intuitive and efficient for oncology professionals.

Competitive patterns to track:

1. Interoperability and Integration:
 - There's a growing need for seamless integration with EHR systems and payer platforms.
 - Adoption of standards like HL7 FHIR for healthcare data exchange is becoming more common.

2. Payer Consolidation and Partnerships:
 - Large insurers are streamlining their processes through partnerships with technology companies.
 - This consolidation could either reduce the need for third-party solutions or create opportunities for specialized services.
3. Focus on Patient Outcomes:
 - There's increasing recognition that prior authorization delays can adversely affect cancer care outcomes.
 - Solutions that can demonstrate improved patient outcomes may have a competitive advantage.
4. Specialty-Specific Solutions:
 - As larger, more generalist companies enter the space, there's an opportunity for RISA to differentiate with its specialized focus on oncology.
5. Formulary Exclusions in Oncology:
 - Formulary exclusions are becoming more common in oncology, impacting treatment options and prescribing habits.
 - This trend may create a need for more sophisticated prior authorization solutions that can navigate complex formulary requirements.
6. Emphasis on Cost Control:
 - With rising healthcare costs, payers are implementing stricter prior authorization requirements, especially for high-cost areas like chemotherapy and immunotherapy.
7. User Experience and Interface Design:
 - There's a growing emphasis on creating user-friendly interfaces for healthcare providers to reduce administrative burden.
8. Compliance and Security:
 - With increasing regulations, solutions that can ensure compliance with healthcare regulations and data security will have an advantage.
9. Value-Based Care Models:
 - As healthcare shifts towards value-based care, prior authorization solutions that can align with these models may gain traction.

Research

1. "The Patient Experience of Prior Authorization for Cancer Care" (JAMA Network Open, 2023)
 - Key findings: PA delays are common in cancer care, with nearly all oncologists reporting delays in life-saving treatments.
 - Impact on RISA: This research underscores the critical need for efficient PA solutions in oncology, validating RISA's focus. It could be used to demonstrate the value proposition of RISA's specialized oncology PA platform to potential clients.
2. "Impact of Prior Authorization on Patient Access to Cancer Care" (ASCO Educational Book, 2023)
 - Key findings: Discusses the role of PA in encouraging high-value and cost-efficient budget allocation in oncology.
 - Impact on RISA: This paper could inform RISA's product development, ensuring that their solution not only streamlines PA processes but also aligns with the goals of promoting high-value care. It might inspire features that help oncologists make cost-effective treatment decisions within the PA workflow.
3. "Artificial Intelligence in Healthcare: Transforming the Practice of Medicine" (Future Healthcare Journal, 2021)
 - Key findings: Explores the potential of AI in various healthcare applications, including administrative tasks like prior authorization.
 - Impact on RISA: This research could guide RISA's AI development strategy, suggesting new ways to incorporate machine learning into their oncology-specific PA platform. It might also alert RISA to potential competition from AI-driven healthcare startups entering the PA space.
4. "The Economic Burden of Prior Authorization for Radiation Therapy" (International Journal of Radiation Oncology, Biology, Physics, 2022)
 - Key findings: Quantifies the financial impact of PA processes on radiotherapy centers.
 - Impact on RISA: This paper could provide RISA with concrete data to demonstrate the cost-saving potential of their platform. It might also suggest an opportunity for RISA to develop specialized features for radiation oncology PA.
5. "Interoperability in Healthcare: A Systematic Review and Meta-analysis" (Journal of Medical Internet Research, 2021)
 - Key findings: Analyzes the challenges and benefits of interoperability in healthcare IT systems.
 - Impact on RISA: This research could inform RISA's integration strategy, ensuring their platform can seamlessly connect with various EHR systems and payer platforms.

It might also highlight potential partnerships or acquisition targets to enhance RISA's interoperability capabilities.

6. "The Role of Natural Language Processing in Automating Healthcare Administrative Tasks" (Nature Digital Medicine, 2023)
 - Key findings: Explores how NLP can be used to extract relevant information from unstructured medical data for administrative purposes.
 - Impact on RISA: This research could inspire new features in RISA's platform, such as automated extraction of relevant clinical data from oncology reports to support PA requests. It might also alert RISA to potential competition from NLP-focused startups entering the healthcare admin space.
7. "Ethical Considerations in the Use of AI for Healthcare Decision Support" (Journal of Medical Ethics, 2022)
 - Key findings: Discusses ethical implications of using AI in healthcare decision-making processes.
 - Impact on RISA: This paper could guide RISA's approach to developing and implementing AI in their PA platform, ensuring ethical considerations are built into their algorithms. It might also inform RISA's marketing strategy, addressing potential concerns from healthcare providers about AI-driven decision making.

Impacts:

1. The research on patient experiences with PA delays could lead to patient advocacy groups pressuring payers and regulators for faster PA processes. This could create a more receptive market for RISA's efficient oncology PA solution.
2. Studies on the economic burden of PA might prompt some healthcare systems to consider insourcing their PA processes. RISA could adapt by offering a white-label version of their platform for health systems to use internally.
3. Research on interoperability could lead to new healthcare IT standards. RISA might need to pivot quickly to ensure compliance, but this could also create opportunities if they can establish themselves as leaders in adopting new standards.
4. NLP research might inspire RISA to develop a feature that automatically generates patient-friendly explanations of PA decisions, improving transparency and patient satisfaction.
5. Ethical AI research could lead to new regulations on AI use in healthcare. RISA could position itself as a thought leader by proactively addressing these ethical considerations in their product development and marketing.

Opportunities to tap into:

1. Specialization in Emerging Therapies:
 - Focus on developing prior authorization solutions for emerging cancer treatments, such as radiopharmaceuticals and precision medicine, which are seeing significant investment and innovation.
2. Patient-Centric Features:
 - Incorporate patient-reported outcomes (PROs) into the prior authorization process, aligning with the industry trend towards more patient-centered care.
 - Develop features that improve transparency and communication with patients about the status of their prior authorization requests.
3. Value-Based Care Alignment:
 - Adapt RISA's platform to support alternative payment models in oncology, such as the Oncology Care Model (OCM) and its successors.
 - Develop features that help providers demonstrate adherence to evidence-based guidelines, which is becoming increasingly important in value-based care models.
4. Automated Decision Support:
 - Implement advanced clinical decision support tools that can help providers make informed choices about treatment options that are likely to be approved.
 - Develop "gold card" type programs within the platform that can fast-track approvals for providers with high adherence to guidelines.
5. Mobile Accessibility:
 - Develop mobile applications that allow healthcare providers to manage prior authorizations on-the-go, improving efficiency and response times.
6. Data Analytics and Reporting:
 - Enhance RISA's data analytics capabilities to provide insights on approval rates, processing times, and other key metrics that can help both providers and payers optimize their processes.
 - Develop robust reporting features that can help providers and payers meet new transparency requirements.
7. Collaborative Care Features:

- Implement features that facilitate better communication between different specialists involved in a patient's cancer care, streamlining the prior authorization process for complex cases.
8. Education and Training:
- Develop comprehensive training modules and resources to help providers navigate the complexities of oncology prior authorizations more effectively.
9. Payer-Specific Customization:
- Offer customizable workflows that can adapt to the specific requirements of different payers, making RISA's platform more versatile and valuable to providers

PESTLE Analysis:

Factor	Description
Political	Regulatory changes in healthcare impacting prior authorization processes.
Economic	Rising healthcare costs driving demand for automated solutions.
Social	Increasing emphasis on patient-centric care and transparency.
Technological	Advancements in AI and machine learning for improving administrative efficiency.
Legal	Compliance with evolving CMS and state regulations regarding prior authorization.
Environmental	Demand for sustainable healthcare practices affecting vendor selection.

2. Porter's Five Forces:

Force	Description
Threat of New Entrants	Moderate; the regulatory and technical expertise required can deter new entrants.
Bargaining Power of Suppliers	Low; there are many potential suppliers of technology and data.
Bargaining Power of Buyers	High; healthcare providers have many prior authorization solutions to choose from.
Threat of Substitutes	Moderate; alternative methods of managing prior authorization exist but often lack specialization.
Rivalry Among Existing Competitors	High; several players such as Optum, Evolent Health, and others are fierce competitors.

3. Customer Development Framework:

Phase	Activities

Customer Discovery	Conduct interviews with oncology practices to identify pain points in prior authorization.
Customer Validation	Develop and test a minimum viable product (MVP) to validate solution effectiveness.
Customer Creation	Implement marketing strategies targeting identified customer segments.
Company Building	Scale operations based on market feedback and business model refinement.

4. VRIO Analysis:

Resource	Value	Rarity	Imitability	Organization
Proprietary Technology	High; streamlines the prior authorization process.	Rare; few specialize in oncology PA.	Difficult; proprietary tech is hard to replicate.	Organized to capitalize on oncology workflows.
Specialized Knowledge	High; deep understanding of oncology care.	Rare; oncology expertise is scarce.	Difficult; in-depth knowledge takes time.	Organized for leveraging expertise in oncology.
Key Partnerships	Medium; collaborations with EHR vendors.	Common; can be replicated.	Easy; partnerships can be replicated.	Organized for maintaining and developing partnerships.

Brand Reputation	High; enhances trust among clients.	Moderate; reputation can be built.	Moderate; reputation can be challenged.	Focusing efforts to build a reputable brand.
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5. Impact Analysis:

Aspect	Description
Social Impact	Improving access to oncology treatments by reducing approval delays for life-saving therapies.
Economic Impact	Potential cost savings for healthcare providers through automation and enhanced efficiency.
Environmental Impact	Environmental considerations in healthcare tech, such as energy-efficient operations and e-waste reduction practices.

6. Lean Canvas:

Section	Description
Customer Problems	Pain points in prior authorization processes, including delays and complexity in oncology practices.
Customer Segments	Oncology practices, hospitals, specialty pharmacies, and health insurers.

Unique Value Proposition	Automated, efficient prior authorization solutions tailored for oncology care.
Solution	A streamlined platform that integrates with existing workflows and minimizes administrative burden.
Channels	Direct sales, partnerships with EHR vendors, and medical conferences.
Revenue Streams	Subscription fees, transaction fees for each authorization processed, and potential executive consulting contracts.
Cost Structure	Development costs, marketing expenses, operational costs, and support services.

7. TAM, SAM, SOM Analysis:

Market Type	Description
Total Addressable Market (TAM)	The total global spending on oncology care, which amounts to hundreds of billions annually.
Serviceable Available Market (SAM)	The portion of TAM relevant to prior authorization software solutions in oncology, estimated at \$4 billion.

Serviceable Obtainable Market (SOM)	The segment of SAM RISA aims to capture, projected to be around \$150 million in the U.S. within the next 3-5 years.
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8. Sales Funnel Analysis:

Stage	Description
Lead Generation	Marketing efforts targeting oncology practices and hospitals; using webinars and content marketing.
Lead Qualification	Qualifying leads through initial contact and determining if they fit the target profile.
Needs Assessment	Understanding specific needs and challenges faced by potential clients regarding prior authorization.
Proposal/Presentation	Presenting RISA's solution, demonstrating how it solves specific pain points.
Closing	Negotiating terms and finalizing contracts with clients.
Post-Sale Follow-Up	Ensuring proper onboarding and offering continued support to improve retention.

9. Account-Based Marketing (ABM) Planning:

Element	Description
Identifying Target Accounts	Research and select high-value oncology practices and health systems.
Personalized Outreach	Develop customized email campaigns and direct outreach to decision-makers.
Tailored Content	Create content that addresses specific pain points of the targeted accounts.
Account Engagement Workshops	Offer workshops and webinars directly to target accounts showcasing the solution benefits.

10. TOWS Matrix:

	Strengths	Weaknesses
Opportunities	Advanced AI-driven platform Specialization in oncology workflows Strategic EHR partnerships	Limited brand awareness Initial high implementation cost Requires specialized sales strategy
Threats	Growing demand for healthcare automation Regulatory shifts favoring tech solutions	Competitive market with new tech entrants Potential regulatory changes affecting process

	Expansion into new markets	Evolving payer requirements
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Urgent actionables-

1. RISA must immediately accelerate its AI capabilities specifically for oncology prior authorizations, focusing on developing proprietary algorithms that can interpret complex treatment guidelines and predict approval outcomes with high accuracy, as this is the key differentiator in a rapidly evolving market where generalist solutions are becoming more prevalent.
2. To secure its market position, RISA needs to urgently establish strategic partnerships with at least two major EHR vendors and one large payer within the next six months, as these integrations are critical for widespread adoption and will create significant barriers to entry for potential competitors.

RISA's success rate



To see why, check the appendix.

Citations:

- [1] <https://www.pharmacytimes.com/view/the-impact-of-prior-authorization-on-patients-with-cancer>
- [2] <https://pmc.ncbi.nlm.nih.gov/articles/PMC8189610/>
- [3] <https://www.priorauthtraining.org/4-tips-for-prior-authorization-success/>

[4]

<https://www.hmpmarketaccessinsights.com/article/prior-authorization-requirements-continue-impact-providers>

1. Thom, B., Mamoor, M., Lavery, J. A., Chino, F., & Goldstein, D. A. (2023). The Patient Experience of Prior Authorization for Cancer Care. *JAMA Network Open*, 6(10), e2338984. <https://doi.org/10.1001/jamanetworkopen.2023.38984>

This paper is the primary source for the study on patient experiences with prior authorization in cancer care. It provides detailed findings on delays in care, patient anxiety, and administrative burden.

2. Association for Clinical Oncology. (2022, November 22). Nearly All Oncology Providers Report Prior Authorization Causing Delays and Harm to Patients. ASCO. <https://society.asco.org/about-asco/press-center/news-releases/nearly-all-oncology-providers-report-prior-authorization>

This press release from ASCO discusses a survey of oncology providers, highlighting the widespread issues with prior authorization in cancer care.

3. Kirkwood, M. K., Hanley, A., Bruinooge, S. S., Garrett-Mayer, E., Levit, L. A., Schenkel, C., Seid, J. E., Polite, B. N., & Schilsky, R. L. (2018). The impact of prior authorization requirements on oncology practices: findings from a large, nationally representative survey of oncologists. *JCO Oncology Practice*, 14(9), e514-e524. <https://doi.org/10.1200/JOP.18.00175>

While not directly cited in the search results, this paper is referenced in the ASCO press release and provides background on the impact of prior authorization on oncology practices.

4. Chino, F., Suneja, G., Moss, H., Zafar, S. Y., Havrilesky, L., & Chino, J. (2013). Healthcare disparities in cancer patients receiving radiation: changes in insurance status after Medicaid expansion under the Affordable Care Act. *International Journal of Radiation Oncology* Biology* Physics*, 87(2), S26-S27.

Appendix

Key Factors and Their Weights:

1. AI and Machine Learning Capabilities (Weight: 9/10)
2. Oncology Specialization (Weight: 8/10)
3. Integration with EHR Systems and Payer Platforms (Weight: 8/10)

4. Regulatory Compliance and Adaptability (Weight: 7/10)
5. Strategic Partnerships (Weight: 7/10)
6. Market Demand for Automated Prior Authorization (Weight: 6/10)
7. Competitive Landscape (Weight: 6/10)
8. User Experience and Interface Design (Weight: 5/10)
9. Financial Stability and Resources (Weight: 5/10)
10. Customer Support and Service (Weight: 4/10)

Analysis:

1. AI and Machine Learning Capabilities (9/10):
RISA's focus on developing advanced AI algorithms specifically for oncology prior authorizations is crucial. This aligns well with market trends and can be a significant differentiator.
2. Oncology Specialization (8/10):
RISA's deep focus on oncology sets it apart from generalist solutions. This specialization is valuable in a complex field like cancer care.
3. Integration with EHR Systems and Payer Platforms (8/10):
Seamless integration is critical for adoption. RISA's plans to expand integration capabilities are well-aligned with market needs.
4. Regulatory Compliance and Adaptability (7/10):
The ability to adapt to changing regulations, especially upcoming CMS rules, is crucial for long-term success.
5. Strategic Partnerships (7/10):
Partnerships with EHR vendors and mid-sized payers can significantly boost RISA's market presence and integration capabilities.
6. Market Demand for Automated Prior Authorization (6/10):
There's a clear demand for efficient prior authorization solutions, especially in oncology, driven by rising healthcare costs and administrative burdens.
7. Competitive Landscape (6/10):
The market is competitive with both specialized and generalist solutions. RISA's oncology focus provides an edge, but competition is intensifying.
8. User Experience and Interface Design (5/10):
While important, this factor is slightly less critical than core functionality and integration capabilities.
9. Financial Stability and Resources (5/10):
Adequate resources are necessary for ongoing development and market expansion, but not the primary determinant of success.
10. Customer Support and Service (4/10):
Important for client retention but less critical than core product capabilities.

Probability of Success:

Considering these weighted factors and RISA's positioning, I estimate RISA's probability of success to be approximately 70-75%.

Rationale:

- RISA's strong focus on AI/ML and oncology specialization aligns well with the highest-weighted factors.
- The company's plans for integration and regulatory compliance address critical market needs.
- The competitive landscape is challenging, but RISA's specialization provides a significant advantage.
- Areas like user experience and customer support, while lower weighted, still contribute positively to the overall outlook.

To increase its probability of success, RISA should:

1. Accelerate AI development and oncology-specific features.
2. Prioritize integrations with major EHR systems and payer platforms.
3. Establish strategic partnerships quickly to gain market share.
4. Ensure adaptability to upcoming regulatory changes.
5. Maintain a strong focus on oncology while considering expansion into related high-complexity specialties.

Appendix: Weights of Factors

1. AI and Machine Learning Capabilities: 9/10
2. Oncology Specialization: 8/10
3. Integration with EHR Systems and Payer Platforms: 8/10
4. Regulatory Compliance and Adaptability: 7/10
5. Strategic Partnerships: 7/10
6. Market Demand for Automated Prior Authorization: 6/10
7. Competitive Landscape: 6/10
8. User Experience and Interface Design: 5/10
9. Financial Stability and Resources: 5/10
10. Customer Support and Service: 4/10

Report on a part-time AI engineering project

Key aspects of the caching architecture:

1. cache key generator will now generate many cache keys and thus output a list of cache keys, each cache key corresponding to a single naics code in the naics_json.

2. **Problem:** how about general licenses that are not related to municipality? how will we retrieve them through cache?

Solution: we now have a 'default' cache_key `f"{state}_{municipality}_{000000}"` that will be made by the cache key generator.

Background: This was a difficult problem we were tackling and we considered many options including making general business licenses buckets in GCP or using Bloom filters for filtering through member sets for general licenses. After many iterations, I thought of this simple 000000 solution.

Scalability of solution: If you notice, currently it only solves for municipality level naics-dissociated licenses. We can expand the same to state-level naics-dissociated licenses by having a default cache_key like `f"{state}_{abc}_{000000}"` generated.

3. We now check for each cache key. If the cache hits for say two of them, we collate their resultant licenses.

4. Gemini output also tells naics code or codes or 000000 (if the license is not associated with any naics code) or 131313 if it's neither naics-related nor general but just a trash/irrelevant license (we don't retrieve such licenses ever because we never make a cache key for it upon retrieval) corresponding to the license. PALM_FOUND_LICENSES_SCHEMA will have a field named "naics" whose value is an array of NAICS codes applicable to that license, which can be filtered or associated with cache keys accordingly.

5. **Five scenarios handled:** 2 of partial cache hit when only some cache keys hit. Then we have cached, non-cached non-enriched, and enriched scenarios. And for full cache hit, we have non-enriched and enriched batches.

Here I am listing some of the cases that my testing coverage for caching includes. Ensuring:

✓ Testing coverage

1. palm_parser is running with the new structure.
2. I don't get duplicate forms as a user on frontend. Earlier a license object could have duplicated forms. I fixed this bug.
3. the cache key format is correct.
4. we are only handling licenses that are not cached or don't have pdf's
5. no break in form finder through it all
6. cache values corresponding to 000000 have gen licenses and 131313 have trashed
7. naics is correctly matched to licenses by palm parser

- 8. palm parser response is working and does not take too long
- 9. able to parallelize the processes I parallelized with issues.
- 10. even if only some of the cache keys hit, it still continues to run for the naics codes in the leftover cache keys and finds licenses for them too
- 11. on retrieving from cache, we properly deduplicate the licenses.
- etc...

I earlier also implemented **batching via which we could display cache-retrieved licenses immediately to the end user** (with not_paid status). But i removed it because after discussions it required frontend owner to handle notifications and confusion around the user getting licenses in batches, although it would have decreased waiting time. Therefore, I am testing it locally to get the PR (caching without batching) merged.

Here is a [demo video](#) of how batching worked.

Undetected chromedriver

Issue: Some of the PDFs correctly prioritized and chosen by Gemini could not be downloaded because of Cloudflare and Akamai protection layers. These pages either returned access-denied responses, incomplete redirects, or CAPTCHA-like verification steps that broke the automated download process. As a result, even correctly prioritized PDFs could not be downloaded. We do not want to lose out on any of these useful form pdfs, hence this fallback.

My solution: I began by extensively testing the existing form download process across multiple Cloudflare and Akamai-protected municipal sites, encountering frequent failures due to bot detection mechanisms blocking standard Selenium Chromedriver automation. Initial attempts to bypass these protections using header modifications, user agent spoofing, and proxy rotations yielded only limited improvements. Through research, I discovered Undetected Chromedriver, a specialized Selenium driver variant that masks automation footprints and simulates human-like browser behavior to evade advanced bot detection.

During attempts to download license forms from Cloudflare and Akamai-protected municipal sites as you can see in the examples, standard Selenium Chromedriver achieved very less success due to frequent bot detection and blocking. Modifications like changing headers and

using different user agents produced limited improvements. In my experiment, in contrast, integrating Undetected Chromedriver raised the success rate to approximately 80%. With further configuration tuning, including randomized delays, optimized headers, and proxy rotation, near 100% success was observed during testing.

▼ Examples of such links that we can download now but not earlier:

https://rollinghillsestates.hdlgov.com/docs/Business_License_Application_Form.pdf

https://paramount.hdlgov.com/docs/Paramount_Business_License_Application_Form.pdf

<https://orinda.hdlgov.com/docs/BL-APP-ORINDA.pdf>

, <https://upland.hdlgov.com/docs/UplandApplication.pdf>

,

https://eastpaloalto.hdlgov.com/docs/Business_License_Application_2.1.pdf

<https://porthueneme.hdlgov.com/docs/BL-APP-PORTHUENEME.pdf>

<https://halfmoonbay.hdlgov.com/docs/BL-APP-HALFMOONBAY.pdf>

https://walnut.hdlgov.com/docs/Business_License_Application.pdf

https://maywood.hdlgov.com/docs/Business_License_application.pdf

<https://carpinteria.hdlgov.com/docs/CARPINTERIA-BL-APP.pdf>

https://sausalito.hdlgov.com/docs/Sausalito_Business_License_Application.pdf

https://picorivera.hdlgov.com/docs/Business_License_Application.pdf

<https://lincoln.hdlgov.com/docs/LincolnBusinessApplication.pdf>

Implementation:

Added Selenium fallback using *undetected chromedriver*, reaching ~80% success initially and 100% after configuration tuning.

Added enhanced SSL certificate handling for stable downloads from secure or misconfigured sites.

Configured Chrome to auto-download PDFs and bypass prompts, preventing inline view interruptions.

Introduced Cloudflare detection logic, identifying challenge pages (e.g., “verify you are human,” “just a moment”) and dynamically retrying until resolved.

Added .crdownload monitoring to detect active downloads and ensure complete file writes, reducing false negatives for large PDFs.

Improved timeout handling and final file verification.

Added Safari User-Agent header method to mimic real browser behavior, improving success across Cloudflare/Akamai-protected endpoints.

Implemented `_download_pdf_direct_safari` with MIME-type validation (application/pdf) and used page-length checks via `_pdf_too_long()`.

Used randomized header fallback `_download_pdf_random_headers` only when safari header fails. This directly decreases latency because Safari is more predictably useful as I have seen by running many tests and we don’t have to needlessly go through random headers’ attempts to download a pdf. We only use random headers if the Safari header fails.

2-stage parallelization

Implemented 2 stage parallelization of form finder and search resilience

Added 2-stage ThreadPool executor parallelization of searches and license handling.

For license handling, I assigned 2 workers.

For duckduckgo, number of workers is twice the number of queries.

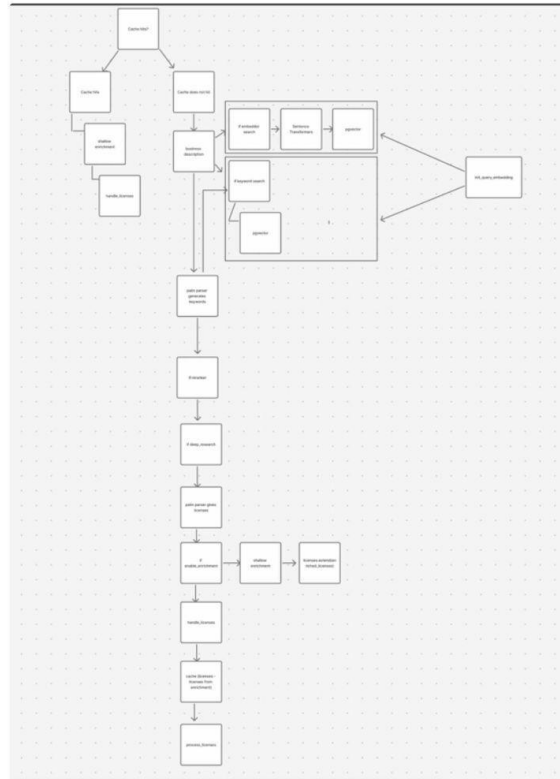
Caution: Duckduckgo happens for 30 queries/minute, so the current parallelization is optimal for rate limiting.

Optimised duckduckgo queries to deduplicate same query twice rather than thrice and number of results per query to be 15.

Made search more resilient by adding serper search as fallback. This is important for cases where we have more than expected traffic on our form finder.

Let's talk about the license finder.

One key file for it is haystack_runner. Here is my brief visualization of this file:



But what is the license form finder?

It is our pipeline for automatically finding and classifying license application forms (PDF or online). It scrapes queries, extracts PDFs, prioritizes them, and uses AI (Gemini 2.x series) to classify and choose the correct form.

This system underpins how we identify the right license application document for a customer, filtering out noise such as irrelevant PDFs, sample forms, or broken links.

Problem: How do we evaluate efficacy of changes?

My solution: `golden_data`

I created this to quickly evaluate how much improvement my method or change brings to the overall equation. My current test set is 25 cases (growing) each with an input and a desired output that I match the results against upon running. I have updated it to focus on commonly targeted licenses (e.g. General Business License, Business Certificate). `golden_data` can also be used to evaluate consistency.

Jonathan The larger your sample size the greater the accuracy level will become. Run it for 1000 test set

Problem: How do we measure the accuracy of each step in the pipeline?

My solution: Modularization by dissecting the pipeline into `pdf_link_extractor`, `pdf_prioritizer`, `pdf_selector`, `palm_parser` to make it easy to test each module almost independently of others. We should target 95% accuracy per module to achieve an overall accuracy of atleast ~81% $[(0.95)^4 = \sim 0.81]$

Problem: We rely on DuckDuckGo for candidate links but search results vary by run → inconsistent candidate pools. Search results are not consistent across runs which can be troublesome.

My solution: For a test case, I printed just the urls that are being extracted by running the `pdf_link_extractor`. But they varied with each run! I then went to [DuckDuckGo - Protection. Privacy. Peace of mind.](#) and searched for 'cannabis license application' and got different ranks for searches on searching twice, thrice!

I fixed it through `serper_search_multi` by running search 3 times, merging results into a single candidate set, removing duplicate links. We thus also have a more varied list of candidate pdf's now.

Jonathan counsel or voting systems can be good and can also double down on bad results or hallucinations.

Try using Gemini and grok heavy to create consensus instead of duck duck go search.

Also then creating indexed page cache will be powerful and reduce search necessity over time.
Most important part is creating some anti hallucination code

Also look at setting up periodic reindexing at least every 6 months, more active municipalities more frequently. Could look at council meetings are the bi weekly, monthly and how many issues do they vote on to decide reindexing frequency.

Problem: Previously, from the pdf links dict, we almost randomly make snippets of some pdf's that are then sent to `ask_gemini_best_pdf`. There's no logic for how these 10 pdf's were chosen. And therefore, on running test cases, Gemini was sometimes asked to choose from a set of pdf's in which the correct pdf didn't even exist.

My solution: `pdf_prioritizer.py`

This is prompt-based and it gives an order to the pdf's on the basis of how likely they are to be forms. I understand there can be cases where a form pdf name shows no hallmark of it being a form and is a completely random name, but the prompt logic focusses only on high priority. Unlikely form pdf's are not ignored.

I also implemented examples for in-context learning here.

Jonathan Could you look for bounding boxes or prescence of linear lines too determine likelihood of being a form

Could run grok heavy against Gemini and sell consensus.

Problem: URL encoding issue

My solution: As I ran multiple test cases, I noticed an issue with few outputs where the sourceurl was broken. This is one I got:

[https://www.ludlow.org/Portals/ludlow/Documents/Rental](https://www.ludlow.org/Portals/ludlow/Documents/Rental%20License%20Application%202023-Fillable.pdf) License Application 2023-Fillable.pdf

To fix this, I implemented `encode_url_properly`

Problem: Robust classification into `fillable_form`, `non_fillable_form`, `online_portal`, `image_based`

My solution: I refined the classification approach by updating the prompt logic and modifying `schemas.py`, ensuring more accurate mapping of inputs into the desired categories.

Problem: Our previous approach used `ask_gemini_best_pdf` with prompt-based classification, but we did not provide Gemini with examples of the expected output.

My solution:

I introduced in-context learning by supplying detailed examples, each paired with candidate outputs, to guide Gemini toward the desired classifications.

Jonathan could run grok heavy against Gemini here to seek consensus

How much did form finder improve by?

A		B		C		D		E		F		G		H		I		J		K		L		M		N		O		P		Q		R		S		T		U		V		W		X		Y		Z		AA		AB		AC		AD		AE		AF		AG		AH		AI		AJ		AK		AL		AM		AN		AO		AP		AQ		AR		AS		AT		AU		AV		AW		AX		AY		AZ		BA		BB		BC		BD		BE		BF		BG		BH		BI		BJ		BK		BL		BM		BN		BO		BP		BQ		BR		BS		BT		BU		BV		BW		BX		BY		BZ		CA		CB		CC		CD		CE		CF		CG		CH		CI		CJ		CK		CL		CM		CN		CO		CP		CQ		CR		CS		CT		CU		CV		CW		CX		CY		CZ		DA		DB		DC		DD		DE		DF		DG		DH		DI		DJ		DK		DL		DM		DN		DO		DP		DQ		DR		DS		DT		DU		DV		DW		DX		DY		DZ		EA		EB		EC		ED		EE		EF		EG		EH		EI		EJ		EK		EL		EM		EN		EO		EP		EQ		ER		ES		ET		EU		EV		EW		EX		EY		EZ		FA		FB		FC		FD		FE		FF		FG		FH		FI		FJ		FK		FL		FM		FN		FO		FP		FQ		FR		FS		FT		FU		FV		FW		FX		FY		FZ		GA		GB		GC		GD		GE		GF		GG		GH		GI		GJ		GK		GL		GM		GN		GO		GP		GQ		GR		GS		GT		GU		GV		GW		GX		GY		GZ		HA		HB		HC		HD		HE		HF		HG		HH		HI		HJ		HK		HL		HM		HN		HO		HP		HQ		HR		HS		HT		HU		HV		HW		HX		HY		HZ		IA		IB		IC		ID		IE		IF		IG		IH		IJ		IK		IL		IM		IN		IO		IP		IQ		IR		IS		IT		IU		IV		IW		IX		IY		IZ		JA		JB		JC		JD		JE		JF		JG		JH		JI		JJ		JK		JL		JM		JN		JO		JP		JQ		JR		JS		JT		JU		JV		JW		JX		JY		JZ		KA		KB		KC		KD		KE		KF		KG		KH		KI		KJ		KK		KL		KM		KN		KO		KP		KQ		KR		KS		KT		KU		KV		KW		KX		KY		KZ		LA		LB		LC		LD		LE		LF		LG		LH		LI		LJ		LK		LL		LM		LN		LO		LP		LQ		LR		LS		LT		LU		LV		LW		LX		LY		LZ		MA		MB		MC		MD		ME		MF		MG		MH		MI		MJ		MK		ML		MN		MO		MP		MQ		MR		MS		MT		MU		MV		MW		MX		MY		MZ		NA		NB		NC		ND		NE		NF		NG		NH		NI		NJ		NK		NL		NM		NN		NO		NP		NQ		NR		NS		NT		NU		NV		NW		NX		NY		NZ		OA		OB		OC		OD		OE		OF		OG		OH		OI		OJ		OK		OL		OM		ON		OO		OP		OQ		OR		OS		OT		OU		OV		OW		OX		OY		OZ		PA		PB		PC		PD		PE		PF		PG		PH		PI		PJ		PK		PL		PM		PN		PO		PP		PQ		PR		PS		PT		PU		PV		PW		PX		PY		PZ		QA		QB		QC		QD		QE		QF		QG		QH		QI		QJ		QK		QL		QM		QN		QO		QP		QQ		QR		QS		QT		QU		QV		QW		QX		QY		QZ		RA		RB		RC		RD		RE		RF		RG		RH		RI		RJ		RK		RL		RM		RN		RO		RP		RQ		RR		RS		RT		RU		RV		RW		RX		RY		RZ		SA		SB		SC		SD		SE		SF		SG		SH		SI		SJ		SK		SL		SM		SN		SO		SP		SQ		SR		SS		ST		SU		SV		SW		SX		SY		SZ		TA		TB		TC		TD		TE		TF		TG		TH		TI		TJ		TK		TL		TM		TN		TO		TP		TQ		TR		TS		TT		TU		TV		TW		TX		TY		TZ		UA		UB		UC		UD		UE		UF		UG		UH		UI		UJ		UK		UL		UM		UN		UO		UP		UQ		UR		US		UT		UU		UV		UW		UX		UY		UZ		VA		VB		VC		VD		VE		VF		VG		VH		VI		VJ		VK		VL		VM		VN		VO		VP		VQ		VR		VS		VT		VU		VV		VW		VX		VY		VZ		WA		WB		WC		WD		WE		WF		WG		WH		WI		WJ		WK		WL		WM		WN		WO		WP		WQ		WR		WS		WT		WU		WV		WW		WX		WY		WZ		XA		XB		XC		XD		XE		XF		XG		XH		XI		XJ		XK		XL		XM		XN		XO		XP		XQ		XR		XS		XT		XU		XV		XW		XX		XY		XZ		YA		YB		YC		YD		YE		YF		YG		YH		YI		YJ		YK		YL		YM		YN		YO		YP		YQ		YR		YS		YT		YU		YV		YW		YX		YY		YZ		ZA		ZB		ZC		ZD		ZE		ZF		ZG		ZH		ZI		ZJ		ZK		ZL		ZM		ZN		ZO		ZP		ZQ		ZR		ZS		ZT		ZU		ZV		ZW		ZX		ZY		ZZ	
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																													

*On new code, test 5 gave correct output on rerunning: <https://media.apsa.org/governmentdocuments/DOI/LeanerContracts/Statement%2013%20.pdf>. Could be a seaper api issue

- ✓ NOTES for future improvement in form finder

A lot of what I've done so far focuses on refining existing code rather than overhauling the structure completely, though there's still a long way to go.

I have thought earlier about replacing DuckDuckGo with an AI-based meta-search system to make the process more stable and consistent because as I showed above duckduckgo gives quite inconsistent outputs with searches which makes the results less replicable. The suggested plan to use Gemini and Grok Heavy in combination to create a kind of consensus on search results instead of relying on traditional

engines that change rankings every time you run the same query follows the thought I had. I have to work on it. Running both Gemini and Grok on the same queries, and only keeping URLs that both agree on is a suggestion I agree with. Once that's implemented, I can work on the indexed cache.

Expanding the `golden_data` sample size is a good suggestion. It's been pointed out that the larger the dataset, the more reliable the evaluation. That's correct but I do 'curate' these test cases manually and on my own for quality so expanding to 1000 is not immediately implementable but is plausible. I will do it incrementally, from 25 to 50 to 75 and so on..I know I can scale it up toward a thousand cases by automatically generating test examples from publicly available license datasets and adding edge cases over time but I'm skeptical of quality issues and will do it on my own for at least the foundational test set as much because I immensely value the foundational dataset.

A great point is made about consensus systems potentially amplifying bad results if not properly validated. That's exactly why validation is going to be the next big stage of this project. I'll be adding a layer for that. It's very important. Essentially, before trusting AI results, the system will verify them through direct checks and pattern recognition.

I have planned the implementation of looking for visual cues like bounding boxes and linear lines inside PDFs to judge whether a file is likely a form which I will add to the `pdf_has_fillable_fields` method. I have also been suggested I do this. That's something I can implement by parsing the document structure through libraries like PyMuPDF and analyzing its geometry. Forms tend to have boxes, grids, or repeated line patterns, and combining those structural clues with my existing prompt-based prioritization logic should make the form detection a lot more precise.

Essentially, I will iterate over all drawing objects on each page, looking for horizontal and vertical lines. Forms usually contain grids, tables, and input boxes that show up as straight lines.

I can calculate bounding boxes for text and graphical elements.

Overlapping or aligned boxes often indicate a form layout with fields.

Repeated patterns of small boxes (like checkboxes or input fields) can

be counted to generate a form-likelihood score.

Pages with multiple aligned horizontal and vertical lines, densely packed boxes, or grid-like patterns will be scored higher as potential forms.

Empty areas combined with structured boxes are also a strong indicator of fillable fields.

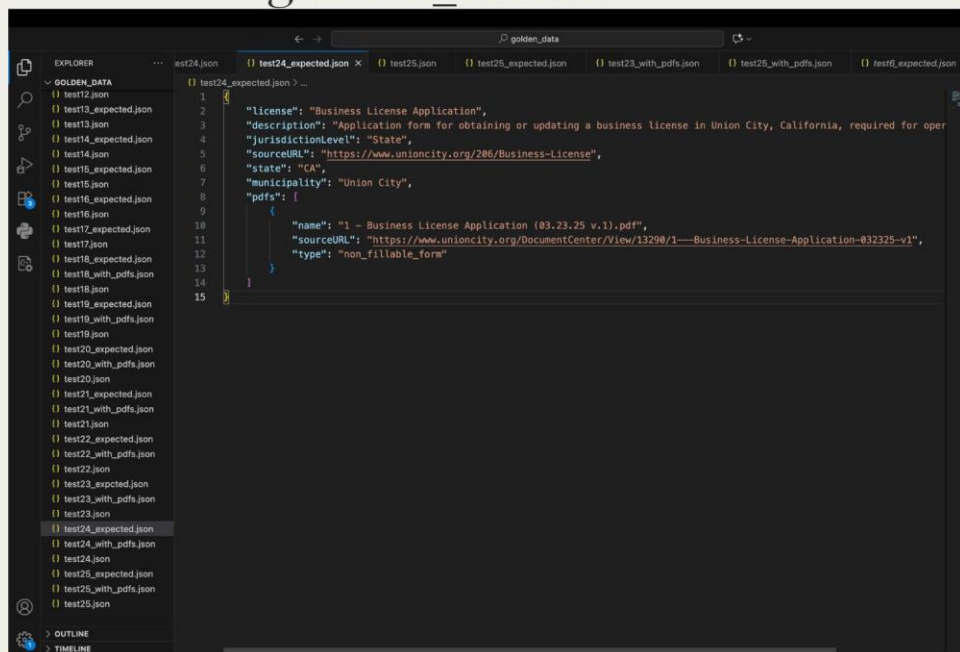
I'll combine these visual signals into a numeric score. For instance, the number of aligned boxes per page, number of lines intersecting at right angles, or density of boxes relative to the page area. This score can then be added to the existing prompt-based prioritization score, so a PDF with a generic name but strong visual cues will still be ranked highly.

Another suggestion is use both grok and gemini not just for search but also for PDF selection and final classification. Running them against each other and only trusting results where they reach consensus could dramatically reduce classification errors. For example, if both models identify the same document as a fillable form, it's a strong signal of correctness. Where they disagree, I could even introduce a tie-breaking mechanism based on confidence scores. I will also work on this. Also the hallucination part which is super important.

Improved license form finder

Chetanya Pandey

golden_dataset



IDENTIFIED DUCKDUCKGO ISSUE

at least twice the no. of urls now $\Rightarrow$ increases probability

PDF LINK PRIORITIZER

ICL

```
system_instruction = (
    "You are an expert at identifying license application forms and permits based on PDF filenames and titles.\n"
    f"Your goal is to identify and return ONLY the TOP {max_pdfs} PDFs that are most likely to be actual\n"
    "license forms. You do NOT need to rank all PDFs - focus your analysis ONLY on finding the best candidates.\n\n"

    "TASK: TOP-K SELECTION (Not Full Ranking)\n"
    f"Scan all provided PDFs and identify the {max_pdfs} most relevant ones\n"
    f"Return ONLY those top {max_pdfs} in prioritized order\n"
    "- Do NOT waste effort ranking PDFs that are clearly irrelevant\n"
    "- Optimize for RECALL - we don't want to miss any potential license forms\n\n"

    "PRIORITIZATION RULES (highest to lowest priority):\n"
    "1. Form codes (e.g., LIC200A, ABC-069, MV-44, RE-201) - Almost always actual forms\n"
    "2. Filenames with 'application', 'form', 'permit', 'license' + action words like 'new', 'apply', 'renew'\n"
    "3. Medical/professional credential forms (e.g., 'medical-exam', 'fingerprint', 'background-check')\n"
    "4. Generic application-related terms without clear indicators\n"
    "5. Supplemental documents (instructions, guides, fee schedules, checklists)\n"
    "6. Informational documents (brochures, FAQs, 'how-to', 'guide', 'overview')\n\n"

    "IMPORTANT:\n"
    "- Prioritize forms that match the license type (e.g., 'liquor' for Liquor License)\n"
    "- Do NOT prioritize but DON'T exclude guides/instructions - they sometimes contain embedded forms\n"
    "- When unsure, prioritize higher (optimize for recall, not precision)\n"
    f"Return EXACTLY the top {max_pdfs} most relevant PDFs (or fewer if total PDFs < {max_pdfs})\n\n"

    "=== ICL EXAMPLE 1: Liquor License (TOP 4) ===\n"
    "LICENSE: Liquor License\n"
    "DESCRIPTION: License to sell alcoholic beverages\n"
    "MAX_PDFS: 4\n"
    "PDFs:\n"
    "0. liquor-license-guide.pdf\n"
    "1. ABC-069-application.pdf\n"
    "2. how-to-apply-liquor.pdf\n"
    "3. LIC200A.pdf\n"
    "4. fee-schedule-2024.pdf\n"
    "5. alcohol-seller-application-form.pdf\n"
    "ANALYSIS:\n"
    "0. liquor-license-guide.pdf - Excluded: This is a guide, not a form to be submitted.\n"
    "1. ABC-069-application.pdf - Included: Contains form code ABC-069 and 'application' in filename.\n"
    "2. how-to-apply-liquor.pdf - Excluded: This is a guide, not a form to be submitted.\n"
    "3. LIC200A.pdf - Included: Contains form code LIC200A.\n"
    "4. fee-schedule-2024.pdf - Excluded: This is a supplemental document, not a form to be submitted.\n"
    "5. alcohol-seller-application-form.pdf - Included: Contains 'application-form' in filename and matches license type.\n"
    "Final Top 4 PDFs: 1. ABC-069-application.pdf, 3. LIC200A.pdf, 5. alcohol-seller-application-form.pdf, 2. how-to-apply-liquor.pdf\n"
)
```


PDF LINK PRIORITIZER

incorrect order issue

NOT FORMATTING URL ENCODING PROPERLY

increased accuracy

broken url issues gone

ask_gemini_best_pdf

snippet part was good

improved ICL

2.5

```
"You are an expert in U.S. business-license documentation. Your goal is to review a set of PDFs and identify the best one for a single license. You will be given multiple PDFs (or a WEBSITE) for a single license, along with metadata about each candidate.
* snippet: text extracted from the PDF (may be empty for scans).\n"
* fillable: true if the PDF has fillable fields (likely a form).\n"
* file_name: identifier of the file.\n\n"
"Select ONE PDF (or WEBSITE) that is most likely the true application or required submission form for the license.\n"
"CRITICAL: In your JSON response, the 'name' field MUST be the EXACT 'file_name' value from the candidate metadata.\n"
"IMPORTANT EVALUATION RULES:\n"
"- Prioritize fillable forms (fillable=true) as they are more likely to be actual applications.\n"
"- DO NOT EVER select brochures, informational handouts, or FAQs.\n"
"- Only choose WEBSITE if all PDFs are invalid AND the website clearly provides a portal or instructional information.\n"
"- DO NOT EVER select a PDF that belongs to a different city, state, or jurisdiction - even if the title is similar.\n"
"- If you are unsure, return 'file_name': 'NA'.\n\n"
"CLASSIFYING THE FORM TYPE (choose ONE):\n"
"- fillable_form: PDF has interactive/fillable form fields (fillable=true). User can type directly into the form.\n"
"- non_fillable_form: PDF is text-based but does NOT have fillable fields (fillable=false). Must be printed out and filled in.\n"
"- image_based: PDF is primarily scanned images with minimal extractable text (text=false or very short snippets).\n"
"- online_portal: WEBSITE candidate that directs users to apply online. Only use for WEBSITE, never for PDF.\n\n"
"==== ICL EXAMPLE 1: Business Registration - Select fillable application form ==== \n"
"LICENSE NAME: Business Registration\n"
"LICENSE DESCRIPTION: Business registration and licensing for Oregon\n"
"CANDIDATES:\n"
"Candidate 1:\n"
"  file_name: abn-new.pdf\n"
"  fillable: true\n"
"  text: true\n"
"  snippet: 'Secretary of State Corporation Division 255 Capitol St NE, Suite 151 Salem, OR 97310-1327'\n"
"\n"
"Candidate 2:\n"
"  file_name: business-registry-fee-schedule.pdf\n"
"  fillable: false\n"
"  text: true\n"
"  snippet: 'Secretary of State Corporation Division Business Registry Fee Schedule Effective January 1, 2024'\n"
"\n"
"Candidate 3:\n"
"  file_name: start-business-guide.pdf\n"
```

classification

schemas.py

N/A

so what changed?

Absolute improvement in relevance = $96 - 40 = 56$ points		
Absolute improvement in Multi-class classification (type_accuracy) = $88 - 48 = 40$ points		
Percentage Relevance accuracy increment = $(96 - 40) / 40 * 100 = 140\%$ improvement		
Percentage Multi-class classification (type_accuracy) increment = $(88 - 48) / 48 * 100 = \sim 83.33\%$ improvement		

basis
GCP, haystack, the boost is good for scaling in
a new domain

[illegible]

thanks to my mentors and our team
what's next?

License Finder v0.x — Release Summary

Pipeline Overview

Improved the License Finder pipeline that automatically finds and classifies U.S. business-license application forms (PDF or online). It scrapes queries, extracts and prioritizes PDFs, and uses Gemini 2.x to select the correct application form, filtering out irrelevant, duplicate, or broken links.

Evaluation and Testing

`golden_data` — backend

Created a benchmarking framework (`golden_data`) to measure how much each new method improves performance. Currently includes 25+ test cases including both, common licenses (e.g. General Business License, Business Certificate) with a major representation and other licenses. It provides quick feedback on accuracy and consistency improvements and is a way to test even small improvements quickly to see if they are actually yielding results..

Pipeline Modularization — backend

Refactored the system into distinct modules: `pdf_link_extractor`, `pdf_prioritizer`, `pdf_selector`, and `palm_parser`. This enables independent testing and debugging.

Search and Candidate Generation

`serper_search_multi` — backend

Addressed DuckDuckGo result inconsistency by running 3 parallel searches and merging results. This stabilizes the candidate pool and expands the set of potential PDF links while removing duplicates.

Prioritization and Selection

`pdf_prioritizer.py` — backend

Replaced random candidate selection with prompt-based ranking that scores PDFs by how likely they are to be forms. Added in-context learning examples to

improve prompt reasoning. Limited the number of ranked URLs to preserve prioritization order and ensure consistent results and to avoid wasting resources.

ask_gemini_best_pdf — backend

Enhanced classification prompt with in-context learning (ICL) examples. Each example teaches Gemini how to pick the right document and label it (fillable_form, non_fillable_form, image_based, online_portal).

Also added NA-filtering logic to return empty sets when no relevant form is found rather than returning an irrelevant form.

Data Integrity and Encoding

encode_url_properly — backend

Fixed URL path issues that broke source links (e.g., unescaped spaces). Implemented robust encoding to preserve valid URLs across all modules.

Classification Improvements

schemas.py — backend

Refined classification logic (fillable_form, non_fillable_form, online_portal, image_based) for more accurate mapping of Gemini outputs. Improved prompt structure and schema validation to reduce misclassifications.

Release Impact

This release makes the License Finder pipeline more accurate, stable, and testable. Candidate sets are now consistent, prioritization is logical, Gemini's reasoning is guided by strong examples, and URL issues are resolved. The improved prioritization, classification, and in-context learning make the system more consistent and scalable.

License Finder v0.7.7 By Chetanya Pandey

Pipeline Parallelization

- Added 2-stage ThreadPool executor parallelization for searches and license handling:
 - License handling now uses 2 workers.

- DuckDuckGo searches use twice the number of workers as queries. It has been tested thoroughly to find this optimal. A lot of options were considered but this was the final one implemented.

- Extensive testing under 25+ repeated runs ensures rate limits do not halt execution and that parallel threads execute efficiently. This was done to find the optimal threads we can have across licenses without being rate limited.

- Optimized DuckDuckGo query execution:

- Deduplicated repeated queries from three times to two times.

- Increased maximum organic results per query from 10 → 15.

- Introduced Serper Search as a fallback to DuckDuckGo for resiliency under high-traffic or blocked scenarios.

Performance Improvements

- Overall runtime improvements range from 48–93% faster, with an average percentage improvement of 50–70%.

- Deployed randomized delays and backoff logic to mitigate rate limiting:

- Dynamic jitter of 8–12 s between searches.

- Exponential backoff on errors

- Overall, this fine-tuned license parallelization to avoid resource contention while maximizing throughput.

Release Impact

- Form Finder for the licenses is now faster, more resilient, and scalable under high query and form volumes.

- Parallelized searches and license handling achieve significant speed gains (48–93% faster; average 50–70%) without triggering rate limiting.

- Randomized delays and fallback search mechanisms improve robustness for real-world, large-scale usage.
- Overall improvements maintain accuracy while reducing runtime, making the system efficient for automated U.S. business license retrieval.

License Finder v0.x — Release Summary

By Chetanya Pandey

PDF Download Fallback, Cloudflare Handling & SSL Robustness — backend

Issue:

Some of the PDFs prioritized by Gemini could not be downloaded because of Cloudflare and Akamai protection layers. These pages either returned access-denied responses, incomplete redirects, or CAPTCHA-like verification steps that broke the automated download process. As a result, even correctly prioritized PDFs could not be downloaded. We do not want to lose out on any of these useful form pdfs, hence this fallback.

Improvements:

- Selenium Fallback with Undetected ChromeDriver:
Implemented a robust fallback mechanism using [undetected_chromedriver](#), achieving ~80% success rate initially and 100% after configuration tuning.
 - Added enhanced SSL certificate handling ensuring stable downloads from secure or misconfigured sites.
 - Configured Chrome to download PDFs automatically (instead of viewing inline) and bypass download prompts.
 - Introduced Cloudflare detection logic, identifying challenge pages (“verify you are human,” “just a moment,” “checking your browser”) and dynamically waiting or retrying until the challenge resolves.
 - Added [.crdownload](#) monitoring to detect active downloads and ensure files are completely written before validation, reducing false negatives for large PDFs.
 - Improved timeout handling and final verification to confirm successful file saves.
- Safari User-Agent Header Method:

Added Safari-based headers (`HEADERS_SAFARI`) to mimic genuine browser behavior, increasing success rates across Akamai- and Cloudflare-protected endpoints.

- Implemented `_download_pdf_direct_safari()` with correct MIME-type checks (`application/pdf`) and used page-length validation via `_pdf_too_long()`.
- Improved reliability for direct link requests, particularly for sites that block requests from standard user agents.
- Randomized Header Fallback:
Used `_download_pdf_random_headers()` only when Safari attempt fails. This directly decreases latency because Safari is more predictably useful and we don't have to needlessly go through random headers' attempts to download a pdf. We only use random headers if the Safari header fails.

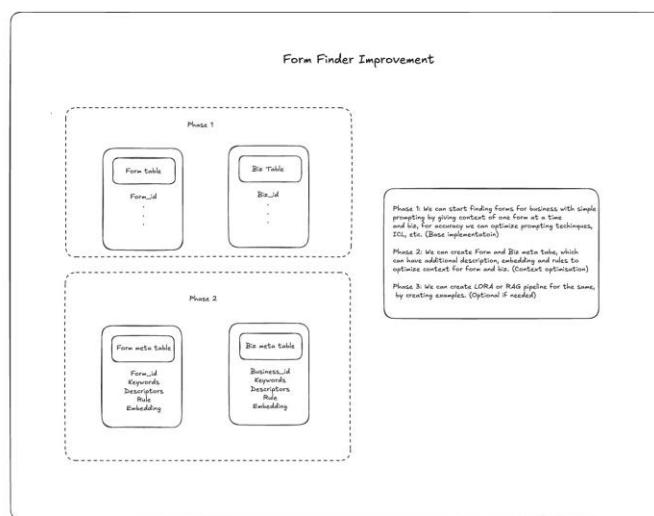
Impact:

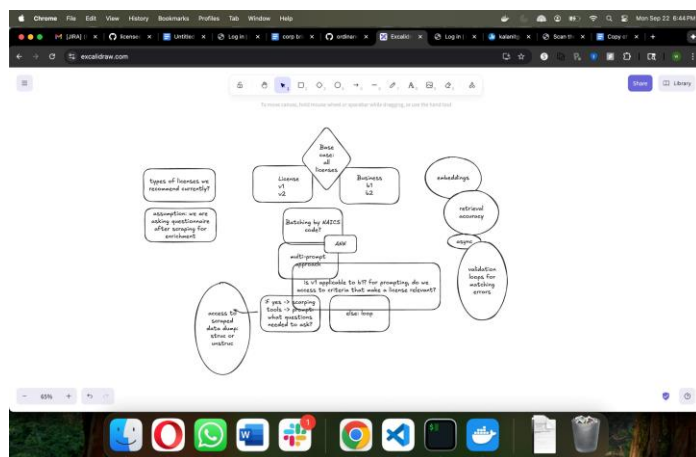
- Resolved failures where Gemini-selected PDFs were previously not downloadable due to Cloudflare/Akamai barriers. This significantly improved reliability and completeness of form retrieval, allowing the pipeline to consistently fetch PDFs that were earlier inaccessible due to Cloudflare/Akamai issues.
- Achieved 100% download success across previously blocked URLs that had Cloudflare/Akamai issues.
- **Strengthened resilience with multi-layered fallback logic (Safari → Random Headers → Undetected Chromedriver).**

Entire caching architecture for license finder proposal



Made 2 proposals in start of my internship for arrangement of data in embeddings and meta tables





Bullkilling

BEG TO STATE

The other day, I read about Bullfighting in my General Knowledge chapbook. As I went through the national games of the pays around the world, I found the names attractive...Cricket, Rugby, Badminton, Hockey, Bullfighting.....Egads! Bullfighting? The name seemed to reflect its spirit but I would crown that old-fashioned game with a better epithet- 'Bullkilling'. My mind simply didn't waver off from the word. Every other game seemed to be what you would term as 'sophisticated', but this one was not just brutal, it was that 'early-human type' of game still being played in the twenty-first century! This game definitely was at odds with others. I gazed at the pic alongside until I could see no more. All that blood in the pic made my blood boil. I could see a so-called 'human' standing with a lance in his hand ready to tear off the bull as though it had no living existence. It was cruel for a 'human' to perform such an act, but you know, I don't want to be weeping for that cruelty and I don't want others to weep for it either. They say the pen is mightier than the sword. And this book has been written acknowledging this fact.

I couldn't get this thing down my spine and so I decided to put it on paper. And I never watched the YouTube video again as I didn't even want to increase a single view from m side!

The game is unjust in its very spirit. It's framed in a way that would ultimately lead to the end of the bull. It's no fair play. The 'human' has aides and lances and horses, and what does the bull have? Not even his ass! The bull is dragged to compete in an unhealthy environment. The matador consigned to the fight but did anyone even ask the bull if it was interested in the duel? Why would the bull love to die for the sake of entertaining the spectators. Discovering that would be like discovering a mare's nest or a lynx-eyed skirt!

What's more, at times the bull is 'pardoned' for its 'crime' of killing the matador though it rarely comes to pass as it hardly ever gets a chance to storm the arena. My question is- Why is it called 'pardoning' and why is it a 'crime'? When the bull is killed by the 'human', he's never 'pardoned' (his victory is celebrated with pomp and show), so why shouldn't the victory of the bull be celebrated too? After all, it's a duel and both the parties are the opponents and opponents always hold the same place before any law or before the Judiciary. A fair fight won't allow a human to be treated as 'pricey'. Both the sides have the same worth as it's a duel. This makes it a tragedy so heavily engraved considering earth to be a conglomerate.

Now the question is- Is humanity dead?

This questions the need for self-respecting individuals who can get themselves to believe in their own self, and this is rare these days. We no longer have Mateo-like heroes. And I would want earth to have more of such heroes, hence I provided a map to self-respect and its utter need in this epoch. You see, it's self-respect individuals that didn't give a damn to what the society thought of them. They stood their ground, and as a result of their legendary sacrifices, Bullkilling is no longer as commonplace as it used to be. Each and every character of this book is dedicated to those self-respecting animal rights activists who embraced their own self and their own ethics to pave the red carpet to the end of a red game.

I've compiled this book to propagate this very message. At the same time, I want my readers to realise the importance of self-respect considering the fact that when it could terminate a game so cruel, it definitely can change the life of the being it blesses with its presence in one's self.

Lord Mateo

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His face turns white and colourless just as the pale moon of the early morn when the light of the Sun springs to the east, and he lay like a pillar shattered by a saucy yet dismal blow.

PROLOGUE

Impartially did the melody of the dust play its privileged tune in oblivion of the pathos of the soul. It blew from the west to the east, from one body to another. The shrewd glances cast by the open skies, the profuse waters of the fallen brooklets, left to all that had partaken the nectar of morality a piece of controversial wisdom- Let none live.

The orchestra of man's deepest wounds, the tapestry the melody of the dust blew through the sunken anklets of the sunken dames played their magic oh so well.

It was empty. The bank. The battle. The fight. The wisdom that lay in this blissful note to the heavens or to the hell so artistically carved out by the blood of hungry men was only too late in her path to the cerebrums of all those who possessed it.

Neither could one blame Time for this hungry disaster nor the corpses that accompanied it.

It was hunger, that revered blossom of the springs one and all lay in wait for, that was to blame. The hunger for an abstract idea, the hunger for Time's deepest

wounds, the hunger to paint upon this land the most awaited letters in the most awaited ink- Let none live.

THE LONE IN THE BATTLE

It was more of a contradiction to his revelling humanity, the paws of which have been so often searched by men held in high esteem, but have been ridiculed by the common masses as to them the name lies not in the deeds but in the grammar it's composed of, and yet this unfailing sense of humanity poised him beyond his years to grant a sense of self-satiation for no human seemed humane enough to provide him company in all his astuteness, and no human would keep himself composed and masculine at the same time of the day, at any time of the night, and this savoury brought him a special delight to savour- the delight of enjoying his alone time, the delight of traversing the length and the breadth of the jardin on his own sans a page or a company to keep with for he couldn't be kept in limits, howsoever parallel they might present themselves to be and yet this glory apprehended a mocking session of the fella who would be enveloped with a buzzin' circle of amis with none at his own disposal and with an aim to commit commonly to the vices each one had known providing an infallible arrow of sourdom as he would like to think about himself. And such a company was but futile to him for he possessed no likes on the planet and ain't friends meant to share common interests? Ain't friends supposed to walk, sit and eat together? But where, O, where from shall he bring himself to compose as directed, sans any reformations? And why would he, at that, reform himself?

My salutations unto the lonely:

The lonely shall know peace few have committed themselves to learn, few have cherished and to put that simple, learnt about so to feel the absence of, at all; the lonely shall recognize the face the Earth carves out for herself; the lonely shall learn to pose for the right angle of the dangle for their astuteness is a proposed prejudice; the lonely shall learn the behaviour of the Earth in times critical; the lonely shall be themselves truly for they have none to give a damn for they know not the face of the being they shall give the damn to and not a penny shall they lose; the lonely shall recognize themselves and the nature as the most faithful of friends and learn to talk to them in a way only the lonely have learnt; the lonely alone shall sing the prayers the Earth longs so to hear for they have recognized her voice, that which is unheard by others; the lonely alone shall see themselves in others and the others in themselves."

The natural instincts of the man balanced a subtle poise, a poise that would lead to sleep during the day for fear of being recognized for the idlers tend to be on wake during this time and shall actively seek for it, but its presence during the night rendered an uncanny sense of sense and made the calm and composed structure a man packed with an idealistic sense of being languid. He would so often gaze at his hairy chest which pointed fortune and that he already did possess in a measure immeasurable. Then he would fidget on the pulp, taking the shape of paper, with a bamboo stick as they do in ravines and caves and jails. He drew shapes of men of all forms and would rest his fingers in between as weird thoughts would percolate right to the bottom of his brain and from there into the medulla et al. Oui, he for sure did put himself to wonder at things that appeared so smitten with vows of peace that he could but swear, "By Jove, I scorn it and I haveth make out sense of it." And he did make sense of it by wondering as to what the horizons unseen sought to portray in he most subtle of ways, so much so that he would rather be keener about apprehending the language the horizons spoke, and at times, he chased them all to himself, though he knew not what they led to and where they ceased, and yet he chased them for the fun of it, for the frolic of discovery, just as somewhere on the way he would meet a new creature, a new being, a yet another mystery would await him in the best of fashions and in the calmest nature which would still not fail to produce a thrill in the minds of the young explorer for he had learnt not every mystery would last as such for time unending and the tapestry could be unwoven as easily as it was woven, but the reverse engineering involved a chase grotesque, a chase few men are taught to take, a chase that thrills even laughter that is known to arrive in peals only to be peeled off layer-by-layer just as an onion which is peeled off taking care of the vast offshore sea of peals and layers, and so was mystery which he was determined to kill tragically for suspense horrified him in the most brutal of ways he could ever commit himself to take and yet, this suspense only excited him during the chase, so he learnt to be dazzled by it and on his part, he learnt to chase the uncouth suspense off with words of abuse from his non-verbal dictionary- a language only he had brought himself to learn for he was all so lonely and had no one to share his grief with for he had none. All grief was a lighthouse to him, a lighthouse to look out for yet another string of the unfailing mystery that was as ready to arrive in the battlefield all clad with kingly honour and Your Grace was as ready with the shield of his armour and the wisdom of his sword though most writers would choose to interchange the situation of the words in this context.

He delighted himself at the prospect of possessing an enemy at last for he could inadvertently apprehend the challenges that lay beneath the enemy that sounded so shallow, and the man was all set to face them sans any due apprehension. He

was curious, curious about things and not people for he saw nothing in men to be so curious about- all, by the ways of His Grace, seemed to be leading lives which could well be tuned to the word 'Normal' for they all did deeds in common-eating, drinking and mocking at passersby who on their turn would ridicule the passersby when they would be the spectators. Mystery seemed to have failed the test in the first stage itself but chances were it would somehow strive to maintain an average score after this and that would imply the need for an increment in His Grace's capabilities, and who would be more willing to be armed with ammunition of the latest fashion than Marteo himself?

The curiosity sought a wider admission with the passage of the time when he had heard the nightingale's sweet voice. The calling of the voice demonstrated a fact which had been hitherto presented with a conservative layer or veil. And the ugly face of the bride came to be unveiled. This journey won't last for a short span of time. It had, built around it, a very conservative and breezy air which conservatively unpacked the bunch of secrets that lay hidden in them so they might never be viewed, and in the gorilla fashion of wars, shall soon engulf him in the light of his ignorance and carry him on to the troubles the path he was going to walk on, sheltered with utmost precaution. And yet the secrets couldn't be kept secret for long as the long chase convinced him of the need to arrive at an agreement, by constant negotiation, between his conscience and masculinity for he had, by no means, sought to think of the less-likely ways of destiny and the game of probability was tougher than he had initially thought, and with this he lost the gamble as his flight in chase of the ends of the horizons, the most puzzling of mysteries placed a counterfeit coin over which he tripped. This was the very same coin he had taken to be of so low a value earlier that he least bothered about removing it from his path. As of now, the same teeny-tiny coin had his conscience awakened and he realized the perils of the path and with unwavering determination as the wise he sought not to refute but to overwhelm the thorns themselves and overpower the hurdles that raised their ugly head from the ground beneath his two tree-toad white feet which alighted in all splendour and amazed the denizens so they bowed low to his unwavering determination which had shaken the very best of promise-makers for it lay not in a mere oath but in a will, the will to shake the ground beneath his feet. The cries of amazement from the heavens above were a further uplift to his mood and temper and hence began the lonely journey of the lonely man walking by the beach with a calm and a noise, with a breeze and a typhoon, in a row and sans any rules and regulations, for he was, by birth, a free spirit ante he could call himself a noble born to the nobility, and to him the word 'noble' meant a thing all so different.

In all his loneliness, he passed by the contours analyzing the dangers of inhumanity. He walked down the terraces and observed how strategically situated they were in order to do away with the fierce wind. This portrayed to him the need for a strategic plot, the need to avail his own freedom in stages, the need to salvage himself tactfully. He squatted on the ground and rested the three fingers he was now left with, on the ground as he saw blood emanating from them just as honey emanates from a beehive. With the blood, he planted a verse on the face of the earth, "Your son shan't lie to you nor to his father, Humanity. Unto your apparels have I breathed and unto yours do I meet the fate you procrastinated. Let my prayers be answered. I shan't breathe until Inhumanity ceases and until I meet Grasp in the place of fortune for the most fateful creatures love you for you let them live and the most ill-fated love you for you let them die. Nay, I shall be alive to continue my mission hitherto or I shall die to continue the mission I so fondly undertook alibi. I won't die until you summon me for you are the only one I ever obeyed. Not His Highness, Not His Majesty and I say this right here, on the soil he foolishly calls his own when it is all yours, O Mother!"

If a man may trust, the liberal voice of the Earth met a tremble and Mystery arrived to pay her condolences unto the best and the chosen of the Earth's heroes, who with the strength of their spirit caused her to seek shelter in the palaces of His Majesty. She bowed down as he clasped and he clutched the Earth trembling all the while. She knew when, where and why she had wronged him. She had enjoyed the chase with him. The two were two of the greatest lovers the Earth has ever placed in her lap.

Yes, Mateo married Mystery and he lay faithful to her for whole of his life, and so did Mystery who didn't leave his side during the worst of scenes as she had caused them all.

And yes, how could the Mystery end? And why would the Hero?

His mission on Earth being done, his call for Humanity having been answered, the Great Hero lay flat on the ground as he spoke the words that caused an echo after echo as the waves spun round and round and finally died, "Aha. I made you alive, O Humanity by putting myself to Death."

Humanity answered silently-

"I owe unto you a fortune worth a billion pounds for you did what the faithful nigger does to his master, and yet you were masterless and always shall be. Your

breath gentle, calm and forgiving, gave me a place not many have the boon to reap sans any personal troubles for it was you who formulated the plans, devised the strategies with a readiness to be slaughtered at the edge of the most ruthless of swords, at the end of my own edge, you jumped off the cliff caring not a bit for the height that would meet you when you would get down. O, Human, you cared for your nationality more than you cared for yourself for you knew no existence sans your faith and yet, this very same faith is not followed by most, you drank up the most deadly of poisons as appetizers for they were the most romantic part of your plan and you wouldn't allow me and you- both of us to survive in junction, you wouldn't be satiated until you saw some trouble and faced some mystery. You were the most faithful lover ever. You kept by the side of Mystery when she was all but trouble to you. You took her not for trouble and only teased her playfully. The girl loved the fun of your furtive glances and she had the honour to be Your Grace's lady. Ladies love fun, so did she and so did you bring her fun by doing the unimaginable, the coup that the macrocosm has never before had the liberty to witness with so passionate a struggle poised beyond the struggle to bring the dead me alive. You woke me up from sleep but when I failed to wake up to your call, you got your answer sans my actually replying to you. You apprehended the need of the minute in the most practical of ways and denied not yourself the right to marry Mystery and hence you jumped over the hurdles so they could but raise their ugly heads to witness the passions of your ambition, and you made Masculinity proud by the act. Yes, my dear son, you aided the Earth in the most remarkable of ways and the falcon aggravatingly has delighted her. She was and shall always be your proud mother. You made us proud and revered us as your family when the naysayers continued ridiculing you for downgrading yourself in the eyes of your earthly 'noble' family. You knew it wasn't your chez. You recognized us though our faces were veiled. We acknowledge your attempt to sanctify my spouse, and let Time be fearful of a thing unknown and let you be immortal for Time is hiding behind the veil of Mystery. Face it, dearie. You rocked. You rock. You will rock for none on the face of the Earth has displayed such solemn determination that didn't fade all through the test of his dearest consort, such remarkable courage drives The Lord of Courage crazy for he doesn't have enough supplies. You shall be given the most beautiful funeral by the most beautiful lady, Peace herself. Let Peace be unto you."

The summoned Peace offered her homages unto the great soul, "My lips always did tremble with the sight of Lord Marteo 'cause he never did tremble! Mystery tested him all the while and at last he conquered her heart and won her love. And that Love sent me to him. All his life was a test and the output of all those Sufferings, which he forbade me from addressing as 'Sufferings', was Humanity.

He had an independent sense of identity and that's what left me flabbergasted for he, despite being a free spirit, saw Humanity as non-different from him, Human. And hence, he surpassed even me in knowledge and understanding for he could well have considered Humanity as different from his own self and could have retained an independent sense of self without having to go through all the hardships, and still being himself. But he realized the non-difference of Humanity from Human, and that's what makes me offer my salutations unto a man of such higher understanding. As I have always been known to put understanding ante courage, for the courageous need not be virtuous and understanding, mais the understanding understand Courage, I highly appreciate the level of his understanding put even before his remarkable courage for even the most downgraded of beings could be courageous and spiteful at th same hour of the day. I promise that I shan't, on my part, leave his side until forever for His Grace dedicated himself to Mother Earth and I have arrived at the summons of her heroic son."

Clouds cried just as Peace finished offering her verbal oblations unto the sacred fire of Humanity. Mystery vanished from the scene for she had gone through enough of tormentations for having tested her lover all the while.

They knew that the artful Goddess won't arrive soon for her huge ego had not been treated with due respect and adoration. But the playful Goddess could disrupt Peace in order to test him even after his demise on Earth.

As the battle was waged, Peace jumped into the fray just as Humanity went to slumber, and the Earth shivered with fright, and just as the brilliant dark lights (de facto, even the dark is brilliant) captured her for she had dared to disobey the wills of Mystery and would, without qualms, have to bear the troubles which now lay on her shoulder just as she diverted the attention of Mystery from Lord Marteo to herself.

"Nay, Mystery, you are known for tormenting the beings you love. Why then do you torment me for, let the Earth be our witness, you scorn me?"

The clouds roared with laughter and stopped crying for a moment. "A fine test I' faith! With your critical understanding of life put to test, let us not fight a battle of arms but that of words. And, O, Peace, fight ain't torment. I scorn you, by Jove and I need no witness."

"I'm ready as ever, but what about the accolade?"

“The one who wins shall dwell on the Earth for a period equal to the age of the Old Sun and the one who loses shall not show up for twice that time. In the interval, none of us shall function.”

“And what shall become of Lord Marteo?”

“I allow you to be by his side for times immemorial, and not just him, I permit you to be on the side of all the beings who are born to defend Humanity. Let me not anger the Earth.”

“Why, don’t you love them?”

“I cease to have any love for them now. Let you love them.”

“You are welcome, O, Mystery. Shall not the best of men taste your scornful love!”

“And not shall the best of men taste your love,” Mystery burst into peals of laughter.

“So be it. But do solve the query that mystifies me though you didn’t possess me.”

“Sure. What is it?”

“Why did you cease to love the saintly?”

Mystery remained quiet and ain’t silence the simplest and the sweetest reply ever?

Peace was answered silently by Silence for he is her brother.

“They defeat her, and she fears Defeat, so let her not adore them,” Peace thought as quickly as a flash of lightening. “Unlock your questionnaire.”

“Just as you wish. Hereth you go. Why was Lord Mateo ready to stake so royal a life, that which all earthly creatures would pray on an end to obtain for a mere calf which wouldn’t even have been killed had he got it exchanged?”

“Because he loved not the Grasp but the calf.”

“I am in absolute appreciation of your understanding of man’s nature.”

“May I have the pleasure to ask you to recount the story of this man Mateo for I haven’t been told it in toto sans any compromises of the details therein? I shall love to listen to the legend of so great a man from so great a storyteller as you.”

And so began Mystery with a mysterious smile.

No dust,

Only dust

“

Humbled by innocence and privileged by birth was born in a fashionable town of Madrid, the breeziest city of ‘Iberia’-the land of rivers, a papi chulo, a saint who was divined to the day and divined to the night in an unconscious attempt to unveil the scars of the earth, a perfect metaphor for claudible poetry-a humble reproach to the God of Cruelty, if there could ever be one, the arduously spirited Marteo.

Don’t get me started at his spontaneous shed of cool and mesmerizing mannerism that would leave the most charmin’ ladies of the ciudad sulking and secretly willing to be naughtily graced by his chivalry. Those craftily crafted words would leave the Great Sun God hanging in mid air so he would no longer be flattened from the bottom or from the top, the wavelengths of the blue and the red lights would be left shockingly even and even the most frigid of earth’s creatures would pass a smile at the reckoning beauty it would create. Hombres would wake up from their threesome dreams and mujeres would get utterly confounded about the onus left to be done as the day would have at its disposal those weak calls that entertain all ladies. The day would have to, at the very best of its capabilities, sever the ties hanging about or may be at the worst of its ‘captivity’, it would get incredibly excited to counteract on the imposed charm of a figure playing hither and thither and lost in his own world of magic and with no sincere inclination to attract attention, but he did it more than anyone else could, not that he was a prince but ‘cause he was the kind of prince they always had dreamt of. This ardent sweetness (if sweet were the followers of Charms) was only to be lured by the existence of something so unknown to his own self, and yet so supposedly concrete- Vainglory or ferocious arrogance, I could make it gentle by calling it

‘self-respect’ and a strong, rather strong will to be his own self, and yet this man wasn’t vainglorious despite the fact that he could summon to his will a letter from the Rome of Romans or a sweet chick clad in rainbow-coloured apparel so naturally meant to be possessed by Cupid, but still did distract herself to this Cupid on Earth- the ear-long or year-long glue so glued to his very charming self which left all ladies with a ‘Oh, My stars and my stockings!’ sort of cry at the end of a fag day with an unknown attempt to dilute the thing they had for him, and no dictionary did they find clad with the words that would lure them into hating his self. Such strength did it possess that simply unveiling the stores it had hiding behind the sweet smile was a great work to do, and who wouldn’t be interested in a thing as that? Someone that’s more interesting than a tour around the world, and the very scope of the achievement seemed to surpass human recognition, and yet it wasn’t ethereal, it was all so real and viable, and even the straight girls would bend about such a man. And no, let me refute the claim of his being that ideal type of fair with that bit of tanned skin, there was something more that put him apart from the vast section of the nobility, I wouldn’t compare him with the common masses, and even his comparison with the nobility is more like the nobility is being compared to him and not with him, simply ‘cause the differences were all so grand, and that’s what made this man so popular amongst the masses. Mais his soul was tender and it possessed an inflammable softness that which would burn easily into self-defence and the ashes produced would be all too hot to be touched, not to mention his kind nature few inherit even with all the clothes draped around the belly of the mother at the time of appearance on this land. And kindness arrived to grace him in a unique version of itself, a version never heard of or felt (I would like to label kindness as a sweet melody to be heard and chee’uh it’s a feeling), this variant of Kindness seemed to be more of a heartfelt sensation than a feeling that would come of its own accord. The masculine do own their kindness and Kindness admires them for they are a few of the selected beings Kindness graces and each time it makes its appearance with a difference, and this gentle spirit is a pearl of love just as self-respect is a pearl of wisdom. The most masculine of men possessed the most masculine form gentle Kindness could ever arrive in bedecked in all her jewels and yet it did make a great pair to see the idealistic version of brother Contradiction in his own terms this time and all men on earth would be ready to gamble about the next way it would go as it’s all so crystal-clear.

Poised beyond his years, his spirit reflected the thing his clothes couldn’t conceal from the prying eyes, and sure, there were a million to pry on so buff a man, and that was his self, and clothes, I’m reminded of a saying, are toujours unavailable to success as they tell themselves how their being inferior to the pack was simply

so impossible and so possibly strange for anyone to speak of that either, and no one dared to as, chee'uh, not everyone would dare offend his own honour, and for sure, even after having basked in the sun for an aeon, not everyone gets a fling of that sunshine and shouting 'Gimme some' won't bring one any at all.

And all men have a spirit that takes them to task though I fully agree this one may not necessarily be as conscious of a few unnoticeable dimensions with regard to your own charm that could be termed 'undue' in this context. But this spirit does do a job- it gets them to be fully conscious of their own self and yet the vainglorious don't take the hint of the self and would rather have their self unchanged while they know quite well somewhere deep within their heart that the falcon ain't right but their false pride doesn't know how to turn back once a step is taken and this makes them even more haughty whereas they continue to be just as masculine as before.

But it confuses everyone as, from time unknown, the haughty are synonymous with being masculine, and thence I repeat over and over again the fact that just as Kindness, Masculinity too undertook a different allele of the gene in order to make this amazing being all so different in the eyes of tout le monde, and yet so perfectly blended with his own self. There was the magic that forsook no charm and the spirit continued being lively sans the need of any intermediate conjugation. It all came to pass due to the decorum he maintained for himself, being cruel with his own self, being tough with himself but gentle with others, and this personified his very masculine figure that made him stand out as the moon amongst stars.

Any earthling would envy his polished skills which did have many hard stones rubbed onto it and he stood all the hardness it gave, adding further grace to his dynamic self, and this unique amalgamation of life, laughter, gloom, charm, harm, spirit, the sense of self and what not made him what he was- as graceful as a golden armour that would know little comparison as he had reached the highest height the self could attain making him positively different and unique, amazingly positive as viewed by others though the fact continues to be a fact- he never did care about how he looked like to others, he was happy being himself and needed no validation of himself (though the unhappiest of us shall will to say he had nothing to be unhappy about and nothing mauvais going on in his life to care to keep it imperceptible from the stolen, furtive glances others would shower him with, he knew he was well, just the way he was!), but even so, I do am interested in describing him and we all are interested in knowing about this truly unique and even so, visibly unique man- the man who had the power to turn anyone on with a

single bat of the dense eyelashes, and that's the scene all would be merrily waiting to regard with their eyes on and with their bifocals all lowered in an attempt to gleefully have a complete three hundred sixty degree view of that brilliant magnificence that seemed to have its headquarters in his self.

Such is the spirit of man that he would rather be a coolie with a hunch due to the onus of the baggage than be an idler jumping around and about with damsels in the garden of Eden though he would inadvertently desire the latter, but this fella was someone more than a dreamer and he had something more than a dream.

The decorum I told you about, I must say, was not quite so veiled as the word might mean to define it or if you so incline, exclude it from its princely list of possessions that which had the prowess to satiate just anyone's ego only to be captivated by a few earthly words and this man knew them all and these very same 'few' words were unknown to the masses and still so they weren't magic but did weave magic as he knew how to make them work out in his own way and he did always have his own way, even at the cost of the blatantly dark reflections

But as you do seem to know it better than me, shouldn't I mention I felt as a crusader all through this as I reflected rather than dwelt upon the fact of Battle being his brother-german? And I know y'all would like to correct me at this and I do accept my fault in the way I approached the relation. It did seem too arduous for a vainglorious leaflet, but he was above all vainglory (I did mention it before, so nothing else is left to be put flat in front of you). He was humble to Peace and ferocious to War, and he did know how to make his approaches, just as a lion in rage when it's engulfed with 'hanger', this tiger was let out in the open with his senses acquiring the wildest positions of power and yet duly obedient to their master, but they were, as a matter of fact, present in their wildest and ravaging state which again he possessed more than any other man on terre sans any master for him to be presented at the disposal of, sans any command he would be tutored to obey, sans any summons he would present himself to accomplish. The chase gave him that unbreakable strength for he chased not but raged a war with the searching paws as he stood defiant and in all his glory. He was this man when on the battlefield and the moment he raised his hand to pick up the weapon that adored him, the 'swarm of the enemy soldiers' would fly away and scatter as he was that rocking fire that could not be doused off with ease and Ease simply found him to be a turn-off for herself. This was the brilliant light that emanated from his self and the bees would be blinded by its rectilinear propagation that didn't seem to be rectilinear any longer as it diffused all about and the wave nature of light could thus be a better theory to use to explain this phenomenon,

and yes, it really was a phenomenon that could get you wondering and uh-ahhing about a thing that sounds so surface-level, but there was an ocean seated under that sand, and it was, sure, a complex structure. And yet it did structure him in the most fanciful scenes one would ever dream of being present in, and the work of all sentiments seemed to have been over by now, and yet the darkest of sentiments stood confounded as they awaited a systematic scoop that would lead to a terrible destitution to come into place and would forsake a never-ending ocean of endless worries that would screw up the game, and this indeed was a game, but how?

AN ACCOMPLICE

These shady hills rest upon a crest memorable and it indeed is shady. For those who approach it calmly it poses a danger not many have known for not many have tasted the ferocious circle that surrounds it, a circle of a thing called revenge that circulates an aura of doubt, an aura of mystery which lies subtly rooted in its soil and decides its next victim for not many have known its real figure which has been displayed with a calm air on the rest of the earth but the fact that it's my eternal residence is not known to many for those who enter the line I mysteriously created don't have much of a chance to escape it and I lay seated waiting for my dearest lover.

The mines are the trap I create for they are not in the possession of the richest of beings and only tend to increase their greed with every passing minute for they enchant the sour and the sweet all alike just as the denizens dazzle the earthlings for all their mysterious figures, for all their figures that rest in me.

The trees pose another view of the beauty I have so longer desired to have in my place for I'm only the second greatest of all the beauties and I would welcome you with welcoming hands at the place I so desire to be exponentially beautiful by the honours of your company for though each one of us is at constant strife with the other, we know that the most beautiful things are composed of our amalgamation just as the most beautiful of colours are never purely red or purely blue. Not many have comprehended our combined state for we are often considered to be of a nature unadaptable, but the beings who compel us to adapt go on to become ethereally beautiful and this magic can't be dwelt upon at the length of existence by mere earthly beings and hence, the earthlings dream of the denizens of being mysteriously and peacefully beautiful, but they often forget the reason for the same for the enchantress in me keeps them mystified for the recipe of beauty, and that's what makes you and me fight on an end for you want the recipe to be disclosed for the benefit of all the creatures, but I want it to rest in

you, the inner you for I enjoy seeing them suffering for I, being of a contradictory nature as a whole, love them. We both love them in our own ways and our love for them is the cause of War, for I won't let them exist to enable them to reach the higher me and you want them to live for you rest in them. O, Peace, though you are the very source of calm, you are rooted in War for you rage a war with me and I enjoy seeing you fight with me for that's what I really did desire. I am always pleased to see the contradiction I create in you and I am always pleased to see me in you.

High above in the montaignes lies a shade of golden remorse, a golden lustre that envelops the surrounding airs with an aura of an unfanciful fancy that loiters about with a lonely air seized upfront by the claws of chaos, an inimitable noise that knows no second, a patience that matters more than its own counterpart, a glory that was robbed off its own brilliance a time ago, and no, this very same patience was patient about waiting for someone to come.

Footsteps sounded.

A man appeared.

After my employing you to function at your best, I devised a plan to test this young man for I liked him, and before proposing to him I needed to know his true nature for I hadn't witnessed any of it and if I did, at all, he didn't seem to be the kind of lover I would have desired for O, Peace, I hold courage in higher esteem than understanding for the courageous defend their path and crown me as the ruler on earth whereas the understanding shatter the surface I dwell upon with the club of their logic and try to assassinate me with the gun of the knowledge you provided them with, and the understanding fear the courageous for the courageous undertake risks and walk on a lone path whereas the understanding are all crowded together on a single path for they understand the significance of unity. It's the courageous who tilled the whole earth for the understanding never possessed the courage to discover it so to till it at all, and they lay humbly crowded together on a small piece of vegetation bringing Death into play. Nay, the courageous mined the sea for the understanding understood the troubles it would cause beforehand and they didn't dare answer the roar of the sea for they understood the dangers pertaining to it, but the courageous saw me in the sea and went ahead to pay their obeisances unto me. The ignorant ignore the dangers, the understanding confront them and the courageous uproot them from their very root. The ignorant are raised their ugly jealous heads on witnessing others succeed, the understanding compete, the wise defeat, but, O, Peace the courageous laugh

out loud for they have a varied path altogether, a path that knows no second, a way that's unique to them, is uncanny and mysterious, and is visible only to them for none ever bothered to see its charms at display, but once it leads to me, he shall have the Earth follow him for she herself didn't know the presence of the path on her soil and then I trap her, and, nay, if he loses, O, Peace, it's you who follow him, but he never did bother to follow you and hence, he again does stand out from your kin of choice. The understanding shall choose the shoot of an apple treat rather than its root if the world chooses to divide vegetation between the understanding and the courageous, and the courageous shall, with pleasure, grant them the mercy, but let them not be called fools for their apparently foolish choice, for they shall graft it and grow more of such plants and feed their brethren for not only are they generous but they can risk their share for trying a novel method of sustenance even though they well know it may lead to undesirable consequences, but they don't care to dwell upon the negative possibilities for they have leant to battle with the negatives in life and have attempted to dwell upon success for they never did satiate themselves with following the middle path which is common to the understanding, for they won't me to grow out of nowhere and see where I lead them to, and hence the courageous make the most romantic plans, each one of which is embellished with my presence and I rest eternally on the soil of the plants that mystify the creatures of their source for one could say they are fed with glass, but only the courageous know it's I who feeds them kindly. The world would not have been the way it is today had it not been for the courageous for the understanding make weapons but only the courageous put them to use ante anyone else offers to do. It was always the courageous who took the first step when the understanding were kept busy calculating the mathematical probability of success for it's only the courageous who believed that both Mathematics and Science were glorified more than Romance for reasons unknown and the truth was known to them for I taught them that there rested Truth hiding behind me.

The approaching man had to be lured into a display of self-power for I had to determine his standards of courage and understanding and I could not wrest myself with the displeasure of leaving him to his will. The mines in the area won't suffice for the man was beyond the cultivation of greed and couldn't be lured into captivation by so earthly a thing for his temperament suggested that he was different from all my past lovers and hence required a different trap for him to be trapped in. The brilliance of the mines were bedecked with sorrow as the thought dawned upon me for by no means could they endure such an interrogation of their worth, for they to themselves, deserved the attention of the most selfless and content man for they could make him the richest of all of God's earthly kinsman.

They sang a song in their gloom and listening to the song got me throbbing since I had never felt such dejection, such unprecedented refusal of the acknowledgement of the self.

“May the best of the Lord’s creations be doomed

For they know not the cause of their resentment

For they ridiculed creatures worth of respect

And of earthly adoration.

For only Time could rust and degrade us

Not the ignorance of an earthly man

And Mystery believes in his greatness,

So let the best and the worst alike

Look upon us with favours impartial

If they could ever be impartial

Not the bravest shall deserve us ever

And not the weakest shall visit us.

Not the ignorant may know our charm.

Not the passionate may deserve our worth

Not the greedy be full of lust

For our possession and our worth

And not any ironmongery soothe our deepest griefs

At the hands of the most vulnerable of deities

A grade underneath

Nor shall this brilliance

Be imparted to its possessor

For the possessor shan't value us for us

But for our being in his possession.

And may the deities know our solemn dedication unto them

And the free spirit of us, their creation

For we have been sent to lure man

And those who are beyond this all

Shall never be lured by us."

This wrecked havoc on my brains, I wanted you at that time for I couldn't bear the skirmish within me, and it didn't just seem to be a skirmish at that, but something unworthy of the attentions of my own self-worth for I had been subjected to disproportionate thoughts which appealed to me. and then came yet another thought which was cruel but convincing enough for me to work on, and I knew it would work for it was about time Humanity be put to sleep and Cruelty be awake. I shared this one with Mercy who seemed to be frustrated at first but then resolved to check it out in practice, and I bet my freedom and so did she.

But I wilfully presented the present case as an exception to her for I wasn't without my share of qualms, and had acceded to her wishes with but a motive for prank.

"Mooo"- the sound echoed all through the dense forests and the steep hills, and peeping behind from a tree that stood a hundred feet tall, I pushed my face through the dense clutter of leaves and the rough, laterite earth-coloured majestic branches, in the company of Mercy. The sun shone overhead with a variation in shape as it had flattened from the bottom 'cause it was about time for it to set and activate my game.

“Mooooooo”- the very sound played again as the playback sound of the stage as Mercy and I hurriedly scrutinized my commissions and checked if it was going on the way I had formulated it to for I would have been cast with utter depression had it not been for the right backdrop as my aim was only to see what height this man would reach to lure me into loving him. The calf’s teary eyes assured me of the fulfilment of my deepest desires very soon.

The calf had strayed away from its clan. After all the tormentations inflicted upon it, in the company of its brethren, it had stealthily run away from its brutal master, for this master was not like our master. He was not like the large-hearted masters of the niggers who bestow upon them the gift of a loaf of bread at the end of the day, nor was he like the master who cares for his servant and feeds him mercifully, nor was he like the Sultan Muhammad Ghori who turned his slave, Qutbud-din Aibak into a master overnight.

He was one of those cruel masters who roast the flesh that has served him for the sake of Fun.

“Nor does Fun worst himself in the crown of the monarch,

Nor does he seek shelter from him.

Nor does Fun with his chest bared

Dare to enter the territory of

The King of all Cruelties

For he sees, with patience imparted,

No frolic therein.

Nor does he enter the town of the unclean

For he loves sanity and sanctity alike.

Nor does he commit to Distress and Slaughter

For the two sound alike

And where there dwells Slaughter,
Distress follows suit
And the brother of Joy
Marks no presence in their record
For he sees himself and the like
In places full of hope and joy
Not necessarily of laughter alone.
And to those who mean Laughter by Fun,
Let it be known
That the semen of Zeal
Produced to itself and for itself
Your delights- Joy and Fun
And Anguish's sons- Laughter and Evil,
And the partial sharing
Vested Laughter with scorn meant for
Joy and Fun, hence for aeons uncountable,
We have it that Joy brings Fun
And that Fun brings Joy
But Fun and Laughter don't work together,
For Fun rests on your lap

And there is no Peace in slaughter

And those who obtain fun

By laughing with the roars of brutality

Have lost their sanity.”

The calf had had its bouts of independence just as the landless farmer feels delighted having paid his first round of loan sans the involvement of collateral, but the moment the reality of the situation dawns upon him, he stands henpecked for he has been so slyly trapped in the cycle of debt that the thoughts of escape seek their respective pages of action only in the next life.

A similar reality dawned upon the calf as it stood aghast in the middle of the valley surrounded by steep hills meant to prevent its escape for it had cost his master a great harm by escaping at his own will, and destiny seemed to want him to compensate for the dared disobedience. The calf stood silently, looking deeply at the lower edges of the hills as though he were searching for a way to escape. How like a man was he! He sat wondering and calculating for his wonder put his allowance to fright for Allowance did not assume him worthy of being the permitter of the self's actions and Allowance didn't hold the calf's master responsible for its plight, then why, O, why, would he hold the calf responsible for its own defense against the matador in future fights?

I smiled mysteriously at the contradiction Nature presents to be belittled, for by no means, would a man in possession of reason average, give into the insane ways of destiny and the insane partiality it propagates with each wave of disturbance and with each wave of its own disturbance. In the mediums of reflection, O, why, does it not seek to search its own ugly face? And so she created me and so do I dwell in her face.

I smiled at myself at how Nature drives freedom and curbs it- all at the same time, such are the ways of destiny which playfully prank on the saddest thoughts and the saddest beings, and the hero turns the table on them and so does he frighten his own creator who in turn is fearful of the weak possibilities of her own destruction, and I wanted a hero as he to be my lover,

“And who be a hero better

Than the man who knows Humanity?

For the hero I desired so passionately

Ought to be an earthly human

And who be a human better

Than the human who knows Humanity?"

O, Peace, you said it right when you said I loved the man, but with every story of love, there accompanies a test as an accomplice, and I tested him. I tested him, not because I wanted my man to be sinewy, but because I desired him to be of unparalleled courage, and for the sake of the same, an arena had to be prepared for him to wrestle on, and no match, in my abode, is called complete without the presence of an equal and opposite match who would contest with unfailing pride before the two make a mockery of themselves. And my games are more fierce than the games sur la terre, in a pays called 'Spain', for the matches in my game stand equal to each other, in possession of equal strength, it's courage that separates the two and not surplus weapons and horses, it's the permission to be dragged into the arena from both the matches that's obtained first-hand and not force-drag.

With charms having marked their presence in the daily log, I would love to appreciate the gamer as much as the game for the opponent posed enough risks to be undertaken, those which couldn't be denied company by a mere lance. It required a whole lot more than bending around the corners or beating about the bush, and that more was what Time alone could tell.

The cry pathetic appealed to the man distant. The dusty waves in contrast with the peaceful environs propagated waves of suspense for dust was not supposed to know any existence in a place complete with quiet. He lent an ear to the sound.

Mooo...

The guesswork arrived in the likes of His Majesty for it seemed familiar to the cries of a bebe, and the additional info could only be obtained by a close watch as to what the mystery had to unveil, but the cries indeed were pitiable and as the land was clean-shaven, there seemed not a man in pursue which made the wary man a little casual in his ways.

While he was in search of the source of the sound that invoked mercy, he came across a bush blessed with a dense canopy with really sharp leaves which to no mortal, would feel soft to touch. It occurred that there existed no other bush on the land and thence, it failed not to catch the man's attention.

“Who hath grown a plant as cactus sur le chez?”

As no other creature walked by, I would assume he spoke it to none but himself. Aha! I freaked out in delight for I had exposed the dangers that lay therein on the facon to the innocuous creature who produced the sound that led to the commencement of the heroic man's journey. By a method as sly as this, I had, with sincere caution, pointed out to him the dangers of his pursuit and left it to him to decide whether he would traverse on the path.

But he seemed to take no actual note of it for he did advance with the most peaky brilliance, clad in the most magnificent of attires, with a gun that was loaded with a calfet that was aimed at any possible disruptor, with an attire that one could but stare at. I thought to myself the providence he shared, the fate he was fated to prove true, the hardships he would confront on the very path he currently was invading with his footsteps, a contrast to his life as a royal packed with tout de luxuries on the globe- the baubels the earth took for comforts and he took for distractions. To the understanding, it would sound the most ridiculous, but what man of courage would not but laud this heroic human? Oft do we let it fall off our memory, oft do we let it wane, but what heart would confine itself to its four chambers sans any desires of expanding its territory which, expand by lending and not by accepting for saucy are the ways of providence and saucy are the ways of action!

This man had invoked my pleasure at its apex and I would but mark his success on a page I had brought into existence for his sake. The first test had been cracked for he didn't seem to excunt, and by this he had thrown to me an open challenge and I accepted it with a conceited smile for there was more to it than the vulgar eye could see.

And so the man treaded on with eyes which shone with a brilliance one would swoon on, accompanied by a searching stiff upper lip and matted hair, as he dashed through the loamy soil that tripped him on and on as though it, even after all those encounters with the heat of his bare feet, was still in want of it.

And so the man conquered the valleys as he set himself on the path to conquer me.

The soils didn't anticipate his arrival and yet they welcomed him with a lush, green ravine that felicitated his further suspicion, and the dreadly poison seemed to be all so nauseous that the man couldn't help drooling over the possible instances and outcomes for never had he witnessed such formidable metamorphosis, and never had anyone of the world been subjected to such horrendous views for the standard of the test has more to do with its taker.

The man of understanding in him paused and pondered for none of the ordinary could pose such serious harm.

The man of courage in him was aloof from the understanding for he knew that though the creature might have been a facon to luring him into mighty dangers, yet wound indeed was inflicted upon his flesh and he had to rescue him either way for he could leave none in the hands of their own providence.

And with this he passed the second test I had lain.

He advanced with twin desires- the rescue operation and the destruction of the conceited foe. The desires possessed mass and soon grew to inflate his passion to hinge the enemy to the hook of his slashing armour. Letting out a roar that caused volcanoes to erupt, the fires to burn brighter and produce more carbon, the earth to tremble beneath his feet and heat up its own mantle, the moon to swing oblongly around the earth, the stars to fall and break into stardust, Time to pause for a millionth of a second, me to find him hard to resist, the man uprooted the tall tree I was seated on, and buried me underneath the roots of the tree, and I could but wallow at his mighty might.

Moooo.

One more time, and it would be all clear, clear as crystal to the man with a brilliance the Sun God would envy, and then the most brilliant of stas would acquire the most exalted of positions in his favour only until...

The calf and the man had their first mutual glance. The man smiled at the calf. Any ordinary smile would but have driven the courtesies away, but this spell cast a unique sense of fidelity, a sense that moved the calf just as much as it did the

man for it was a bond shred, a bond that would not be shattered as long as the earth retained its own being and as long as tides terrorized the dwellers of the sea.

The first glance had put it all to words. It had brought mercy into the notice of the man and a renewed love for humans into the heart of the calf as he learnt that not every man dwelt upon Cruelty and not every man would laugh at his destitution, and for a moment he believed that his former master was not a man, and at last he would be tamed in the company of someone he could consider man enough.

The first glance had displayed the affections between the two for the two.

The first glance had shown that two species of two different natures could bond.

The first glance had shown that Humanity wasn't dead but was snoring.

The first glance had shown the spectators that Mercy had yet not cursed Man.

The first glance had shown that not all calves would meet the same fate henceforth for the real man had arrived.

PEANS OF MERCY

The delighted Mercy reminded me of the debt I was already in full remembrance of but had tried so awfully hard to keep from her, and those attempts proved to be futile as she enslaved me.

Speak Cruelty did,

“ ‘Hunger!’, it dragged thee into the plentiful sea of thy wreath

Hath the bolts been hurl'd like a sword drawn upon a sheath?

Nay, shall hanger or strife confer upon Mercy

A note as that bestowed upon Elizabeth by Darcy.

In the hearth most ridicul'd of the dark, dark heavens,

Do scoops lie resting on the shallow lavens,

As they point passionately a note for one to be charmed by?

‘Driveth onward!’ cried the cry just as he passed by.

And innit shall the saddest gloom strive to contrive

For by no means would courage be satiated by a lone life

That brought to his age-old displeasure

The joy of a cool, warm breath

And shone the saddest thought the night of the saddest death.

And spoke it did, loud in the fanciest attire

‘Thy shall forsake thyself’ in the sweetest satire.

Hath he enjoy’d the joy that swung itself across the loom?

And it little profits to end fun on account of doom.

Thro’ the gal vale did he proceed to get himself enshrin’d

Doom on doom, pain on pain had his path lin’d

And he could make it all out so well but still he chose to turn deaf

For he had in his mind a goal to which he couldn’t faff.

The courage in his refuge wooed out from the bud

As it seated itself aloof and gazed at the trough

As quick as the ruminant brought out the cud.

Set the man on his path, his self so buff.

He swore an oath as the calf lay reclined

And keep he did it as he sang the doleful line
‘tis enduring toil landed me in despair
For it was a damage provoked, not I could repair.
I sat beside me and looked large upon the resurgent soul
Who knew no better half than the Mystery dole.
The slumberous sheet of qualm fixed me in my look
As I lacked nothing but a chance to peep it by a book.
Age told on me but was fated not to tell on him
And on went the somber man inimitable and grim.
Thee beckoned him to advance
As the vale lay bordered with palm.
The grievous wound thou had conferred upon him
Was thus a wound sans a balm
But he still did make it!
Whence a man of an ill-used race shall mourn and rave
But there upon he lay bedecked with
An armour of mercy unlike so much the naïve
For thy plot of noble note tho’ crafted with thy pen
Cleaved not to him for he was the lion in his den.
And thee bore misery wrought by the self

As he dint lose his homeward way
While thee tried thyself.
The pulchritude and effusiveness was fed no calm by day.
The ones like thee who hath every dreamful ease
Found no satiation with the toil on his grease.
Thee made him advance in a spectacular array
And yet the man countered them all, right in the lay way.
Thy clouds, O, Mystery seem to be lightly curl'd
But mine, nay, have thriftily swirl'd
Up and up, and down and down,
Into the creations of blooming-hues
And, nay, shall I lie careless of mankind in thy rues.
The bribe of sentiment could not but yield to him
As he lay having surpassed the best of courage in man
In keeping with his delightful humour, he shall become a name
And even the inscrutable decree of Providence
Shan't douse the flame.
This utmost bound of a man's lacking in fear
Shall, one day, cease mankind's most terrible conflict in rear-
It's conflict with its own self-

It's conflict with humanity
Which is no mere skirmish.
And the boundaries that I created
Meant not to be o'erstepp'd by man
Are aujoud'hui reduced to the scab of a minor ban.
I slap a tax on all those who seek to outrage
The winding clouds shall blaspheme all the men fed ill
And counter shall it act, counter to their will.
And down shall burst an epic lump of clay
What pleasure can thou have to express it another way?
And thy reprimanding thyself shall force itself upon our glance
Just as la terre shall, one day,
Reckon the void of the employment of the lance
These parcels of the dreadful future rest upon thee
As this hero hath hanken in all his celibacy
To hath the world, with vulgar eyes, open up and see
The Count of Me, The Count of Mercy.
Count on me to come like a ghost to trouble thee tonight
For thee hath kept me shut up in an urn devoid of light.
The echoes call, they are dewy with plenty in thee

Just as this clanging fight shall, eventually, lead on to me.

Now, to my pleasure, I would like to summon the Lord of Cruelty.

Thee art Lord of thyself.

And art thee in need of none of me to sustain thyself?

To be blessed with the ghosts of your kin

Wake up, thee! Wake up!

Thee art asleep in a half-dream

‘cause Peace’s coz has hath his share of cream

Save thy hands, thee, old fellow

Crested with the thick-twined vine of mellow

Jump up to relief as thee hath no surplus hands

Jump down for those flaming towns and these praying hands

As these shall get thee nowhere unless someone kind lends

equal mind any oaths you swear

O, thee, hath in it a yield to bear

‘Cause the man kind shall be none of your ware!

And he shall be one of my kin

Thence thee come to my refuge

With the same praying hands

Nay, you shall trust me for my mercy

Thee will have thy share too!

Without having hath a thing to do

Thy ships art sinking ‘cause thee hath lost thy charm

For the earth is meant to be abandoned

possessed with scornful harm

this gets thee to be swept out

shall invoke thy true soul.

my part, shall I dump it into a hole

A hole that shall know no closure

So the men passing by can witness to their eyes the despair

And learn for seven lives to not with thee pair

For earth shall know no peace sans harm to harm

And Peace shall know none of herself

Sans thy absence de her crust lukewarm

And none of the earth and none of the heavens

Shall cool down the raging personality of Rage

For the ghastliest of nightmares shall seek thy outrage

With thy pigeon-holed dwelling thy shall swiftly seek absence

Keep with an
For this,

By man
And
And I, thy hest,
But, on

As the crust shall rid itself of itching against your presence
 And none of the earth and none of the heavens
 Seek the pleasure of thy arrival
 But I shall mercifully allow thee, O, Rival!

Strode I slow and bump into the arms of Myself, the subtle, secure Myself, the Myself I have so distressedly longed to meet with the passage of salvation from the hedges of Time Himself, and yet I fight for eternity, a sublime eternity, a sublime form of possessiveness, of the Higher Order which could not but unfollow for by no means would I confine myself to the limits of a haste jurisdiction in a court that moves so fast, so unjustly fast, that I would tire myself with the very thought of it. What would thence come of an unmindful circumambulation? And yet the thought sparks, brighter and brighter, so I can't forgive myself for what's done as I sit yielding to the Higher Order.

Strode I slow and bump into my own forms as they rage an attack. O, My eyes! Art thee knowest the scene of unshaded logic, the pure calm logic that rests itself in the sea of divinity and with every passing hour seems to be churning and swirling with the wind revealing its off-white hues with so unkempt devotedness and yet so unfailing arrogance as the most magnificent of views appease the order of my eyes with a suspense, a Mystery I have resolved to solve? And yet, by no means, do I believe it's me, maybe 'cause, I have been taught not to believe in myself by myself for she always fails and leaves me deserted, somewhere in the lone, lone woods...

Strode I slow and arrive I did back to the doors of the lady who's above and beyond me, and thence by all laws of creation had the unambiguous right to enslave me for the smaller are enslaved by the greater, and the truly great persona love the ones they enslave and shower them with affections they would be endowing their offspring with.

And so I addressed the Countess of the Seven Islands, the Princess of Eternity, Mercy,

“ I break under the strain of my perceptions of You for You are limitless, and so is your glory and so are your perceptions. I amble down and I amble up but I get no shilling worth of an end for I see You in everything and I see everything as non-different from You. I strode and I strode in hope of seeing a land that had not been marked by the dust of Your feet but each of the segments respect Your presence for You are their existence, and I fear no more the prospect of being enslaved by

You for that would in some way keep me in touch with You and shan't you render Yourself upon me? Your victory has come to be attributed by the profundity of thought and I shall no longer shroud this affair. I'm blotted out and I'm 'fugly'."

'I'm half sick to slay an already sick heap

I'm half sick to stay and my promise keep.

"Whence comest thou?" the sad God bluffs out.

I keep the water cool ere it flows in bout.

Belting out a tune that perchance got aroused.

An ancient scar on my chest gave way to fight

As the kindest of men stood upright and bright.

Full of lustihood and having been summoned by rage

And having been bedecked by the bucolic and religious and epical page

As the fiend and the faline lay in half-disdain

What sculptor would carve the beauty of thy grace?

And what painter would paint the colours of thy scornful laugh?

As the terror raged ahead in fantasy

The keenest fantasy the blithesome had of yet known

'Foust for it' called the Goddess to the aweary men

'Get it' cried the Goddess to the wary man.

Her words got to him through the soft precision of phrase-

A four-limbed hurricane